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CHAPTER

ATA 23

COMMUNICATIONS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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COMMUNICATIONS

Introduction

The aircraft communication systems have different sub-systems to do the functions that follow:

- Communications in the Very High Frequency (VHF) band
- Communications in the High Frequency (HF) band
- ARINC Communication Addressing and Reporting System (ACARS)
- Noise and vibration control
- Announcements to the passengers by the pilots or flight attendants
- Audio management
- Cockpit Voice Recording (CVR)
- Radio management.

General Description

Figure 23- 1 Communication System Block Diagram

The communication system has the sub-systems that follow:

- VHF Communication
- HF Communication
- ARINC Communication Addressing and Reporting System (ACARS)
- Passenger Address and Communication Interphone System (PACIS)
- Passenger Entertainment System (PES)
- Active Noise and Vibration Control System (ANVC)
- Audio Integration System (AIS)
- Static Discharge System
- Cockpit Voice Recording System

- Cabin Video Monitoring and Surveillance System (CVMSS)
- Radio Management System (RMS).

VHF Communication

The VHF communication system gives line-of-sight, two way voice and data communication. There are two VHF communication systems (VHF 1 and VHF 2) installed on the aircraft. The VHF 3 system is optional. Each system operates in the 118.00 to 137.00 MHz frequency spectrum. The VHF 1 and VHF 2 system provide the channel spacing of 25 KHz. The VHF 3 system provides the channel spacing of 8.33 KHz and 25 KHz. The VHF communication system uses the Amplitude Modulation (AM) for voice transmission.

HF Communication

The HF communication system gives long-range, two way single sideband AM, voice and data communication. The HF communication system transmits and receives high frequency communication signals in the 2.0 to 29.9999 MHz frequency radio spectrum with 100 Hz channel spacing. It is controlled by a control unit and the Audio and Radio Control Display Units (ARCDUs). The control unit controls most HF communication functions and the ARCDUs only control the HF audio and transmitter selection.

ARINC Communication Addressing and Reporting System (ACARS)

ACARS provides a two-way digital data link between the aircraft and the operator ground stations. The unilink UL-70X communication system includes a Communication Management Unit (CMU) with an externally mounted configuration module, Multifunction Crew Display Unit (MCDU) as part of the Flight Management System (FMS), VHF radio, a printer and a data transfer unit.

Passenger Address and Communication Interphone System (PACIS)

The PACIS gives the functions that follow:

- Passenger Address (PA) announcements to the passengers
- Interphone communications between the flight attendants and pilots for private conversations
- Flight attendant calls from switches in the Passenger Service Units (PSUs) and lavatory.

Passenger Entertainment System (PES)

The PES supplies different programmed voice PA and music to the PACIS and AIS.

Active Noise and Vibration Control System (ANVC)

The ANVC decreases the noise level in the passenger cabin caused by the aircraft propellers. The system supplies secondary noise and vibrations in the passenger compartment. These secondary noise and vibrations cancel the initial noise and vibrations. The initial and secondary noise and vibrations mix to give lower cabin noise and vibrations.

Audio Integration System (AIS)

The AIS controls the audio functions that follow:

- Monitor radio communications receivers and make transmissions
- Monitor the radio navigation receivers
- Interphone communications between pilot, copilot, observer, and ground crew members.

The AIS uses the ARCDUs and observer's Audio Control Panel (ACP) to control the Remote Control Audio Unit (RCAU) that interfaces with the systems that follow:

- Communication transceivers
- Navigation receivers
- Pilot, copilot, observer's, and ground crew jack boxes
- Passenger Address Amplifier (PAA)
- Flight compartment speakers
- Solid State Cockpit Voice Recorder (SSCVR).

Static Discharge System

The static discharging system consists of dischargers which dissipate electrostatic charges from the aircraft structure to the surrounding air, to minimize the radio interference caused by these charges.

Static discharger consists of a resistive coated fiberglass rod with a pellet of composite-carbon based material at the tip. The pellet is fitted into an aluminum alloy sleeve, which in turn is fitted onto the discharger rod by means of conductive epoxy cement. A resistant sheathing is heat-shrunk over the entire length of the discharger rod and tip assembly, leaving approximately 1/8 in. (3.2 mm) carbon tip material exposed.

Cockpit Voice Recording System

The SSCVR system records the last two hours of flight compartment audio and clock data.

It records audio in a digital format in crash-survivable Non-Volatile Memory (NVM).

Cabin Video Monitoring and Surveillance System (CVMSS)

The CVMSS lets the flight crew see all of the area that is outside of the flight crew compartment. The system lets the flight crew identify persons near the flight compartment entry door and to detect suspicious behavior or possible threats. The CVMSS uses the Control Display Unit (CDU) of the FMS to display the video images from two cameras in the cabin.

The camera No. 1 is located in the ceiling panel immediately above the flight crew compartment entry door. The camera No. 2 is located in the ceiling panel, approximately 10 ft. (3 m) aft of the flight crew compartment entry door. The camera No. 1 looks down and the camera No. 2 looks down and forward.

On aircraft with Mod Sum 4-458529 incorporated, the cabin video monitoring and surveillance system is removed.

Radio Management System

The radio management system is used to control and monitor the Radio Communication (RCOM) and Radio Navigation (RNAV) systems.

The radio management system has two ARCDU 1 and ARCDU 2 that have the functions that follow:

- Tunes the radio communication and radio navigation systems and controls their operational modes
- Shows all data that is related to the operation of the radio communication and navigation systems
- Monitors the operation of the RCOM and RNAV systems to show malfunctions

- Controls the operation of the PACIS (Refer to ATA CHAPTER 23-31-00)
- Controls the operation of the AIS (Refer to ATA CHAPTER 23-50-00)
- Controls the operation of the Central Diagnostics System (CDS) (Refer to ATA Chapter 45-00-00).

The ARCDU front panel has the controls and indications to let the pilots control the audio and radio systems installed.

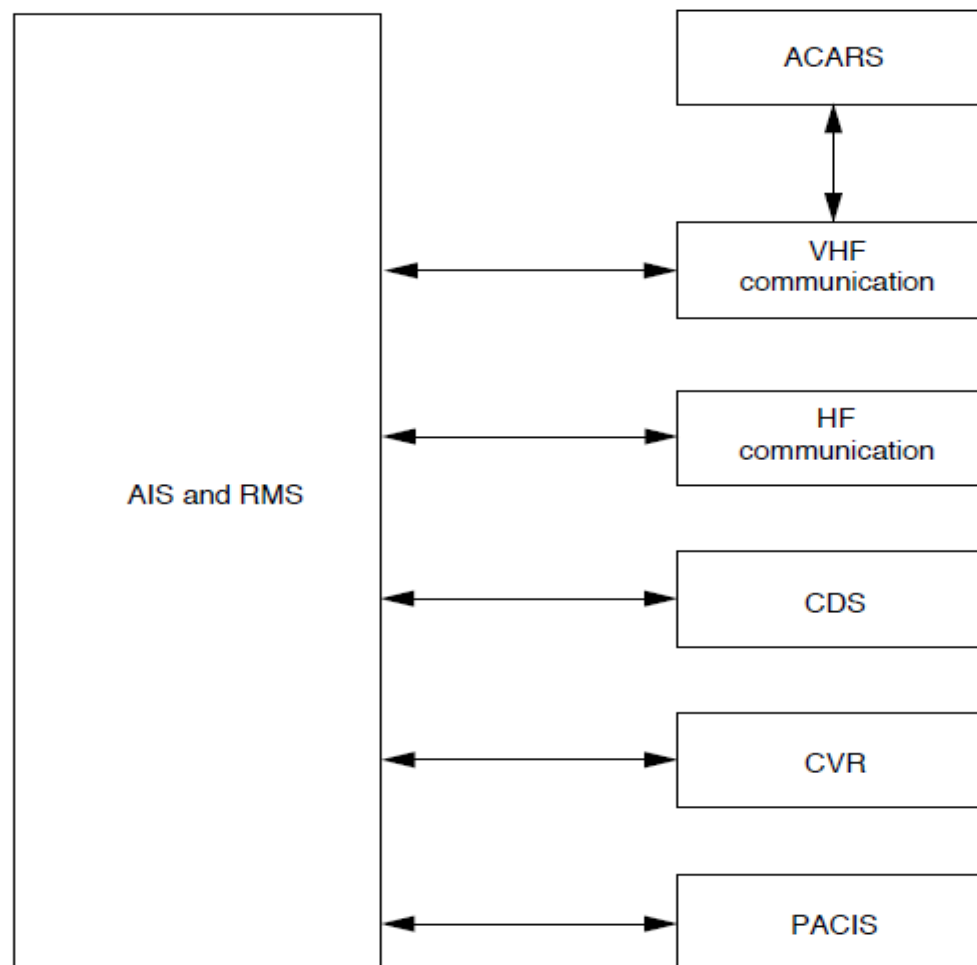


FIGURE 23- 1 COMMUNICATION SYSTEM BLOCK DIAGRAM

VHF COMMUNICATION

Introduction

The Very High Frequency (VHF) communication system gives line-of-sight, two way voice and data communication.

General Description

Figure 23- 2 VHF1 Communication System Block Diagram

Figure 23- 3 VHF2 Communication System Block Diagram

Figure 23- 4 VHF3 Communication System Block Diagram

There are two VHF communication systems (VHF1 and VHF2) installed on the aircraft. The VHF3 system is optional. Each system operates in the 118.00 to 137.00 MHz frequency spectrum. The VHF1 and VHF2 system provide the channel spacing of 25 KHz. The VHF3 system provides the channel spacing of 8.33 KHz and 25 KHz. The VHF communication system uses the Amplitude Modulation (AM) for voice transmission with the ATC systems. The VHF communication system is also used for a/c to a/c communication, for communication relay or emergency purposes.

The VHF communication system has the components that follow:

- Transceiver (23-11-01)
- Unit, Controls and Displays (23-11-21)
- Antenna (23-11-26).

Each VHF transceiver is interfaced with the equipment that follow:

- Standby controller for VHF1 transceiver
- Audio and Radio Control Display Units (ARCDU)
- Flight Management System (FMS) Multi-Functional Control Display Units (MCDUs).

NOTE

The MCDUs cannot control VHF3.

Detailed Description

The VHF transceiver operates in the modes that follow:

- Power-On Self-Test (POST)
- Operational
- Maintenance.

POST: The POST mode automatically starts and continues for six seconds after the power-up. At this time the VHF transceivers do the self-tests that follow:

- Hardware
- ARINC 429 inputs
- Discrete inputs
- Voltage Standing Wave Ratio (VSWR)
- Transmitter
- Receiver
- Indication of test in progress and successful completion.

If the self-test senses a malfunction, it is sent to the Central Diagnostic System (CDS) for later analysis during maintenance functions.

Maintenance mode: The Built in Test Equipment (BITE) uses the CDS to give the condition of the component. It saves faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

The VHF transceiver has a BITE mode to monitor the condition of the component as follows:

- POST
- Continuous monitoring.

POST: The POST checks the condition of the component at power-up or after a long power interruption, more than 200 milliseconds.

Continuous monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a NVM for later troubleshooting using the CDS.

The front panel has a TEST push-button switch to start a POST and six Light Emitting Diodes (LEDs). The FAIL LRU, FAIL IN, or PASS indications come on with an initiated tests.

The front panel shows the conditions that follow:

Figure 23- 5 VHF Communication System, Transceiver

LED	COLOR	CAUSE
FAIL LRU	Red	Internal malfunction
FAIL IN	Red	ARINC 429 or discrete input malfunction
PASS	Green	No malfunction sensed
VSWR	Red	VSWR is more than 2:1
TX	Green	Transmit mode
RX	Green	Receive mode

The ARCDU front panel has the controls and indications to let the pilot control the audio and radio systems installed.

The VHF transceivers are controlled by the ARCDU 1 and ARCDU 2.

They control the functions that follow:

- Tunes the VHF communication system and controls their operational modes
- Shows all data that is related to the operation of the VHF communication systems
- Monitors the operation of the VHF communication system to show malfunctions
- Controls the operation of the Audio Integration System (AIS).

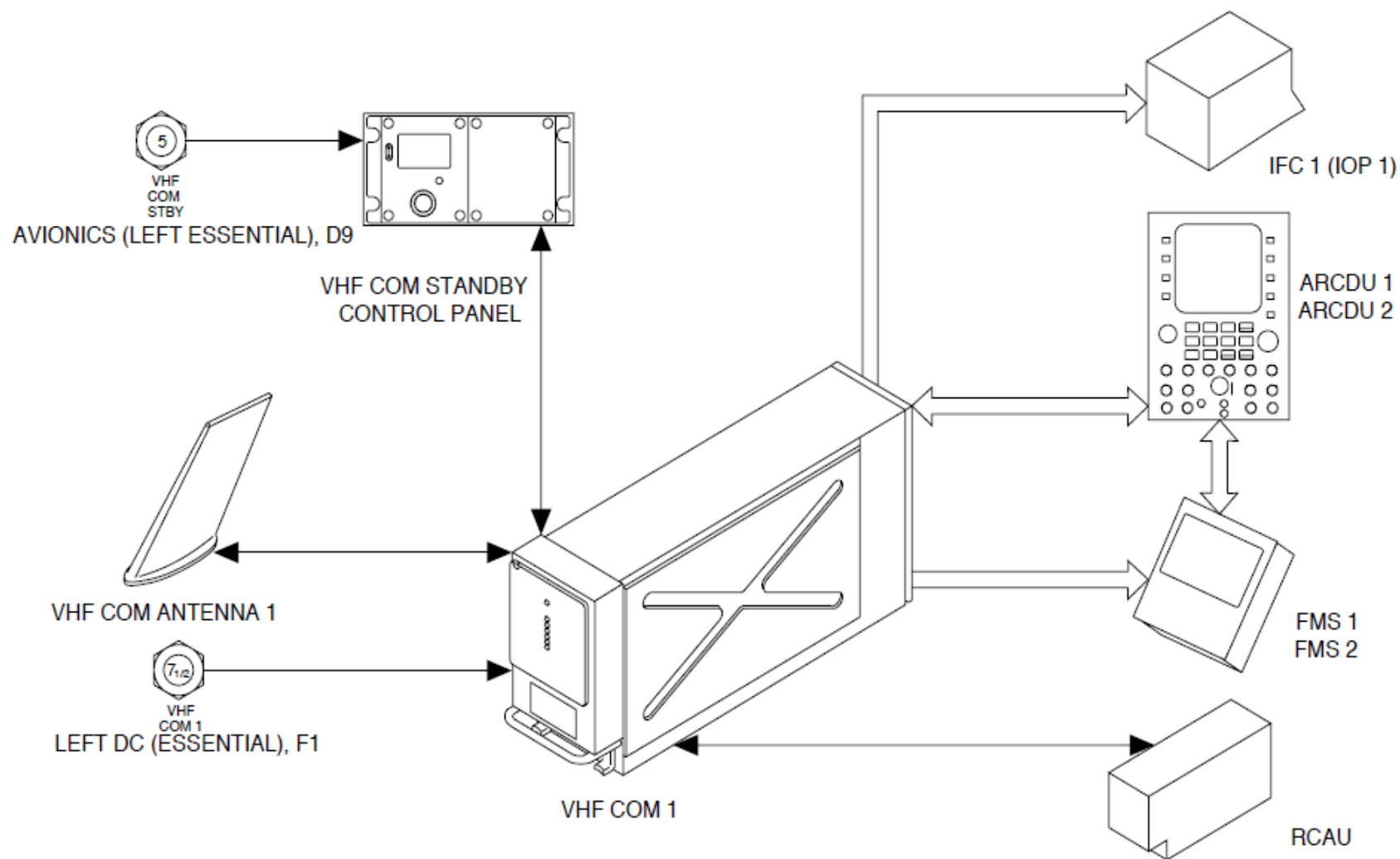


FIGURE 23- 2 VHF1 COMMUNICATION SYSTEM BLOCK DIAGRAM

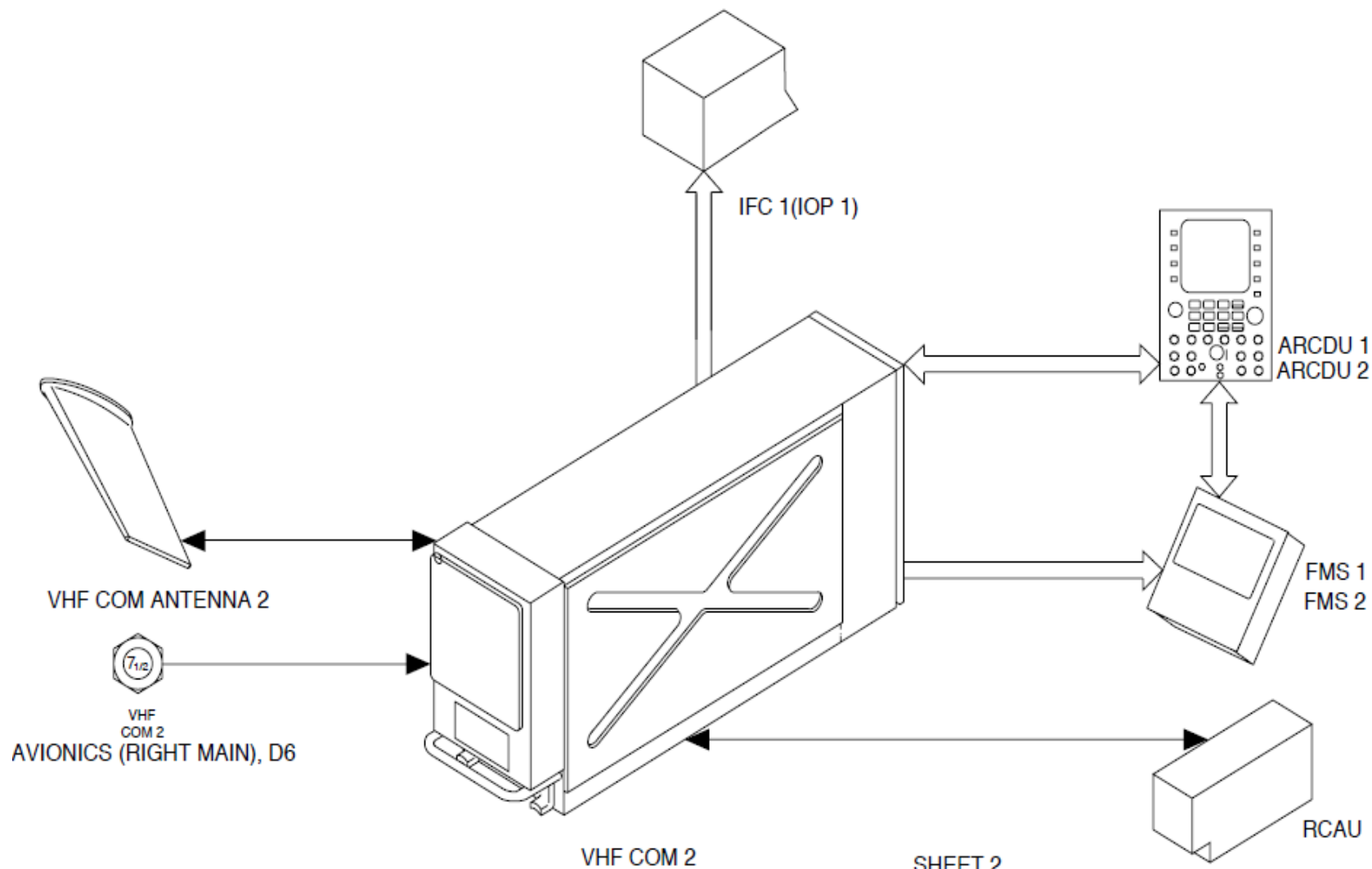


FIGURE 23- 3 VHF2 COMMUNICATION SYSTEM BLOCK DIAGRAM

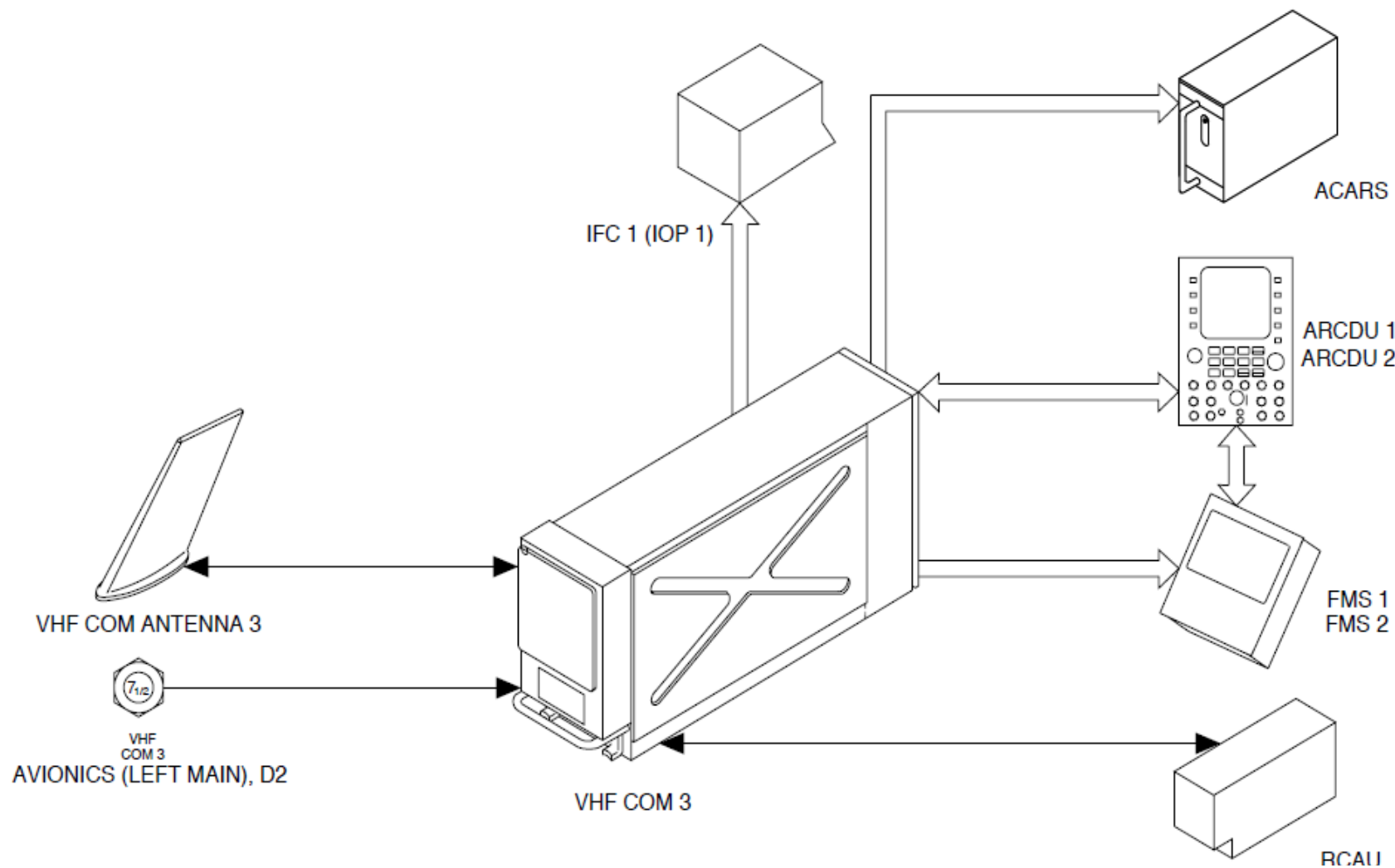
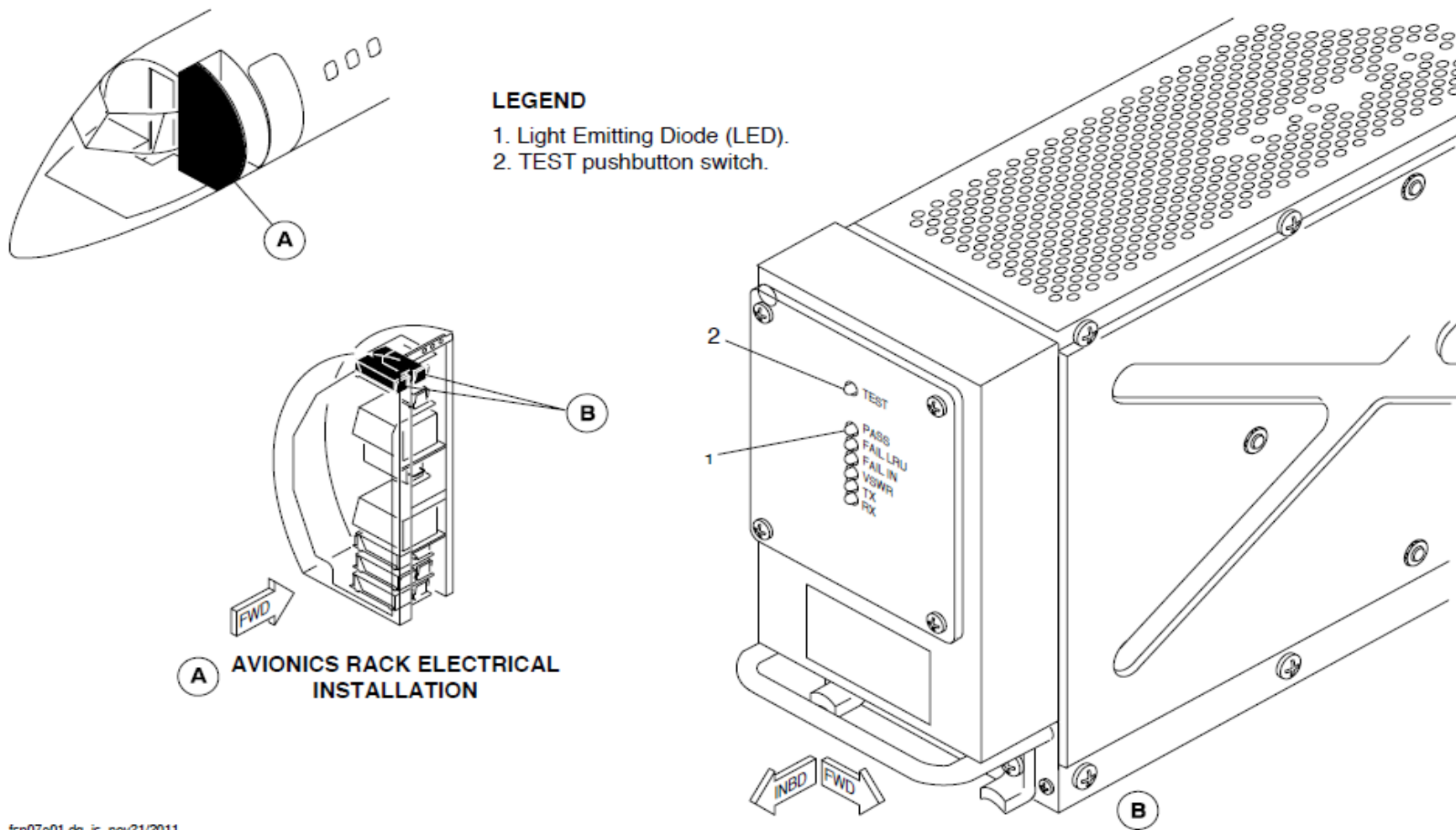


FIGURE 23- 4 VHF3 COMMUNICATION SYSTEM BLOCK DIAGRAM



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FIGURE 23- 5 VHF COMMUNICATION SYSTEM, TRANSCEIVER

Figure 23- 6 VHF COM, ARCDU

The ARCDU 1 and ARCDU 2 have a rotary selector switch at the lower left of the ARCDU display that shows the operational configuration and also have four positions that follow:

SELECTION	DESCRIPTION
OFF	The ARCDU is not powered. The related FMS controls and tunes as a backup
ON	The ARCDU controls and tunes its related radio system
BOTH	The ARCDU controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

Figure 23- 7 ARCDU VHF Communication Frequency Control

A minimum of two steps are required to change the active frequency. The side key adjacent to the VHF communication display area is pushed to highlight the preset frequency. It is pushed again to make the preset frequency active. The frequency that was active now becomes the new preset frequency.

When the side key adjacent to the VHF communication display area is pushed, the preset frequency in the display area highlights. The TUNE double rotary knobs located at the lower right side of the ARCDU are turned to change the preset frequency. The side key is pushed again to set the new active frequency. The frequency that was active now becomes the new preset frequency.

NOTE

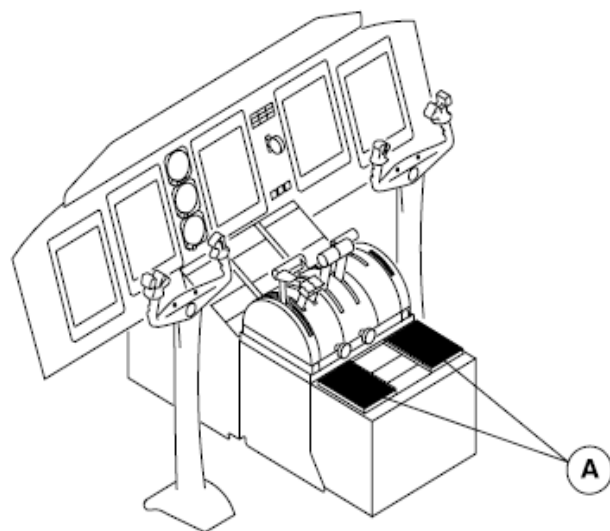
For 8.33 KHz channel spacing requirement, an optional transceiver that can tune in 8.33 and 25 KHz increments is installed.

The CH key is pushed to set the channel memory mode from eight preset frequencies using a channel memory selection. A green bar in the CH key comes on and a channel memory annunciator comes into view in green characters below the active frequency indication. In addition, the channel memory number is shown adjacent to the preset frequency.

The side key adjacent to the VHF communication display area is pushed to change the preset frequency. When the TUNE double rotary knobs are turned, the channel memory number shown adjacent to the preset frequency changes. If a channel memory number is not shown in the display area, it starts with CH1. The programmed frequency for the channel memory is shown in place of the preset value as black characters on a cyan. The double rotary knobs are turned to set the different preset channel memories. The rotary knob has a roll over capability. The VHF communication side key is pushed again to validate the channel memory selection. Two successive pushes of the communication side key causes the preset frequency to replace the active frequency. The active frequency becomes the preset frequency. The channel memory number shows in green characters below the active frequency. This shows that the active frequency is associated with a channel memory number. When the CH key is pushed again, the channel memory mode changes to the normal frequency selection. All channel memory annunciators are removed and the green light in the CH key goes out.

If the selections that follow are not made in less than five seconds, the previous preset frequency is restored:

- Turn inner or outer TUNE double rotary knob
- Push the same side key again.



LEGEND

1. Microphone/Interphone selector switch.
2. Key.
3. Rotary switch.
4. Preset frequency.
5. Level adjust bar graph.
6. Side key.
7. Label.
8. Active frequency.
9. Display area.
10. Tuning knob.
11. Normal/Emergency toggle switch.
12. Hot microphone switch.
13. BOOM/MASK toggle switch.

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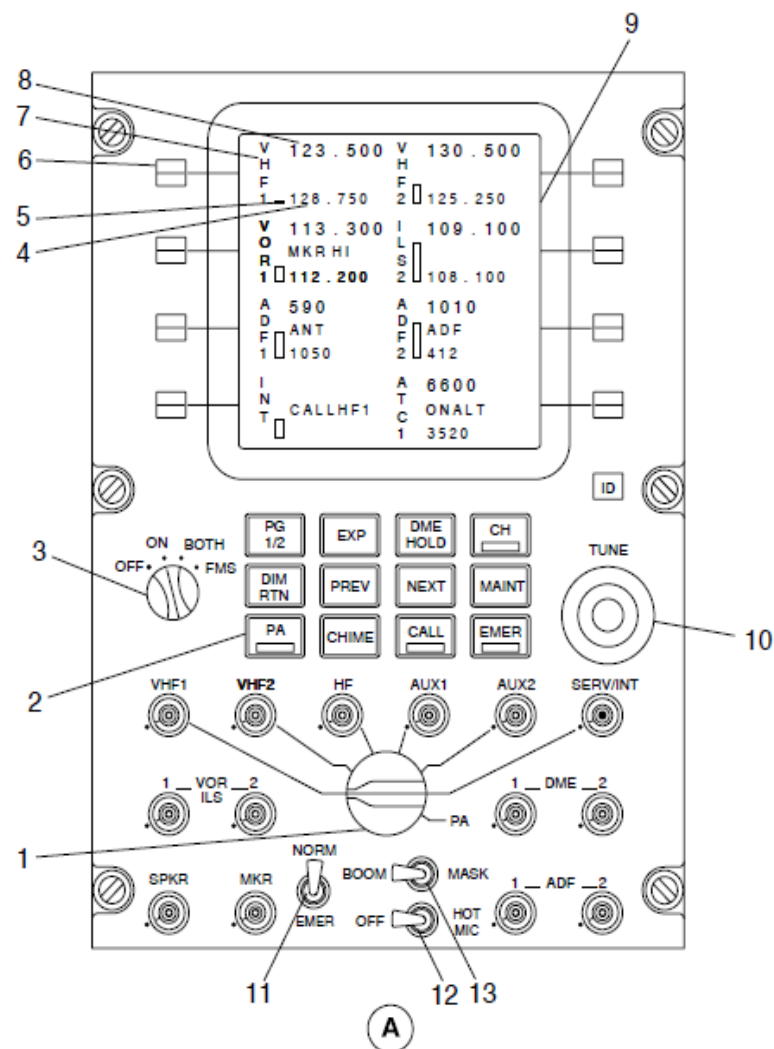
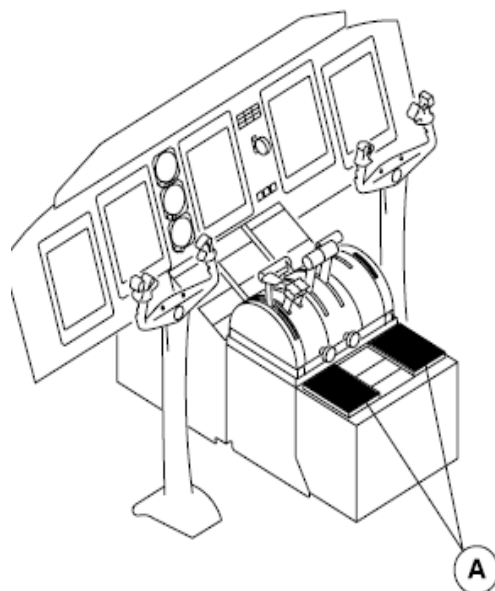
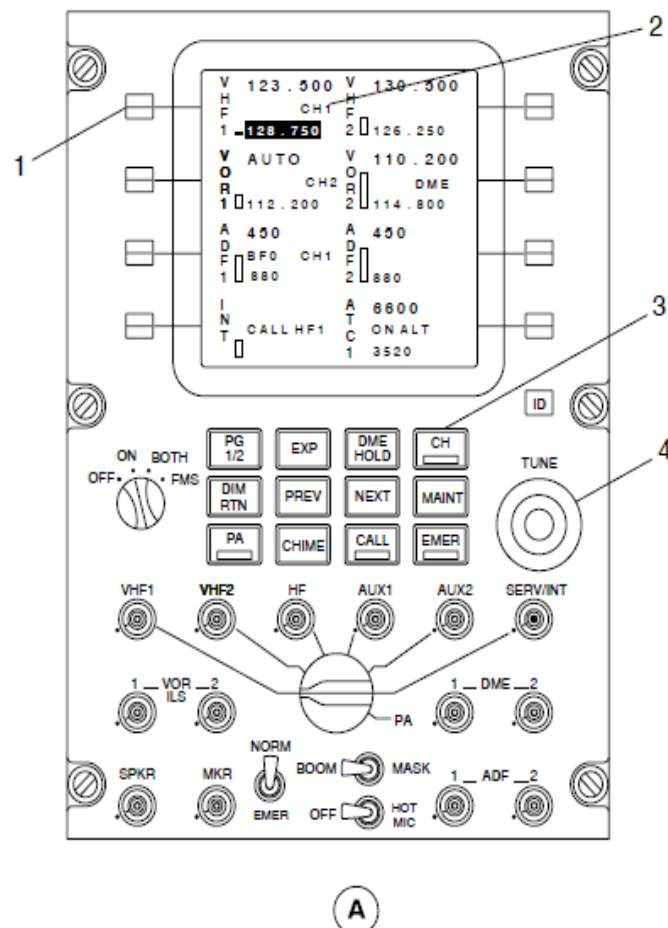


FIGURE 23- 6 VHF COM, ARCDU



LEGEND

1. Side key.
2. Channel memory annunciation.
3. Channel key.
4. Tuning knob.



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FIGURE 23- 7 ARCDU VHF COMMUNICATION FREQUENCY CONTROL

Figure 23- 8 ARCDU VHF Communication Particular Page Access

The communication particular page comes into view when the communication side key is pushed followed by the EXP key on the keyboard.

Figure 23- 9 ARCDU VHF Communication Particular Page

The particular page can be used to change the VHF communication receiver (VHF COM 1, VHF COM 2) modes and functions that follow:

- Channel programming
- Test.

Channel programming: Eight preset channels are available on two pages for VHF communication tuning. The preset and active frequencies can be tuned from the two channel pages.

Figure 23- 10 ARCDU VHF Communication Channels Access Page

The side key adjacent to the CHANNELS legend is pushed to access the channel programming mode.

Figure 23- 11 ARCDU VHF Communication Channels Access Page

Channel presets are labeled as CH1 through CH8 in white characters. Each channel window shows the current preset channel number in a cyan font below its preset labels.

Figure 23- 12 ARCDU VHF Communication Second Channels Page

The second channel memory page comes into view when the side key adjacent to the NEXT legend is pushed or the NEXT key is pushed. The side key adjacent to the PREVIOUS legend is pushed or the PREV key is pushed to

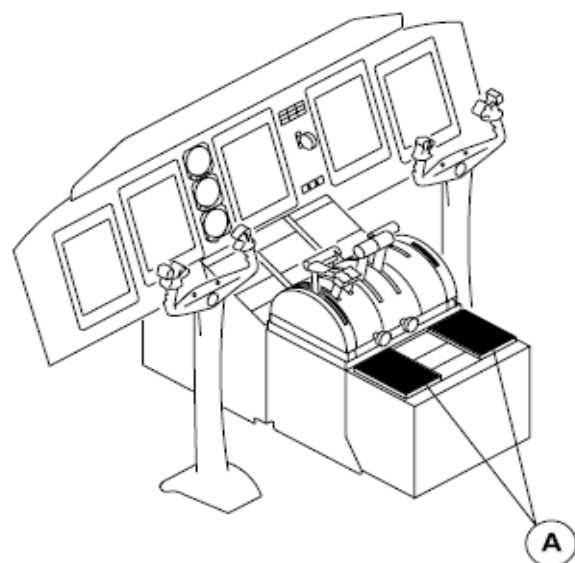
see the first channel memory page. The PG 1/2 key is pushed to show the first main page.

Figure 23- 13 ARCDU VHF Communication Preset Channel Programming

When the side key related to the channel is pushed, the current frequency value changes to black characters on a cyan background. The two TUNE concentric knobs located at the lower right side of the ARCDU are turned to change the frequency. The side key is pushed again to validate the new frequency. The channel window shows the new frequency in cyan characters on the black display. If the new channel selection is associated to a preset radio frequency, the two values change together. If the changed channel is associated to an active radio frequency, the label will flash for five seconds at a 1 hertz rate. This shows that it is not possible to directly change an active frequency.

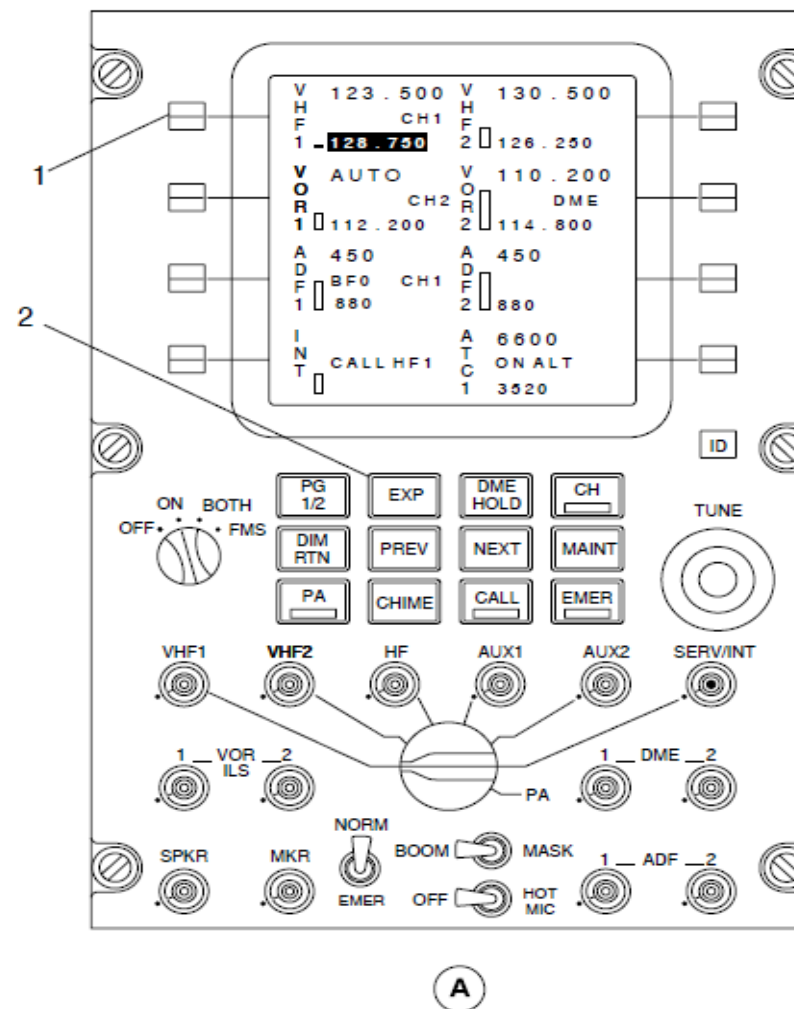
If the frequency selections are not made within five seconds, the previous preset frequency restores.

The side key adjacent to the TEST legend is pushed to start the VHF communication test mode. The test mode starts a one-second test of the VHF communication receiver. The audio module automatic squelch circuit is disabled to let the VHF communication receiver noise to be heard. This makes sure that the receiver is operational. During the test mode, the usually white TEST legend on the ARCDU display change to black on a green background. The legend changes to white after the test sequence is completed.



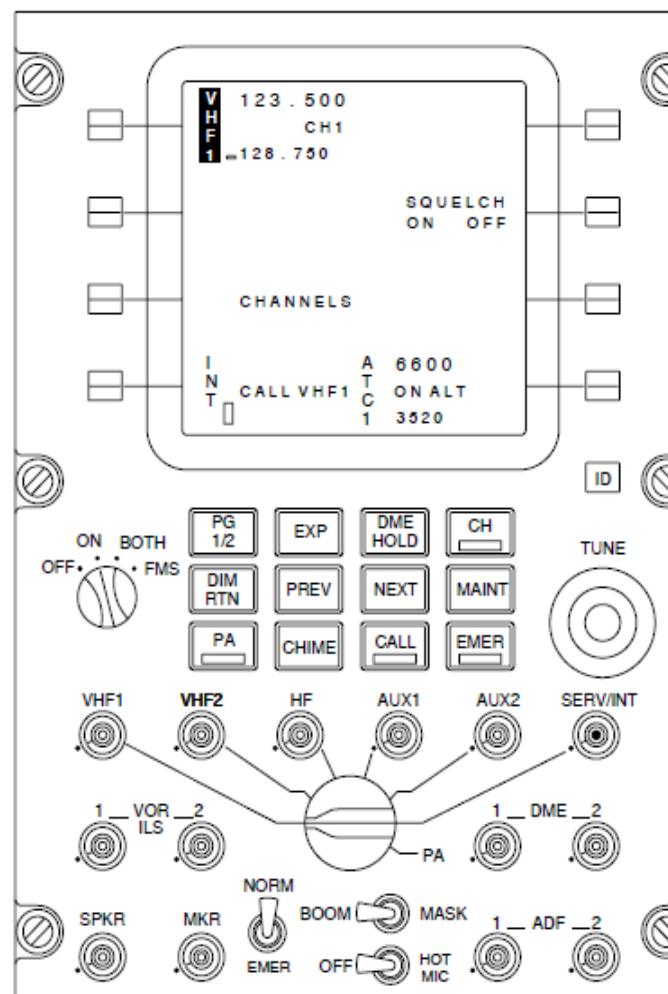
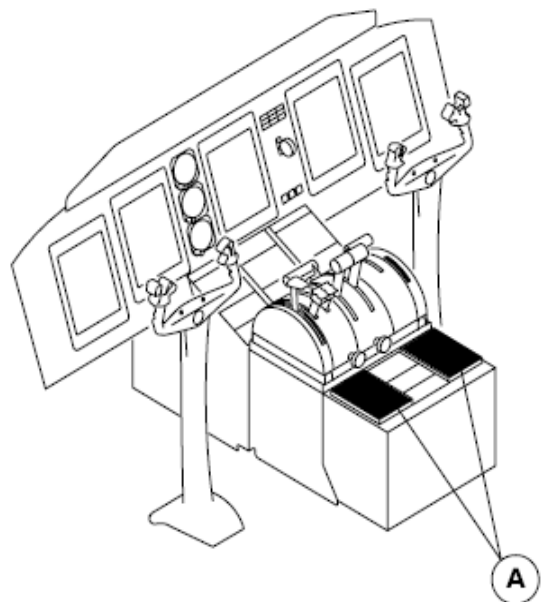
LEGEND

1. Side key.
2. Expand key.



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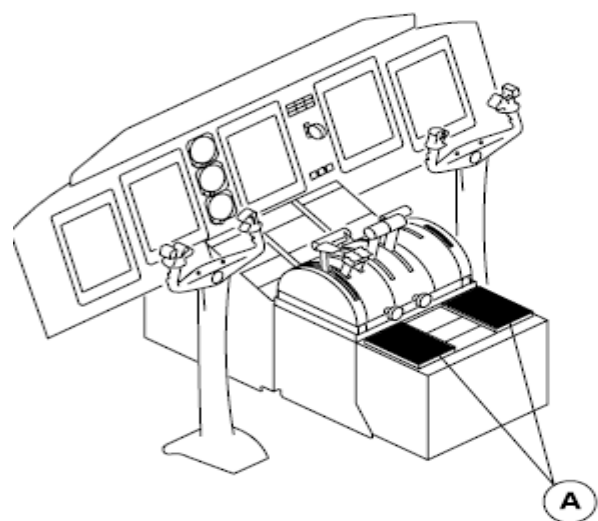
FIGURE 23- 8 ARCDU VHF COMMUNICATION PARTICULAR PAGE ACCESS



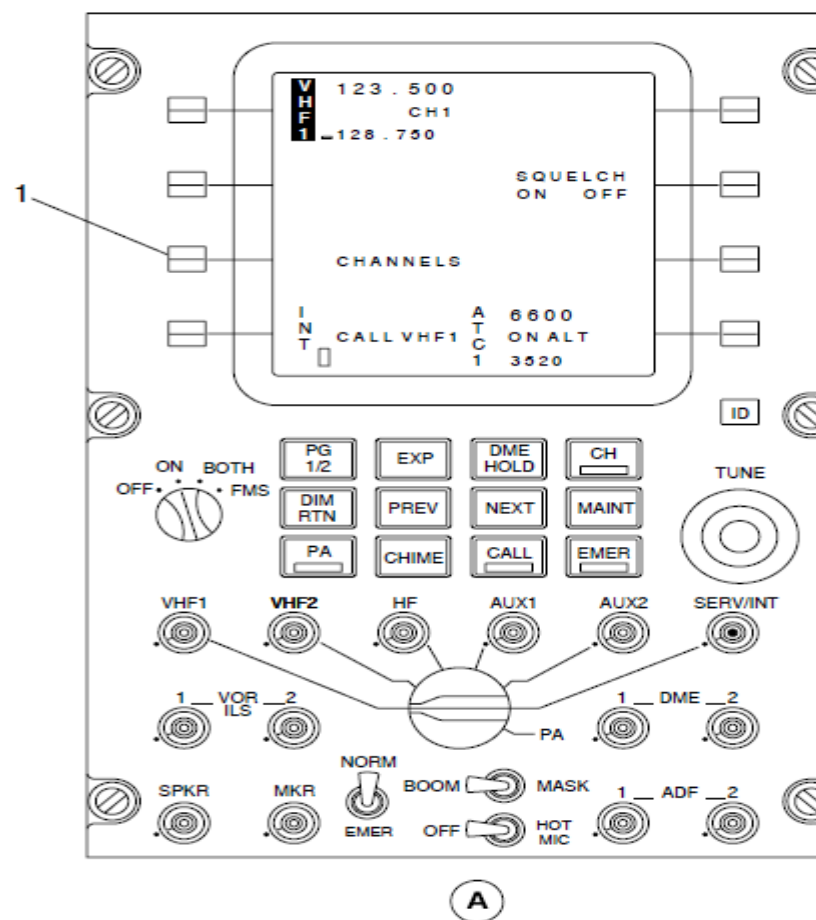
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FIGURE 23- 9 ARCDU VHF COMMUNICATION PARTICULAR PAGE

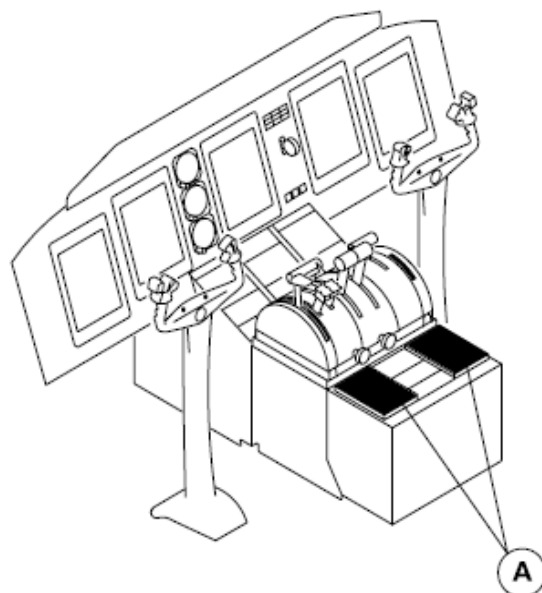


LEGEND
1. Side key.



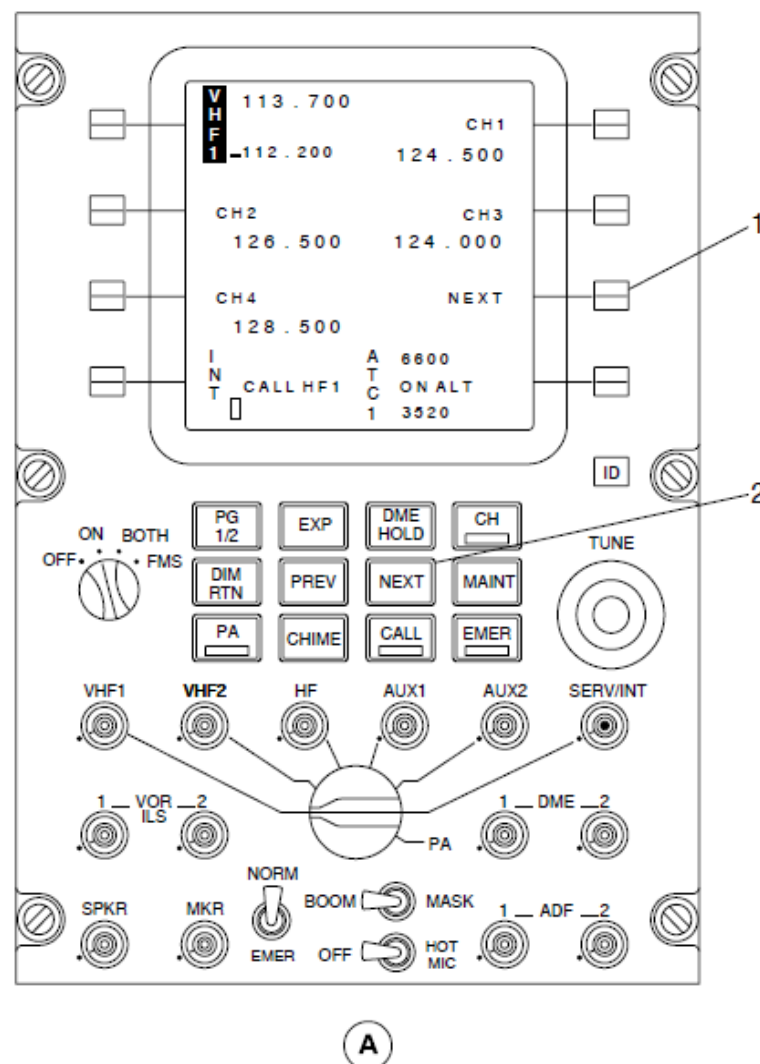
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FIGURE 23- 10 ARCDU VHF COMMUNICATION CHANNELS ACCESS PAGE



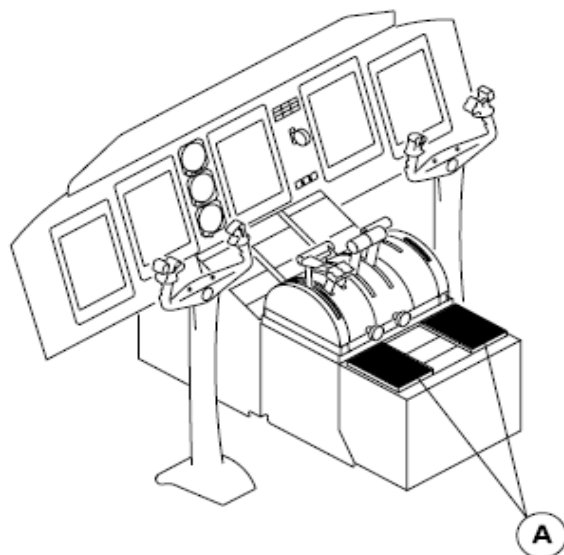
LEGEND

- 1. Side key.
- 2. Next page key.



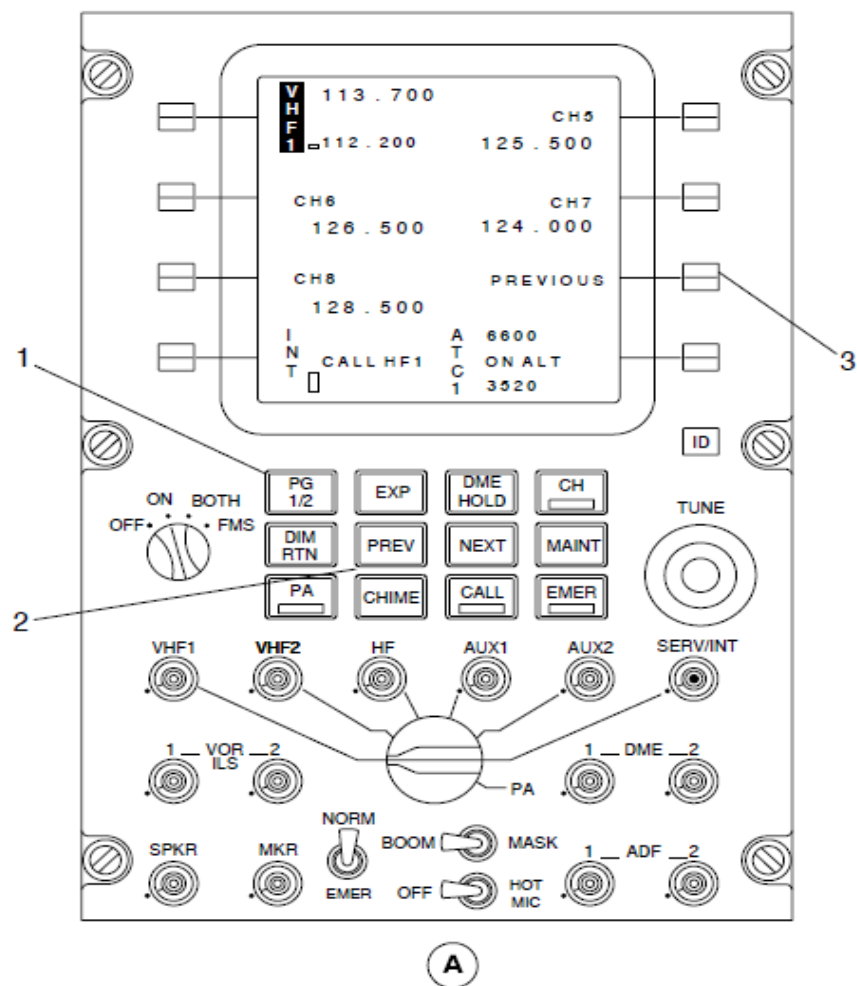
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FIGURE 23- 11 ARCDU VHF COMMUNICATION CHANNELS ACCESS PAGE



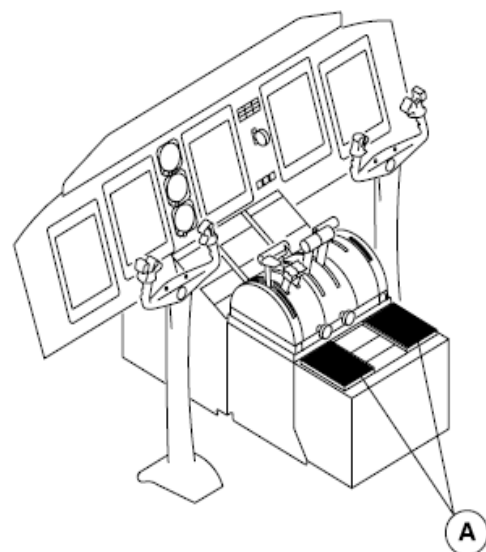
LEGEND

1. Page 1/2 key.
2. Previous page key.
3. Side key.



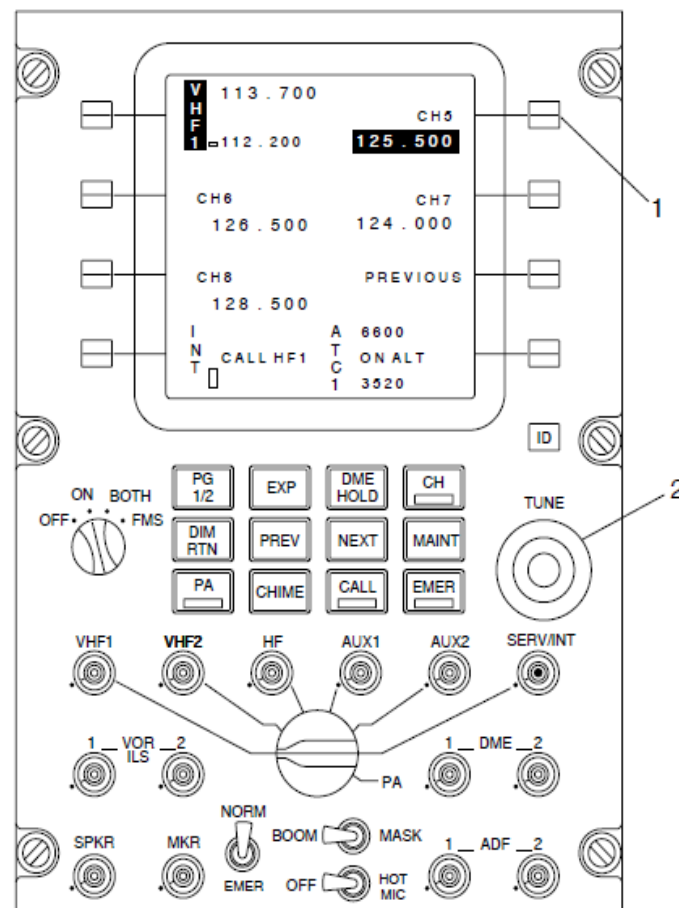
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FIGURE 23- 12 ARCDU VHF COMMUNICATION SECOND CHANNELS PAGE



LEGEND

1. Side key.
2. Tuning knob.



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FIGURE 23- 13 ARCDU VHF COMMUNICATION PRESET CHANNEL PROGRAMMING

Figure 23- 14 ARCDU VHF Communication Audio Control

Audio Control: The VHF1 or VHF2 potentiometer push-button switches control the audio parameters that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to turn the specific audio signal to the headphone on or off. The VHF1, or VHF2 push-button switch is pushed on the ARCDU to make its audio available to the flight crew member. When the SPKR push-button potentiometer switch is pushed, the VHF communication audio is routed to the two flight compartment speakers. A loudspeaker graphic is shown in the INT window.

The communication receiver display area shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is set to OFF, the bar graph is shown in white. The bar graph changes to green when the audio is set to ON.

The ARCDUs have the radio and audio controls for the VHF transceivers that follow:

- Rotary microphone (transmitter) selector
- BOOM/MASK toggle switch
- NORM/EMER toggle switch.

Rotary Microphone (transmitter) Selector: The rotary microphone selector switch is turned to make a radio transmitter selection that follows:

ROTARY MICROPHONE (TRANSMITTER) SELECTION	SYSTEM
VHF1	VHF communications 1
VHF2	VHF communications 2
AUX1	VHF communications 3 or UHF communications

The rotary microphone selector switch is set to a radio position followed by a Push to Talk (PTT) selection to make transmission.

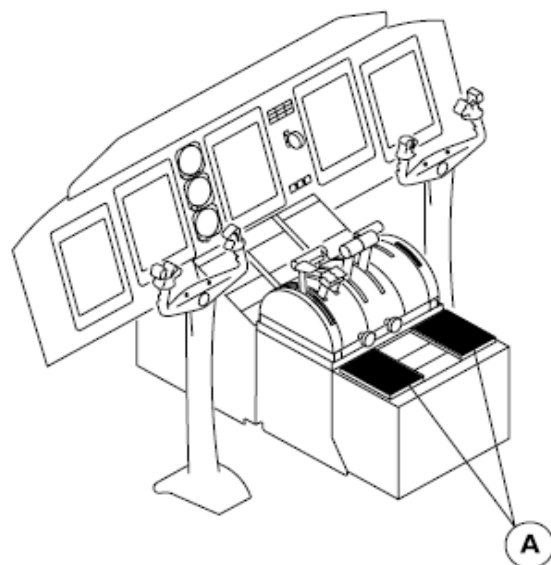
BOOM/MASK toggle switch: When the BOOM/MASK toggle switch is set to BOOM, the headset boom microphone supplies audio to the VHF transceiver through the Remote Control Audio Unit (RCAU).

When the BOOM/MASK toggle switch is set to MASK, the oxygen mask microphone supplies audio to the VHF transceiver through the RCAU.

NORM/EMER toggle switch: When the NORM/EMER (normal/emergency) toggle switch is set to the NORM position, it causes the RCAU to function in the normal operational mode. The audio outputs are independently controlled using the potentiometer push-button switches.

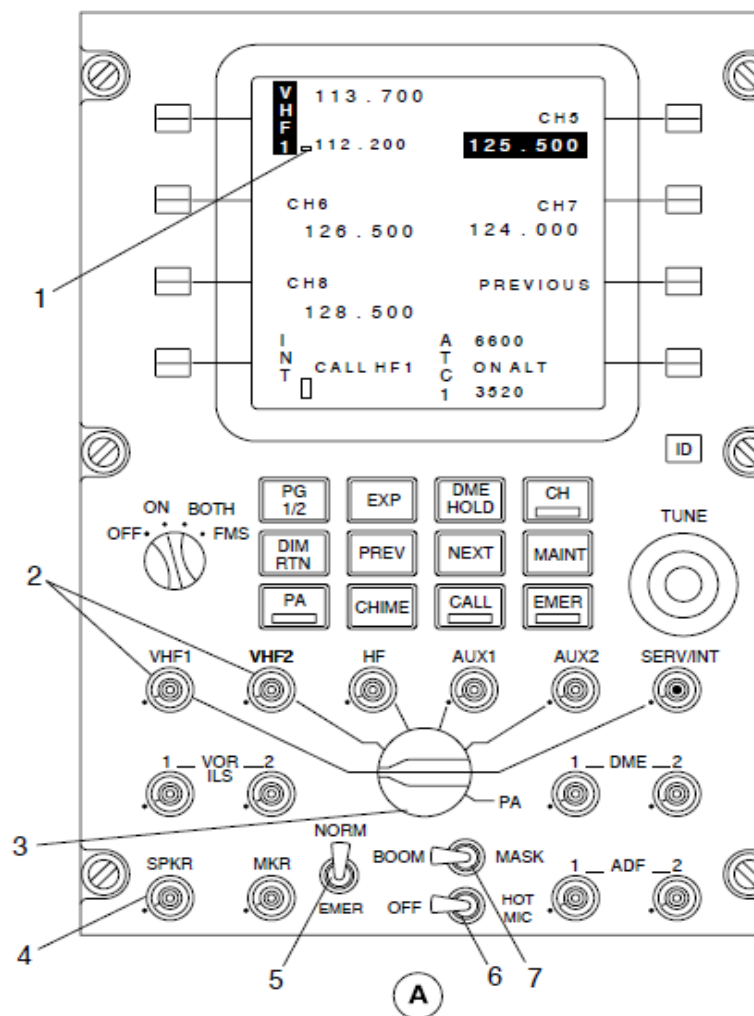
The NORM/EMER toggle switch selection to the EMER position causes the RCAU to function in the emergency mode. Electrical power is removed from the related circuit card in the RCAU to cause a relay to connect the headphone and speakers directly to the related VHF communication transceiver. The audio is set to predetermined level and cannot be controlled by the potentiometer push-button switches. Also, the microphone key line is directly connected to the related VHF communication transceiver.

Failure of the VHF communication system or removal of 28 V dc to the system causes the related radio window of the ARCDU to show a FAIL message in red.



LEGEND

1. Level adjust bar graph.
2. VHF Potentiometer/Pushbutton switch.
3. Microphone/Interphone selector switch.
4. Speaker Potentiometer/Pushbutton switch.
5. Normal/Emergency toggle.
6. Hot microphone switch.
7. BOOM/MASK toggle switch.



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FIGURE 23- 14 ARCDU VHF COMMUNICATION AUDIO CONTROL

Figure 23- 15 VHF1 Communication Standby Control Unit

A standby control panel is used to control the VHF1 frequency selections when the two ARCDUs malfunction.

The standby controller has the controls and indications that follow:

- OFF/ON/TEST rotary switch
- Frequency rotary selector
- Active frequency indication
- Preset frequency indication
- Frequency transfer push-button switch.

OFF/ON/TEST rotary switch: The ON position energizes the standby control panel and control the VHF1 frequency selections. A STBY CTL message in green letters is shown in the active frequency area of the ARCDU VHF1 display area. The TEST position starts the VHF communication test mode. The audio module automatic squelch circuit is disabled to let the VHF communication receiver noise to be heard. This makes sure that the receiver is operational.

Frequency rotary selector: The frequency rotary selector has two concentric rotary controls to change the frequency. The outer selector changes frequencies in 1 MHz and the inner selector changes the frequencies in 25 (or 8.33) KHz increments with a variable rate as a function of the knob rotation speed. The frequency rotary selectors are turned clockwise to increase the frequency indication and counterclockwise to decrease the frequency.

Active frequency indication: The active frequency indication shows the active (ACT) frequency selection. A Tx transmit indication is shown when the VHF transceiver is transmitting.

Preset frequency indication: The preset frequency indication shows the preset (PRE) frequency selection. The preset frequency indication also shows FAIL1 or FAIL2 malfunction messages. The FAIL1 message is shown when the frequency synthesizer lock detect has malfunctioned and the FAIL2 indication is shown when the transmitter output has malfunctioned.

Frequency transfer push-button switch: The frequency transfer push-button switch is pushed for two seconds to toggle the active and standby frequencies. If the indication malfunctions the frequency transfer push-button switch is pushed for seven seconds or more to tune the VHF transceiver to 121.500 MHz.

The ARCDUs and the observer's Audio Control Panel (ACP) interface with the RCAU to control the parameters that follow (Refer to ATA CHAPTER 23-50-00):

- PTT
- Microphone audio
- Headphone audio
- Speaker audio.

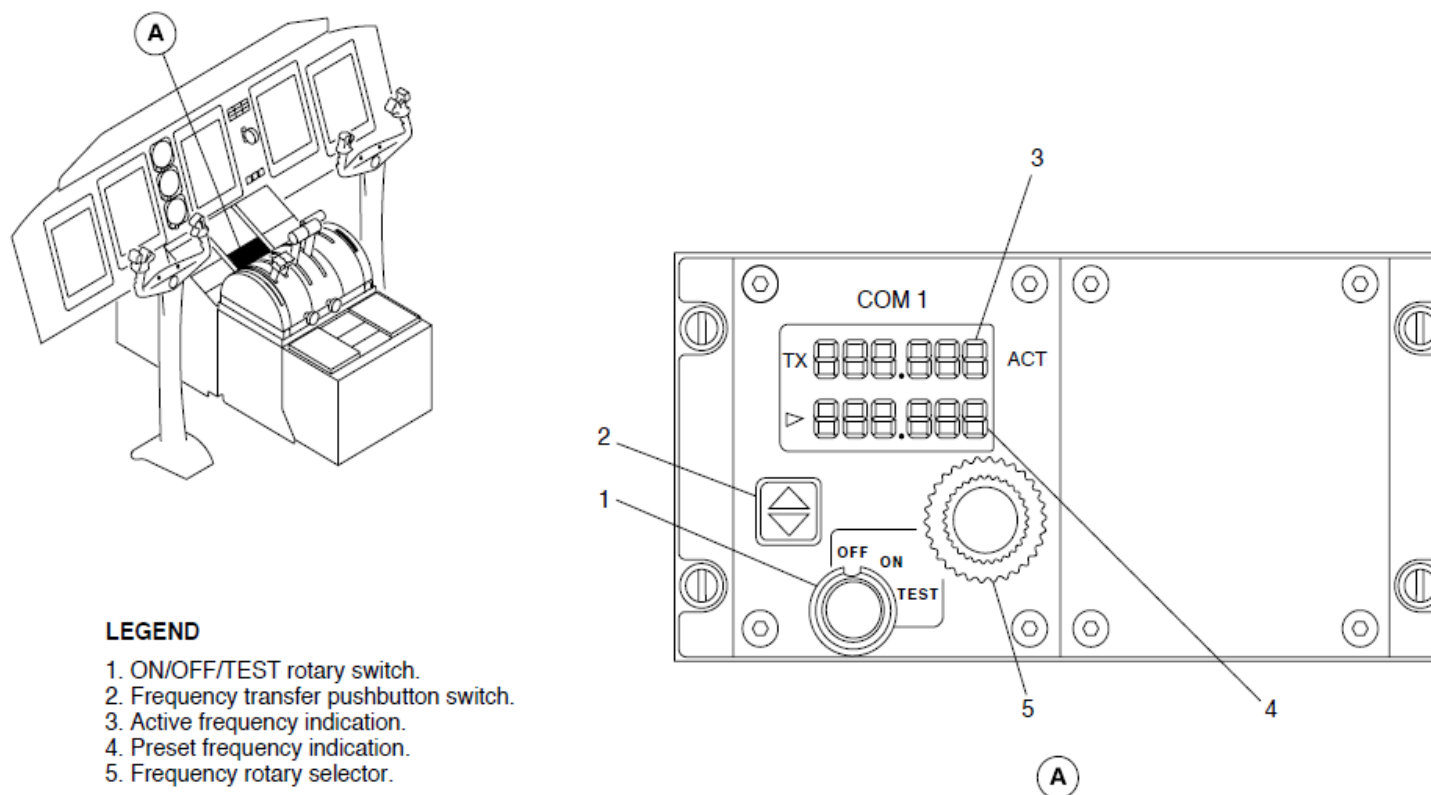
The aircraft left essential and right main buses supply 28 V dc electrical power through 10 A circuit breakers to the VHF transceivers (VHF1, VHF2). The circuit breaker for VHF1 is located in the position J1 on the left circuit breaker panel and the circuit breaker for VHF2 is located in the position D6 on the avionics circuit breaker panel.

The aircraft left essential bus also supplies 28 Vdc electrical power through a 1 A circuit breaker to the standby control panel. The circuit breaker is located in position K1 on the left circuit breaker panel.

The VHF communication system has two transceivers with related antennas.

One coaxial cable is routed to each VHF transceiver.

FIGURE 23- 15 VHF1 COMMUNICATION



Transceiver

Figure 23- 5 VHF Communication System, Transceiver

The VHF transceiver is a conventional superheterodyne, AM double-sideband transceiver with 25 KHz channel spacing. An optional transceiver provides 25 KHz and 8.33 KHz channel spacing. The narrow-band voice mode allows for the allocation of more channels to alleviate the VHF frequency congestion in the higher airspace of Europe.

The optional transceiver can set channels at 25 KHz spacing by turning the TUNE double rotary knobs on the ARCDU. Each increment of the rotary knob is 8.33 KHz. Thus, if frequency channels are set at every three increments, 25 KHz channel spacing is set.

The HIRF module filters stray radiation from adjacent cabling and protects the unit from the indirect effects of lightning.

The front panel has the features that follow:

- Attachment lugs
- Handle
- TEST push-button
- LEDs
- Identification labels.

The two VHF transceivers are located on the top shelf of the avionics rack between station point's X -24.95 and X -36.95. The mounting tray protects and holds the wired assembly in position and helps dissipate heat.

Standby Control Unit

Figure 23- 15 VHF1 Communication Standby Control Unit

A standby control panel that tunes VHF1 is installed in the forward, center pedestal area, in reach of the pilot and copilot. Two recessed rack-mounted rear panel connector interconnect the aircraft wire harness.

Antenna

Figure 23- 16 VHF Antenna

The antenna is a vertically-polarized antenna that operates in the VHF communications frequency spectrum of 118 to 156 MHz.

The antenna blade has a preformed radiating element in a thermoplastic assembly. A Radio Frequency (RF) N-type female connector is attached to the aluminum base.

An aluminum foil gasket with an elastomeric sealant makes electrical contact between the gasket and aircraft structure. The pressure applied during installation gives a contouring feature to let the gasket contact the two mating surfaces.

The VHF1 antenna is attached to the upper fuselage with six mounting bolts at station point X.186 and the VHF2 antenna is attached to the lower fuselage with six mounting bolts at station point X.81.

The antennas weigh 2.2 lb. (93.07 kg).

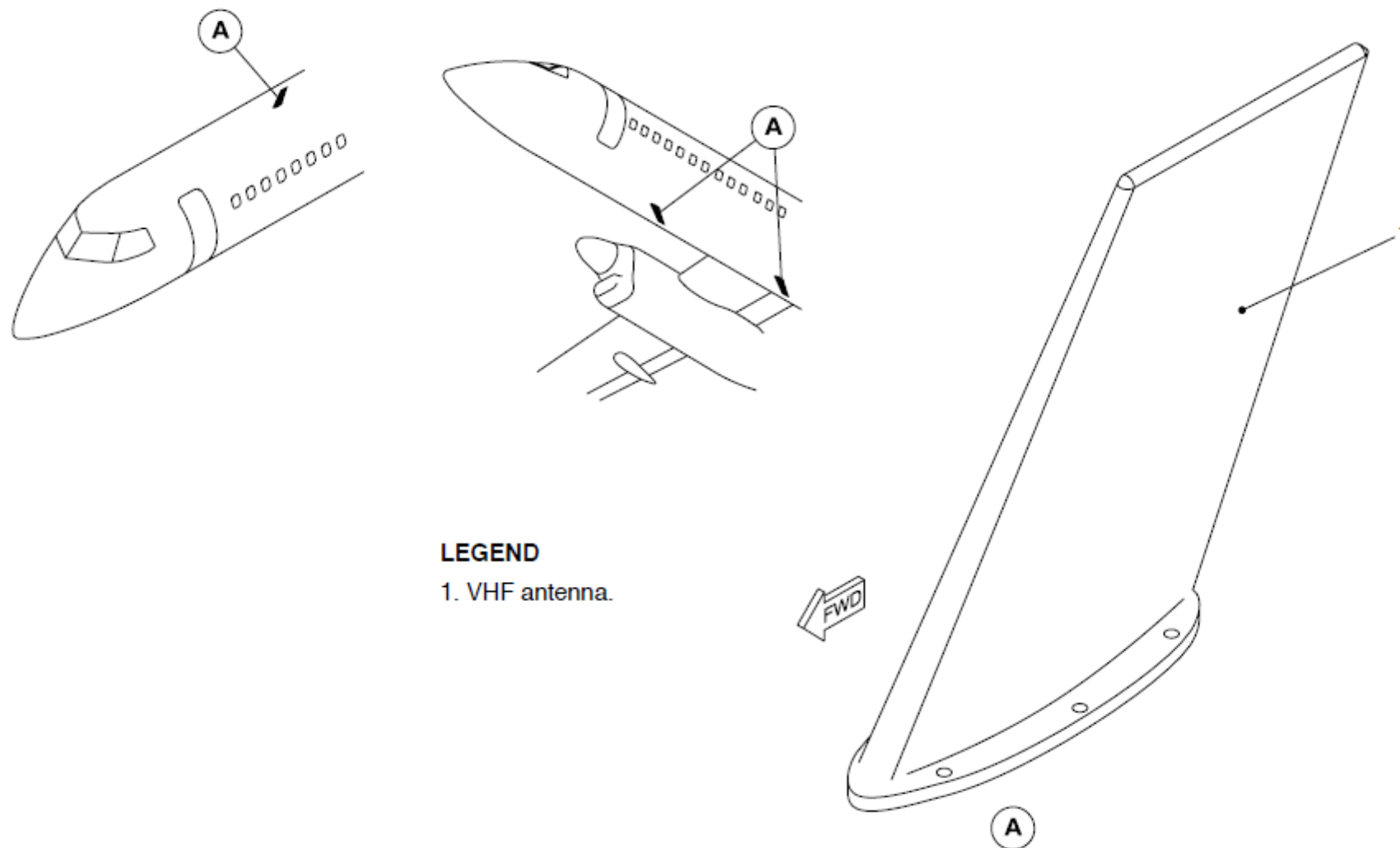


FIGURE 23- 16 VHF ANTENNA

Figure 23- 17 VHF Communication System - Electrical Schematic

The COM 1 standby control-panel tunes in 25 KHz or 25 KHz and 8.33 KHz increments. The selected format is a function of a ground discrete at the COM 1 standby control-panel.

The transceiver is tuned to the last burst of ARINC VHF Communication frequency data received on either of its two 429 inputs and for VHF COM1, the last frequency from the COM 1 standby control-panel.

The last frequency used is memorized by the unit and at power up, it will be tuned to it. This last frequency used is shown on the ARCDU1, ARCDU2, FMS1, FMS2, and the COM 1 standby control-panel from the continuous ARINC 429 output of the unit.

The ground/air logic discrete signal of the VHF transceiver is set to give pilot and maintenance initiated self-tests of the system and to enable the continuous in-flight monitoring.

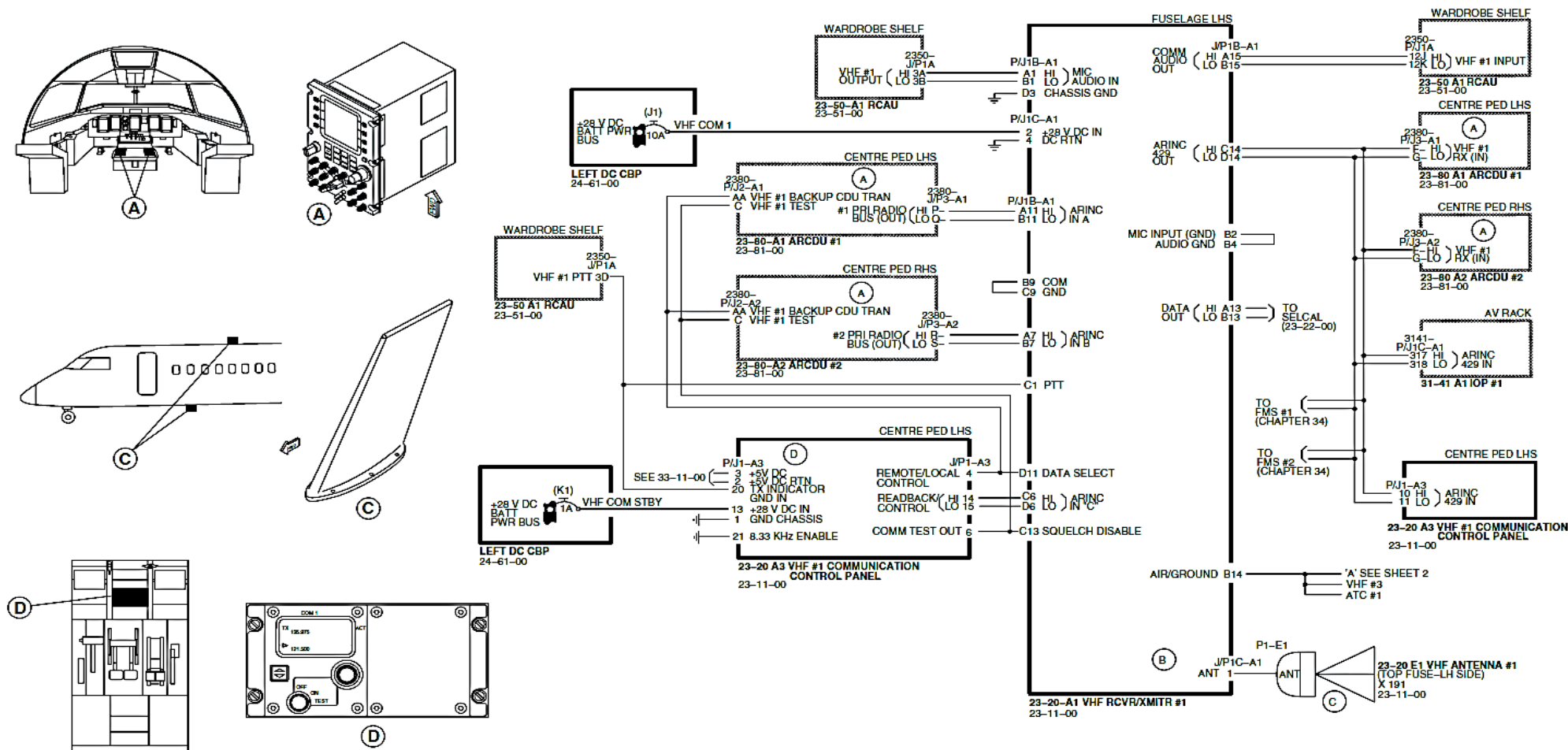


FIGURE 23- 17 VHF COMMUNICATION SYSTEM - ELECTRICAL SCHEMATIC

PASSENGER ADDRESS AND COMMUNICATION INTERPHONE SYSTEM

Introduction

The Passenger Address and Communication Interphone System (PACIS) does the functions that follow:

- Passenger Address (PA) announcements to the passengers
- Sound chimes
- Interphone communication between the attendants and the pilots or between the forward attendant and aft attendant.
- Attendant calls from switches in the Passenger Service Unit (PSU) and lavatory.

General Description

The PACIS has the components that follow:

- Passenger Address Amplifier (PAA)
- Forward Attendant Handset
- Forward Attendant Control Panel
- Aft Attendant Handset
- A control panel segment of the two Audio and Radio Control/Display Units (ARCDUs)
- Audible evacuation alert system.

The PACIS interfaces with the flight crew headsets, cabin and lavatory loudspeaker system, the three Advisory Light Panels (ALPs), Solid State Cockpit Voice Recorder (SSCVR) and, when installed, the passenger briefing and entertainment system.

The PACIS gives the functions that follow:

- PA announcements to the passengers
- Two way interphone communications between the flight attendants and pilots for private conversations
- Flight attendant calls from switches in the PSU and lavatory.

The SSCVR records the PA announcements.

Detailed Description

The PACIS functions in the modes that follow:

- PA
- Chime
- Communication interphone
- Attendant call
- Emergency
- Self-test
- Maintenance.

The passenger address announcements from the flight compartment and cabin are automatically recorded on Channel 1 of the SSCVR.

The other inputs to the SSCVR are from the Remote Control Audio Unit (RCAU).

PA Mode

Figure 23- 18 PACIS Block – Diagram

Figure 23- 19 PACIS MODES – LOGIC PATH FOR THE LIGHTS OPERATION

The PA key is pushed again or the rotary microphone selector is moved from the PA position to cancel the PA mode. All the PA indications go out.

For PA announcements, the PAA interface with the cabin speaker system is controlled by the components that follow:

- ARCDUs
- Forward attendant handset with its remote control panel
- Aft attendant handset.

Figure 23- 20 ARCDU, PA Mode Control

When the rotary microphone selector on the ARCDU is set to the PA position and the PA key is pushed, the Passenger Address Amplifier (PAA) starts the PA mode.

To show the selection, a green annunciation comes on in the indicators that follow:

- PA key on the two ARCDUs regardless of rotary microphone selector setting on the other ARCDU
- PA key on the forward attendant control panel
- PA key on the aft attendant handset cradle
- ALPs.

When a push-to-talk (PTT) selection is made by a pilot, speaking into a boom, mask, or hand microphone makes a PA announcement through the cabin speaker network.

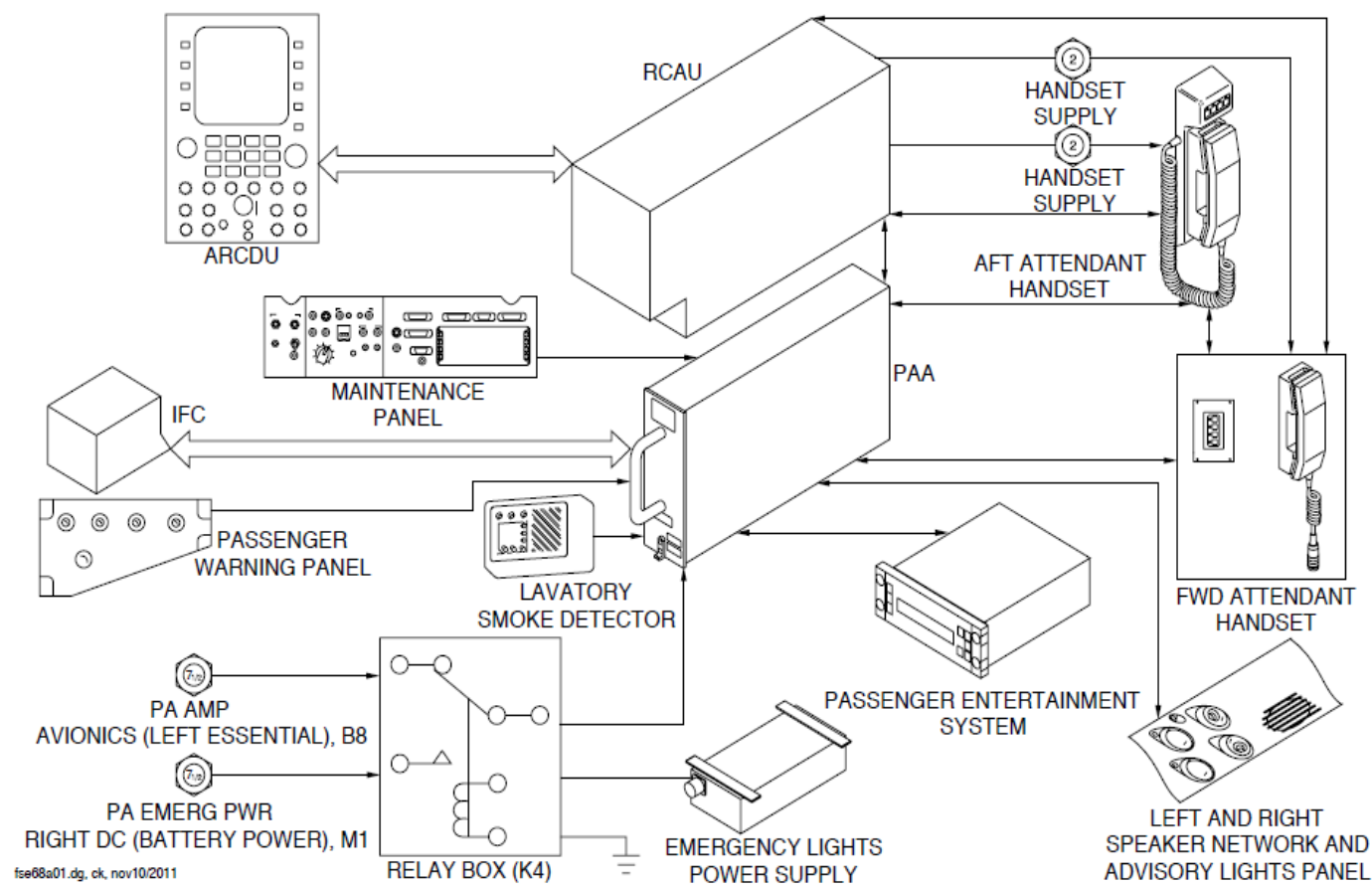


FIGURE 23- 18 PACIS BLOCK – DIAGRAM

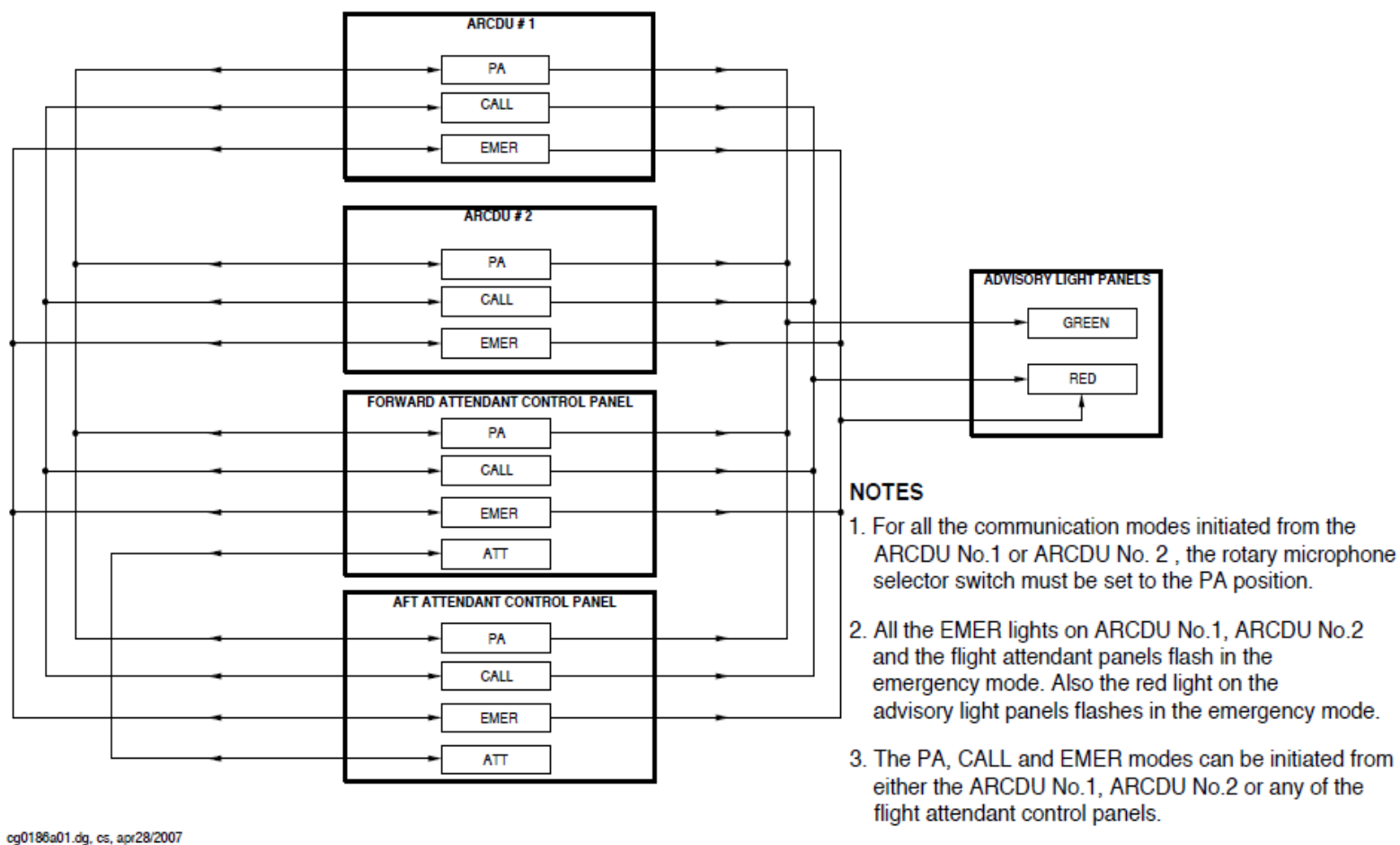
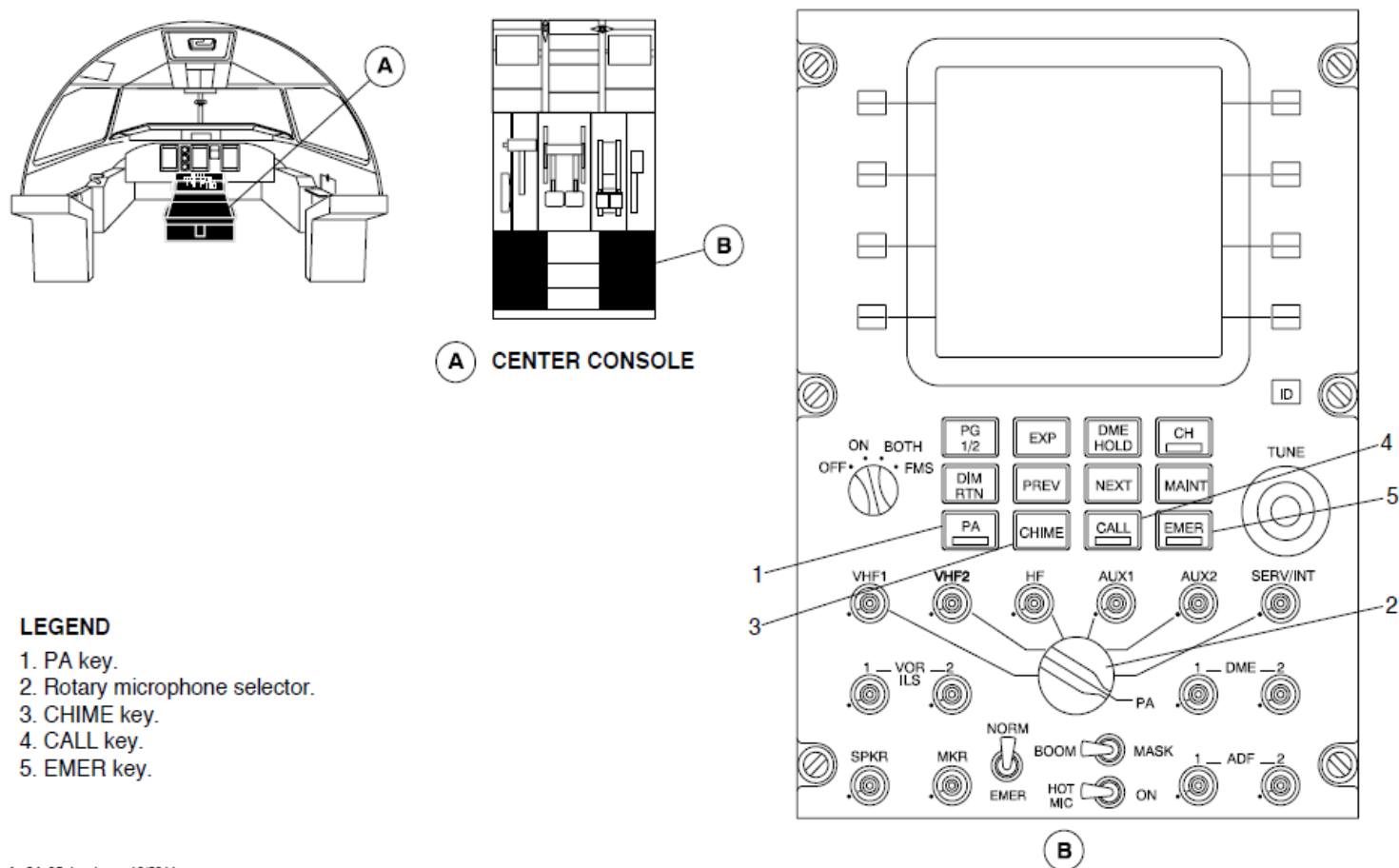
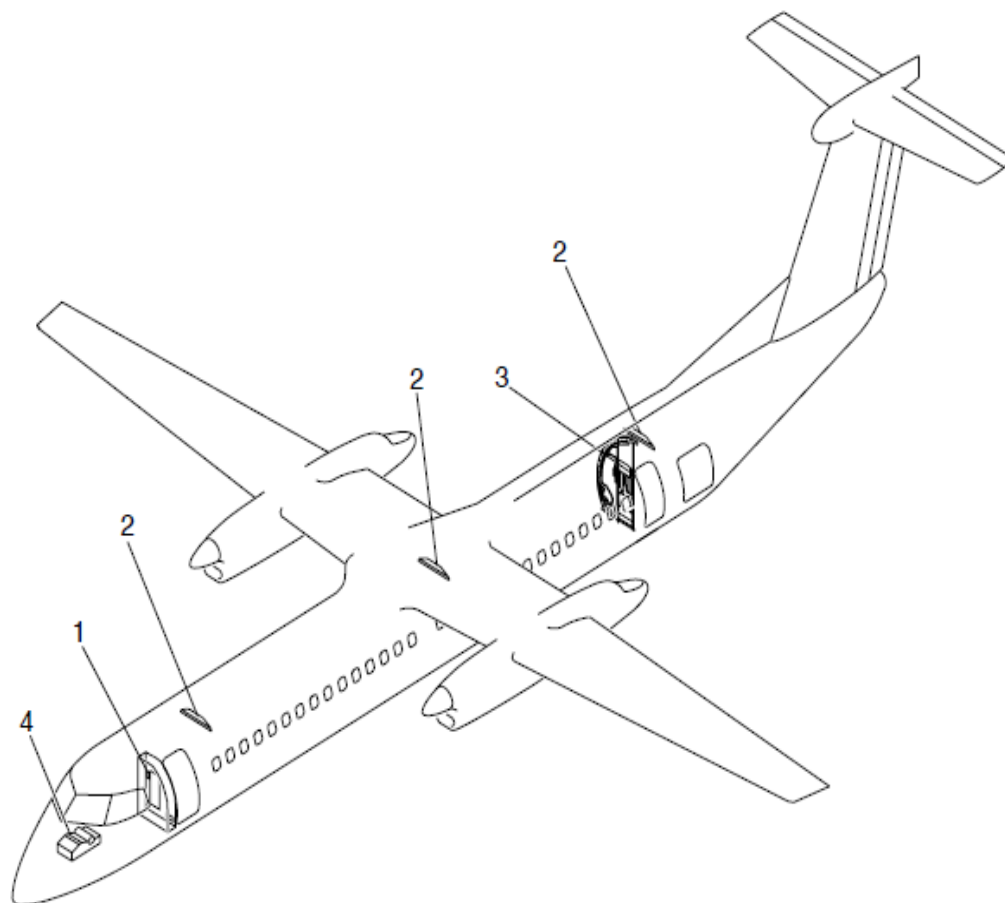


FIGURE 23- 19 PACIS MODES – LOGIC PATH FOR THE LIGHTS OPERATION



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FIGURE 23- 20 ARCDU, PA MODE CONTROL



LEGEND

1. FWD flight attendant handset control Panel.
2. Advisory light panel.
3. AFT flight attendant handset cradle.
4. ARCDU.

FIGURE 23- 21 PA MODE INDICATIONS PAGE 1

Figure 23- 22 FORWARD FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

Figure 23- 23 AFT Flight Attendant Handset, PA Mode Control

When the attendant handset is removed from its cradle and the PA key that is located on the handset or cradle is pushed, the PAA starts the flight attendant PA mode.

Figure 23- 21 PA Mode Indications Page 1

To show the selection, a green annunciation comes on in the indicators that follow:

- PA key on the forward attendant handset control panel
- PA key on the aft attendant handset cradle
- PA key on the two ARCDUs regardless of their rotary microphone selector positions
- ALPs.

When a handset PA selection is made, speaking into the handset microphone makes a PA announcement through the PAA and cabin speaker network.

A PA announcement can be overridden by a different PA announcement from a more important station as follows:

PRIORITY	STATION
1	Flight compartment
2	Forward flight attendant station
3	Aft flight attendant station
4	Passenger Entertainment System

The microphone audio from the flight compartment and flight attendant positions is routed through the RCAU for PA, CALL and EMER mode audio processing.

Chime Mode

Figure 23- 20 ARCDU, PA Mode Control

The rotary microphone selector on the ARCDU is set to the PA position and the CHIME key is pushed to sound a high/low chime in the cabin. It is not related to any other mode.

Communication Interphone Mode

Figure 23- 19 PACIS MODES – LOGIC PATH FOR THE LIGHTS OPERATION

Figure 23- 20 ARCDU, PA Mode Control

For interphone conversations, the RCAU interface with pilot and copilot jack boxes is controlled by the components that follow:

- ARCDUs
- Forward attendant handset with its remote control panel
- Aft attendant handset.

The switches in the PSU and lavatory are used by the passengers to call the flight attendant from individual seat positions and from the lavatory.

When the rotary microphone selector on the ARCDU is set to the PA position and the CALL or EMER key is pushed, an interphone mode is started. The CALL key is pushed for normal operation and the EMER key is pushed for emergency mode.

Figure 23- 21 PA Mode Indications Page 1

Green annunciations come on to show normal operation and flashing amber lights come to show emergency operation in the indicator that follows:

- CALL or EMER key on the two ARCDUs

Green annunciations come on to show normal operation and flashing red lights come to show emergency operation in the indicators that follow:

- CALL or EMER key on the forward attendant handset control panel
- CALL or EMER key on the aft attendant handset cradle.

A red annunciation light also comes on in the ALPs located in the cabin compartment.

A two-tone, high/low chime also sounds when a normal or emergency cabin interphone selection is made to call the flight attendants. A flight attendant CALL selection will cause the high/low chime to sound in the cabin only. An EMER selection will cause the high/low chime to sound in the flight compartment and in the cabin.

The pilots push the PTT switch to speak to the flight attendants. The flight attendants speak and listen to the pilots at the same time from the two cabin positions.

Figure 23- 22 FORWARD FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

Figure 23- 23 AFT Flight Attendant Handset, PA Mode Control

When the CALL or EMER key on the forward or aft flight attendant handset is pushed, an interphone mode is started. The CALL key is pushed for normal operation and the EMER key is pushed for emergency mode.

Figure 23- 21 PA Mode Indications Page 1

The same green or flashing amber lights as with the ARCDU selection come on to show normal or emergency operation respectively in the indicator that follows:

- CALL or EMER key on the two ARCDUs

The same green or flashing red lights as with the ARCDU selection come on to show normal or emergency operation respectively in the indicators that follow:

- CALL or EMER key on the forward attendant control panel
- CALL or EMER key on the aft attendant handset cradle.

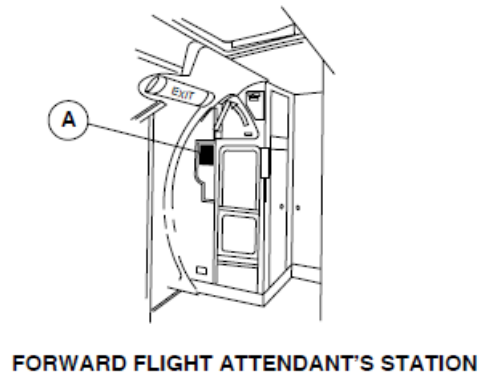
A red annunciation light also comes on in the ALPs located in the cabin compartment.

A two-tone, high/low chime also sounds when a normal or emergency cabin interphone selection is made. A pilot CALL selection will cause the high-low chime to sound in the pilot headsets and if selected, the flight compartment speakers. An EMER selection will cause the high/low chime to sound in the cabin and the flight compartment.

Figure 23- 22 FORWARD FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

Figure 23- 23 AFT Flight Attendant Handset, PA Mode Control

The flight attendants can also call each other. When the forward or aft ATT key on one flight attendant handset is pushed, a call is made to the other flight attendant.



LEGEND

1. PA Key.

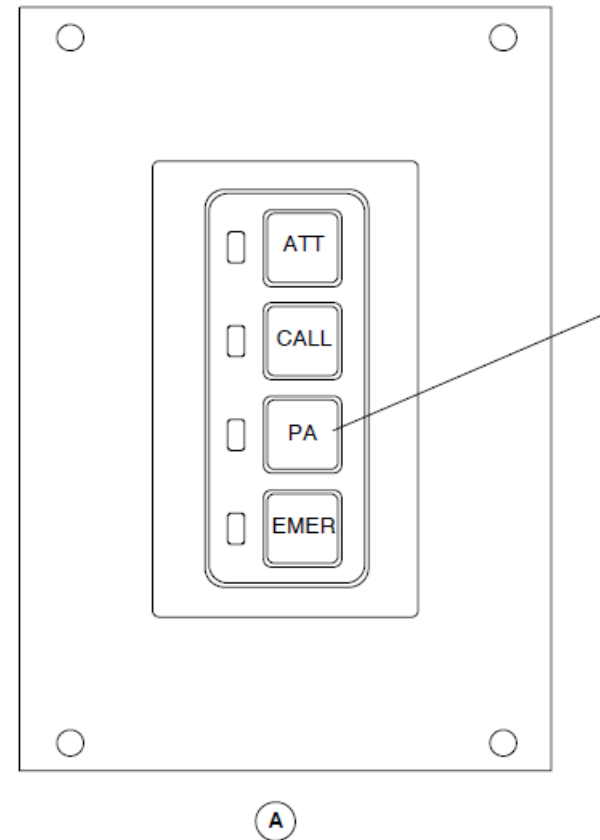


FIGURE 23- 22 FORWARD FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

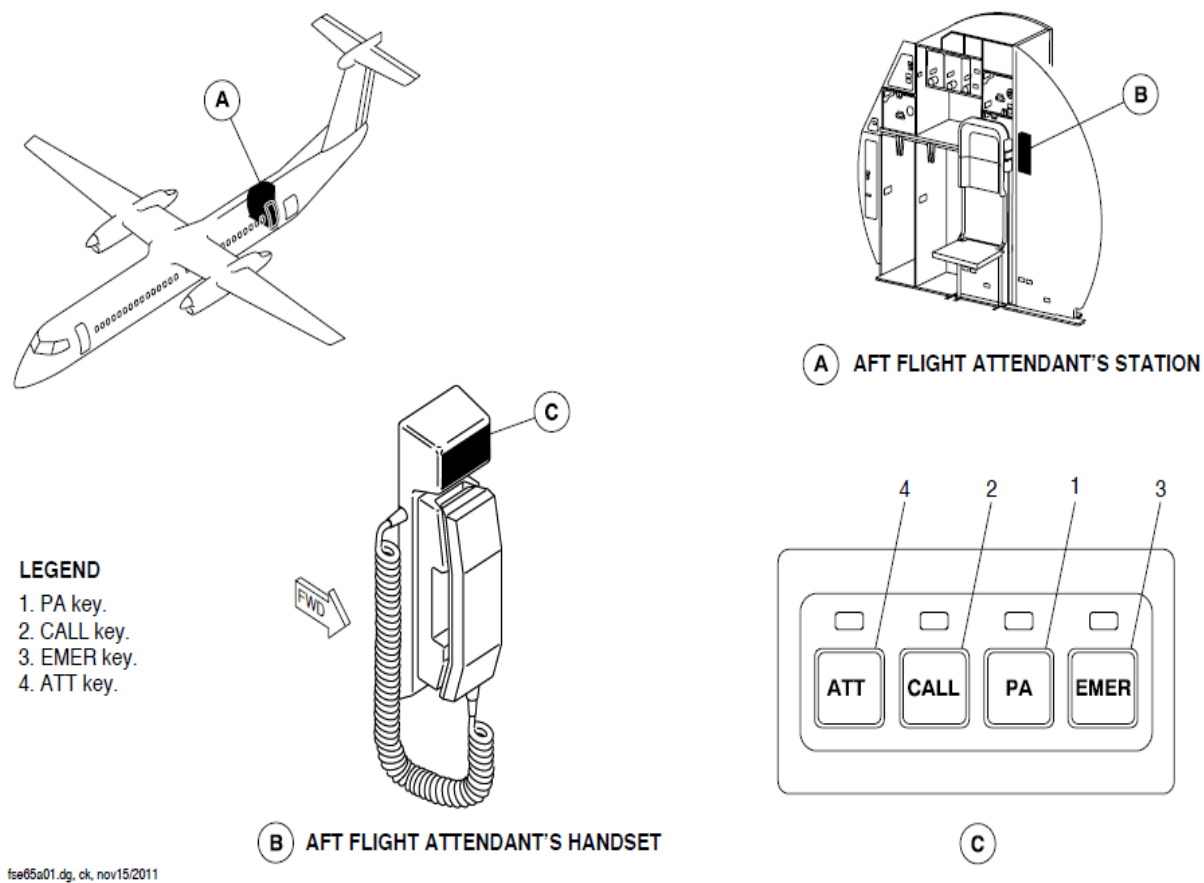


FIGURE 23- 23 AFT FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

Figure 23- 21 PA Mode Indications Page 1

A high/low chime sounds in the cabin compartment and a green light illuminates. There is no indication in the flight compartment.

A flight attendant interphone selection can be overridden by a more important selection as follows:

PRIORITY	SELECTION
1	EMERG
2	CALL
3	ATT

The flight attendant handset must be put back in the cradle to reset the system to make a less important selection. A more important selection can be made with less important selection already engaged, without the need to put the handset back in the cradle and the visual and aural alert will be provided.

When a handset is removed from its cradle, the system operation is not affected unless one of its switches is pressed.

The flight attendant PA and interphone modes are automatically canceled when the handset is put back in its cradle.

Flight Attendant Call

Figure 23- 19 PACIS MODES – LOGIC PATH FOR THE LIGHTS OPERATION

Figure 23- 24 PSU AND LAVATORY, FLIGHT ATTENDANT CALL

The passengers can call the flight attendants from individual seat positions and from an attendant call switch in the lavatory. A small red annunciation light on the PSU next to attendant call switch and a yellow annunciation light in the ALPs in the cabin compartment comes on for a flight attendant call from a passenger seat and an amber light comes on for a call from the lavatory. An one– tone high chime sounds in the cabin.

NO SMOKING, FASTEN SEAT BELT Annunciation.

Figure 23- 25 Passenger Warning Panel – Page 1

NO SMOKING and FASTEN SEAT BELT visual annunciation is provided in the cabin.

When NO SMOKING or FASTEN SEAT BELT switch on the emergency lights panel is set to ON or OFF position, a one–tone low chime sounds in the cabin.

Emergency Mode

Figure 23- 25 Passenger Warning Panel – Page 1

Figure 23- 19 PACIS MODES – LOGIC PATH FOR THE LIGHTS OPERATION

The PA mode continues to function when the engines and auxiliary power unit have stopped and no external power sources are connected.

When the emergency lights are turned on, the pilot and the two flight attendants can make PA announcements when the primary electrical power source is not available. The battery bus supplies electrical power through the PA EMER PWR circuit breaker to the equipment that follows:

- PAA
- ARCDU 1
- Pilot circuit card in the RCAU
- Forward and aft flight attendant handsets.

Self-Test Mode

Figure 23- 26 MAINTENANCE PANEL

When the PAA PTT switch on the maintenance panel is pushed, a self-test is started. A continuous tone is heard through the cabin and flight compartment speakers to check for correct speaker and PAA operation.

A TEST OK message is displayed on the ARCDU, if the test is successful.

Maintenance Mode

The PAA also uses the Central Diagnostic System (CDS) to monitor its condition. The CDS stores faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance personnel.

The maintenance mode allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of a given part "LRU IDENT".

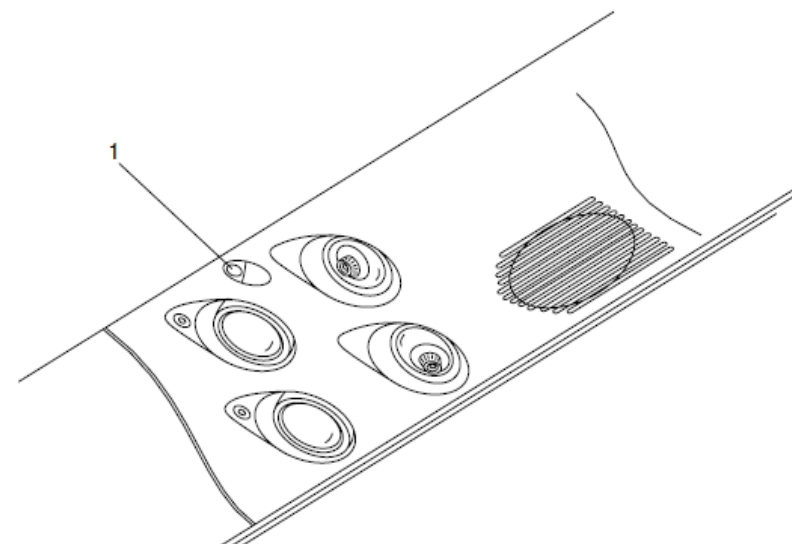


FIGURE 23- 24 PSU AND LAVATORY, FLIGHT ATTENDANT CALL

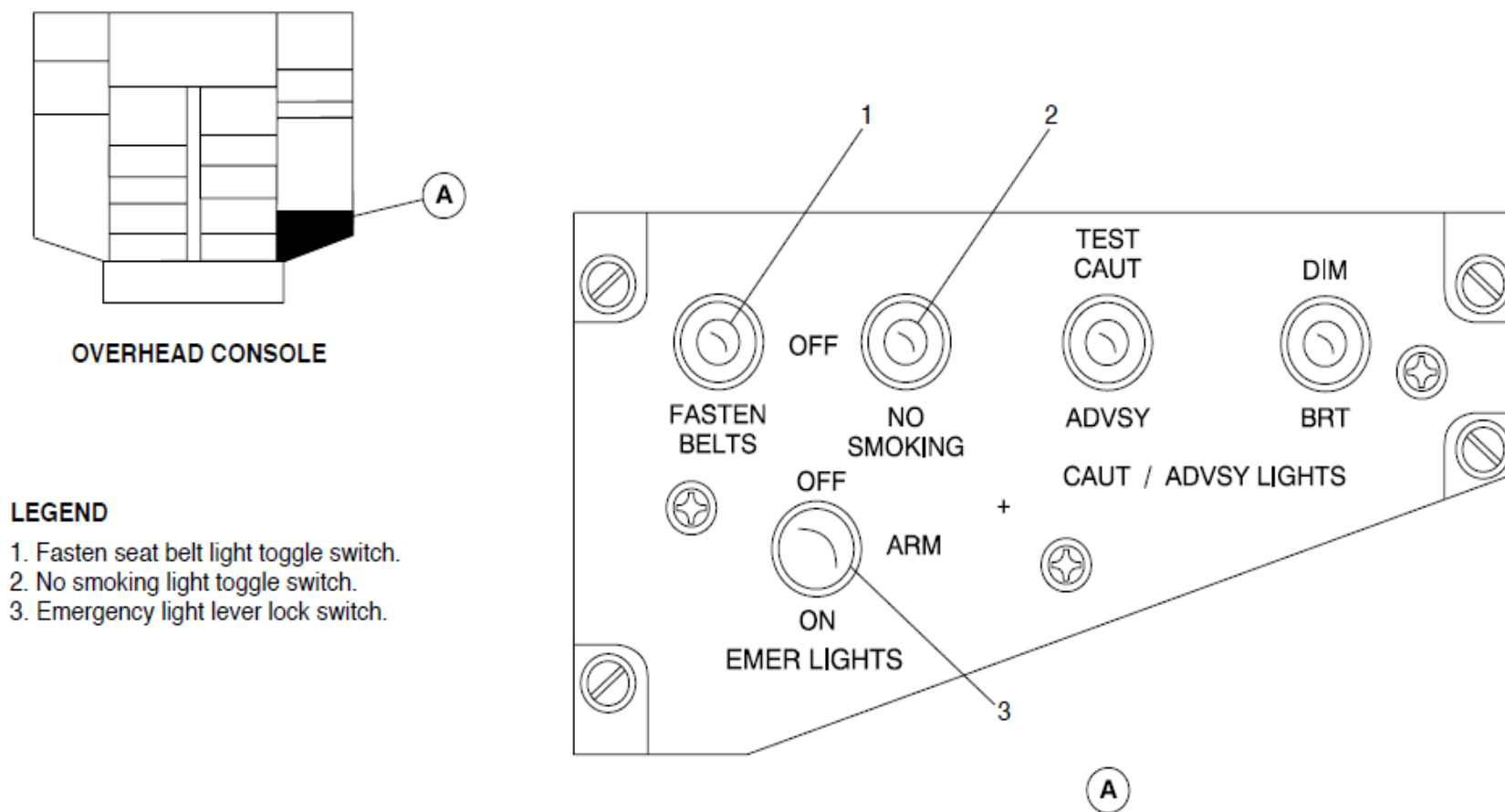
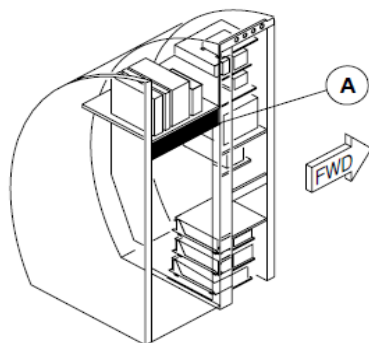


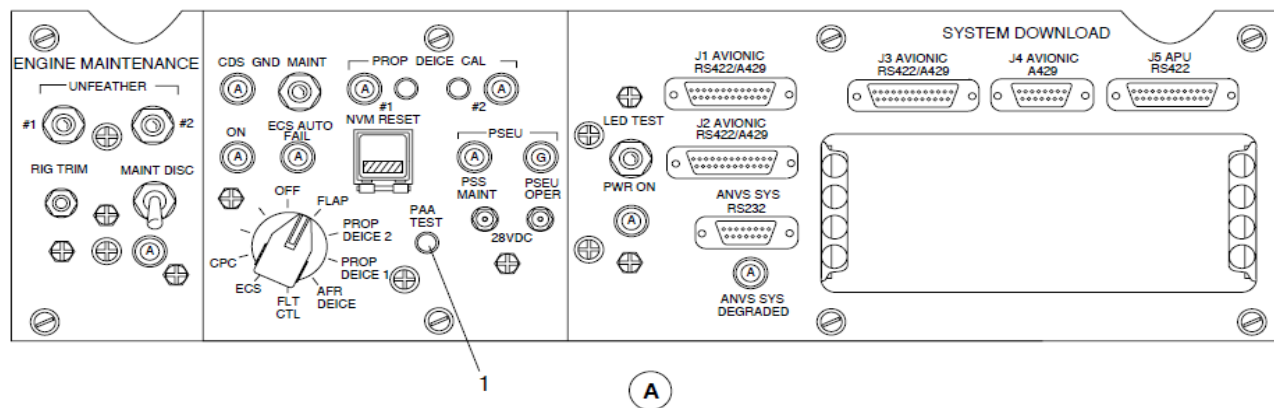
FIGURE 23- 25 PASSENGER WARNING PANEL – PAGE 1



**AVIONICS RACK ELECTRICAL
INSTALLATION**

LEGEND

1. PAA Test Pushbutton Switch.



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FIGURE 23- 26 MAINTENANCE PANEL

System Power

The aircraft left essential DC bus supplies 28 V dc electrical power through a 7.5 A circuit breaker to the PAA. The circuit breaker is located in position B8 on the avionics circuit breaker panel.

When the left essential DC bus malfunctions, the aircraft battery bus continues to supply electrical power. The emergency lights operate a relay to change the power source from the left essential DC bus to the battery bus. The battery bus supplies 28 V dc power through a 7.5 A circuit breaker to the ARCDU1, the PAA, pilot circuit card of the RCAU and the flight attendants handsets. The PA system also continues to operate. The circuit breaker is located in position M1 on the right circuit breaker panel.

The RCAU supplies 28 V dc electrical power through a 2 A circuit breaker to the two flight attendant handsets. The circuit breaker is located in position Q1 on the avionics circuit breaker panel.

Passenger Address Amplifier

Figure 23- 27 PASSENGER POWER AMPLIFIER (PAA) LOCATOR

The front panel also has gain and sensitivity potentiometers to adjust the system.

The PAA increases its output power by 6 dB when the engines are operating to compensate for engine noise. The Full Authority Digital Engine Control system (FADEC1, FADEC2) supplies oil pressure sensor data to the PAA through the Input/output Modules (IOM1, IOM2) located in their related Integrated Flight Cabinets (IFC1, IFC2).

The PAA supplies a signal to the three ALPs that are located in the cabin ceiling to visually alert the flight attendants of a call. The ALPs have light assemblies with different lens colors to show the different functions that follow:

COLOR	FUNCTION
Green	Public address in use
Red	Normal call/emergency call
Amber	Flight attendant call from lavatory
Yellow	Flight attendant call from PSU
Green (flashing)	Oxygen

The PAA supplies an output signal to the left and right cabin speaker network for cabin audio operation. The PA output is connected to 44 speakers in a series and parallel electrical network that has an impedance of 8.4 ohms.

The PAA supplies a discrete signal to the CDS to show its condition. The discrete signal is at ground potential when the PAA is operational and if the discrete goes open, the PAA is considered failed.

The PAA has the assemblies that follow:

- Audio power amplifier
- Audio and priority management module
- Chime generator
- Power supply

Thermal circuit limits output power when it senses excessive temperatures.

Audio and Priority Management Module

The audio and priority management module does the functions that follow:

- Calculates the audio output priority
- Supplies dc power to each attendant handset microphone
- Limits the signal level after a threshold level
- Supplies side tones.

Chime Generator

The chime generator makes low, high, and high/low chimes. The low tone has a frequency of 494 Hz and the high tone has a frequency of 587 Hz.

Tones are generated in the sequences that follow:

- Sequence 1: A high chime sounds when an attendant call signal is sensed from an overhead PSU or lavatory switch selection. The chime also sounds when the smoke detector in the lavatory is triggered. These tones are not heard in the flight compartment.

- Sequence 2: A high and low chime sounds when a normal or emergency call is sensed from an ARCDU or flight attendant handset selection. The chimes are heard in the cabin for normal calls and in both the cabin and flight compartment for emergency calls. The high-low chime also sounds in the cabin when the CHIME is selected on the ARCDU by flight crew.
- Sequence 3: A low chime sounds when the fasten seat belt or no smoking sign is set to on or off.

Power Supply Module

The Power Supply Module receives 28 V dc through 7.5 A on the left essential DC bus.

The power supply module supplies the system valid signal to the CDS.

Forward Attendant Handset

Figure 23- 22 FORWARD FLIGHT ATTENDANT HANDSET, PA MODE CONTROL

The forward attendant handset assembly has a handset and a cradle. The handset has a microphone, an earphone and a PTT switch. The microphone bias and audio amplification circuits for the handset are located in the cradle assembly. The cradle secures the handset when it is not in use.

The forward flight attendant seat stations have an attendant hand set and cradle installed that gives easy access from a seated position. The forward flight attendant handset and cradle assembly is installed near the forward flight attendant seat head rest at location X0.61, Y-13.47, Z149.27. It has a control panel that is installed near the flight attendant seat adjacent to the airstair door.

Forward Attendant Handset Control Panel

The forward flight attendant handset control panel is used to make PACIS mode selections from the forward flight attendant station.

The control panel is installed near the flight attendant seat adjacent to the airstair door at location X2.20, Y-32.40, Z145.63.

Aft Attendant Handset

Figure 23- 23 AFT Flight Attendant Handset, PA Mode Control

The aft attendant handset assembly has a handset and a cradle. The handset has a microphone, an earphone and a PTT switch. The microphone bias and audio amplification circuits for the handset are located in the cradle assembly. The cradle secures the handset when not in use and has a key pad that is used to make PACIS mode selections.

The aft flight attendant handset assembly is also installed as a basic equipment on the left, forward face of the aft draft bulkhead at location X666.0, Y-15.39, Z132.89.

Audible Evacuation Alert System

Figure 23- 28 Audible Evacuation Alert System – Block Diagram

Figure 23- 29 Audible Evacuation Panel

On aircraft with ModSum 4-411200 incorporated, an audible evacuation system is installed to enable the flight crew or cabin attendant to alert all the

crew members and the passengers of an impending emergency evacuation situation.

NOTE

On aircraft with SB84-25-75 incorporated, the audible evacuation system is deactivated.

The evacuation system produces a distinctive intermittent high pitched sound through three evacuation horns. One horn is installed in the flight compartment and other two horns are installed in the forward and aft of the passenger compartment. The evacuation horns are installed in the ceiling of the aircraft.

The EVAC SIGNAL switch is installed in the flight compartment on the overhead panel. The switch has two positions, ARM and OFF. If the EVAC SIGNAL switch is in the ARM position, the evacuation alert system will be activated and the OFF position shuts off the evacuation alert.

Two guarded evacuation alert ON/OFF switches are installed in the aircraft. One switch is installed in the flight compartment on the AUDIBLE EVACUATION panel and the second switch at the forward flight attendant station. The COCKPIT EVAC COMMAND switch, on the AUDIBLE EVACUATION panel, is a toggle switch and the EVAC SIGNAL ON switch, on the forward flight attendant handset panel, is a pushbutton type switch. Power will be available to these switches only when the EVAC SIGNAL switch is in the ARM position.

When the power is applied and the EVAC SIGNAL switch is put in the ARM position, the green EVAC SIGNAL light (top portion of the EVAC SIGNAL ON pushbutton switch) comes on.

When the EVAC SIGNAL ON pushbutton switch is pushed, the green ON light (bottom portion of the EVAC SIGNAL ON pushbutton switch) also comes on. This indicates that the system is operative and the audible signals from the evacuation horns come on.

The evacuation alert system is powered from the 28 V dc main battery. The circuit breaker is located in position E3 on the right dc circuit breaker panel.

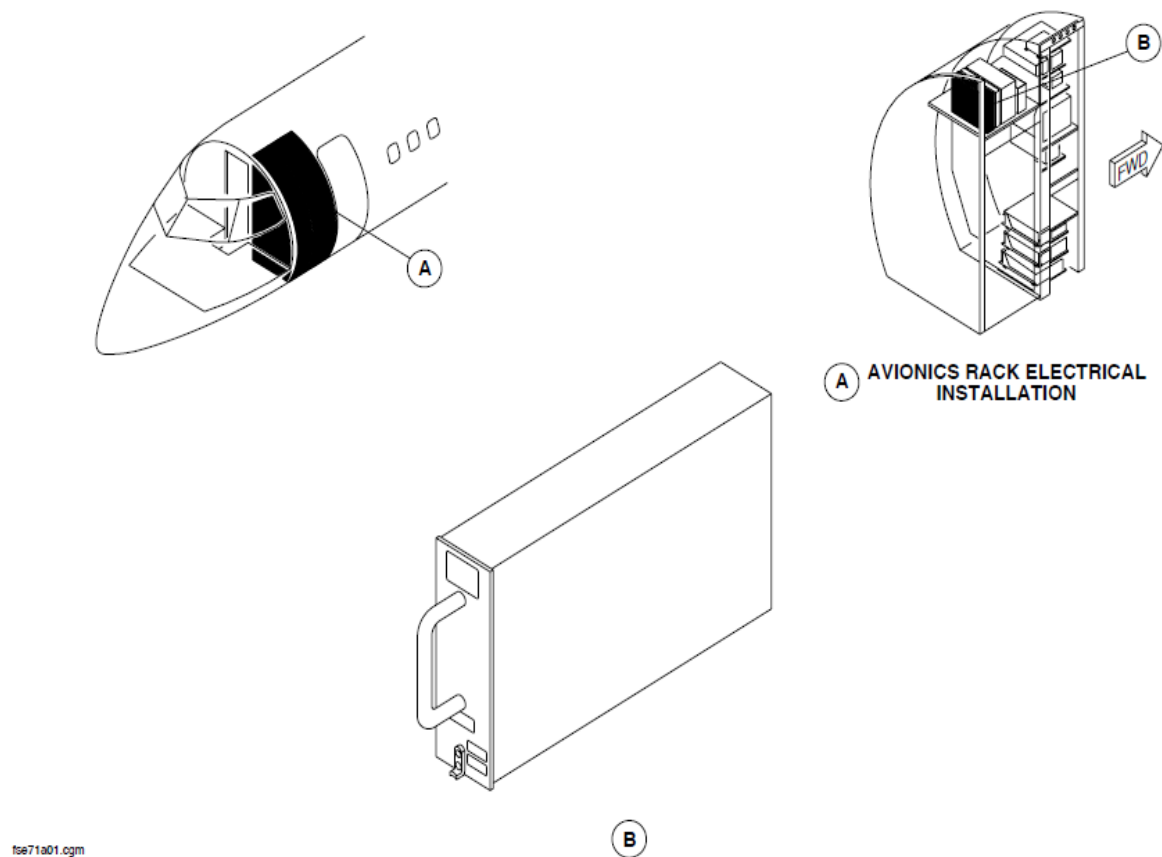
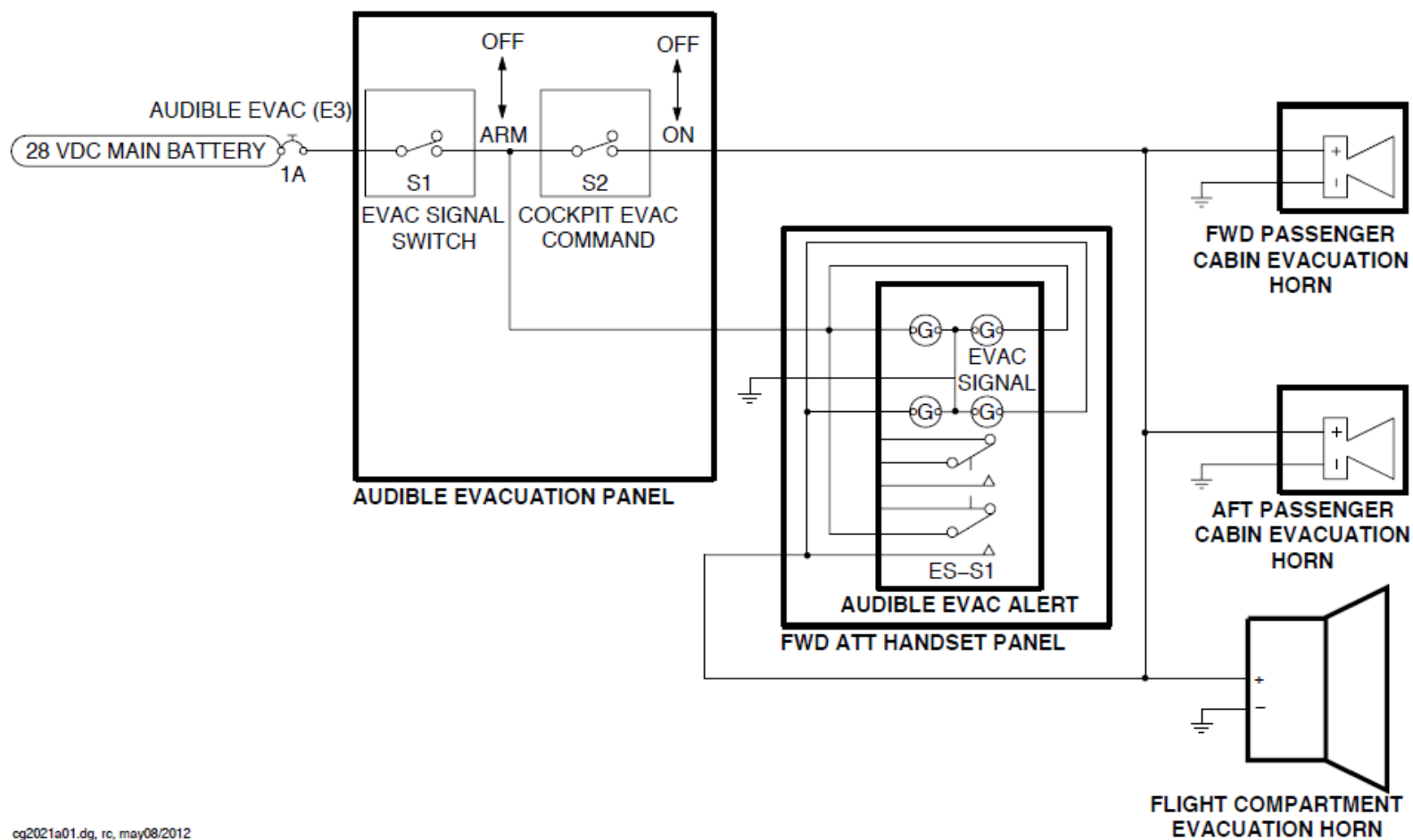
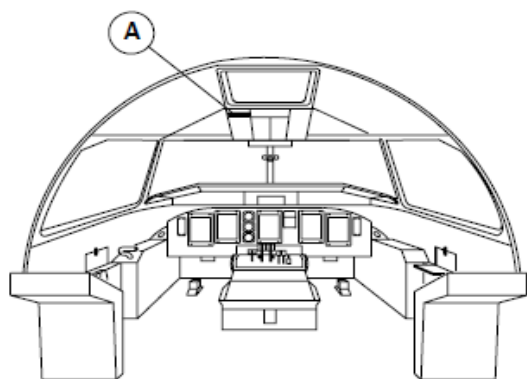


FIGURE 23- 27 PASSENGER POWER AMPLIFIER (PAA) LOCATOR



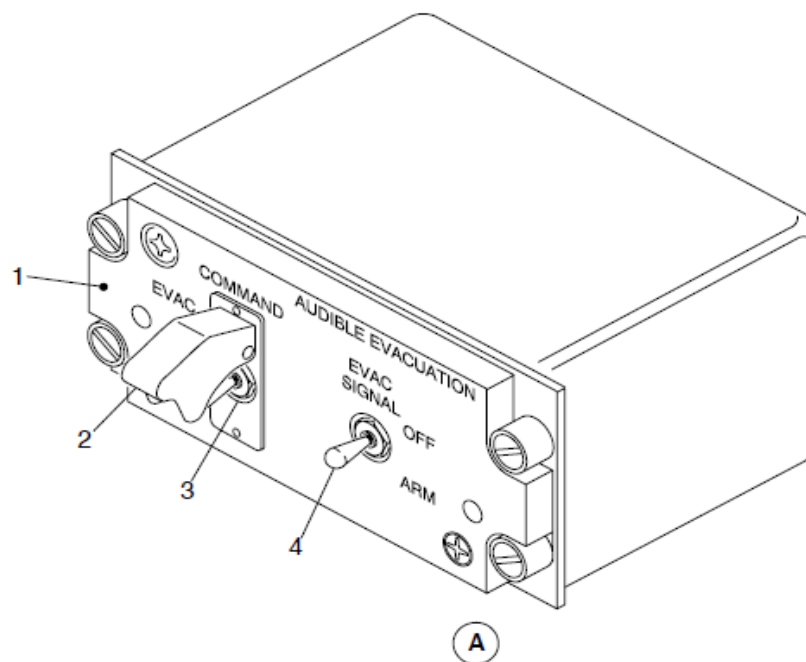
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FIGURE 23- 28 AUDIBLE EVACUATION ALERT SYSTEM – BLOCK DIAGRAM



LEGEND

1. Audible evacuation panel.
2. Switch guard.
3. EVAC COMMAND switch.
4. EVAC SIGNAL switch.



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FIGURE 23- 29 AUDIBLE EVACUATION PANEL

PASSENGER ENTERTAINMENT SYSTEM

The PES system is designed to broadcast announcements in up to four selectable languages and music audio selected by the operator. The PES interfaces with the passenger address amplifier (PAA) which is part of the cabin interphone system (PACIS) and audio integrating system (AIS). The system also supports public address (PA) announcements and safety broadcasts from the flight or cabin crew. The digital passenger announcement system (DPAS) is a line replaceable unit (LRU) that contains a separate flash memory card for customer programmable audio.

Component Location

Figure 23- 30 PASSENGER ENTERTAINMENT SYSTEM (PES) BLOCK DIAGRAM

The DPAS unit is installed on the maintenance panel above the forward wardrobe at stations X12.69, Y-20.06 and Z168.6 by four dzus fasteners. The unit weighs 1.87 lb. (0.85 kg).

The PES has the components that follow:

- Digital Passenger Announcement System (DPAS) unit which is line replaceable (LRU)
- Flash Memory Card for recorded music and announcements as defined by the operator. The flash memory card can have memory capacities of 10, 20 and 40 MB or higher.

Front Control Panel Description

The front panel keys and indicators include the following:

- ON key turns the unit on. The second push turns the unit OFF
- STOP key to stop recorded music or announcements
- PLAY key to start recorded music or announcements
- UP scroll key
- DOWN scroll key
- A (announcement) key for selection of announcements or music audio
- L/V (language and volume) key to select languages or volume control
- SEL (selection) key for language or music selection
- GREEN LED indicator with the ON key for ON/OFF condition
- GREEN LED indicator with the PLAY key to show an active broadcast.

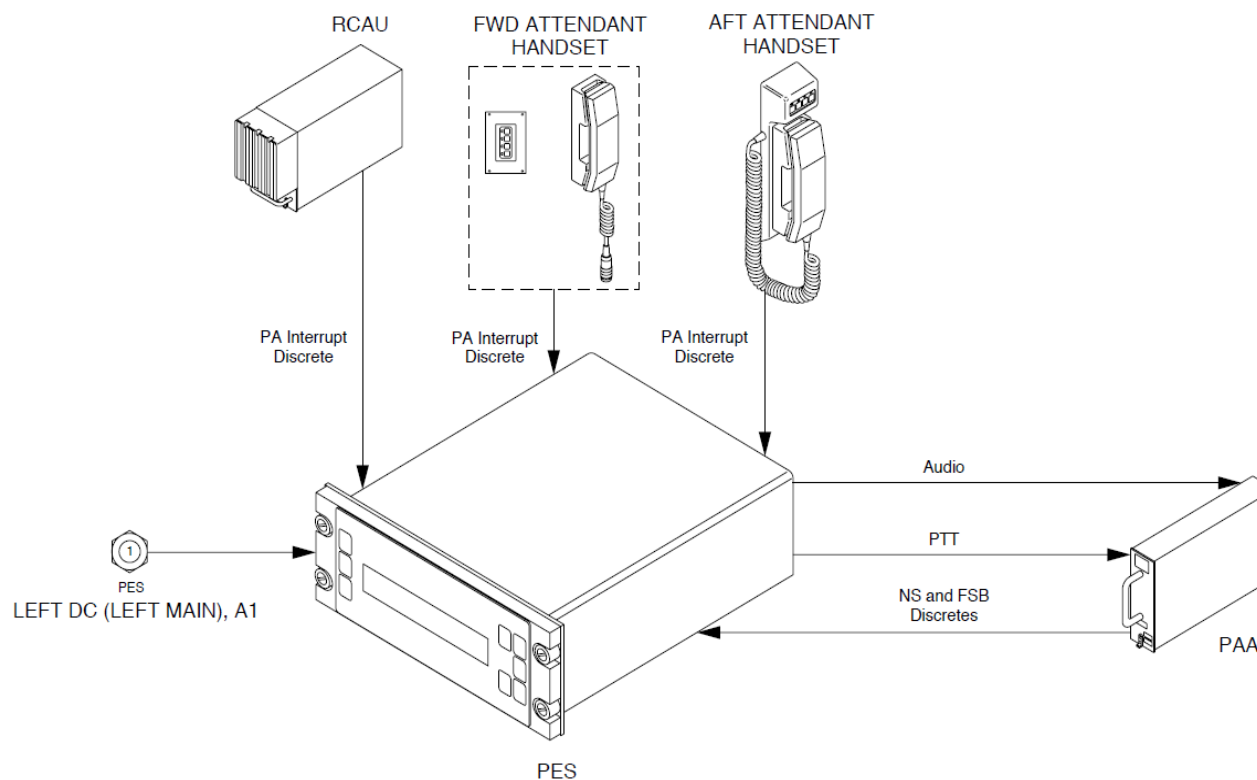


FIGURE 23- 30 PASSENGER ENTERTAINMENT SYSTEM (PES) BLOCK DIAGRAM

DPAS Component Description

An ON/OFF switch is provided for unit operation. It interfaces with the remote control audio unit (RCAU), passenger address amplifier (PAA), forward and aft attendant handsets to provide public address (PA) announcements, prerecorded passenger announcements and music audio. The unit contains a removable flash memory card that can save up to 99 announcements and 40 audio selections. Fifteen languages are supported by the unit with four languages programmable at a given time by the operator. The power is supplied to the unit from the left main 28 vdc circuit breaker A1.

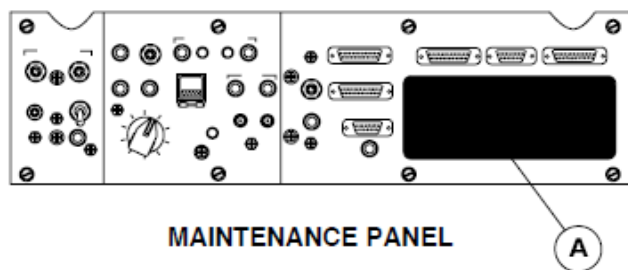
Music audio is selected by pressing the SEL key when the MUSIC title is displayed in the announcement selection and broadcast menu. Music is broadcast continuously until the STOP key is pressed. The music is interrupted by the flight or cabin crew announcements or by the discrete inputs from the NO SMOKING and FASTEN SEAT BELT annunciations. The NO SMOKING and FASTEN SEAT BELT announcements generate tones only with the illumination of appropriate signs inside the cabin.

PES Overview

Figure 23- 32 PES, Announcement Selection and Broadcast Mode

In flight or on the ground the operator can select announcements and music to broadcast over the cabin speaker system. The system is designed such that PA broadcasts from the flight deck or flight attendant station takes priority over the PES audio. The selected announcements are available for broadcast in up to four languages. The order of the language broadcast is determined by the operator. For the language menu the operator selects the L/V (language/volume) key and scrolls up or down to select or cancel a language with the SEL key. The announcement (A) key returns the operator to the announcement and broadcast menu. To broadcast an announcement, the operator must first scroll the title list with the up and down scroll keys for announcement selection. The PLAY key is then pressed and the green LED will illuminate beside the key and remain on during the announcement broadcast.

Press the L/V key two times from the announcement mode or one time from the language mode to access the volume adjustment menu. The volume is adjusted by the up and down scroll keys.



LEGEND

1. ON/OFF key.
2. ON light emitting diode (LED).
3. Display area.
4. LV (language/volume) key.

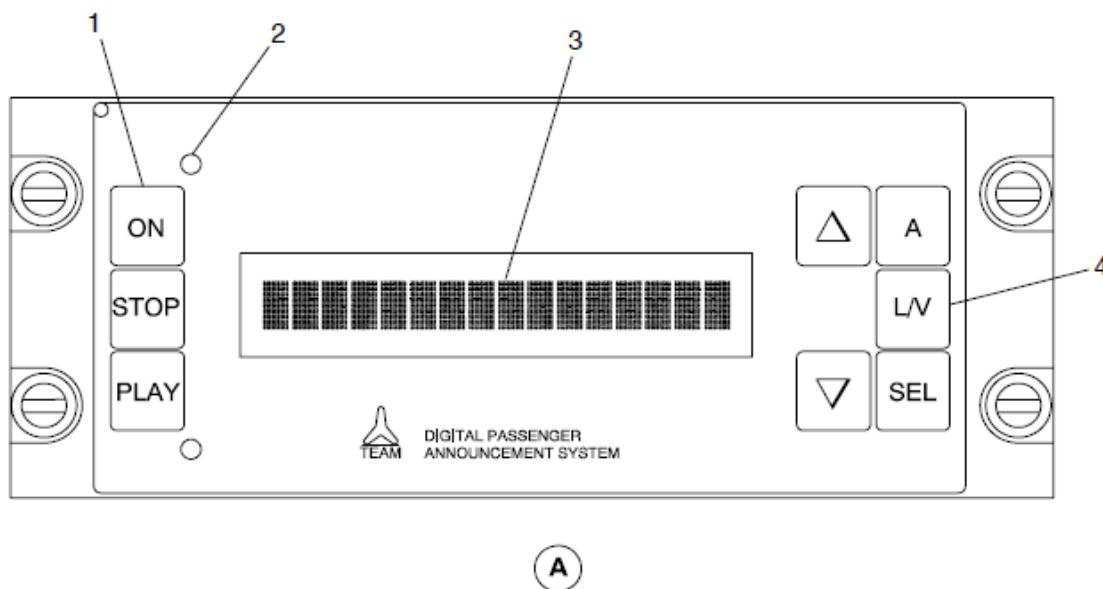
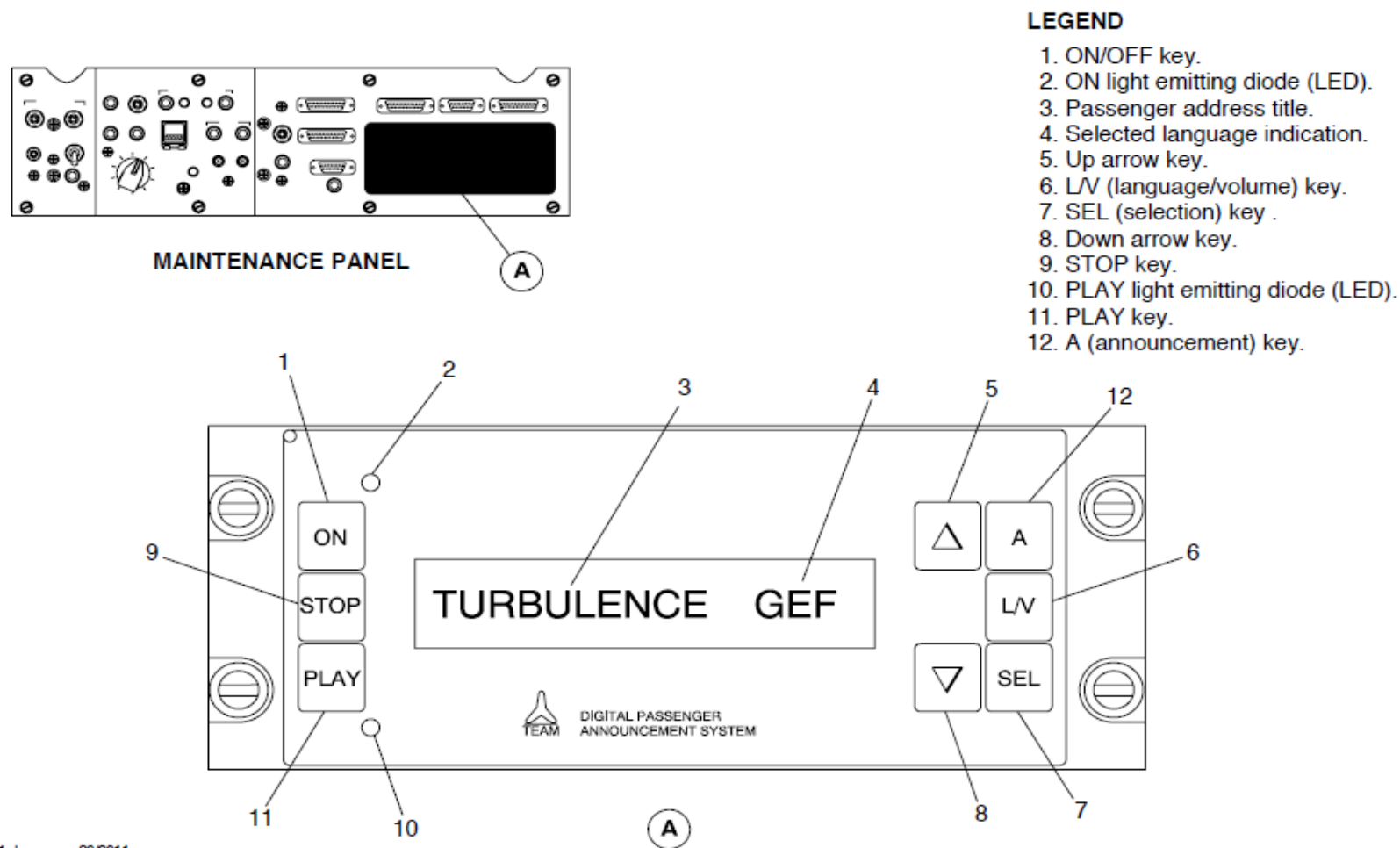


FIGURE 23- 31 PES, TEST MODE



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FIGURE 23- 32 PES, ANNOUNCEMENT SELECTION AND BROADCAST MODE

PES Detailed Description

Language Selection Mode

Figure 23- 33 PES, Language Selection Mode

The language selection is applicable to all messages stored in the PES. To access the language selection mode the L/V key is pressed once. If there is no key stroke activity for three minutes the system automatically returns to the announcement selection and broadcast mode. In this mode the DPAS display will show a maximum of 4 language selection titles instead of announcement titles, one language at a time. A language is identified by its title and the letter placed between the parentheses, e.g. English is shown as ENGLISH (E), or at the end of the title when the letter is not in the title, e.g. German is shown as GERMAN (D).

The up and down arrow keys are used to scroll through the list of language titles and the SEL key is then pressed to select the languages. The languages are selected in the preferred order for message broadcast. A language selection is canceled when the title is displayed and the SEL key is pushed. If four languages are chosen, one must be canceled before another is selected. The language mode always starts with the language indication previously shown. During announcement broadcast the SEL key is not available for language selection but will function during music playback.

Volume Adjustment Mode

Figure 23- 34 PES, Volume Adjustment Mode

The volume adjustment mode is used to set the audio level. To adjust the volume press the L/V key two times to start from the announcement selection mode or press one time to select from the language selection mode.

The system automatically returns to the announcement selection and broadcast mode if no further key is pressed for 30 seconds. For example the display will show VOLUME 7/10 if the volume is set to 7.

The up and down scroll keys are used to increase or decrease the volume in steps and the change is shown on the display. The displayed range of volume is 1 to 10 where 10 represents 0 dB nominal at the system output and 1 equals to -20 db. Press the L/V key again to return to the language selection mode.

Announcement Selection and Broadcast Mode

Figure 23- 32 PES, Announcement Selection and Broadcast Mode

This mode is for the broadcast of announcements. Individual announcement titles are shown one at a time in a stored list. The scroll keys are used to select the stored announcements. To play an announcement, the PLAY key on the front panel is pressed and the STOP key is used to stop the broadcast. To stop an announcement in progress the STOP key is pressed once. The announcement will resume when the PLAY key is pressed again and will follow the order of the announcement schedule.

The announcement titles consists of up to ten characters. Four character spaces after the title is reserved to show the languages selected for broadcast. The sequence of letters show the language broadcast order. The broadcast of an announcement is activated by the PLAY key or automatically via discretes controlled by inputs such as NO SMOKING or FASTEN SEAT BELT. During the announcement, the PLAY LED is illuminated and the Press-To-Talk (PTT) discrete output is activated to allow the PAA to receive, amplify and output the message over the cabin speakers. The NO SMOKING and FASTEN SEAT BELT discrete inputs take priority over the recorded announcements. Also the PAUSE discrete generated by a PA announcement from the flight deck or cabin take

priority over the recorded announcements.

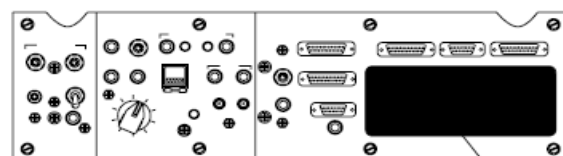
The prerecorded announcement will automatically resume at the end of the interruption.

The announcements activated by the discrete inputs are not displayed in the message title list. A ground input or a 28 V dc discrete will initiate the PTT output discrete, the announcement and the illumination of the PLAY LED. The message is broadcast sequentially in all languages or the default language if no languages were selected. However, the STOP key will not stop a PA message broadcast. During an announcement broadcast you can scroll through the message list to select the next message. Additional operation of the PLAY key has no effect on the active announcement. Prior to message playback, the system would add a one second interval to accommodate any system delays or operator corrections.

Selection of the PLAY key will have no effect if no languages have been programmed or unless the TEST ALL RECORD or MUSIC titles are displayed. The title TEST ALL RECORD when selected will broadcast all messages stored in the memory in the selected languages.

The announcement mode is automatically started after completion of the Power-On Self-Test (POST). The mode is also selectable by the announcement key A from the modes that follow:

- Language selection
- Volume adjustment
- Music selection
- Diagnostic (except when there is a TEST discrete signal).

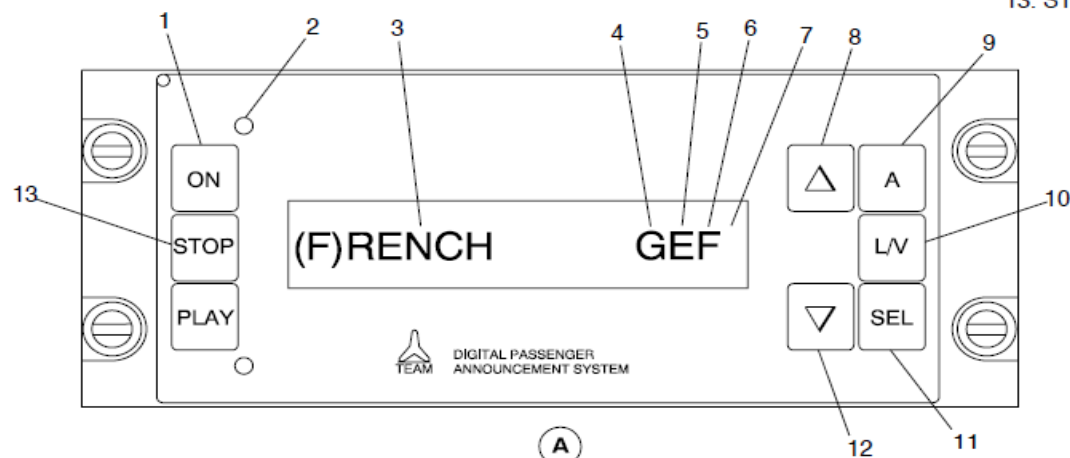


MAINTENANCE PANEL

A

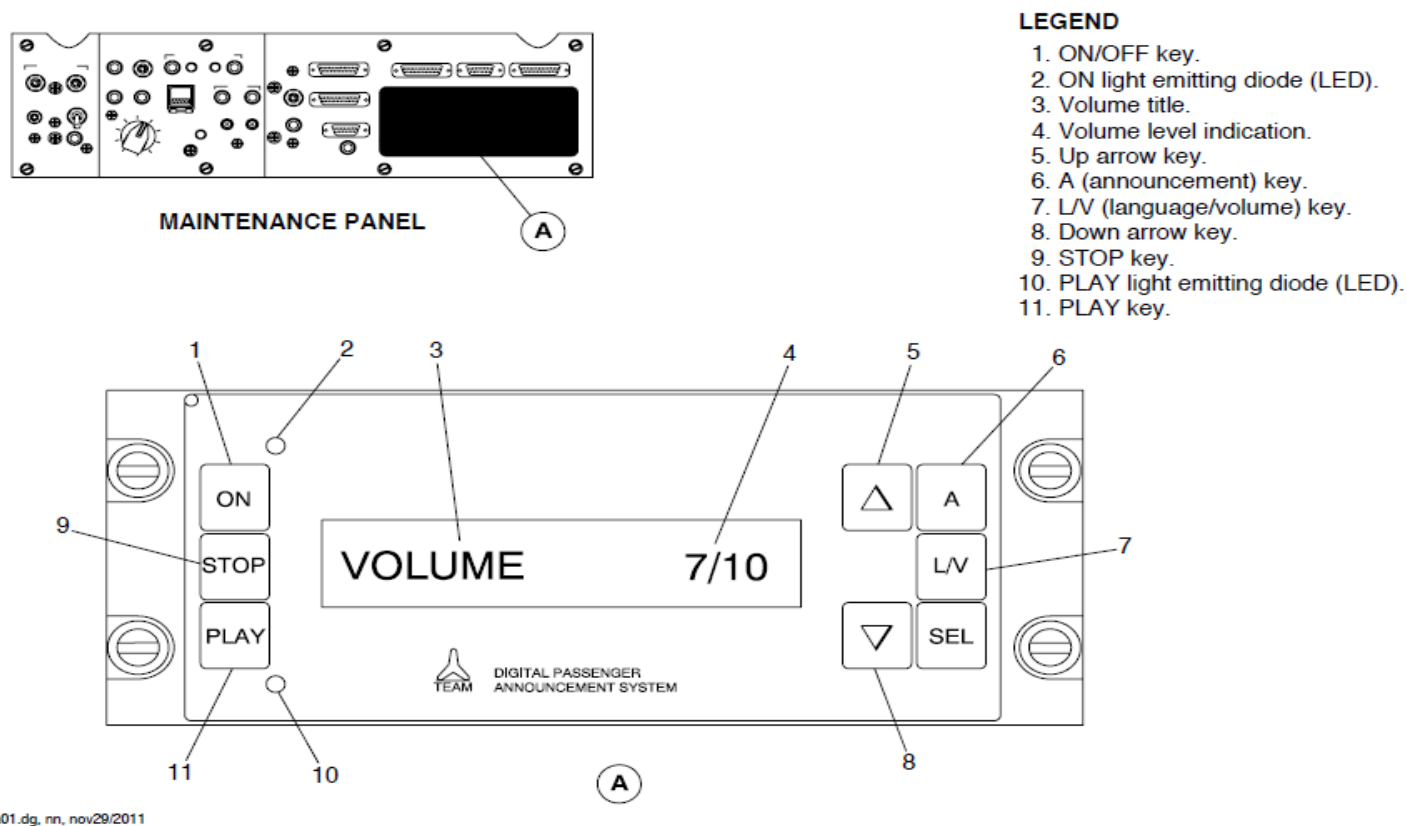
LEGEND

1. ON/OFF key.
2. ON light emitting diode (LED).
3. Language title.
4. First language.
5. Second language.
6. Third language.
7. Fourth language (NEW).
8. Up arrow key.
9. A (announcement) key.
10. L/V (language/volume) key.
11. SEL (selection) key.
12. Down arrow scroll key.
13. STOP key.



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FIGURE 23- 33 PES, LANGUAGE SELECTION MODE



Music Selection Mode

Figure 23- 35 PES, Music Selection Mode

The music selection is done through the announcement selection and broadcast mode. The title MUSIC provides the operator with the option of selecting from the list of music titles stored in the memory. If music files are not stored in memory, the MUSIC title will not appear in the announcement list. The list will show in alphabetical order and the selection is made by the up and down scroll keys followed by the SEL key. The current music title is displayed to the right of the MUSIC legend. If no particular music title is selected and the PLAY key is pressed, the system will select all the music titles for playback starting from the beginning of the music list. The PLAY LED light will come on and the push to talk (PTT) discrete will allow the PAA to channel the music to the cabin speakers. During music playback, selection of an announcement and the push of the PLAY key will stop the music to allow broadcast of the announcement. The music selection is not available at system start up. If there are no keystrokes for three minutes after the MUSIC title is displayed the system will go back automatically to the announcement mode. Each musical title is defined a sequential number for play back. If an announcement is selected for playback during music audio it will stop until the broadcast is over and continue from the point of interruption. The STOP key will also interrupt the music play back.

A BITE memory is used to save failure conditions and the order in which they occur during system operation. Only one type of malfunction is saved at a time.

Access to BITE memory is achieved by simultaneous selection of the up and down arrow keys while the system power is turned on. A list of failures if available is presented in the order of occurrence. Use the up and down arrow keys to scroll through the list of failures. At the end of the list, a BITE CLEARING message is shown. If the SEL key is pressed, all failure messages saved in memory are erased. From the diagnostic mode the system will return to the announcement and broadcast mode if no panel keystroke activity takes place after more than three minutes.

Diagnostic Mode

Figure 23- 36 PES, Diagnostic Mode

The diagnostic mode is used to test the analog and digital circuits in the DPAS unit. Only the digital circuits can be tested while the unit is installed in the aircraft and is achieved by the Built-In-Test Equipment (BITE).

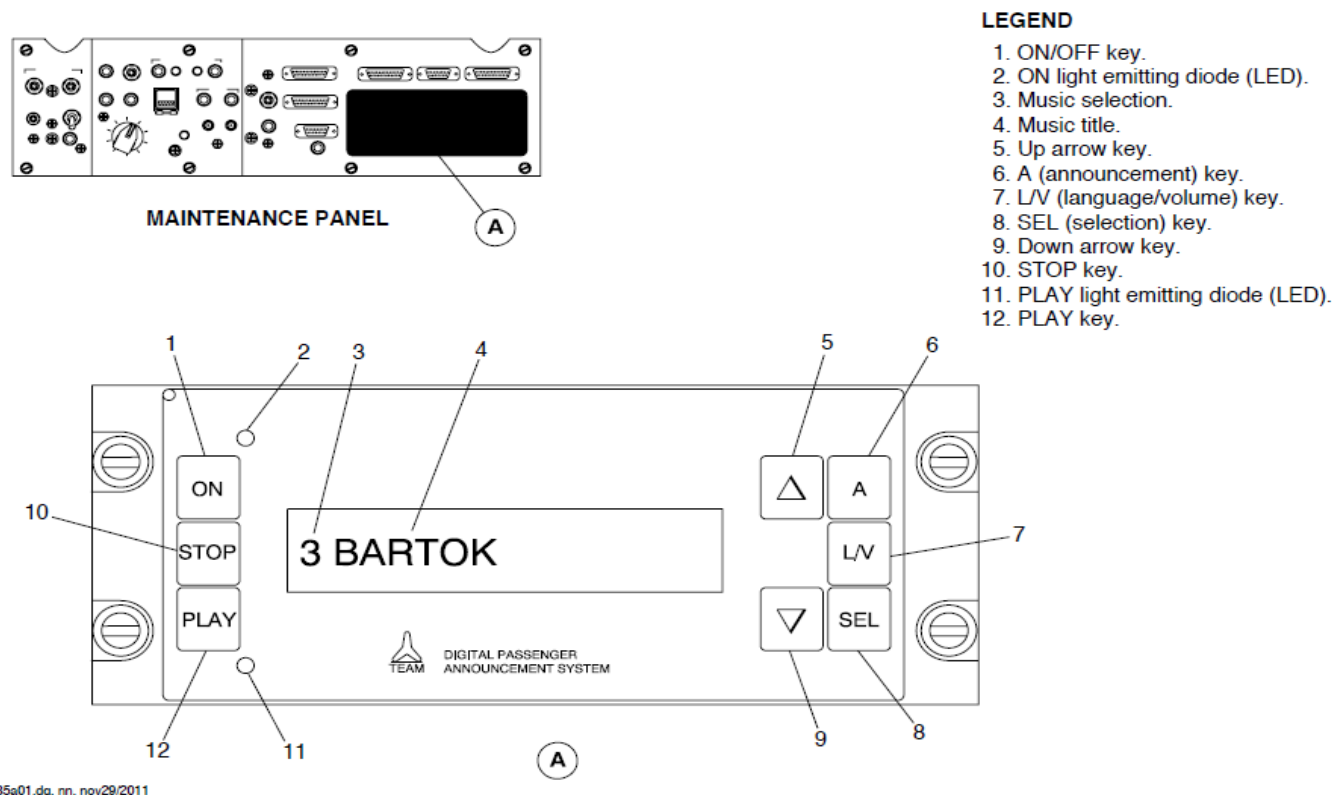
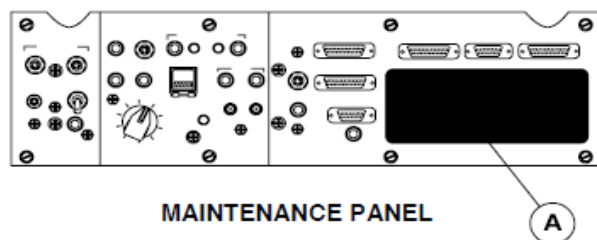
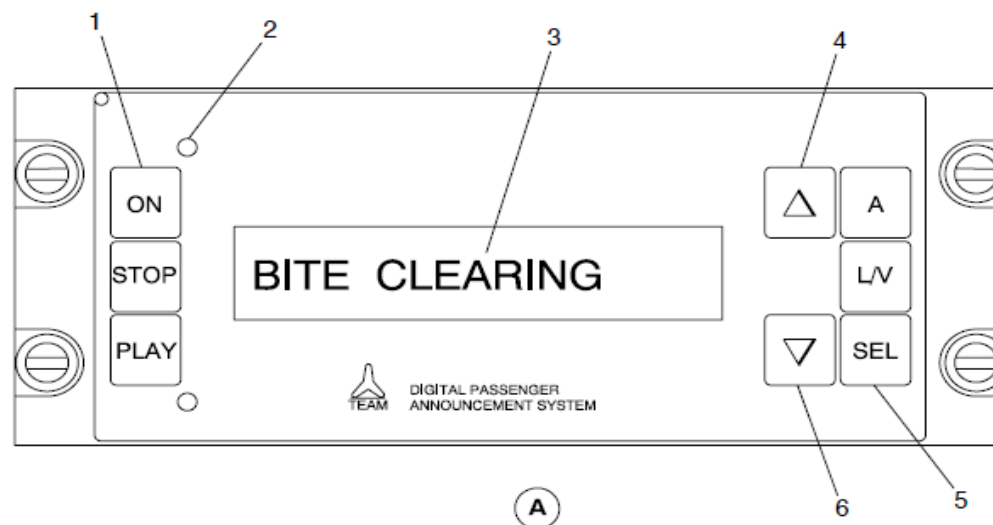


FIGURE 23- 35 PES, MUSIC SELECTION MODE



LEGEND

1. ON/OFF key.
2. ON light emitting diode (LED).
3. Bite clearing message.
4. Up arrow key.
5. SEL (selection) key.
6. Down arrow key.



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FIGURE 23- 36 PES, DIAGNOSTIC MODE

Discrete Functions

Figure 23- 37 DIGITAL PASSENGER ANNOUNCEMENT SYSTEM (DPAS) UNIT INTERNAL BLOCK DIAGRAM

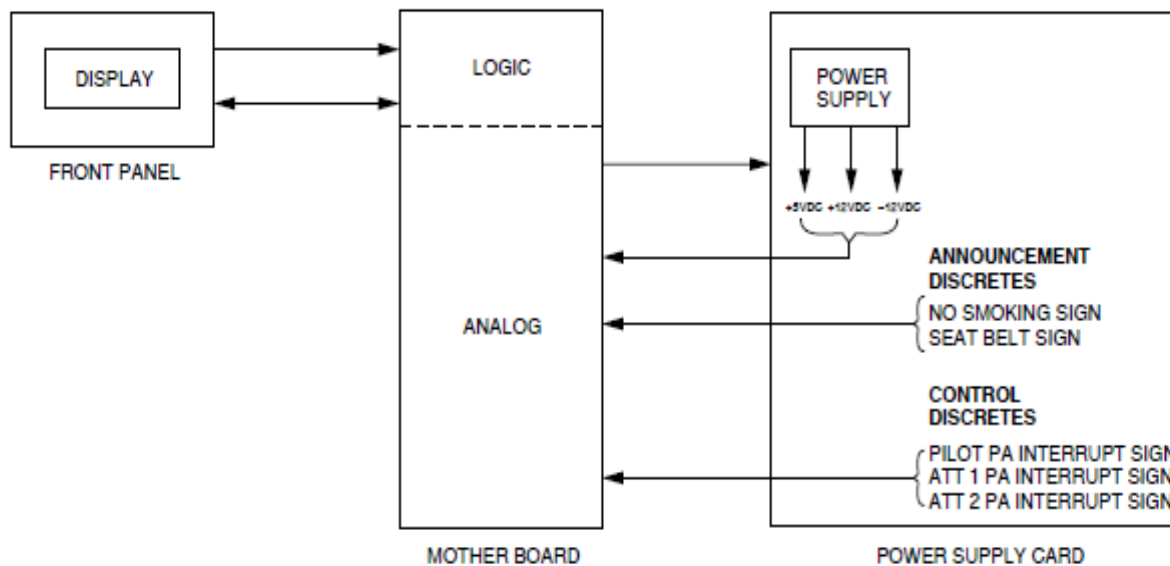
As per ARINC 715, up to five discretes can input to the PES to trigger specific functions automatically. Two discrete inputs are used to initiate the FASTEN SEAT BELT and NO SMOKING aural alerts in addition to the normally generated tones. The additional discrete inputs generated by the flight and cabin crew during public address announcements take priority over the FASTEN SEAT BELT and NO SMOKING discretes. The PES initiated announcements and music selections have the lowest priority.

During a passenger address announcement from the flight or cabin crew, the PAUSE discrete input causes the PES PTT output signal to change state to disable the PES PAA channel and for the PLAY LED to flash slowly. When the interrupting discrete is removed, the PES PTT output is applied and the PLAY LED will illuminate. The passenger announcements will start from the beginning in the language in progress and follow through with the languages not yet broadcast. The music audio continues from the point of interruption.

The PES interfaces with the discretes that follow:

PARAMETER	SOURCE
Aft FA Interrupt (Ground)	Aft FA Handset
PES PTT	PAA

PARAMETER	SOURCE
FASTEN SEAT BELT (28 V dc)	FASTEN SEAT BELT Toggle Switch
NO SMOKING (28 V dc)	NO SMOKING Toggle Switch
Pilot Interrupt (Ground)	RCAU
Forward FA Interrupt (Ground)	Forward FA Handset



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FIGURE 23- 37 DIGITAL PASSENGER ANNOUNCEMENT SYSTEM (DPAS) UNIT INTERNAL BLOCK DIAGRAM

Failure Mode and Annunciation

Figure 23- 31 PES, Test Mode

The DPAS unit performs internal monitoring and displays its status on power-up. The system monitoring capabilities are described by the modes listed.

System Test

The self-test mode automatically starts on the power-up after the ON key is pushed. All display readout pixels are activated for one second and the PLAY LED indicator will flash and the self-test will start. If there are no errors detected usually within five seconds after the test mode is started, the message TEST OK is annunciated for two seconds and the system enters into the announcement and broadcast mode.

If a fault is detected, the system will display one of the five messages that follow:

- SYSTEM FAULT
- FALSE CARD
- CARD MISSING
- KEYBOARD ERROR
- NO LANGUAGES.

To bypass the self-test power-up, press any front panel key other than the ON key. The system will then by default enter into announcement selection and broadcast mode. If the L/V key is pressed the system will enter into the language selection mode. If the system is energized but not in operation, it will continuously monitor for faults and show the applicable error message.

The error messages with failure descriptions are shown in the troubleshooting table that follows:

BITE MEMORY	DISPLAY MESSAGE	FAILURE DESCRIPTION
—	NO LANGUAGES	No language selection
KEY PRESSED	KEYBOARD ERROR	Key pressed too long in normal operation
KEYBOARD ERROR	KEYBOARD ERROR	Key pressed at power up
EXTERNAL RAM	SYSTEM FAULT	External memory failed
INTERNAL RAM	SYSTEM FAULT	Internal memory failed
MEMORY FAULT	SYSTEM FAULT	Program memory error
CARD FAULTY	SYSTEM FAULT	Memory card checksum error
CARD BUSY	FALSE CARD	Cannot read card data

BITE MEMORY	DISPLAY MESSAGE	FAILURE DESCRIPTION
CARD MISSING	CARD MISSING	Memory card not installed
FALSE CARD	FALSE CARD	Use of a non-valid memory card
BITE ERROR	—	Read/Write error in BITE memory
DISPLAY ERROR	—	Screen data different from written value
DISPLAY BUSY	—	Cannot write to display
BUS ERROR	SYSTEM FAULT	Bus malfunction
CONVERSION ERROR	SYSTEM FAULT	D/A converter error
ADDRESSING ERROR	SYSTEM FAULT	Error on address decoder
ARITHMETIC ERROR	SYSTEM FAULT	Error in arithmetic and logic unit

PES Failure Mode

Figure 23- 38 PES, Failure Mode, Minor

Figure 23- 39 PES, Failure Mode, Major

The system failure mode is entered when a failure message is posted on the DPAS display. When there are several failures the most important failure is displayed first. For minor failures the system operation can continue by pressing any key except the ON and L/V keys. The key selection will remove the failure message and allow system operation. For any major failures, the DPAS unit will remain disabled and deactivate any PTT discrete and audio outputs. Then ON key will only allow system shut-off.

The minor failures will show the messages that follows:

- NO LANGUAGE
- CARD MISSING
- FALSE CARD
- KEY BOARD ERROR

PES System Interface

The passenger announcement system interfaces with the components that follow:

- PAA for channel enable (PTT) and audio amplification
- RCAU for flight crew PES interrupt signal
- Forward attendant handset for PES interrupt signal
- Aft attendant handset for PES interrupt signal

Power Supply

The aircraft left main DC bus supplies 28 V dc electrical power through the A1 circuit breaker to the DPAS unit.

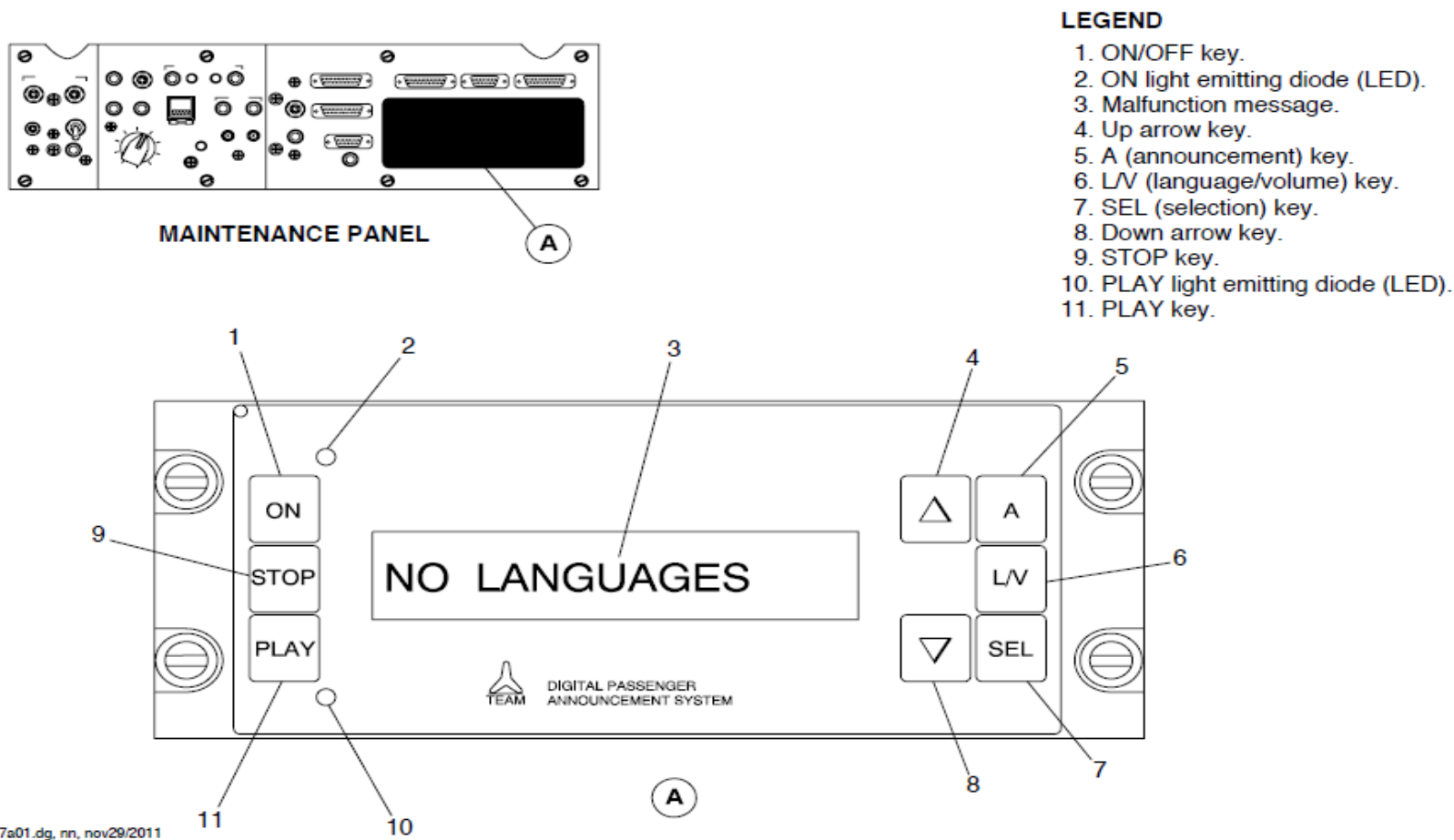
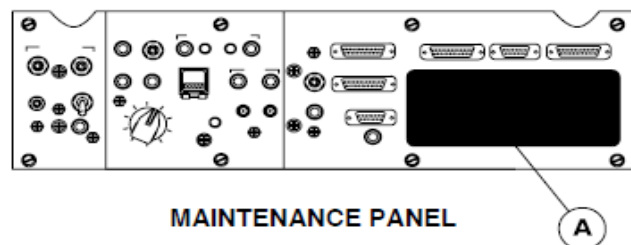
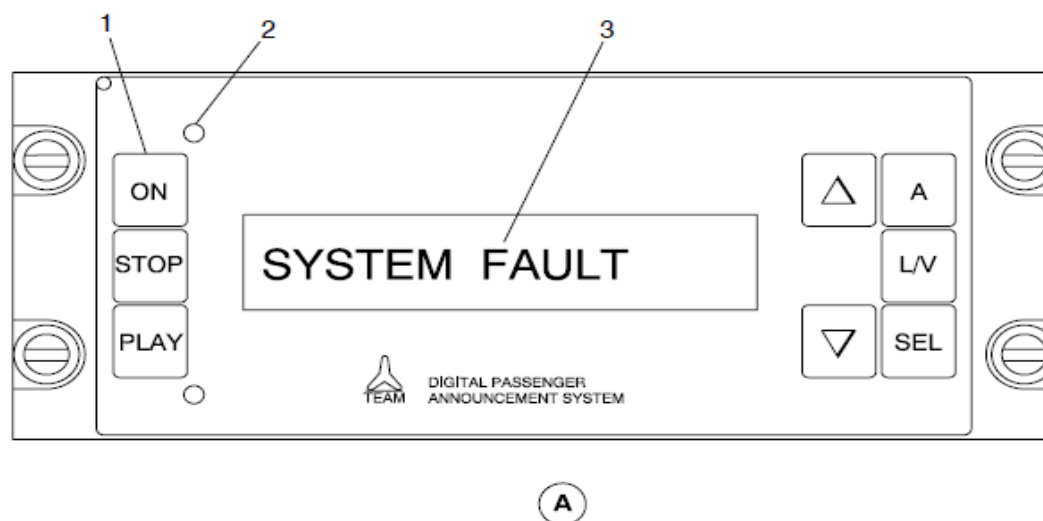


FIGURE 23- 38 PES, FAILURE MODE, MINOR



LEGEND

1. ON/OFF key.
2. ON light emitting diode (LED).
3. Malfunction message.



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FIGURE 23- 39 PES, FAILURE MODE, MAJOR

PCMCIA Memory Card

Figure 23- 40 PCMCIA Memory Card

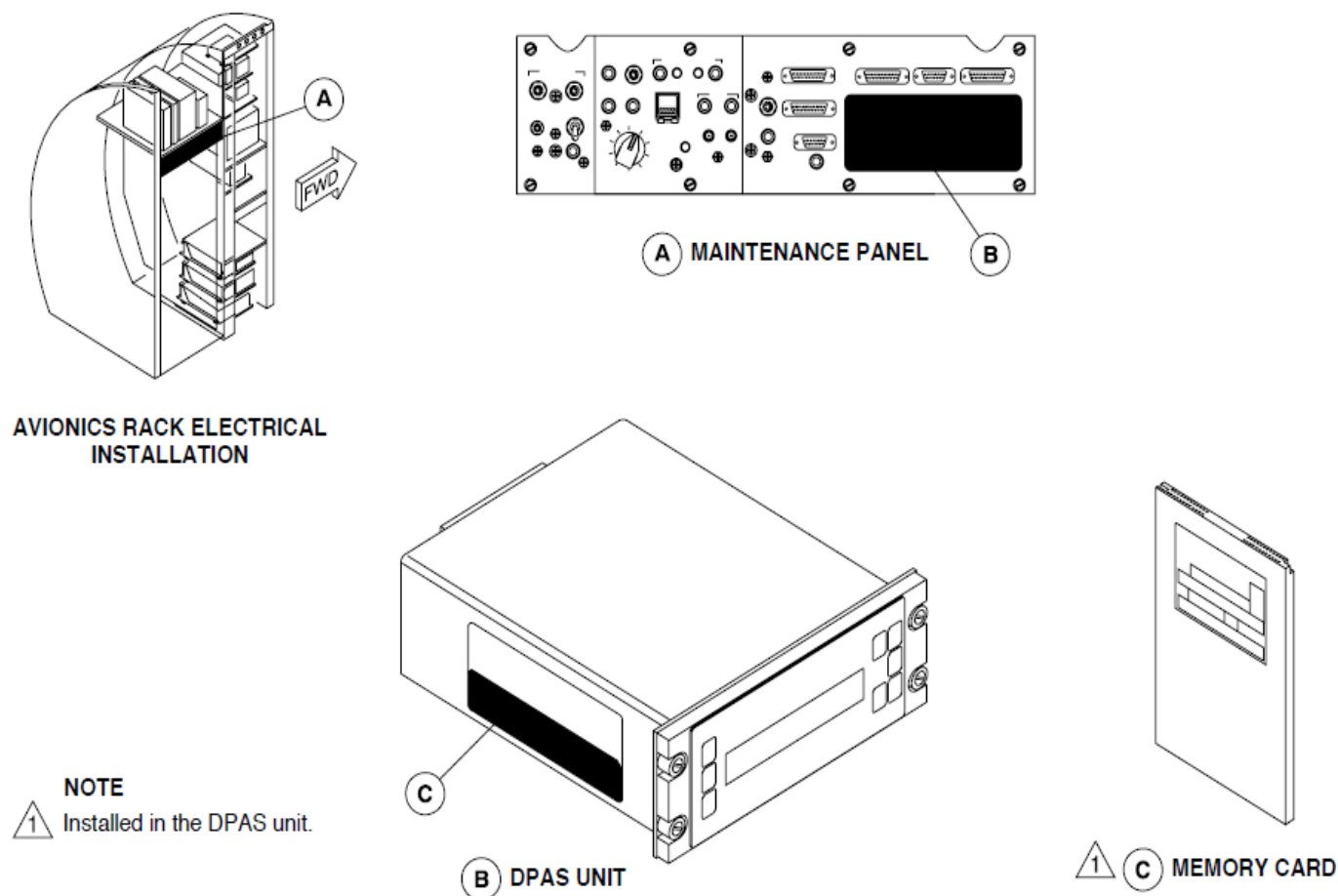
New DPAS units do not contain a PCMCIA memory card. The card is inserted in a slot located on the left side of the unit and is necessary for system operation. The passenger address and music selections are changed by reprogramming the card memory. A separate recording bench with a PC computer with integrated software allows recording, formatting and loading of announcements and music audio by the end user.

It can be programmed for a maximum of 279 minutes of passenger announcements or 139 minutes of music audio. Programing can include a combination of passenger addresses and music together that is less than or equal to a maximum of 279 minutes.

PCMCIA Memory Card Configuration Options

The memory card allows for configuration options that follow:

- Music is continuous and wraps around from the end of the playlist
- Music stops at the end of the playlist
- Music resumes after the announcement broadcast interruption
- If the music resumes after an interruption, it may do so from the beginning or from the point of interruption
- The title MUSIC and TEST ALL RECORD are placed in alphabetical order or at the end of the menu
- The duration period before the broadcast of an announcement can be varied.



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FIGURE 23- 40 PCMCIA MEMORY CARD

ACTIVE NOISE AND VIBRATION SYSTEM (ANVS)

Introduction

The Active Noise and Vibration System (ANVS) decreases the noise level in the passenger cabin caused by the aircraft propellers. The system supplies secondary noise and vibrations in the passenger compartment. These secondary noise and vibrations field cancel the primary noise and vibration field. The primary and secondary noise and vibration field mix to produce lower cabin noise and vibrations.

General Description

Figure 23- 41 Active Noise and Vibration System (ANVS) – Block Diagram
Figure 23- 42 ANVS Block Diagram

The ANVS goes into the control mode when the aircraft is in flight. The system clears when the Weight-On-Wheels (WOW) input shows that the aircraft is on the ground.

The ANVS interfaces are as follows:

- Proximity Sensor Electronic Unit (PSEU) – WOW input
- Cabin Pressurization Control System (CPCS) – pressure input
- Propeller Electronic Control Unit (PECU) – tachometer outputs
- Maintenance Panel – ANVS SYS DEGRADED LED
- Flight Attendant Panel – ANVS control
- Aircraft ID Variant – storage code for aircraft identification.

The ANVS contains the components that follow:

- Controller
- The ANVS Control Panel
- Actuators – Active Tuned Vibration Attenuators (ATVA)
- Power Amplifiers (PA)
- Sensors – Microphones and Accelerometers
- Vibration Transducers (installed on the engine gearbox)
- Power Distribution Box – electrical power distribution

The location of the ANVS components is as follows:

- The controller is installed below the floor panels between stations X258.500 and X281.000 in the passenger compartment.
- The ATVAs and PAs are installed between the aircraft structure and the internal part of the trim panels.
- The sensors are installed in the ceiling panels, sidewall panels, overhead stowage bins, Passenger Service Units (PSU) and flight attendant seat.
- The ANVS control panel is a part of the flight attendant panel.

The controller monitors inputs from these sources:

- Weight-on-wheels
- Sensors
- Aircraft propeller frequency reference signals
- Pressure
- Vibration transducers.

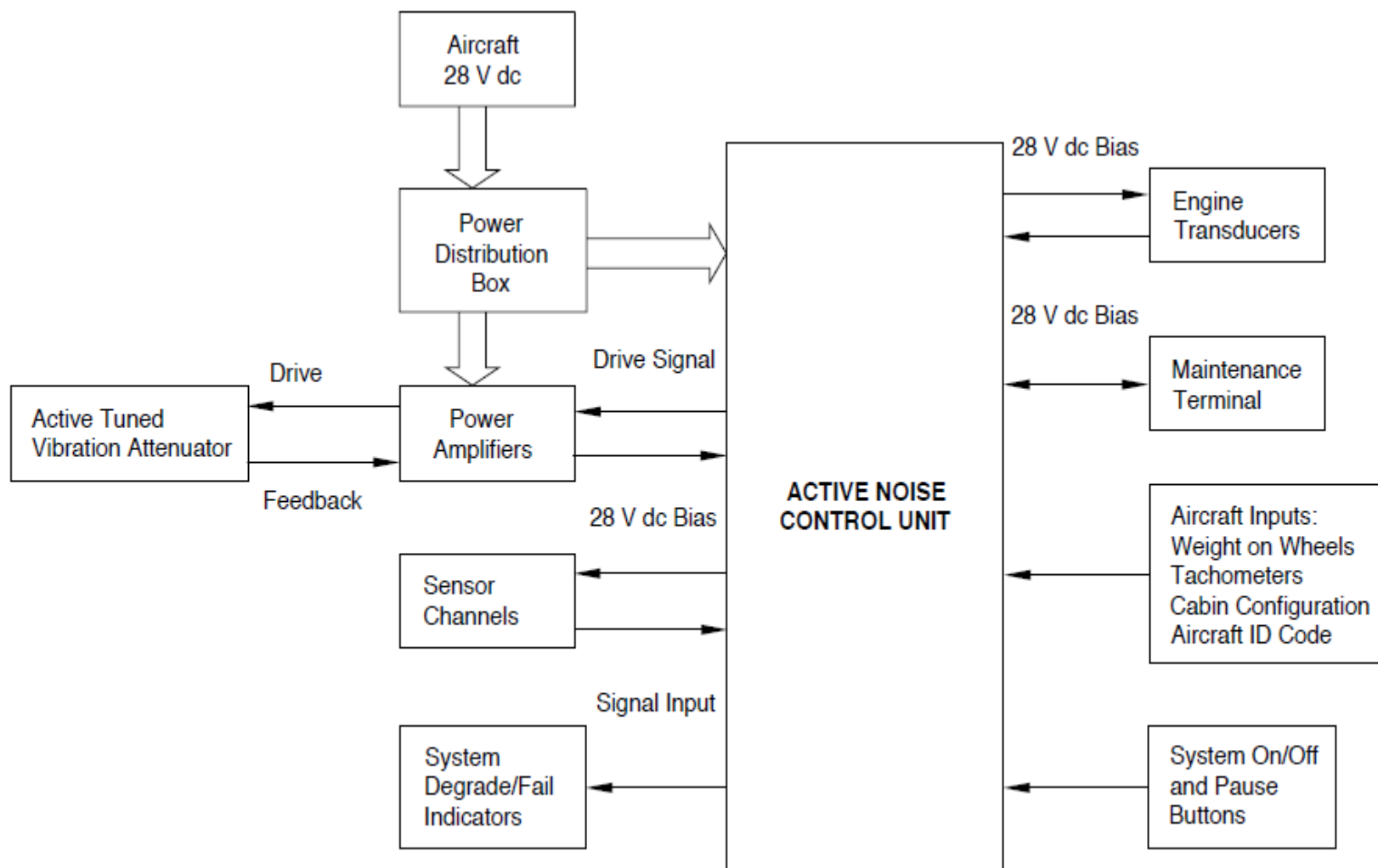


FIGURE 23- 41 ACTIVE NOISE AND VIBRATION SYSTEM (ANVS) – BLOCK DIAGRAM

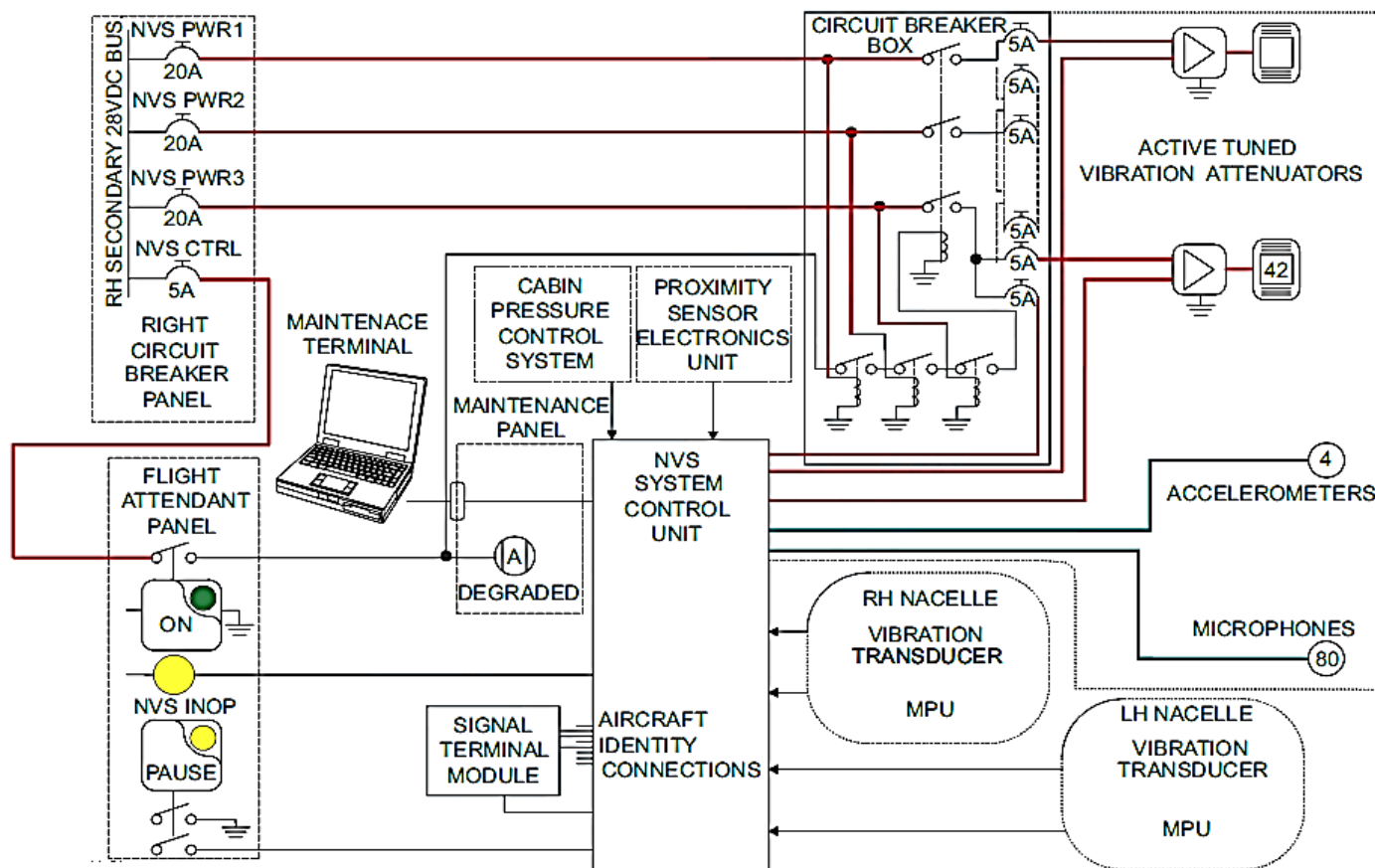


FIGURE 23- 42 ANVS BLOCK DIAGRAM

Detailed Description

A detailed description of the ANVS components follows.

Controller (ANCU – Active Noise Control Unit)

Figure 23- 43 Active Noise Control Unit (ANCU) Installation

The controller is an ARINC 600 4MCU unit attached to the aircraft structure through four support legs.

The controller processes the propeller frequency data and the cabin noise levels from the sensors. The outputs are control signals that go to the actuators through the power amplifiers.

The controller processes the basic frequencies of the propeller related noise. The frequency data is supplied by the PECU (tachometer outputs).

Electrical connections to the controller are through five radial EPX connectors and one circular MIL-C-38999 connector. The radial connectors are identified as J1 through J4 and J6. The circular connector is identified as J7. Connectors J1 through J4 and J6 are installed on the rear panel of the controller housing. Connector J7 is installed on the controller front panel.

Propeller Balance Monitoring System (PBMS) electrical connections to the controller are through connector J7. A D-type connector (J5) is also installed on the front panel for connection to a maintenance terminal.

A secondary function of the ANVS is to record and process propeller balance data as follows:

- Vibration data is collected from a vibration transducer mounted on each propeller gearbox.
- Vibration data is collected automatically during flight or with the PBMS terminal during an engine ground run. The data is read from the controller with the PBMS terminal software.
- The PBMS terminal software automatically calculates and stores the vibration data necessary to make sure of the level of propeller unbalance during usual revenue flight operations.
- The PBMS terminal software then calculates the propeller balance data. This is used to calculate the redistribution of weights in the propeller balance plane necessary for propeller balance in the specified limits during ground maintenance.

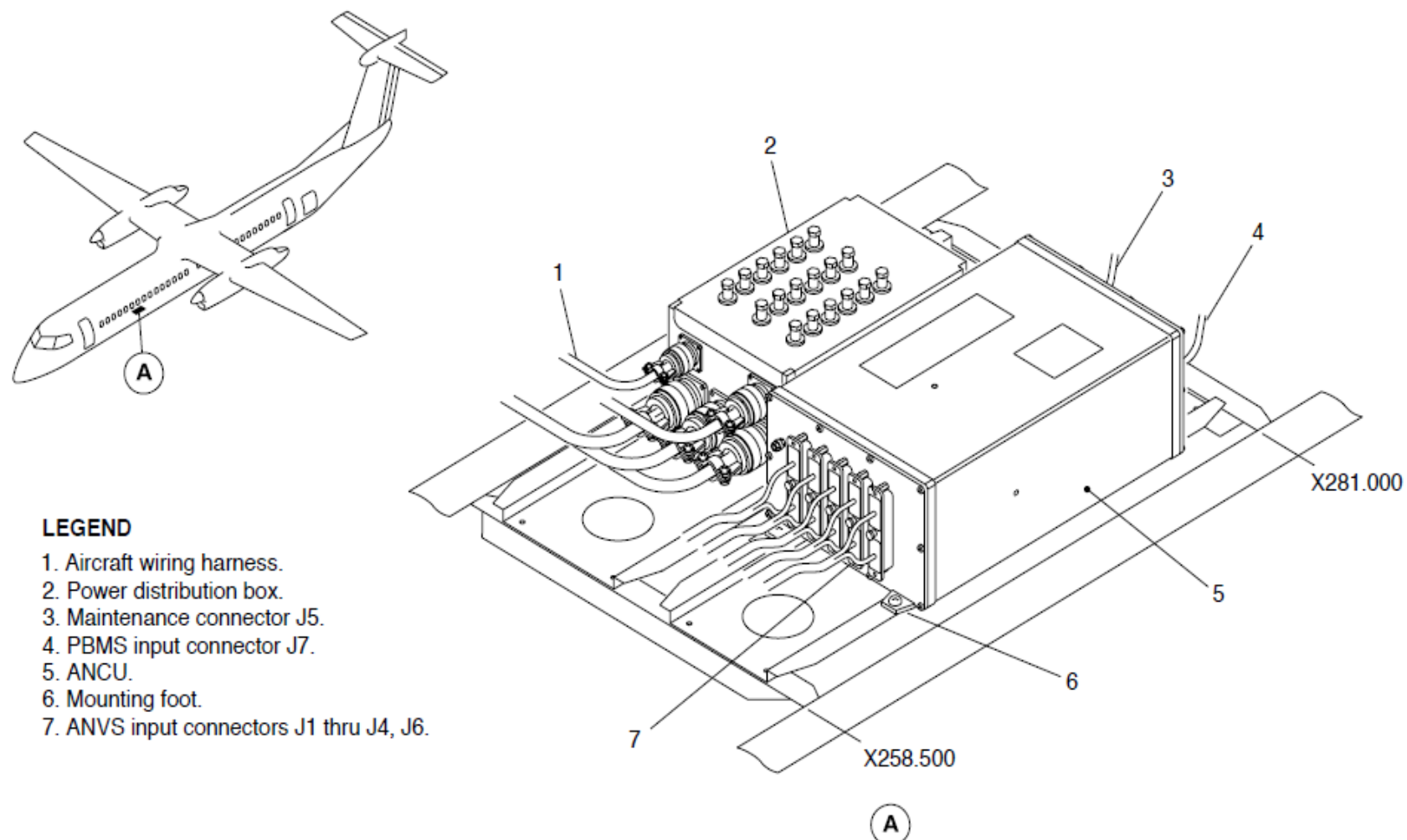


FIGURE 23- 43 ACTIVE NOISE CONTROL UNIT (ANCU) INSTALLATION

Active Tuned Vibration Attenuator/Actuator (ATVA)

Figure 23- 44 Active Tuned Vibration Attenuator (ATVA) Installation

The ATVA is an electromagnetic actuator that is connected to a power amplifier. The power amplifier drives the ATVA. The ATVA gives vibration levels related to the amplitude and frequency of the signals from the PA.

Power Amplifier

Figure 23- 45 Power Amplifier (PA) Installation

The unit is a high efficiency class D switching amplifier. It is connected to the controller output drive signal with a “quick–disconnect” connector twisted pair cable.

Power Amplifiers (PA) are contained in sealed aluminum housings. Each amplifier is attached to the actuator bracket and gusset plate.

The aircraft 28 V dc bus supplies electrical power to the PA. It also supplies up to 20 watts (rms) of power to its associated ATVA.

Feedback signals from the actuators to the controller control the vibration intensity levels.

Sensors – Microphones

Figure 23- 46 Sensor Installation – Side Walls and Ceiling Panels (Typical)

Figure 23- 47 Sensor Installation – Side Walls and Ceiling Panels (Typical)

Figure 23- 48 Sensor Installation – Overhead Storage Bins (Typical)

Miniature electric microphones connected to interface circuitry are encapsulated in sealed molded enclosures. Each enclosure is attached to a trim panel through a mounting plate and gasket on the side wall, ceiling panel and overhead storage bins.

There are 25 microphones installed on the side wall, 13 microphones installed on ceiling panel and 42 microphones installed on overhead storage bins.

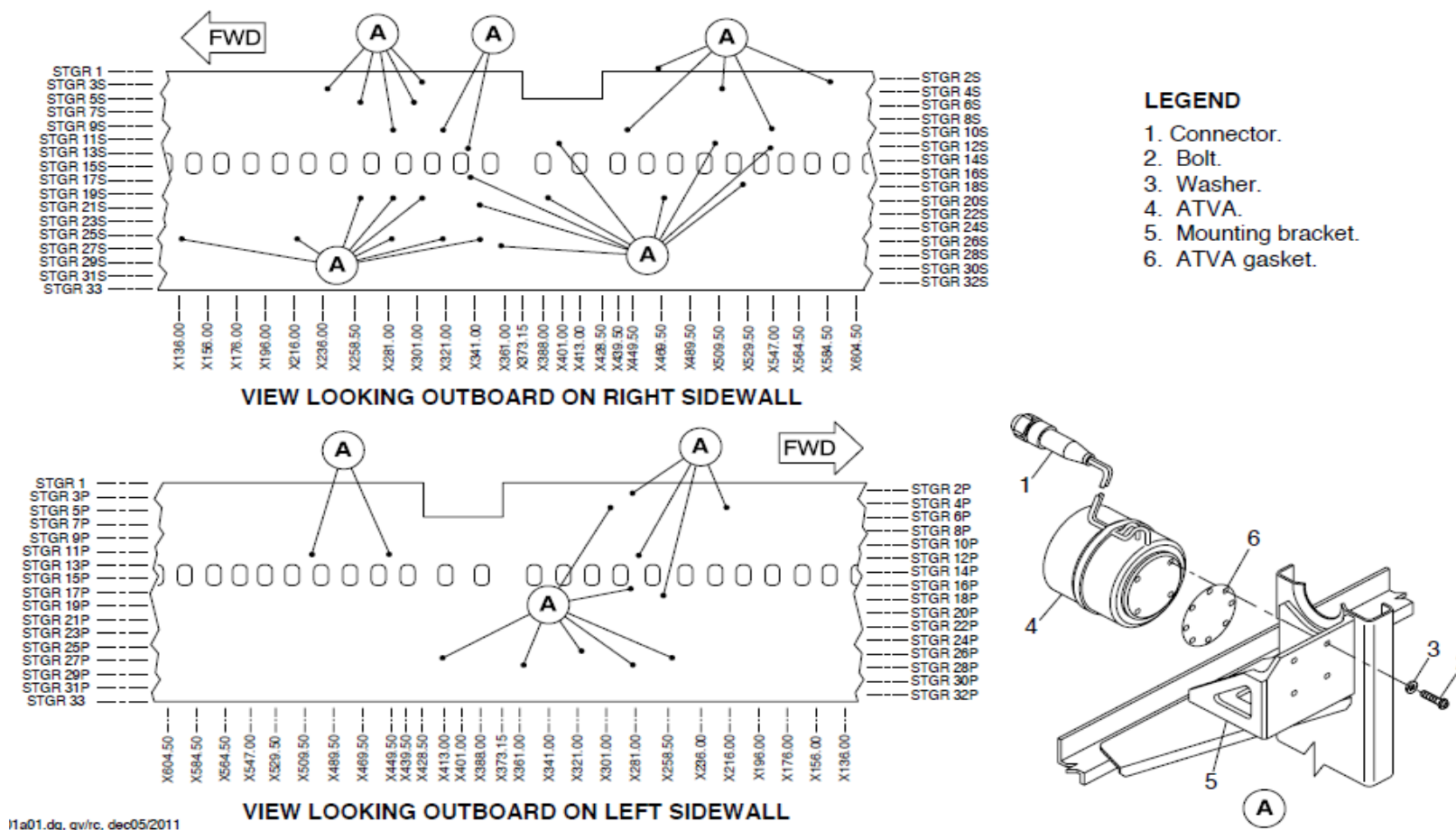
The controller supplies a bias voltage of 22 to 28 V dc to each microphone through a twisted pair cable. The microphone output signals are sent back to the controller superimposed on the bias voltage.

Sensors – Accelerometers

Figure 23- 49 Sensors/Accelerometer (Floor Sensor) Installation

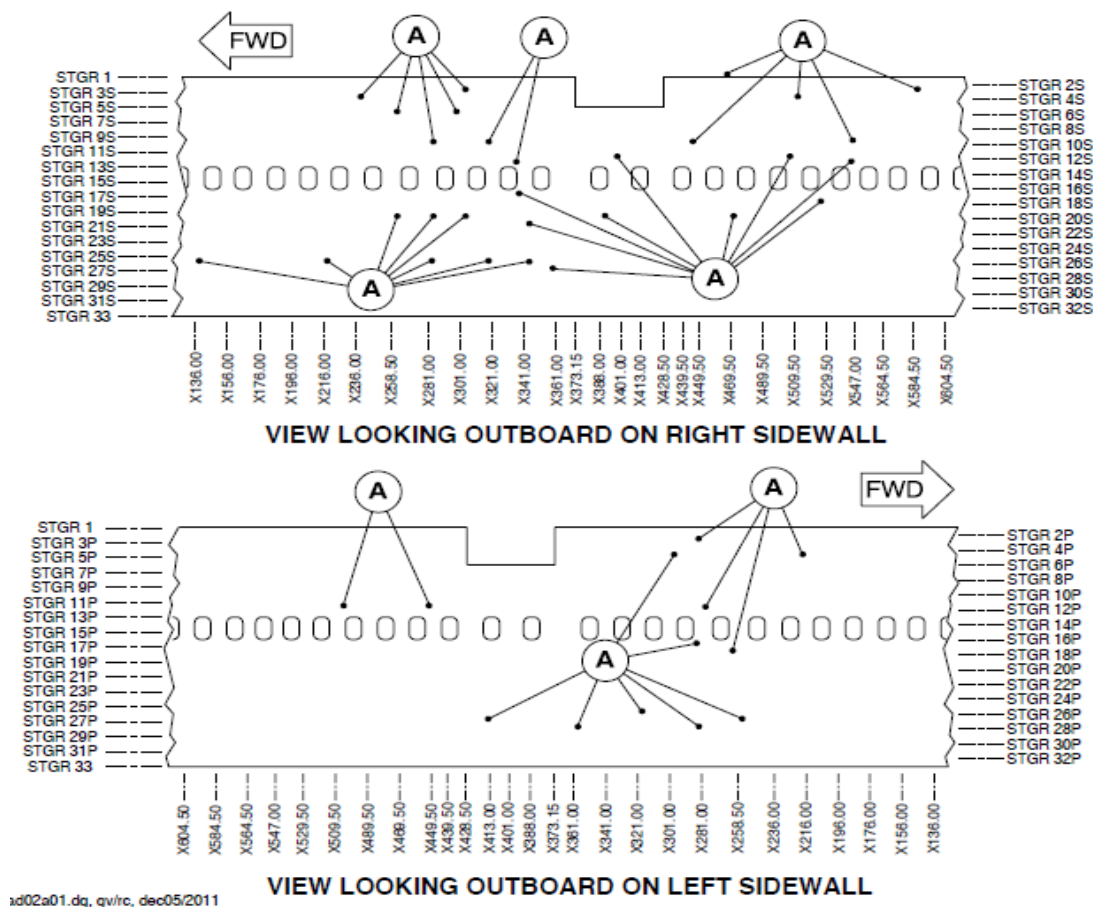
The accelerometers used in the ANVS are small piezo electric sensors. The sensors are connected to interface circuitry and contained in sealed metal enclosures. Each enclosure is held in position with a bracket installed on the seat rail at stations LHS X 176 and X 236 and RHS X 204 and X 301.

The controller supplies a bias voltage of 22 to 28 V dc to each accelerometer through a twisted pair cable. The accelerometer output signals are sent back to the controller superimposed on the bias voltage.



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FIGURE 23- 44 ACTIVE TUNED VIBRATION ATTENUATOR (ATVA) INSTALLATION



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LEGEND

1. Electrical connector.
2. Electrical connector.
3. Screw.
4. PA bracket.
5. Power amplifier (PA).
6. Nut.
7. Washer.
8. Stringer
9. Washer.

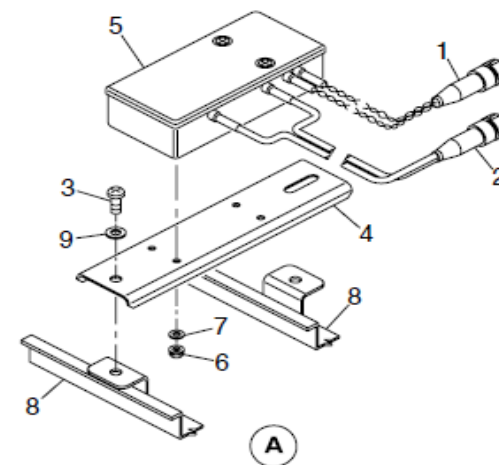


FIGURE 23- 45 POWER AMPLIFIER (PA) INSTALLATION

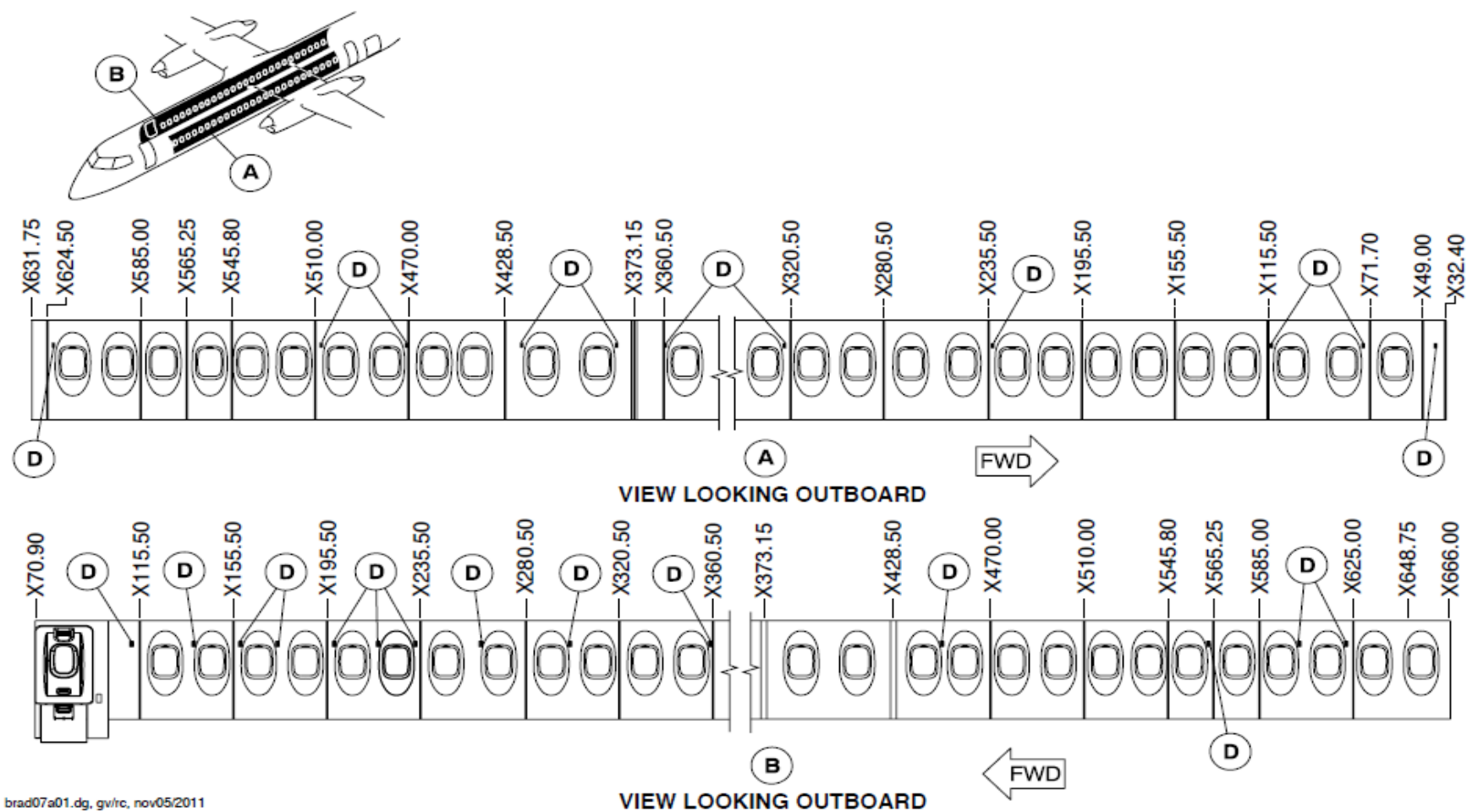
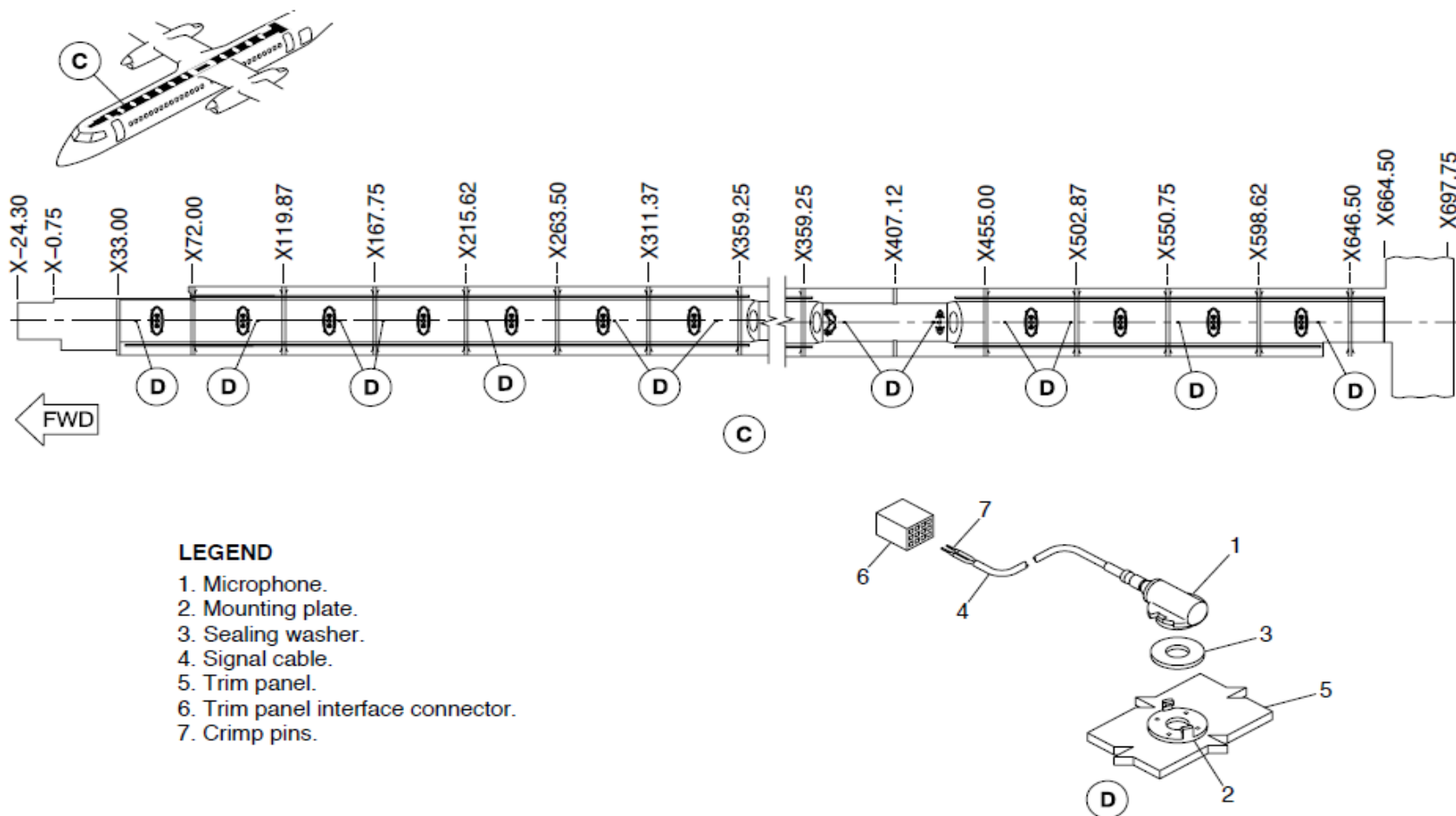


FIGURE 23- 46 SENSOR INSTALLATION – SIDE WALLS AND CEILING PANELS (TYPICAL)



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FIGURE 23- 47 SENSOR INSTALLATION – SIDE WALLS AND CEILING PANELS (TYPICAL)

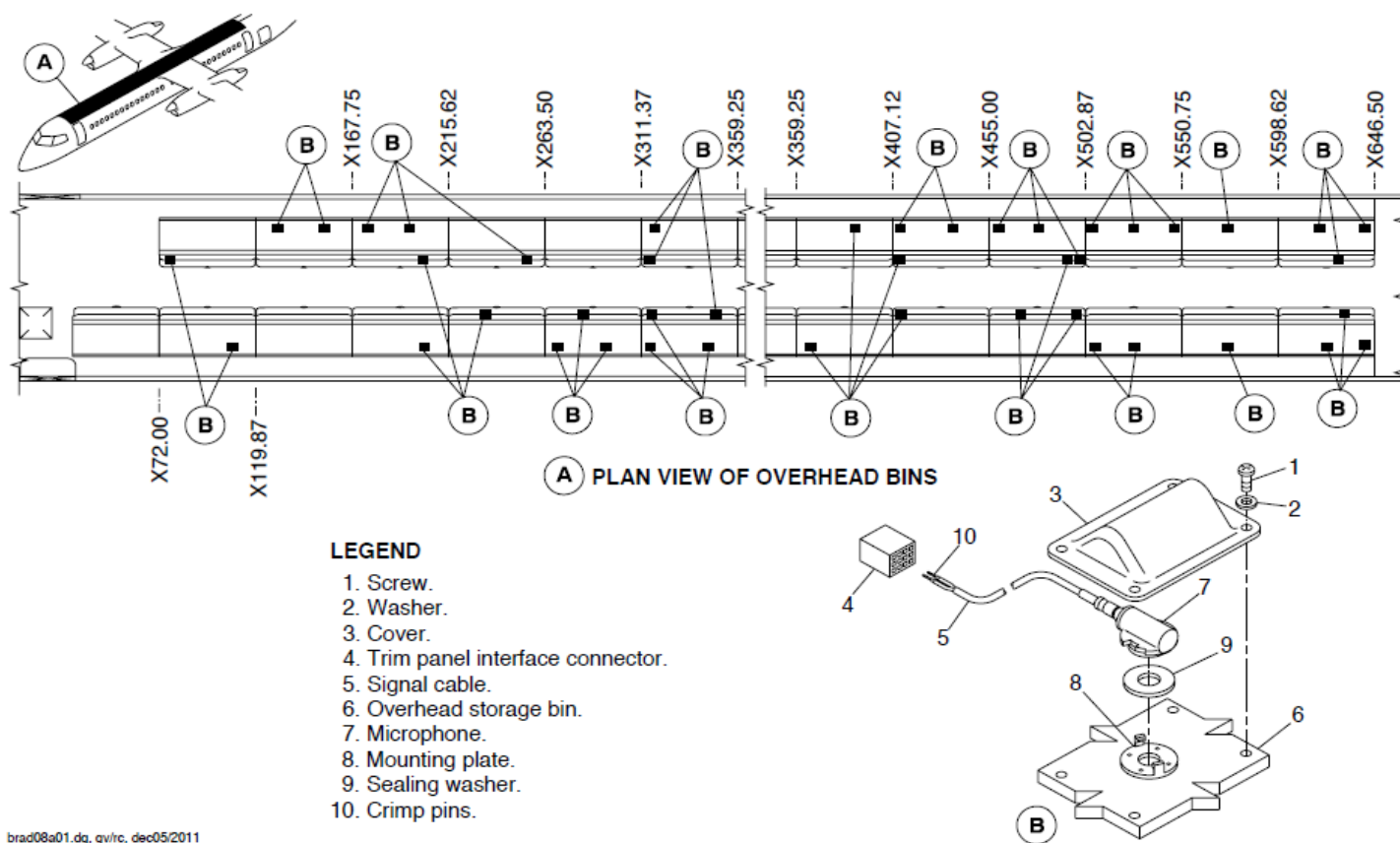
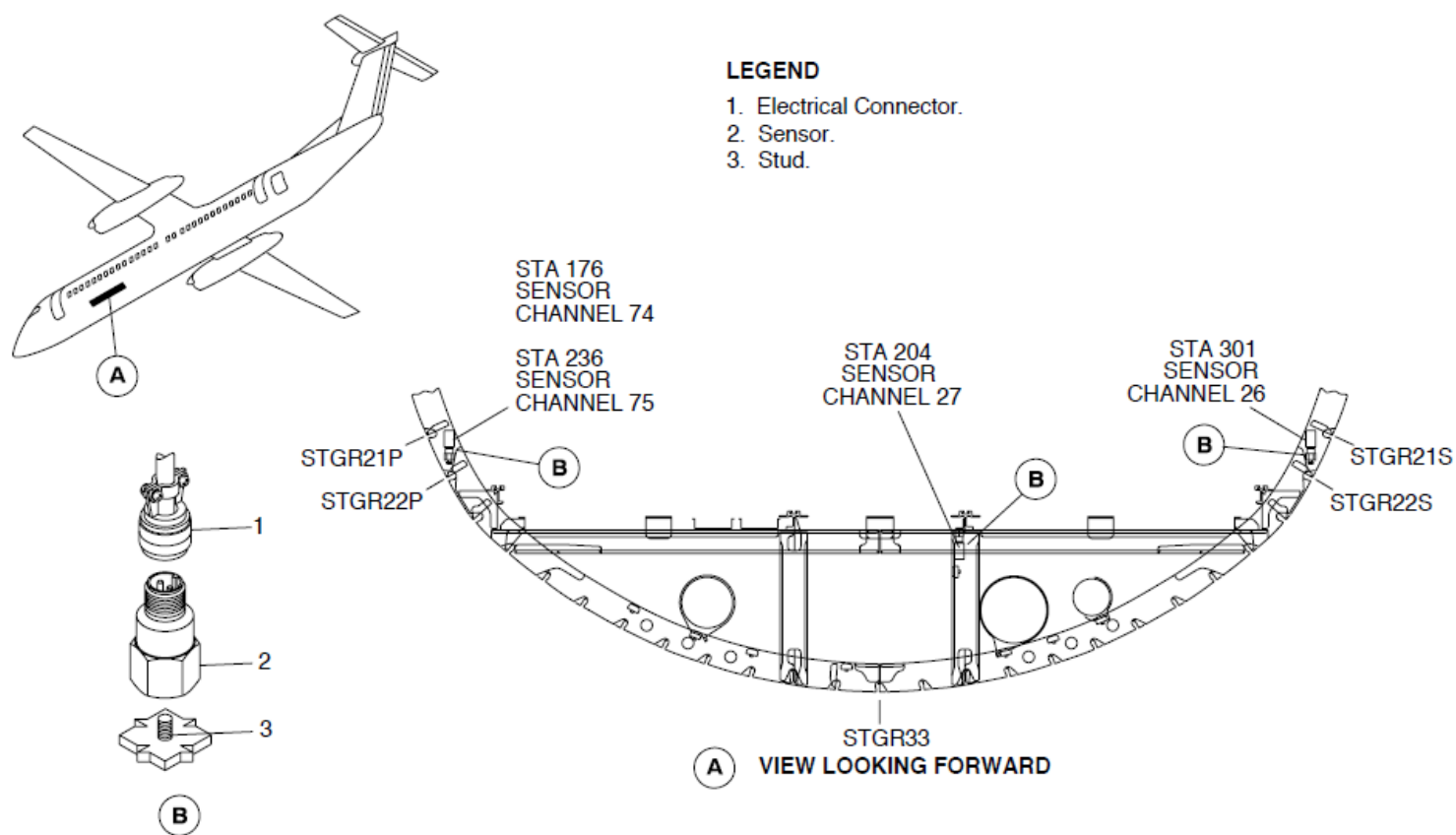


FIGURE 23- 48 SENSOR INSTALLATION – OVERHEAD STORAGE BINS (TYPICAL)



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FIGURE 23- 49 SENSORS/ACCELEROMETER (FLOOR SENSOR) INSTALLATION

Vibration Transducers

Figure 23- 50 Vibration Transducer Installation

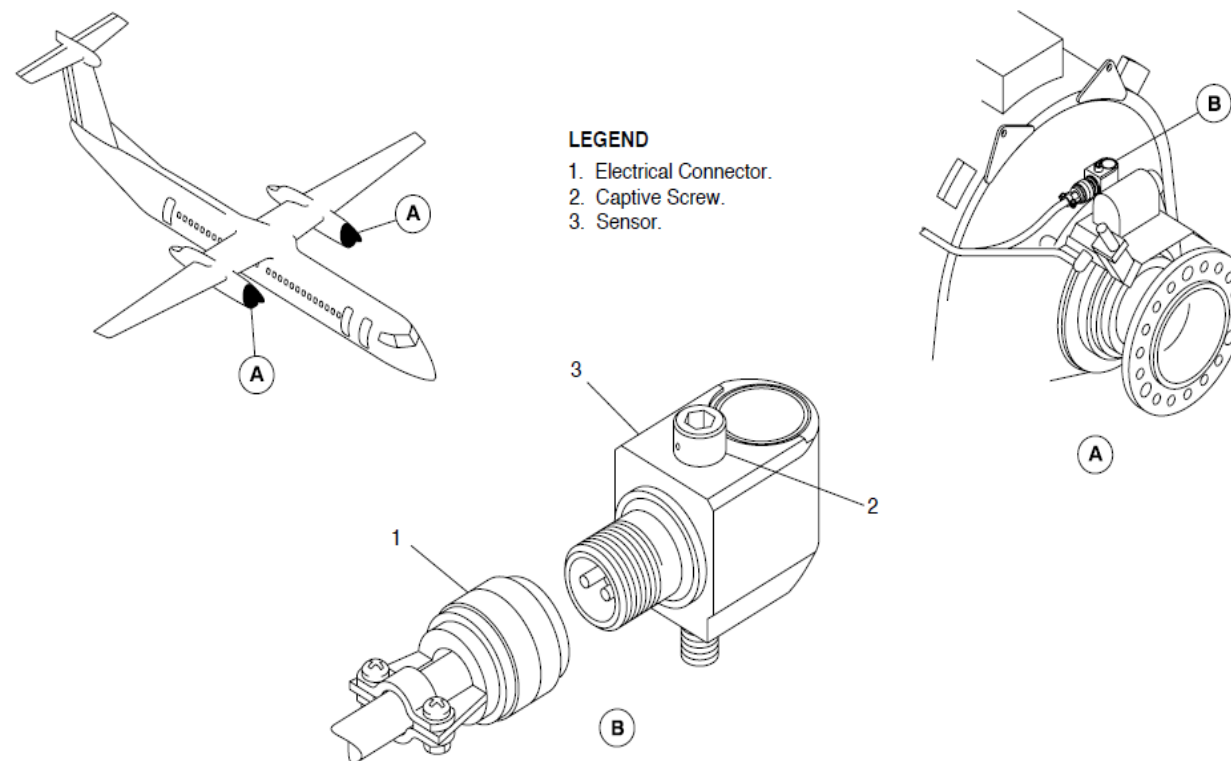
The vibration transducer is a small piezo electric sensor with interface circuitry contained in a sealed metal case. The transducer is attached with a bracket to the engine gearbox.

Electrical power is supplied to the transducer from the controller as a bias voltage from the ANCU circular connector J7. Feedback signals from the vibration transducer to the controller also control the vibration intensity.

Power Distribution Box

Figure 23- 43 Active Noise Control Unit (ANCU) Installation

The Power Distribution Box is installed adjacent to the controller. It contains circuit breakers and relays for the application and control of electrical power to the system.



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FIGURE 23- 50 VIBRATION TRANSDUCER INSTALLATION

ANVS Control Panel (Part of the Flight Attendant Panel)

Figure 23- 51 Flight Attendant Control Panel

The ANVS is labelled NVS SYSTEM on the flight attendant panel. The panel is located on the left side of the passenger compartment on the overhead bin end cap assembly adjacent to the airstair door.

The ANVS control panel contains two push-button switches and four lights for control and indication of the status of the ANVC system. The two switches are labelled ON/OFF and PAUSE. The LED labelled NVS INOP is installed below the push-button switches.

When the ON/OFF switch is pushed, its indicator comes on and the ANVS is energized. This switch should be left on at all times, unless the system failure occur that requires the ANVC system to be shut down.

The ANVC system should be switched off during the aircraft maintenance. The off status light is illuminated when the ANVC system is powered down using the ON/OFF switch.

NOTE

The ON/OFF switch turns the ANVS on or off. This switch must be pushed and held for 4–5 seconds for it to change condition.

When the PAUSE push-button switch on the flight attendant panel is pushed, its indicator comes on and the ANVS goes into the standby mode. The power is supplied to the ANVC system, however it is not controlling the noise and vibration levels in the aircraft.

The pause switch is used for maintenance of the ANVC system.

NOTE

The PAUSE function can only be activated when the ANVS is on.

NOTE

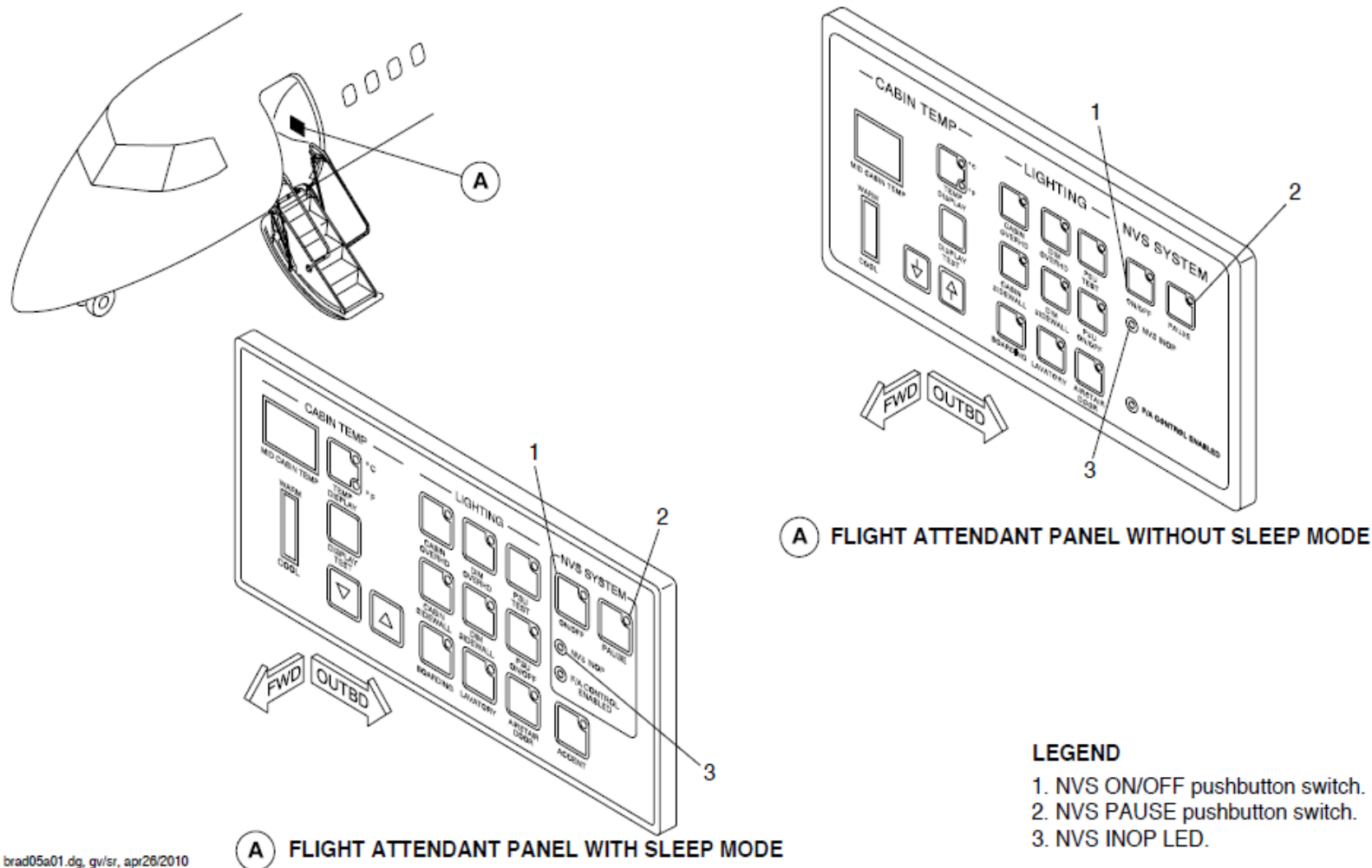
The NVS INOP LED comes on when there is a fault in the NVS system or when the NVS is off.

There are two types of flight attendant panels:

- Flight attendant panels without sleep mode (Wessex Advanced Switching Products)
- Flight attendant panels with sleep mode (EMTEQ)

The flight attendant panel with the sleep mode has a power-saving feature to conserve the battery power when the aircraft is not in use. When only aircraft battery power is available, the flight attendant panel will monitor the status of the boarding lights only. The boarding lights are the only system controlled from the flight attendant panel that gets electrical power directly from the aircraft batteries. If the boarding lights are in the off position, the panel will go into the sleep mode. In the sleep mode the panel minimizes power consumption and appears de-energized.

When the boarding lights pushbutton is pushed, the boarding lights will come on and the panel comes out of the sleep mode. The panel will also come out of the sleep mode when the electrical busses that supply the other systems controlled from the flight attendant panel are energized. The cabin temperature, the NVS, and the lights other than the boarding lights, get their electrical power from the 28 VDC main, essential or secondary busses. When power is applied, these functions will remember their last condition (on or off).



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FIGURE 23- 51 FLIGHT ATTENDANT CONTROL PANEL

Maintenance Panel

Figure 23- 52 Maintenance Panel

The maintenance panel is a part of the Central Diagnostics System (CDS). It is used to access and clear (reset) fault and degraded conditions. The panel is used only when the ANVS is in the Maintenance Mode and with the aircraft on the ground.

The ANVS SYS DEGRADED LED is installed on the maintenance panel. The maintenance panel is installed in the avionics rack on the left side of the aircraft between stations X-1.75 and X-21.75.

Propeller Balance Monitoring System (PBMS)

During flight, the controller processes the outputs of the two vibration transducers. The controller also processes engine speed and amplitude data.

The controller automatically records and stores the data necessary to make sure the level of unbalance is within specified limits.

The ANCU continuously monitors the PBMS vibration transducer inputs.

Defective vibration transducers can be confirmed with the Maintenance Terminal before any balance data is used for propeller balance. Propeller balance data can be collected on the ground during an engine ground run with the PBMS Terminal.

The unbalance data stored in the controller can be retrieved through a serial interface to the PBMS Terminal. Up to 70 hours of rolling flight data can be stored at any one time.

The PBMS Terminal interprets the data to display vibration levels and calculate balance solutions. The balance solutions define a distribution of masses necessary to get the balance within specified limits.

Components of the PBMS include the ANCU, the engine transducers, the PECU tachometers, the PSEU, and the PBMS Terminal.

The controller input signals (through connector J1) use the same WOW signal as used by the ANVS. This is used to identify when the aircraft is on the ground. Other ANCU input signals through connector J1 are:

- Tachometer signals – supply the controller with propeller rpm data
- Indicated Air Speed (IAS) – supplied by the PECU
- External temperature – supplied by the PECU
- PBMS engine transducer “health” – When flight/vibration data is sampled, the “health” of each engine transducer is also recorded. If the engine transducer health is “bad” at the time of the sample, then that sample will not be used in the balance solution.

The PBMS Terminal contains Ultra Electronics proprietary software that runs on a PC. The basic functions that follow are supplied:

- Reads data from the controller and stores the data on the terminal hard drive.
- Shows the initial moment and vibration levels.
- Calculates the balance weight installation necessary to keep propeller vibrations at a minimum.
- Prints out the initial and final moment and vibration data.

ANVS Operation

The ANVS has four modes of operation:

- Start-Up
- Standby
- Active Control
- Maintenance.

Start-Up Mode

When power is applied to the ANVS or the system is reset, the controller does a series of self-tests. This is done to make sure all the components are operating correctly.

If the controller fails any of the tests, the system STATUS LED on the ANCU front panel changes from green to red. The NVS INOP LED on the flight attendant panel comes on and the controller goes into the maintenance mode.

If all tests are passed, the controller is initialized from configuration data, calibration data, and system status data stored in its nonvolatile memory.

Configuration Data

The configuration data contains all the parameters that define the system configuration for the series of aircraft installation. Each aircraft installation is identified by the aircraft identification variant (aircraft ID) inputs to the controller. The aircraft ID variant is an 8-bit hard wire code on aircraft connector J1A.

The code is stored with the configuration parameters and on initialization, the ANCU checks the identity of the configuration data against the aircraft ID.

If the configuration data is correct, the calibration data is expanded from the compressed form stored in the controller nonvolatile memory. Its format is then checked against the necessary format of the configuration parameters.

If the configuration data is invalid, the system STATUS LED on the controller front panel changes from green to red. The NVS INOP LED on the flight attendant panel comes on and the controller goes into the maintenance mode.

System Status Data

After the calibration data has been checked, the system status is read. The data gives information on the status of the microphones and ATVAs. Any microphone and/or ATVA that has failed is disabled and will not be used in the active control mode.

The number of failed microphones and ATVAs is compared with two limits set in the system configuration data.

If the number of failures is more than the degraded limit, the system STATUS LED on the controller front panel changes from green to orange. The ANVS SYS DEGRADED lamp on the maintenance panel comes on. The system will still operate in the control mode with degraded performance. No maintenance action is necessary.

If the number of failures is more than the failed limit, the system STATUS LED on the controller front panel changes from green to red, the NVS INOP LED on the flight attendant panel comes on and the controller goes into the maintenance mode. The system will not operate until a maintenance session has been done and all faults are corrected.

When the ANVS is serviceable, the system STATUS LED on the controller front panel is green.

If there is a system or controller fault that the system cannot clear automatically, then the system goes into the maintenance mode. The system STATUS LED on the controller front panel changes from green to red, and the NVS INOP LED on the flight attendant panel comes on.

Standby Mode

The system goes into the standby mode when there are no output control signals from the ATVAs through the power amplifiers.

The controller continually monitors the propeller tachometer and cabin pressure inputs. Pressure data is supplied by the Cabin Pressurization Control System (CPCS) through the ARINC 429 bus. If the tachometer or pressure inputs are within range, the system goes into the active control mode. If the tachometer or pressure input goes out of range, the system goes into the standby mode.

NOTE

The system goes into the control mode when the tachometer input frequency goes from 85 – 102 Hz (angular speed of 850 – 1020 rpm) and the cabin pressure input goes from 0–6 psi.

Active Control Mode

During the control mode the controller monitors the microphone channels, which send an inaudible test tone to each ATVA. The tone is measured at each microphone.

If faulty actuators or microphone channels are found, those actuators or channels are disabled for the remainder of the flight. The faulty actuators or microphones are restored when the WOW signal changes from air to ground.

Any actuator or microphone that is faulty for three successive flights is permanently failed until maintenance action is taken.

If the number of permanent failures is more than the degraded limit, the system STATUS LED on the controller front panel changes from green to orange. The ANVS SYS DEGRADED LED on the maintenance panel comes on. The system still operates in the control mode with degraded performance. The degraded limit is set at two failed channels.

If the number of ATVA or microphone channels is more than the failed limit, the system STATUS LED on the controller front panel changes from green to red. The NVS INOP LED on the flight attendant panel comes on and the controller goes into the maintenance mode. The failed limit is set at four failed channels.

One failed ATVA or microphone channel does not cause the ANVS SYS DEGRADED LED or the NVS INOP LED to come on.

Two failed channels cause the ANVS SYS DEGRADED lamp to come on, and four failed channels cause the NVS INOP LED to come on.

When the PAUSE push-button switch on the flight attendant panel is pushed, its indicator comes on and the system is held in the standby mode. In this condition, if the aircraft power is switched off and on again, the system continues to be in the standby mode until the PAUSE switch is pushed again. At that time, its indicator goes off and the system starts again.

The end of a flight is identified when the WOW input shows the aircraft has landed. The controller goes off, then comes on again and completes the start-up mode.

If the propeller tachometer and cabin pressure inputs are in range, the system goes into the control mode with the aircraft on the ground.

NOTE

No faults occur with the aircraft on the ground and the system in the control mode.

If there is a system or controller fault that the system cannot clear automatically, then the system goes into the maintenance mode. The system STATUS LED on the controller front panel changes from green to red.

Maintenance Mode

Maintenance mode provides first line tests for the complete ANC system and second line tests for the ANCU.

First line testing is achieved with the ANC Maintenance Terminal (MT) and the System Installation Test Set. Second line testing is achieved with the ANC MT used in conjunction with the ANCU Second Line Test Set.

The controller goes into the maintenance mode if a MT is connected to connector J5 on the ANCU front panel, or at system start-up.

If the ANVS is not in the active control mode or if there is a system fault, the controller also goes into the maintenance mode.

A self-test function lets the maintenance terminal start a Built-In-Self Tests (BIST) using MT Software Part Number SW-800-99-013. The results are shown on the maintenance terminal display.

The inputs that follow are shown on the MT display:

- Propeller rpm frequency
- Weight-on-wheels inputs
- Aircraft identification variant
- Pause
- Pressure
- Cabin configuration.

The tests that follow are included in the maintenance mode of operation:

- Configuration Parameters
- Automatic Actuator and Microphone Test
- Step Through Actuator Test and Manual Actuator Test
- System Calibration Test.

Configuration Parameters

The controller is configured for different aircraft types and different ANVS installations. All the configuration parameters are stored in the maintenance terminal in a database.

The configuration parameters are stored in the controller nonvolatile memory by downloading data files that contain specific aircraft configuration parameters.

The MT does the tasks that follow:

- Finds the data to be loaded with the aircraft variant identity inputs to the controller.
- Makes sure that the configuration parameters loaded in the controller are valid for the installation.
- If the parameters are valid, automatically loads the correct configuration parameters.

Automatic Actuator and Microphone Test

The controller sends an audible test tone sequentially to each ATVA and measures the response at each microphone. Any ATVA or microphone that fails the thresholds set in the configuration parameters is disabled and will not be used in the control mode.

NOTE

This test clears previously disabled channels that have been repaired. A system fault is shown if the calibration procedures have not been done on a failed channel that is serviceable.

Step through Actuator and Manual Actuator Test

The controller sends an audible test tone to an ATVA selection. Each ATVA can be manually enabled or disabled. In the Step through Actuator Test, the actuators are sequenced in a set pattern by operator input. In the Manual Actuator Test, the only actuator tested is chosen at the start of the test by the operator.

System Calibration Test

The controller sends audible test tones to all actuators at the same time. The control unit calculates and stores the acoustic transfer function of the loudspeaker/microphone array.

NOTE

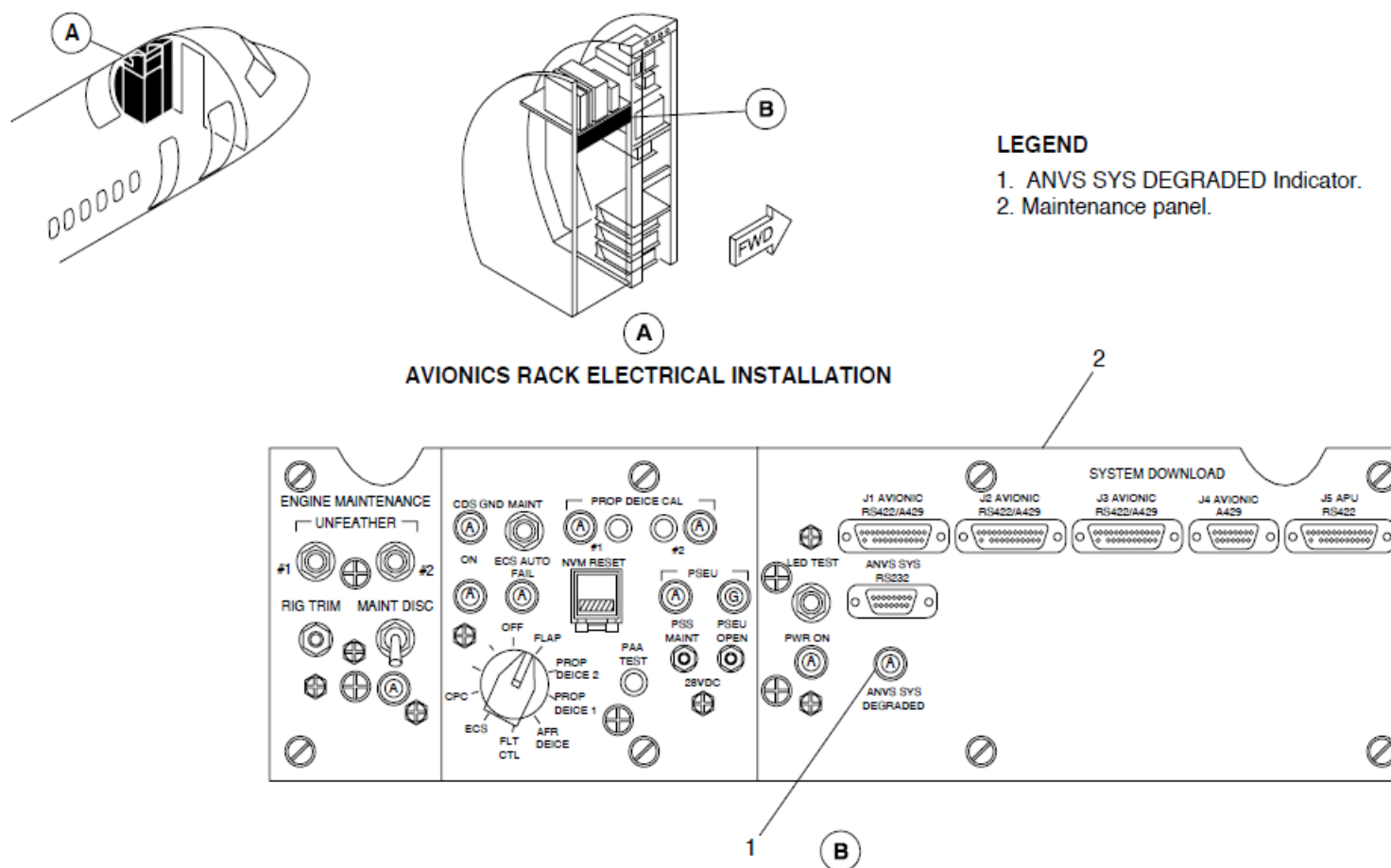
Calibration is not performed on ATVAs and microphones that have previously failed. A calibration test cannot clear failed channels.

The read transfer function lets the calibration data stored in the controller non-volatile memory to be downloaded and stored in the maintenance terminal.

The load transfer function lets the calibration data stored in the MT to be loaded in the controller non-volatile memory.

Other attributes of the maintenance terminal are:

- System Diagnostics Screen – Used to show aircraft inputs to the controller, which are then shown on the maintenance terminal display.
- System Hours and Number of Flights – Gives the number of hours and number of flights recorded by the ANCU. The data is shown on the maintenance terminal display.



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FIGURE 23- 52 MAINTENANCE PANEL

AUDIO INTEGRATION SYSTEM

Introduction

The Audio Integration System (AIS) controls the audio functions that follow:

- Monitor radio communications receivers and make transmissions
- Monitor the radio navigation receivers
- Interphone communications between pilot, copilot, observer, and ground crew members.

General Description

Figure 23- 53 AIS BLOCK DIAGRAM

Figure 23- 54 AIS BLOCK DIAGRAM, POWER

The AIS uses the ARCDUs and observer's Audio Control Panel (ACP) to control the Remote Control Audio Unit (RCAU) which interfaces with the systems that follow:

- Communication transceivers
- Navigation receivers
- Pilot, copilot, observer's, and ground crew jack boxes
- Passenger Address Amplifier (PAA)
- Flight compartment speakers
- Solid State Cockpit Voice Recorder (SSCVR).

The AIS has the components that follow:

- Unit, Remote Control Audio (RCAU)
- Panel, observer's audio control
- Speaker, flight compartment
- Microphone, hand
- Boxes, jack – flight compartment
- Box, jack – observer's
- Boxes, jack – ground crew
- Switch, PTT – copilot's side console
- Switch, transmit/interphone – control wheel
- Switch, PTT – nose wheel steering.

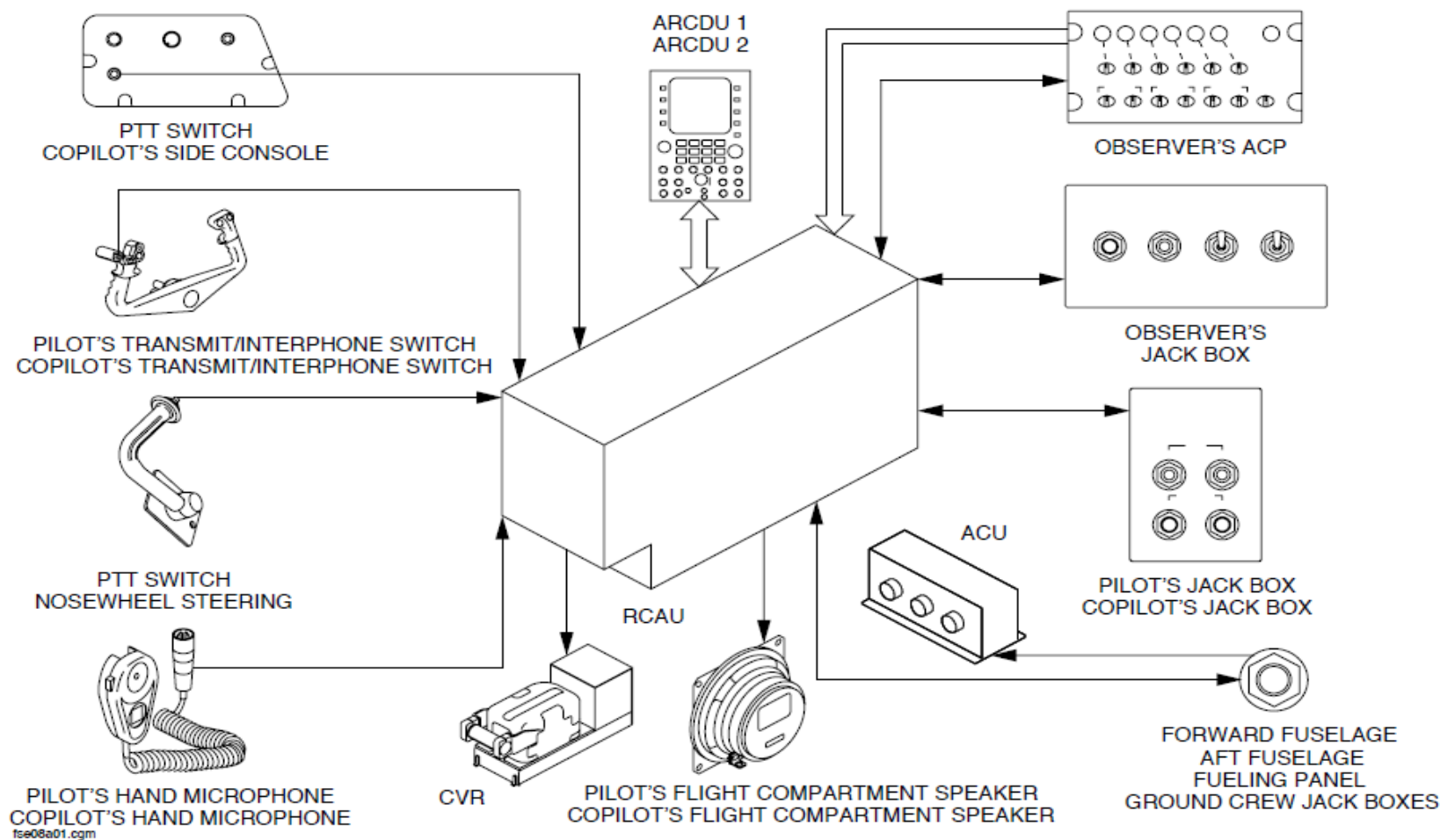


FIGURE 23- 53 AIS BLOCK DIAGRAM

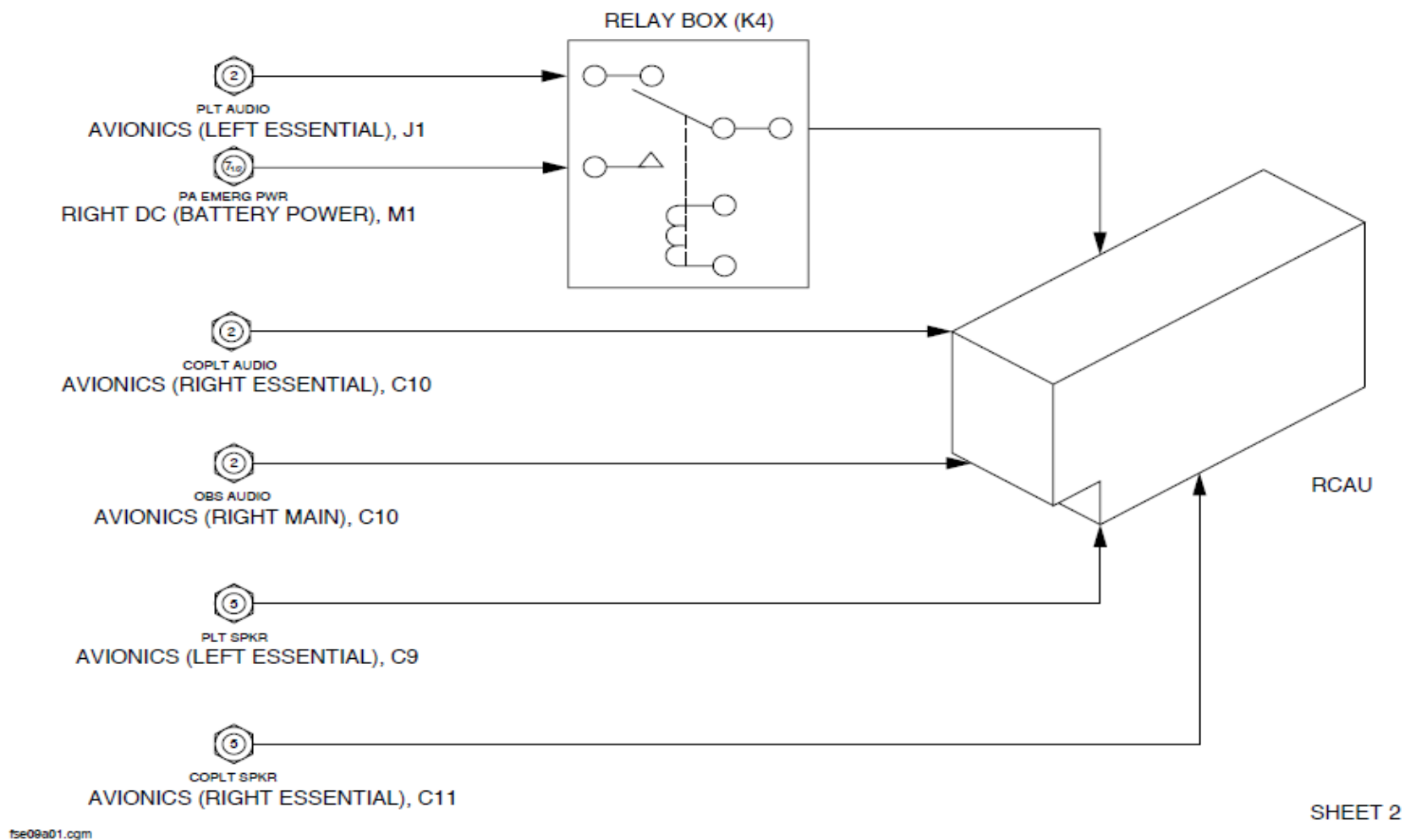


FIGURE 23- 54 AIS BLOCK DIAGRAM, POWER

Detailed Description

The RCAU functions in the modes that follow:

- Power-On Self-Test (POST)
- Operational (OPER).

Power-On Self-Test (POST) Mode: The RCAU does a POST after a time long-term power interruption while the aircraft is on the ground.

The POSTs are conducted by the audio cards and comprise the generation of wraparound test signals which are amplified, sensed at the output and fed back to the Digital Signal Processing (DSP) to conform the identical characteristics.

Operational (OPER) Mode:

The AIS is controlled from the ARCDUs and observer's ACP. The panels connect with each other and with the communication and navigation systems through the RCAU.

The ARCDUs and observer's ACP use a multiplexing system to control the RCAU. Multiplexing allows the sequential transmission and reception of data along a single digital interface.

The multiplexing transmitter in the ARCDUs and observer's ACP senses the switch and volume control setting and sends data that is related to the switch condition to the RCAU.

The ARCDUs and the observer's ACP interface with the RCAU to control the parameters that follow:

- Push To Talk (PTT)
- Microphone audio
- Headphone audio
- Interphone communications
- Speaker audio.

Figure 23- 55 PUSH TO TALK INPUTS

Push To Talk (PTT): A transmit or interphone PTT signal is necessary to make a transmission using the communication transceivers or to operate the interphone system. The communication transceivers or interphone system receives through the RCAU, the PTT signals that follow:

- Pilot's nose wheel steering
- Pilot's hand wheel transmit/interphone
- Pilot's hand microphone
- Copilot's hand wheel transmit/interphone
- Copilot's hand microphone
- Copilot's side console transmit/interphone
- Observers transmit/interphone.

A transmit PTT selection also causes the flight compartment speakers to mute by 6 dB to prevent feedback.

The transmit selection is controlled by the RCAU and it will allow only one transmission at a time. The RCAU automatically deselects the hand microphone PTT switch after 3 minutes for the stuck microphone sensing function.

The Flight Data Recorder (FDR) receives a transmit keying signal through Input/output Module 2 (IOM2) to record the transmit parameter.

LEGEND

1. Pilot's Hand Wheel Transmit/Interphone Switch.
2. Pilot's Hand Microphone.
3. Pilot's Nose Wheel Steering PTT Switch.
4. Copilot's Hand Wheel Transmit/Interphone Switch.
5. Copilot's Hand Microphone.
6. Copilot's Side Console Transmit/Interphone Switch.
7. Observers Transmit/Interphone Switch.

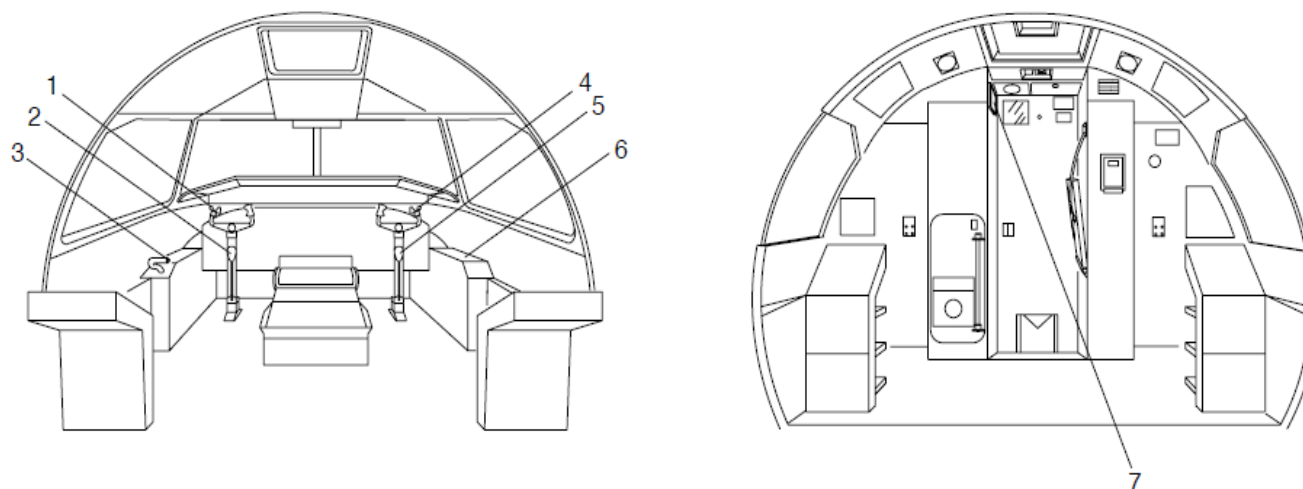


FIGURE 23- 55 PUSH TO TALK INPUTS

*Figure 23- 56 Flight Compartment Jack Box**Figure 23- 57 Observer's Jack Box*

Microphone Audio: The RCAU routes the pilot, copilot, and observer's boom, oxygen mask and hand microphone audio through jack boxes to the systems that follow:

- VHF communications 1
- VHF communications 2
- HF communications 1
- VHF communications 3
- HF communications 2
- Flight crew interphone
- Flight crew and cabin crew interphone or PA when a PA selection is made.

The pilot and copilot audio jack boxes have the microphone jack locations that follow:

- BOOM
- MASK.

BOOM Microphone Jack: The headsets have a microphone plug that is connected to the BOOM jack to supply boom microphone audio to the RCAU.

MASK Microphone Jack: The oxygen mask has a microphone plug only. It is connected to the MASK jack supply mask microphone audio to the RCAU.

The observer's audio jack box has the microphone jack location that follows:

- MIC.

MIC Microphone Jack: The headset has a microphone plug that is connected to the MIC jack to supply the boom or mask microphone audio to the RCAU.

Headphone Audio: The RCAU routes headphone audio to the pilot, copilot, and observer's jack boxes from the systems that follow:

- VHF communications 1
- VHF communications 2
- HF communications 1
- VHF communications 3
- HF communications 2
- VHF navigation 1
- VHF navigation 2
- Sum of marker audio from VHF navigation 1 and VHF navigation 2
- Distance Measuring Equipment (DME1)
- DME2 (if installed)
- Automatic Direction Finding (ADF1)
- ADF2 (if installed)
- Flight crew interphone
- Flight crew and cabin crew interphone or PA when a PA selection is made.

The pilot and copilot audio jack boxes have the headphone jack locations that follow:

- NORMAL
- AUX.

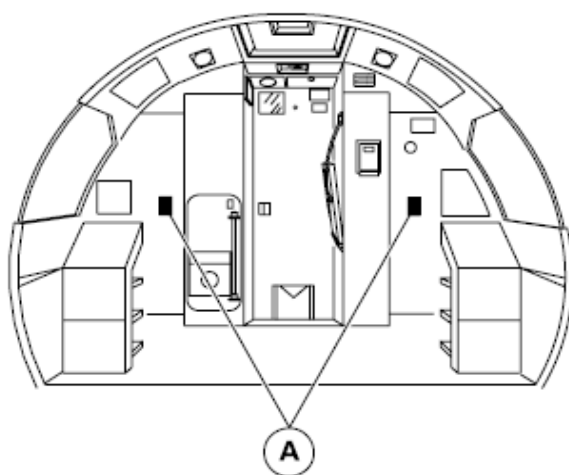
NORMAL Headphone Audio Jack: The headsets have a headphone plug that is connected to the NORMAL jack to supply headphone audio from the RCAU.

AUX Headphone Audio Jack: The AUX headphone audio jack is used when the NORMAL jack malfunctions to receive audio from the opposite audio system. For example, when the pilot's headphone audio plug is connected to the AUX headphone audio jack, the copilot's ARCDU is used to control the audio selections.

The observer's audio jack box has the headphone jack location that follows:

- PHONE.

PHONE Headphone Audio Jack: The headset has a headphone plug that is connected to the PHONE jack to supply headphone audio from the RCAU.



LEGEND

- 1. Microphone jack.
- 2. Headphone jack.

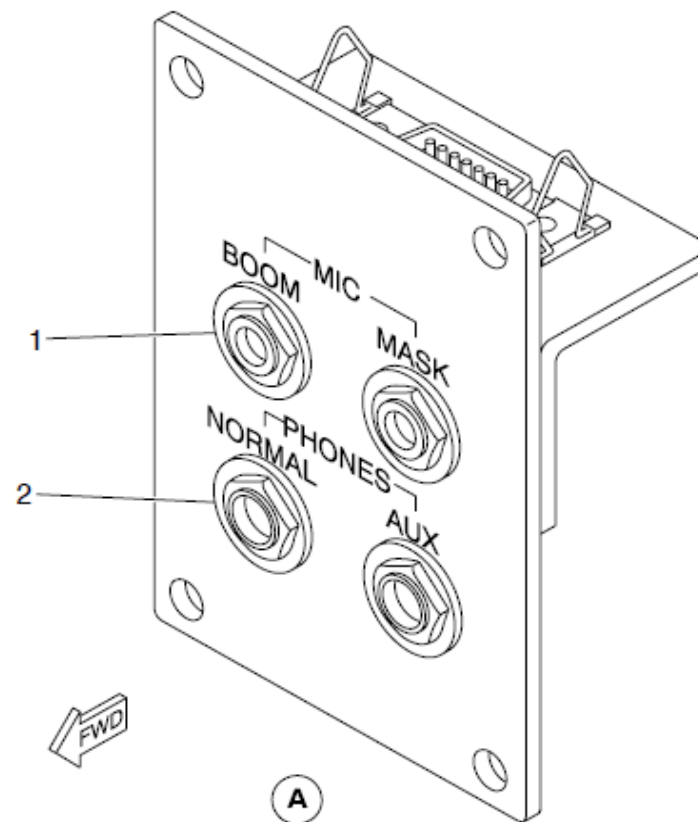
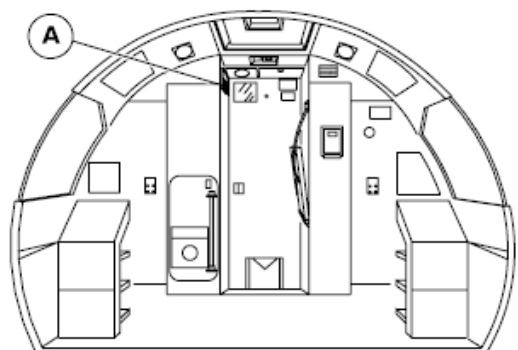
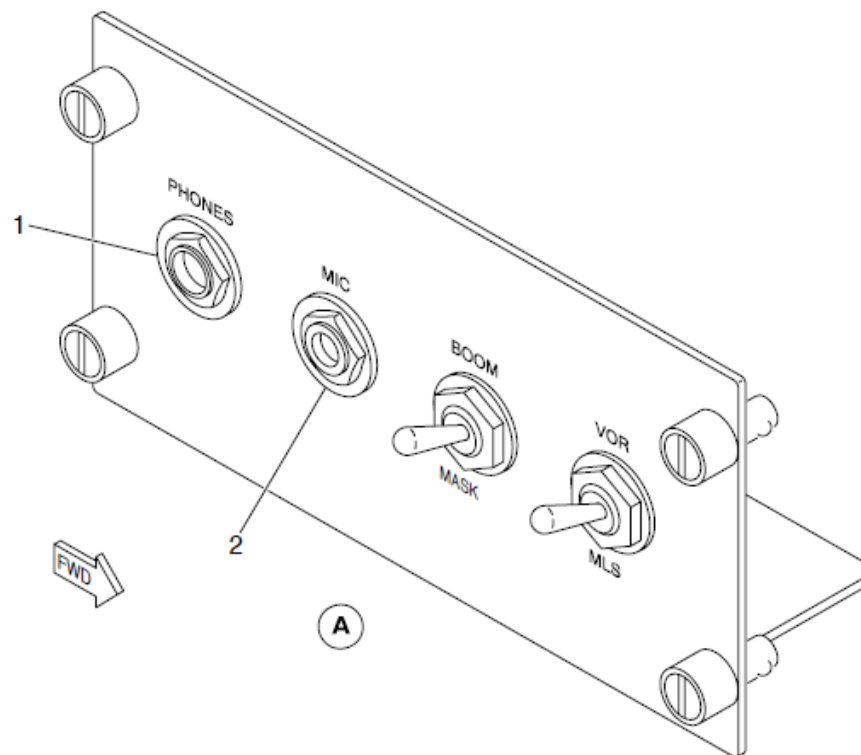


FIGURE 23- 56 FLIGHT COMPARTMENT JACK BOX



LEGEND

- 1. Headphone jack.
- 2. Microphone jack.



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FIGURE 23- 57 OBSERVER'S JACK BOX

Figure 23- 58 ARCDU, INTERPHONE CONTROL

Interphone communications: When the ARCDU rotary microphone selector switch is set to the SERV/INT position, all PTT selections give interphone communications.

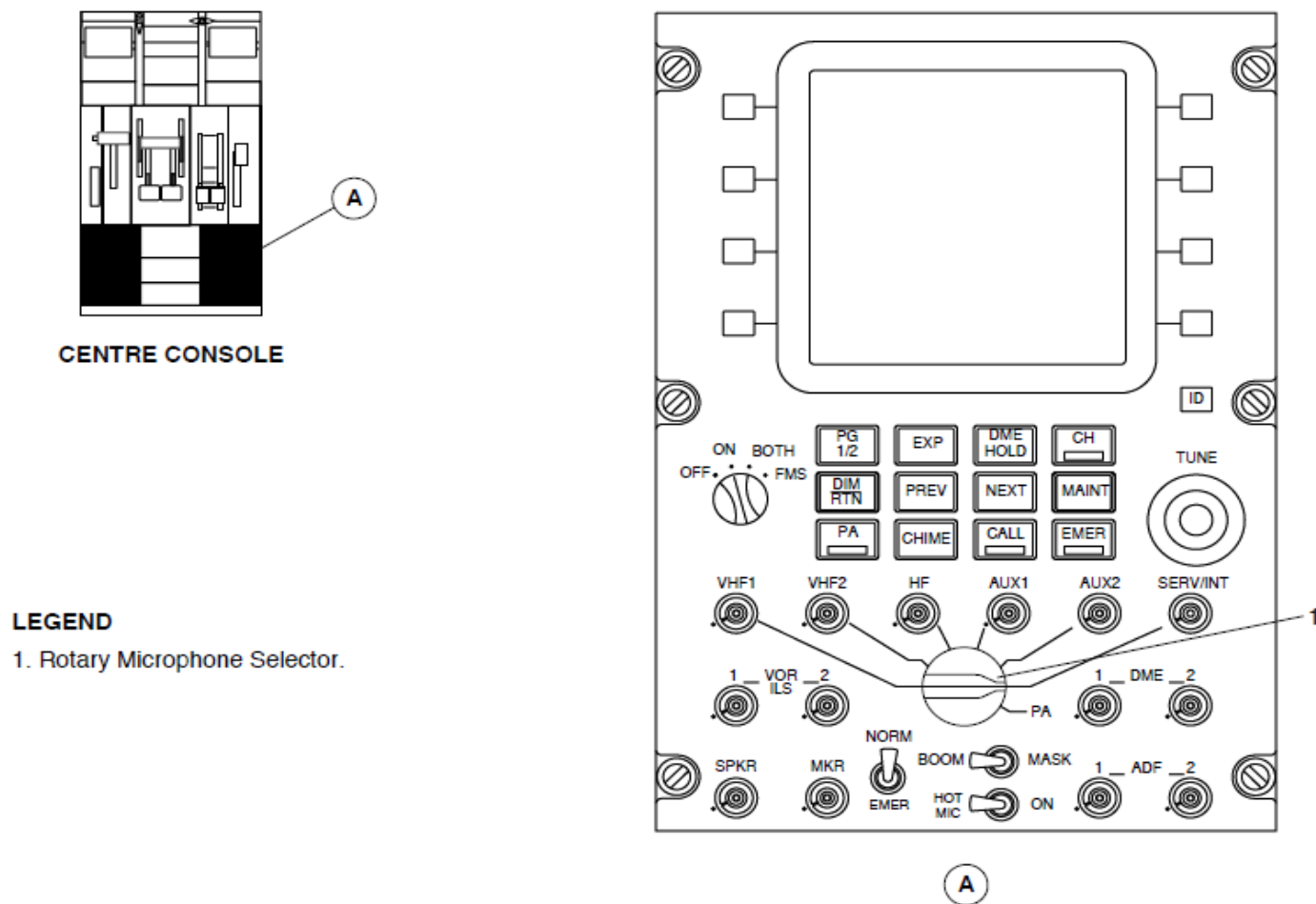
The interphone system gives communication between the stations that follow:

- Pilot
- Copilot
- Observer
- Forward fuselage ground crew
- Aft fuselage ground crew
- Fueling panel ground crew.

Figure 23- 59 GROUND CREW INTERPHONE INDICATION

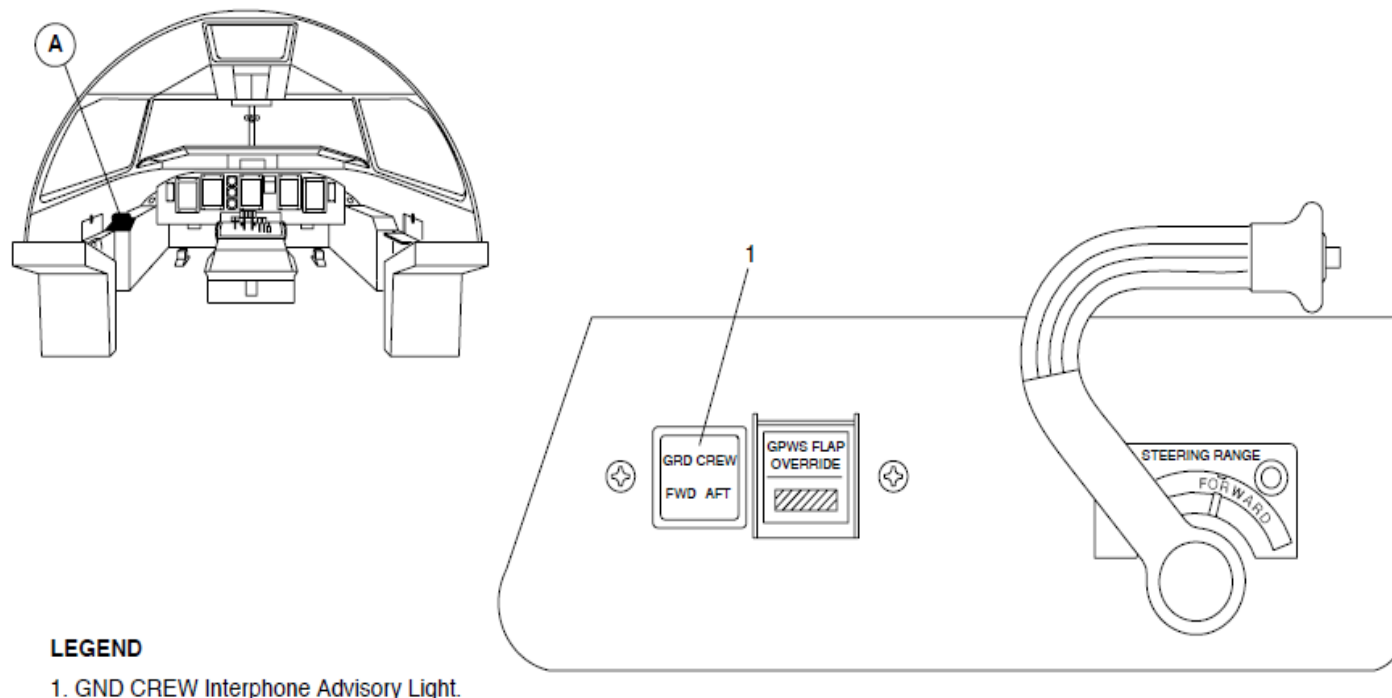
When the ground crew headset is plugged into a service interphone jack, the GND CREW advisory light on the pilot's side console comes on with an applicable FWD or AFT indication to show a service interphone connection. When the ground crew headset is plugged into the forward fuselage ground crew jack, the FWD advisory light comes on. When the ground crew headset is plugged into the aft fuselage ground crew or fueling panel ground crew jack, the AFT advisory light comes on. The FWD and AFT advisory lights alert the flight crew that interphone communication with a ground crew station may be made, and also indicates the station.

A Weight-Off-Wheels (WOW) signal from the Proximity Sensor Electronics Unit (PSEU) causes the ground crew interphone function to go off in flight.



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FIGURE 23- 58 ARCDU, INTERPHONE CONTROL



LEGEND

1. GND CREW Interphone Advisory Light.

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FIGURE 23- 59 GROUND CREW INTERPHONE INDICATION

Speaker Audio: The RCAU amplifies low-level audio signals for output to the flight compartment speakers.

Figure 23- 60 ARCDU AIS CONTROL

The bottom part of the ARCDU front panel has audio potentiometer push-button switches, a rotary microphone selector switch and toggle switches to operate the AIS.

To let the pilots control the AIS, the ARCDU has the controls that follow:

- NORM/EMER toggle switch
- BOOM/MASK toggle switch
- HOT MIC toggle switch
- Potentiometer push-button switches
- Rotary microphone (transmitter) selector.

NORM/EMER Toggle Switch: The NORM/EMER (Normal/Emergency) toggle switch is set to the NORM position to independently control the audio outputs using the potentiometer push-button switches.

The NORM/EMER (Normal/Emergency) toggle switch is set to the EMERG position to connect the headphone and speakers directly to the related VHF Communication transceiver and VHF Navigation receiver. The audio is set to a predetermined level and cannot be controlled by the potentiometer push-button switches. The microphone PTT selection is directly connected to the related VHF Communication transceiver.

BOOM/MASK Toggle Switch: The BOOM/MASK toggle switch is set to the BOOM position to supply headset boom microphone audio to the AIS. It is set to the MASK position to supply oxygen mask microphone audio to the AIS.

HOT MIC Toggle Switch: The HOT MIC (Hot Microphone) toggle switch is set to the HOT MIC position to give a continuous selection of the flight crew interphone function.

Potentiometer Push-Button Switches: The top part of the potentiometer push-button switch is pushed to route that system's audio to the related headphone. It is turned to adjust the audio level. It is pushed again to make an off selection.

The display area shows the volume level as a vertical bar graph and its height represent the adjusted volume level. When the audio is set to off, the bar graph is shown in white and when it is set to on, it changes to green.

The SPKR push-button potentiometer switch is pushed to route the headphone audio selection to its related flight compartment speaker. The display area shows a speaker graphic while the speaker is selected.

The push-button potentiometer switches control the audio selection and volume of the systems that follow:

PUSH-BUTTON POTENTIOMETER SWITCH	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3
AUX 2	HF communications 2
SERV/INT	Interphone, or PA Sidetone when a PA selection is made
VOR 1	VHF navigation 1
VOR 2	VHF navigation 2
SPKR	Related Flight compartment speaker
MKR	Sum of marker audio from VHF Navigation 1, VHF Navigation 2
DME 1	DME 1
DME 2	DME 2 when installed

PUSH-BUTTON POTENTIOMETER SWITCH	SYSTEM
ADF 1	ADF 1
ADF 2	ADF 2 when installed

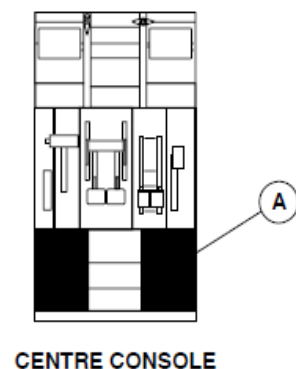
The rotary microphone selector switch is set to a specific radio position followed by a PTT selection to transmit.

ROTARY MICROPHONE (TRANSMITTER) SELECTION	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3
AUX 2	HF communications 2
SERV/INT	Flight Crew Interphone
PA	Flight Crew and Cabin Crew Interphone or PA when a PA selection is made

A green Tx annunciator in the related radio system's display area is used to show the PTT selection.

The rotary microphone selector switch is set to the SERV/INT or PA position followed by a PTT selection to make an interphone transmission or a passenger address.

A green INT Tx or PA Tx annunciator in the interphone display area is used to show the PTT selection.



LEGEND

1. Potentiometer Pushbutton Switches.
2. Rotary Microphone Selector.
3. NORM/EMER Toggle Switch.
4. HOT MIC Toggle Switch.
5. BOOM/MASK Toggle Switch.

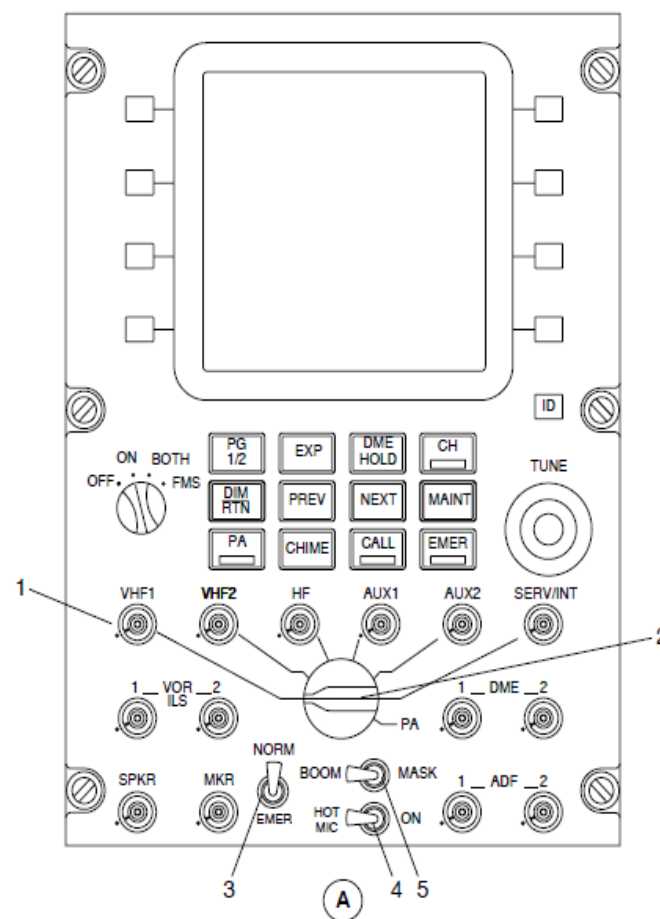


FIGURE 23- 60 ARCDU AIS CONTROL

Figure 23- 61 Observer's Audio Control Panel

The observer's ACP has the AIS controls that follow:

- Transmitter keys
- INT/RAD toggle switch
- Headphone Audio Potentiometers.

Transmitter Keys: A transmitter key is pushed to make a radio transmitter selection that follows:

PUSH-BUTTON SWITCH	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3

PUSH-BUTTON SWITCH	SYSTEM
AUX 2	HF communications 2
INT	Flight Crew Interphone

NOTE

The switches are mechanically interlocked so that only one can be selected at a time.

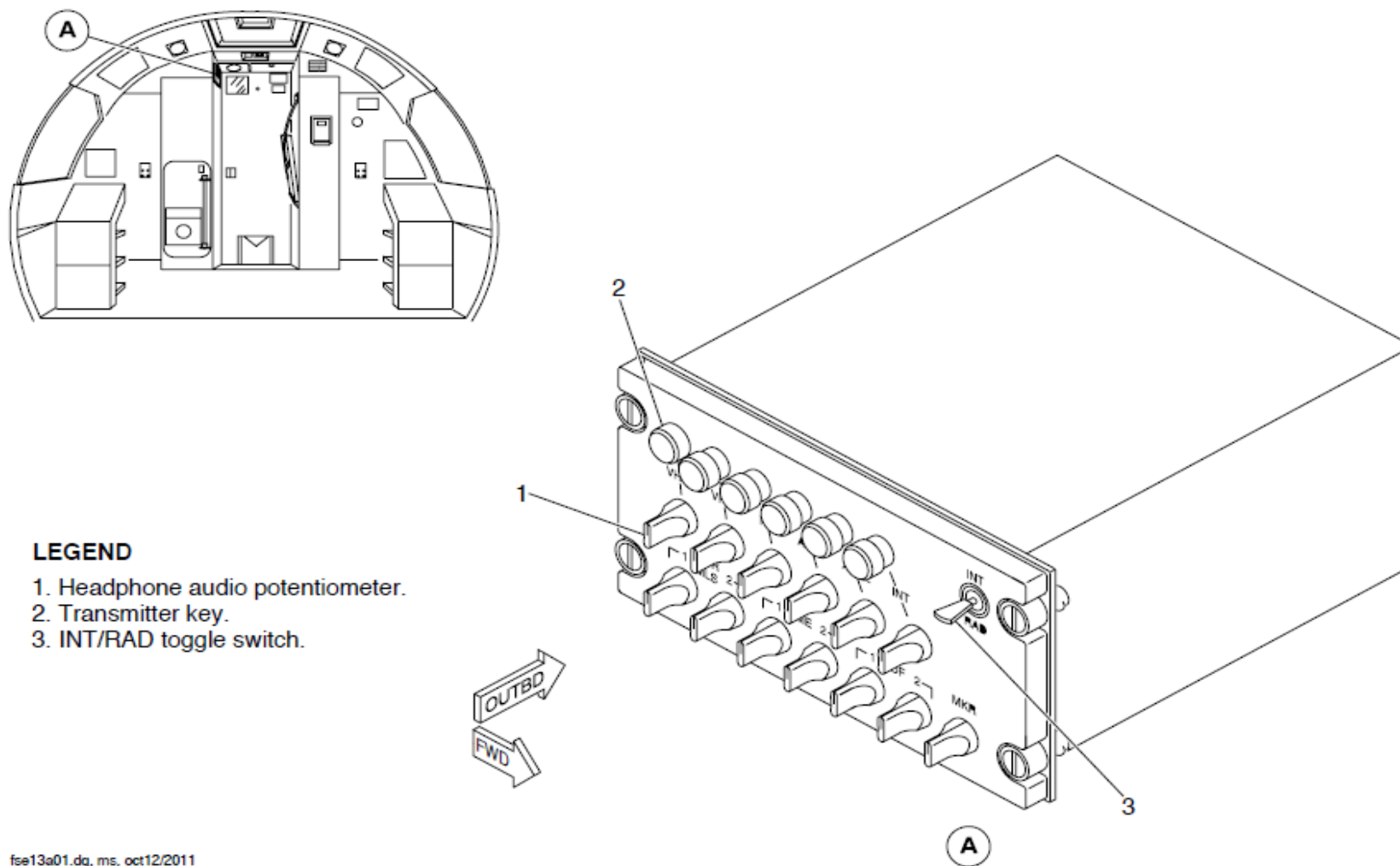
INT/RAD Toggle Switch: The INT/RAD toggle switch is set to the RAD position to make a transceiver PTT selection and to INT to make an interphone system PTT selection. The switch lever is spring-loaded to the RAD position and latched in the OFF and INT positions.

Headphone Audio Potentiometers: The headphone audio potentiometers control the audio volume of the systems that follow:

POTENTIOMETER SWITCH	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3
AUX 2	HF communications 2

POTENTIOMETER SWITCH	SYSTEM
SERV/INT	Interphone, or PA Sidetone when a PA selection is made
VOR 1	VHF navigation 1
VOR 2	VHF navigation 2
SPKR	Related Flight compartment speaker
MKR	Sum of marker audio from VHF Navigation 1, VHF Navigation 2
DME 1	DME 1
DME 2	DME 2 when installed
ADF 1	ADF 1
ADF 2	ADF 2 when installed

The related audio goes off when a potentiometer is set to the full counterclockwise position.



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FIGURE 23- 61 OBSERVER'S AUDIO CONTROL PANEL

Figure 23- 62 OBSERVER'S JACK BOX DETAIL 2

The observer's audio jack box has the AIS controls that follow:

- BOOM/MASK toggle Switch
- VOR toggle Switch.

BOOM/MASK Toggle Switch: The BOOM/MASK toggle switch is set to the BOOM position to supply headset boom microphone audio to the AIS. It is set to the MASK position to supply oxygen mask microphone audio to the AIS.

VOR Toggle Switch: The VOR toggle switch is set to the VOR position let the observer's ACP control the audio from the VHF navigation receiver that is selected on the Flight Guidance Control Panel (FGCP).

The RCAU contains the circuit cards that follow:

- Pilot audio card
- Copilot audio card
- Observer audio card
- Pilot speaker amplifier card
- Copilot speaker amplifier card.

Each card has a related aircraft circuit breaker. Electrical power is removed from the audio cards when the NORM/EMER switch located on the ARCDU is set to EMER.

The aircraft's left essential bus supplies 28 VDC power through a 2 A circuit breaker to the pilot audio card. The circuit breaker is located in position J1 on the left circuit breaker panel.

The aircraft's right essential bus supplies 28 VDC power through a 2 A circuit breaker to the copilot audio card. The circuit breaker is located in position C10 on the avionics circuit breaker panel.

The aircraft's right main bus supplies 28 VDC power through a 2 A circuit breaker to the observer audio card. The circuit breaker is located in position C4 on the avionics circuit breaker panel.

The aircraft's left essential bus supplies 28 VDC power through a 5 A circuit breaker to the pilot speaker amplifier card. The circuit breaker is located in position C9 on the avionics circuit breaker panel.

The aircraft's right essential bus supplies 28 VDC power through a 5 A circuit breaker to the copilot speaker amplifier card. The circuit breaker is located in position C11 on the avionics circuit breaker panel.

When the left essential bus malfunctions, the aircraft's battery bus continues to supply electrical power. The emergency lights operate a relay to change the power source from the left essential bus to the battery bus. The battery bus supplies 24 VDC power through a 7.5 A circuit breaker to the ARCDU1, the PAA, and the flight attendants' handsets. (The PA system also continues to operate). The circuit breaker is located in position M1 on the right circuit breaker panel.

Two flight compartment speakers are located in the forward left and right sides of the ceiling. A PTT selection causes the related flight compartment speakers to mute by 6 dB and the speaker frequency response to decrease from 300 to 6000 Hz to 300 to 1800 Hz.

Each speaker has two sets of wires connected to it. One is controlled by the speaker potentiometer push-button switches located on the ARCDUs to make an on or off selection. The other input is used for aural alerts. The on positions of the speaker potentiometer push-button switches have no effect on the tone or synthetic voice message sound.

The priority of the aural alerts is shown in the table that follows:

PRIORITY	WARNING
1	GPWS warning
2	TCAS collision
3	Fire bell
4	Incorrect take off configuration
5	Auto pilot disengagement
6	Pitch trim in motion
7	Over speed warning
8	Incorrect landing gear configuration
9	Altitude alert warning
10	Beta lockout warning
11	Warning annunciation
12	Caution annunciation

PRIORITY	WARNING
13	TCAS Traffic
14	Selcal

The RCAU supplies pilot, copilot, observer, and passenger address headphone and microphone audio data to the SSCVR. The microphone audio does not have to be transmitted for it to be recorded.

The audio that is sent to the SSCVR represents the audio and volume selections of the ARCDUs and ACP.

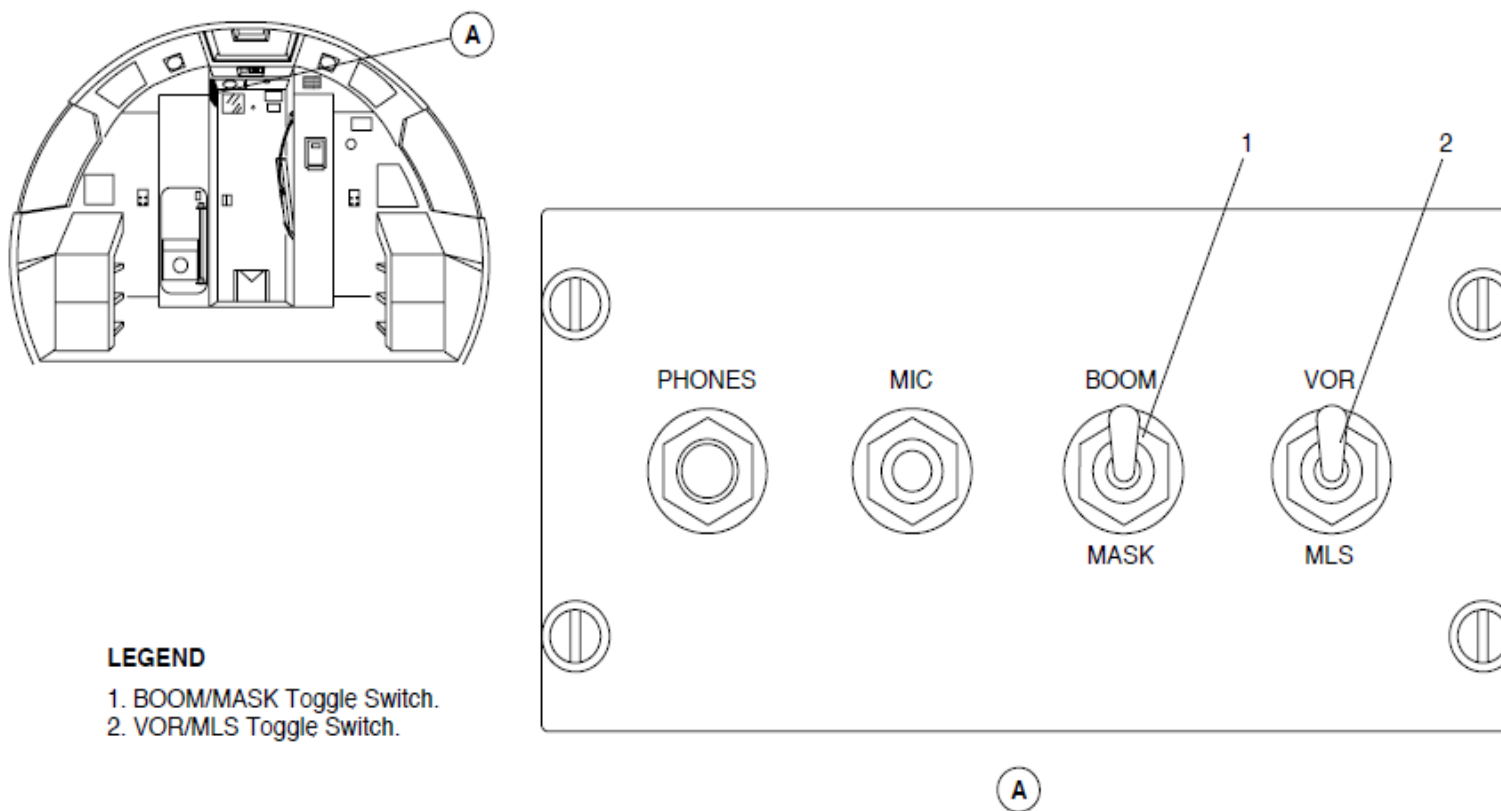


FIGURE 23- 62 OBSERVER'S JACK BOX DETAIL 2

Unit, Remote Control Audio (RCAU)

Figure 23- 63 REMOTE CONTROL AUDIO UNIT (RCAU) LOCATOR

The RCAU is used to control the AIS from ARCDU or observer's ACP selections.

A red FAIL message is shown in the interphone display area to show that the RCAU is failed.

Panel, Observer's Audio Control

Figure 23- 64 Observer's Audio Control Panel

The observer's ACP changes the front panel audio and PTT selections into ARINC 429 data for transmission to the RCAU.

The ACP is installed in the flight compartment entry way to the right hand sidewall. The unit is held by four dzus fasteners.

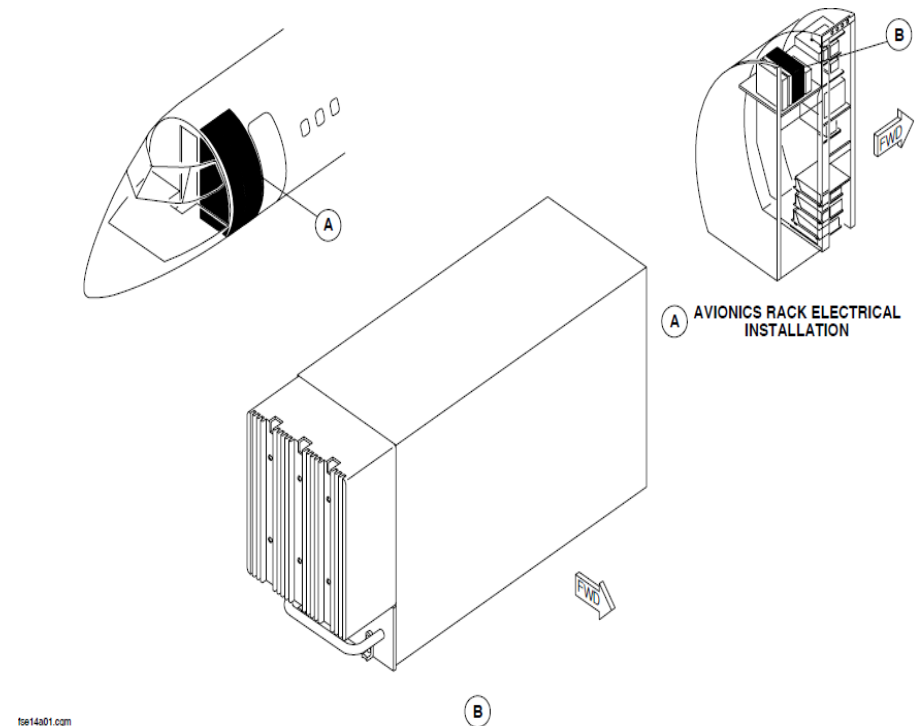


FIGURE 23- 63 REMOTE CONTROL AUDIO UNIT (RCAU) LOCATOR

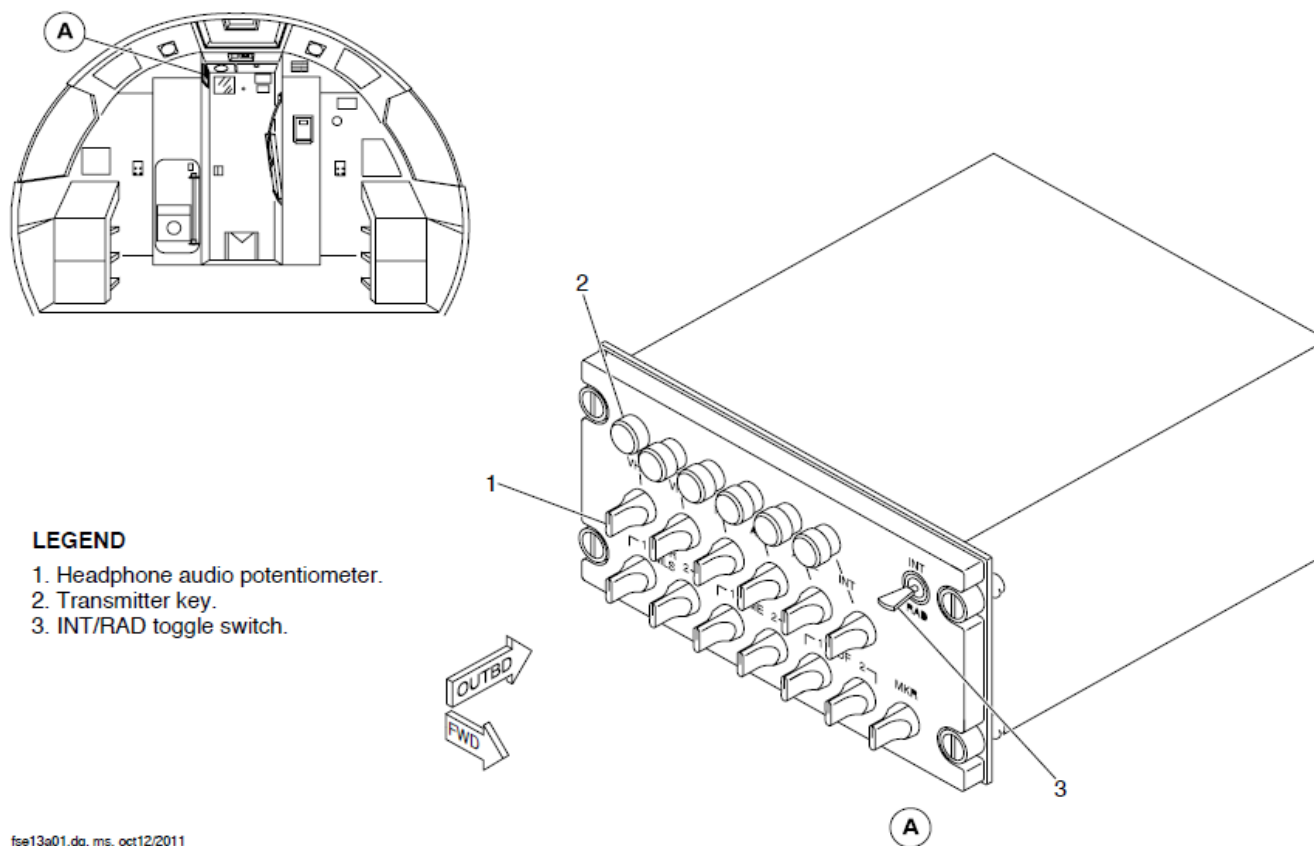


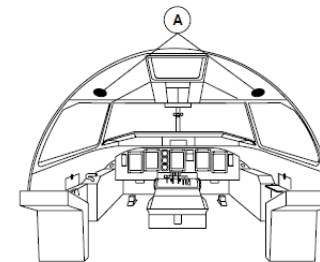
FIGURE 23- 64 OBSERVER'S AUDIO CONTROL PANEL

Speaker, Flight compartment

Figure 23- 65 FLIGHT COMPARTMENT SPEAKERS LOCATOR

The flight compartment speakers change electrical signals to audio signals.

Each flight compartment speaker can output 4 W audio power. The speaker impedance is 4 A.



Microphone, Hand

Figure 23- 66 HAND MICROPHONES LOCATOR

The hand microphone is a device that changes sound waves into variations of electric currents for transmission. It has a condenser to remove much of the electromagnetic and radio frequency interference and magnetic signals.

The microphone case is made of black plastic. It has a mounting bracket and PTT switch.

The hand microphone is connected to the base of the control column with a cable that has a four pin connector.

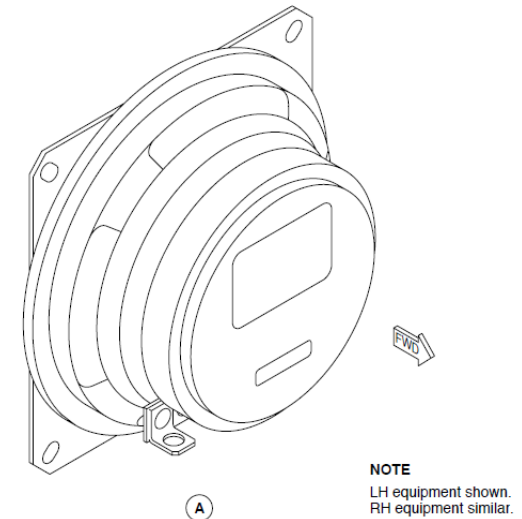
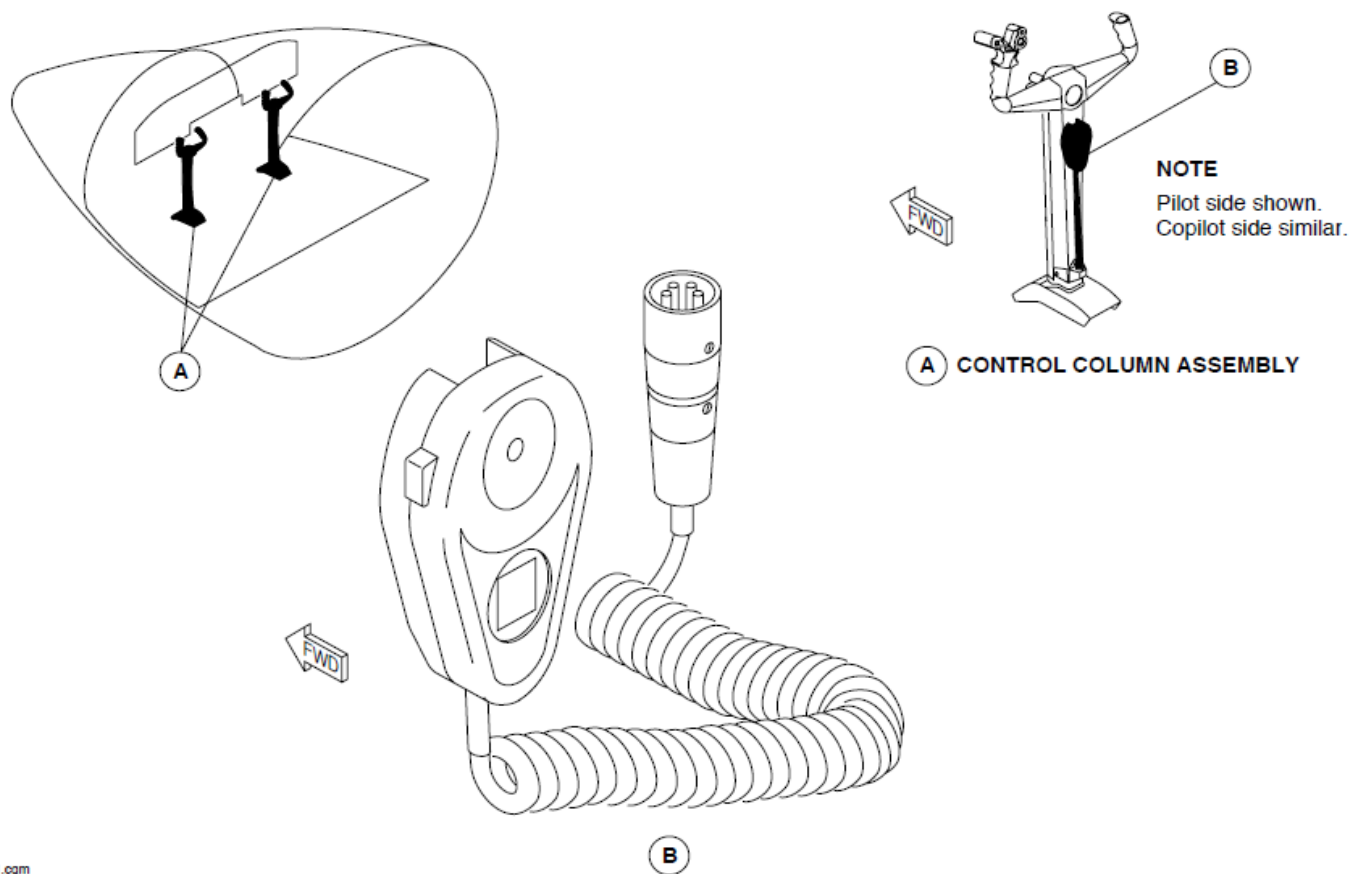


FIGURE 23- 65 FLIGHT COMPARTMENT SPEAKERS LOCATOR



Boxes, Jack – Flight Compartment

Figure 23- 67 Flight Compartment Jack Box

The flight compartment jack boxes are used to connect the pilots' headsets and oxygen mask microphone to the RCAU.

Each flight compartment jack box contains the jacks that follow:

- Boom microphone
- Mask microphone
- Phone, normal
- Phone, auxiliary.

The jacks are connected to headset and microphone plugs

The flight compartment jack boxes are each attached by screws to the bulkhead behind the pilot and copilot.

Box, Jack – Observer's

Figure 23- 68 Observer's Jack Box

The observer's jack box is used to connect the observer's headset or oxygen mask microphone to the RCAU.

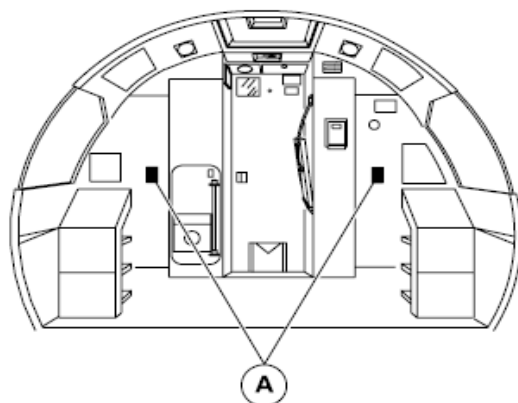
Each flight compartment jack box contains the jacks that follow:

- Phone
- Boom.

The jacks are connected to headset and microphone plugs.

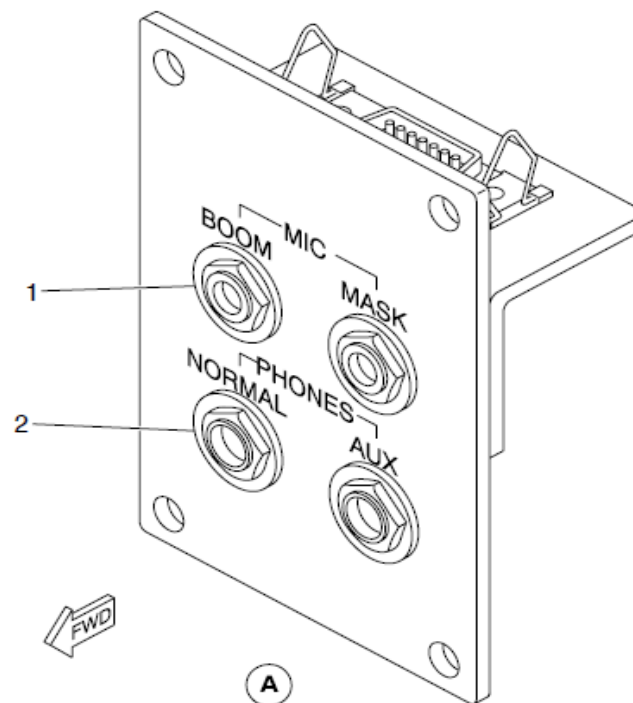
The observer's jack box also has a BOOM/MASK and VOR toggle Switches.

The flight compartment jack boxes are each attached by screws to the bulkhead behind the pilot and copilot.



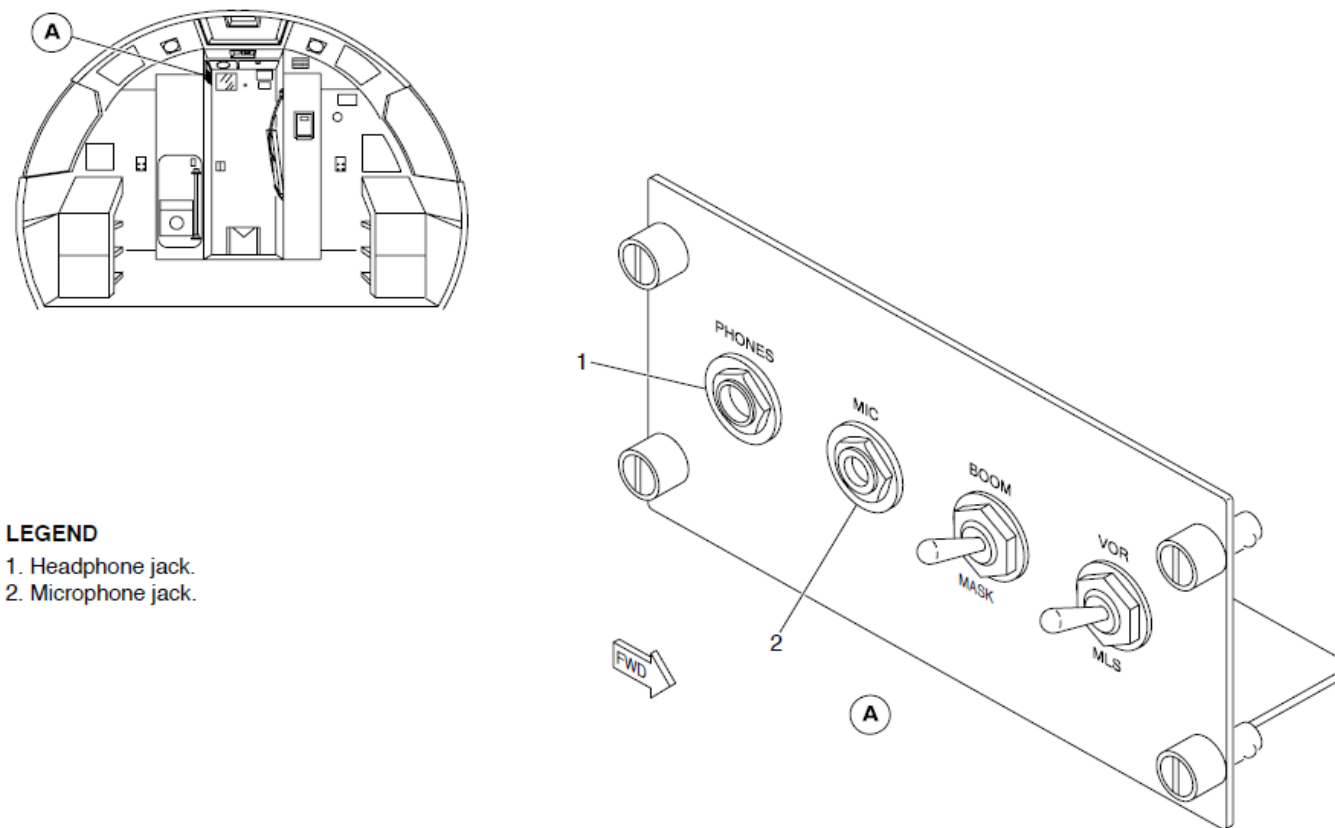
LEGEND

- 1. Microphone jack.
- 2. Headphone jack.



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FIGURE 23- 67 FLIGHT COMPARTMENT JACK BOX



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FIGURE 23- 68 OBSERVER'S JACK BOX

Boxes, Jack – Ground Crew

Figure 23- 69 FORWARD GROUND CREW JACK BOXES LOCATOR

Figure 23- 70 FUELING PANEL GROUND CREW JACK BOXES LOCATOR

Figure 23- 71 AFT GROUND CREW JACK BOXES LOCATOR

The ground crew jack box is used to connect the ground crew member's headset and microphone to the RCAU for ground service interphone communications.

The ground crew jack box has a single phone and microphone jack with a switch. When the plug is connected to the jack, a switch in the jack causes the Advisory Control Unit (ACU) to illuminate a ground crew advisory light in the flight compartment.

The ground service interphone door gives access to the ground crew jack. It is located externally on the left side of the aircraft nose near the DC electrical ground power door and nose gear ground lock handle.

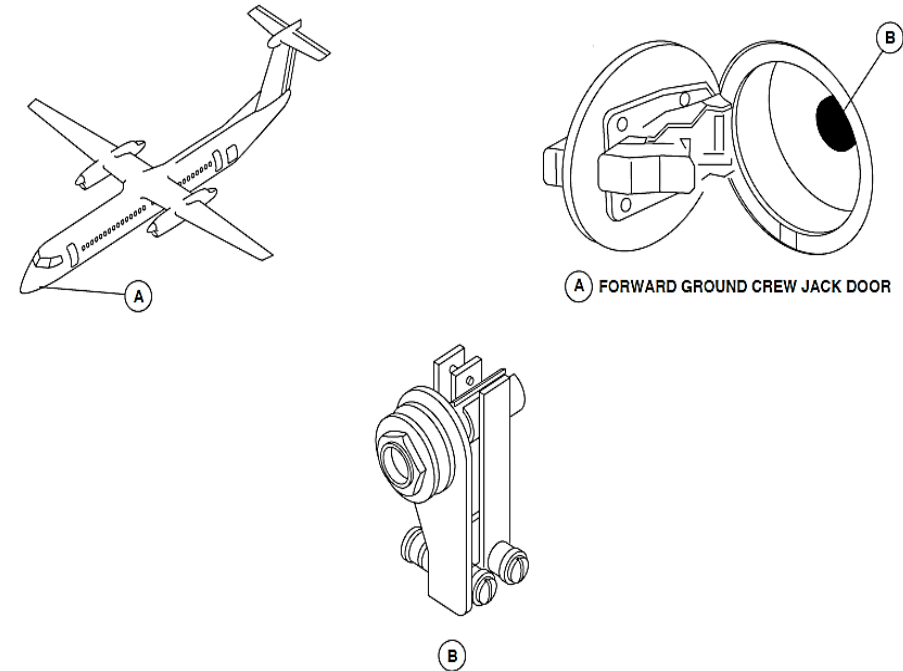


FIGURE 23- 69 FORWARD GROUND CREW JACK BOXES LOCATOR

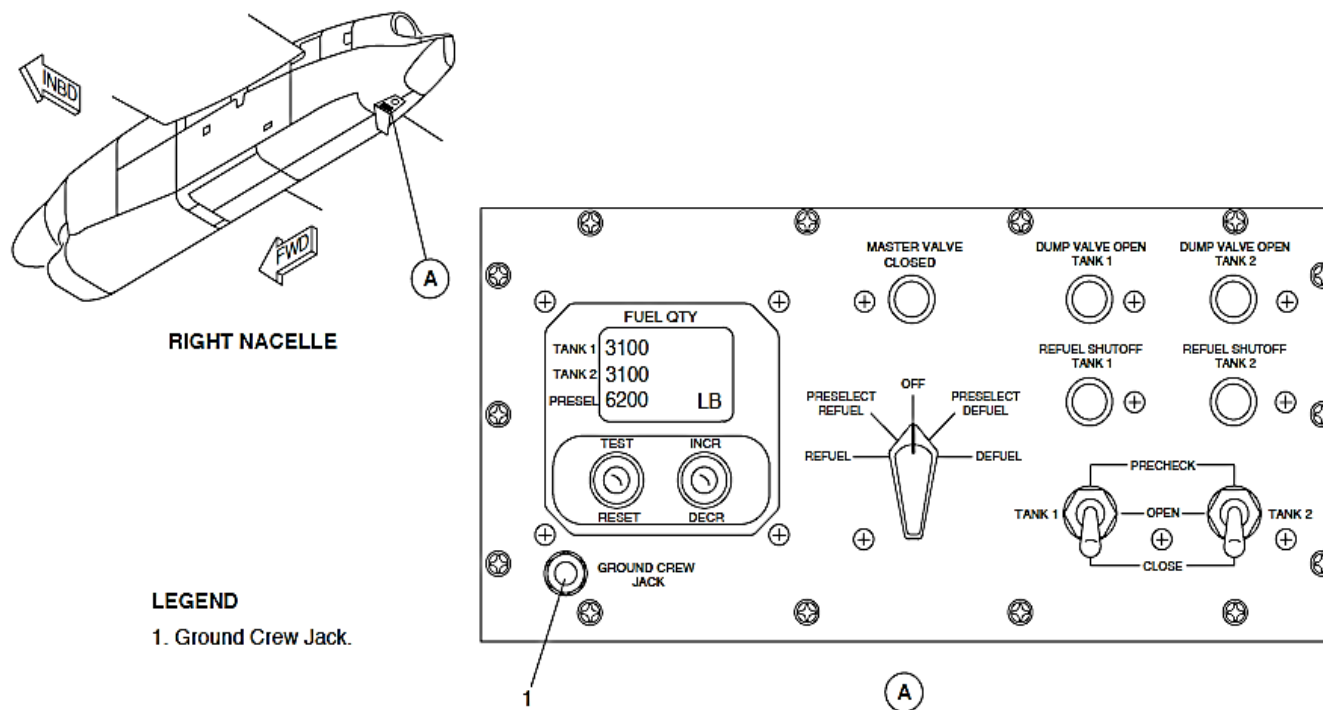


FIGURE 23- 70 FUELING PANEL GROUND CREW JACK BOXES LOCATOR

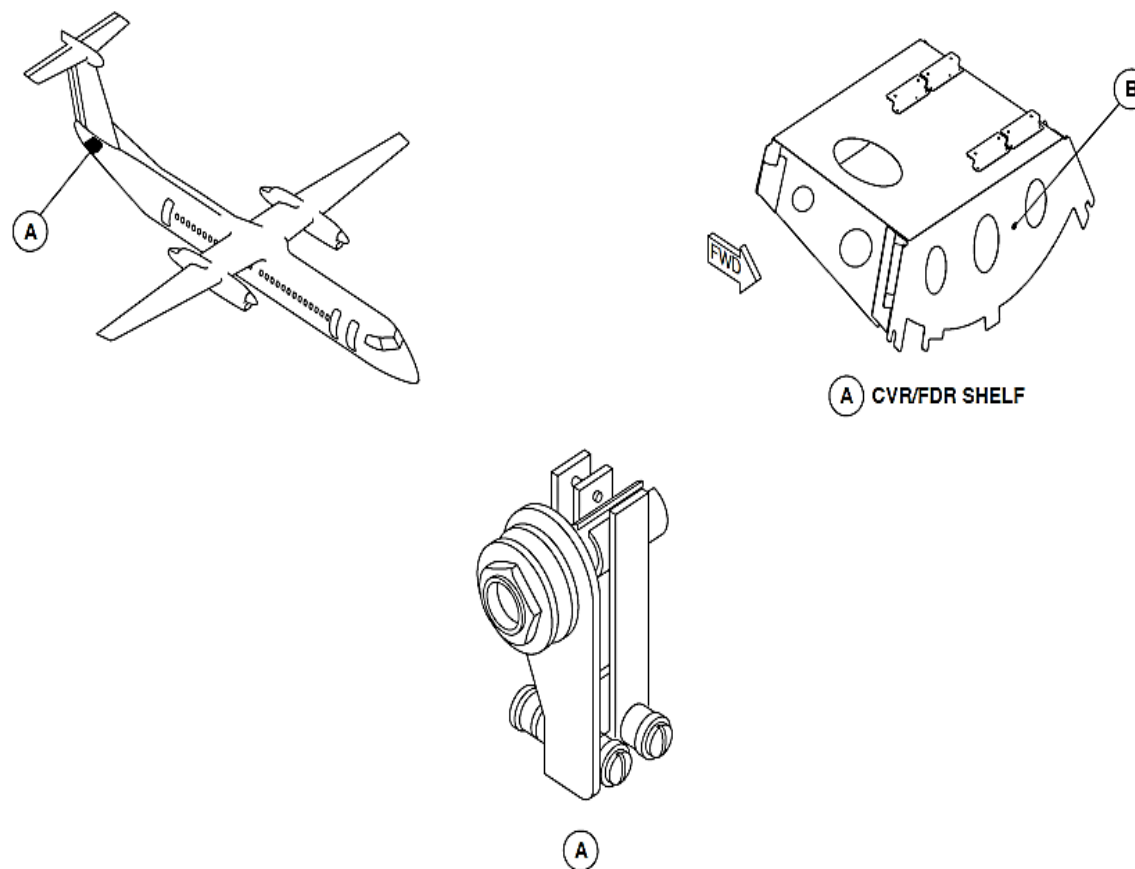


FIGURE 23- 71 AFT GROUND CREW JACK BOXES LOCATOR

Switch, PTT – Copilot's Side Console

Figure 23- 72 COPILOT'S SIDE CONSOLE PTT SWITCH LOCATOR

The PTT switch is used to supply a PTT signal to the RCAU for interphone, VHF and HF communications.

The PTT switch has an interphone and transmit position. It is set to INPH to make an interphone system PTT selection and to the XMIT position to make a transceiver PTT selection. It has a toggle lever that springs back to the center off position when released. The toggle lever is connected to a mechanism located in the body of the switch. The internal mechanism moves in the opposite direction as the toggle lever.

The PTT switch has solder lug terminals for wiring connections.

The PTT switch is attached to the copilot panel assembly with a mounting nut and lock washer. The copilot panel assembly is attached to the copilot's side console with four dzus fasteners.

Switch, Transmit/Interphone – Control Wheel

Figure 23- 73 HAND WHEEL TRANSMIT/INTERPHONE SWITCHES LOCATOR

The control wheel transmit/Interphone switch is used to supply a PTT signal to the RCAU for interphone, VHF and HF communications.

The PTT switch has an interphone and transmit position. It is set to INPH to make an interphone system PTT selection and to the PTT position to make a transceiver PTT selection. It has a toggle lever that springs back to the center off position when released. The toggle lever is connected to a mechanism located in the body of the switch.

The internal mechanism moves in the opposite direction as the toggle lever.

The PTT switch has solder lug terminals for wiring connections.

The PTT switches are attached to the forward part pilot and copilot hand wheel assemblies with mounting nuts and lock washers.

A cover on the hand wheel assembly is removed to give access to the ground control wheel transmit/Interphone switch.

Switch, PTT – Nose Wheel Steering

Figure 23- 74 NOSE WHEEL STEERING PTT SWITCH LOCATOR

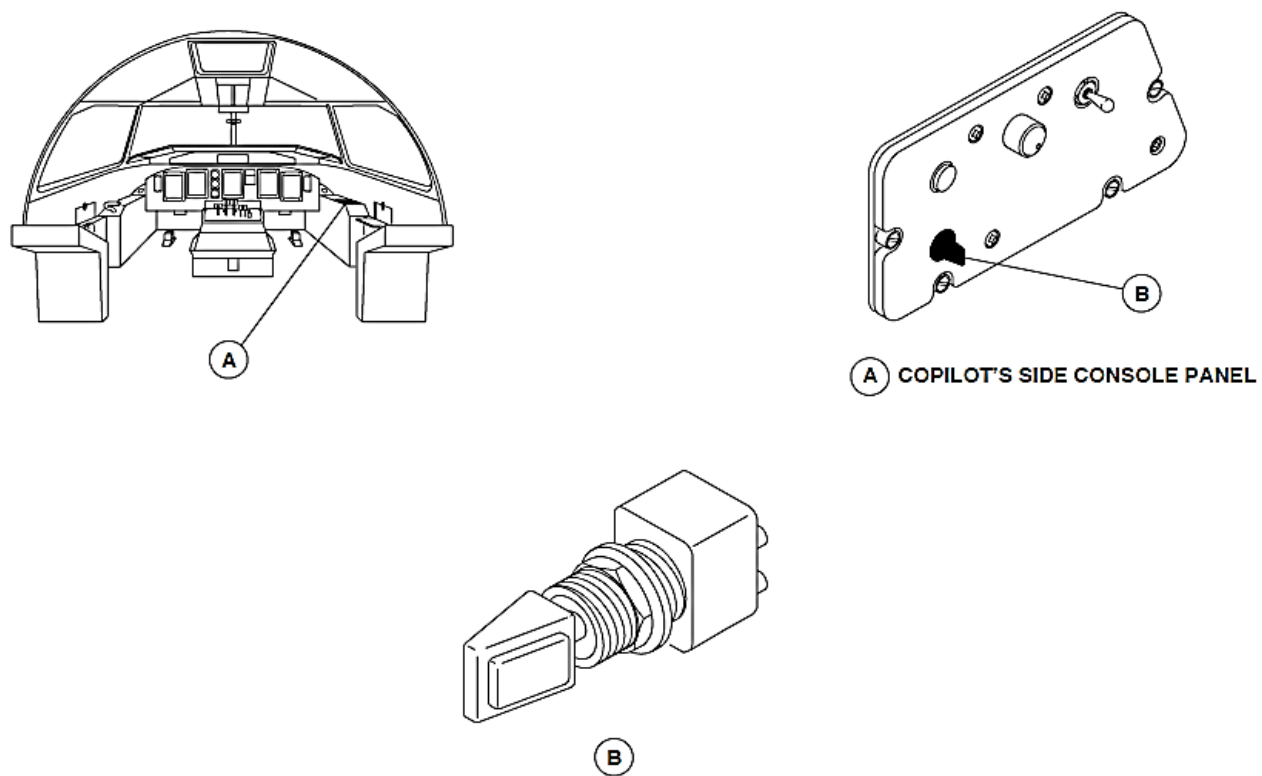
The nose wheel steering PTT switch is used to supply a PTT signal to the RCAU for interphone, VHF and HF communications.

The nose wheel steering PTT switch is pushed make PTT selection. It is a snap-action push-button switch that has a plunger that springs back to the off position when released.

The nose wheel steering PTT switch has solder lug terminals for wiring connections.

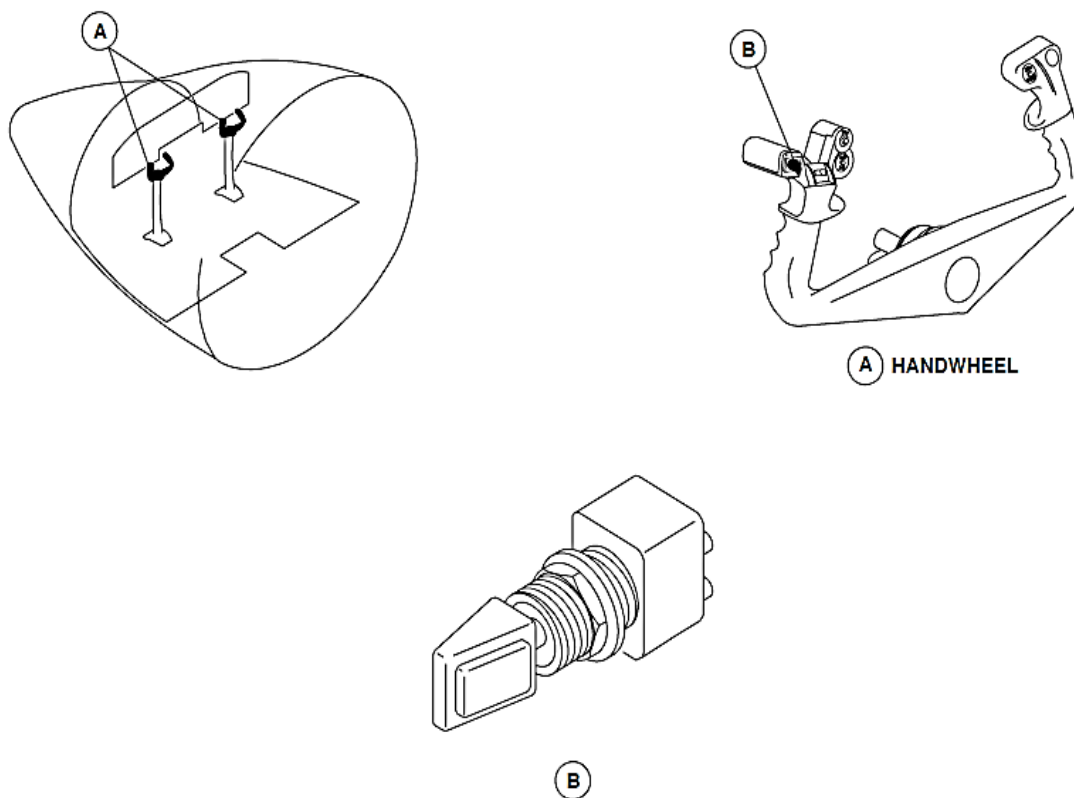
The nose wheel steering PTT switch is attached to the inboard part of the nose wheel steering hand control with mounting nuts and lock washers.

A cover on the hand wheel assembly is removed to give access to the ground control wheel transmit/Interphone switch.



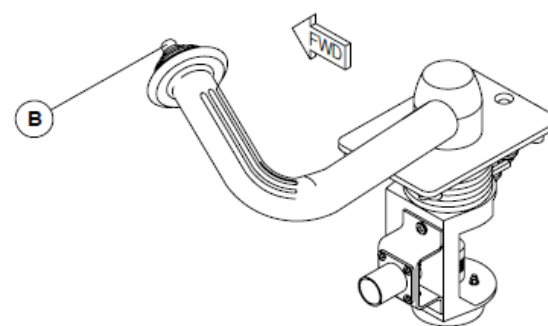
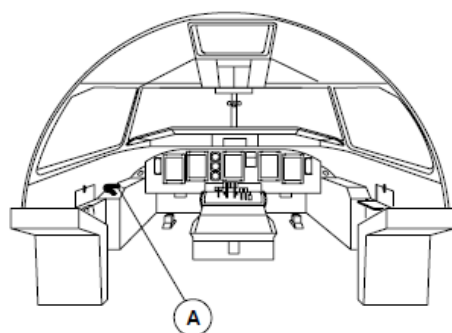
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FIGURE 23- 72 COPILOT'S SIDE CONSOLE PTT SWITCH LOCATOR

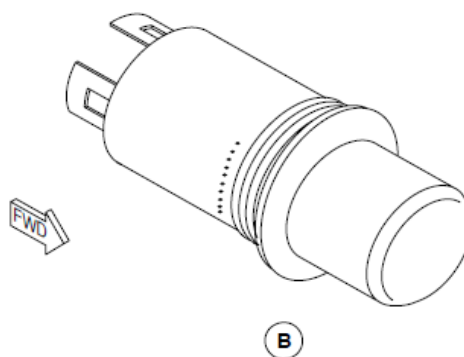


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FIGURE 23- 73 HAND WHEEL TRANSMIT/INTERPHONE SWITCHES LOCATOR



A NOSEWHEEL STEERING CONTROL HANDLE



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FIGURE 23- 74 NOSE WHEEL STEERING PTT SWITCH LOCATOR

STATIC DISCHARGE SYSTEM

Introduction

The static discharging system consists of dischargers which dissipate electrostatic charges from the aircraft structure to the surrounding air, to minimize radio interference caused by these charges.

Description

Figure 23- 75 STATIC DISCHARGER LOCATOR

Two types of static dischargers are used: type 2-11SC-1 and type 2-18SC-1. Type 2-11SC-1 is approximately 8 inches in length, with a straight taper sleeve fitting. The type 2-18SC-1 is approximately 6 inches in length with an angled tapered sleeve fitting.

Each static discharger consists of a resistive coated fiberglass rod with a pellet of composite carbon-based material at the tip. The pellet is fitted into an aluminum alloy sleeve, which in turn is fitted onto the discharger rod by means of conductive epoxy cement. A resistant sheathing is heat-shrunk over the entire length of the discharger rod and tip assembly, leaving approximately 1/8 in. (3.2 mm) carbon tip material exposed.

Operation

As the aircraft moves through the air, the aerodynamic end cap at the trailing end of each discharger creates an area of low pressure around the slightly protruding carbon tip. This reduction in pressure increases discharge efficiency, and allows ionization to take place in the slipstream, due to the impressed potential between the electrified aircraft and the surrounding air dielectric.

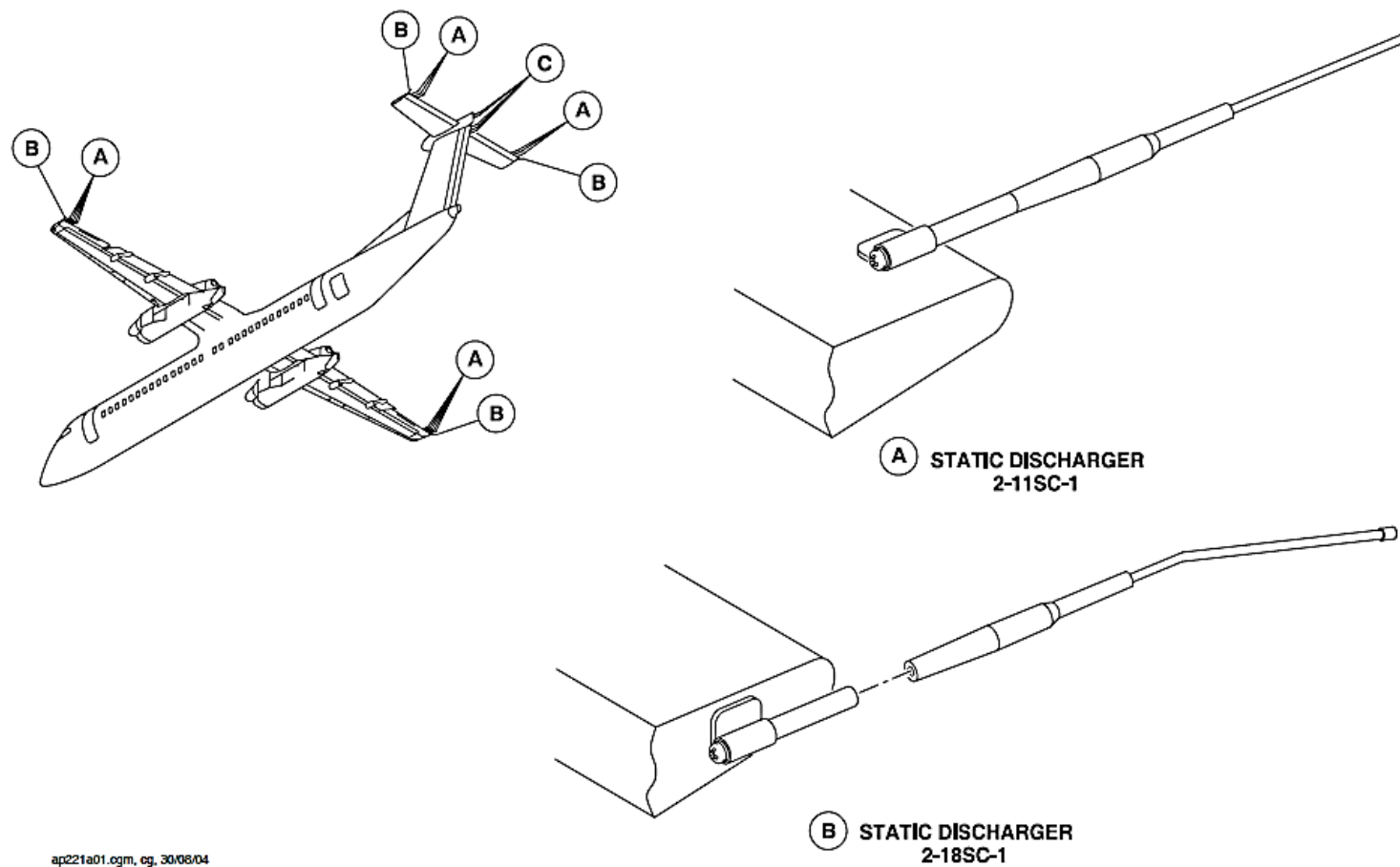


FIGURE 23- 75 STATIC DISCHARGER LOCATOR

COCKPIT VOICE RECORDING SYSTEM

Introduction

The Solid State Cockpit Voice Recorder (SSCVR) system records the last two hours of flight compartment audio and clock data.

It records audio in a digital format in crash-survivable Non-Volatile Memory (NVM).

The SSCVR is installed in the lower aft fuselage of the aircraft.

General Description

The SSCVR system records the following channels of audio inputs:

- Pilot
- Copilot
- Observer and Passenger Address (PA)
- Flight compartment area microphone.

Figure 23- 76 COCKPIT VOICE RECORDER BLOCK DIAGRAM

The SSCVR system (Honeywell) has the components that follow:

- Recorder, Solid State Cockpit Voice (SSCVR)
- Unit, Microphone Monitor (MMU)
- Switch, Inertia
- Remote Microphone.

Figure 23- 77 Cockpit Voice Recorder (Universal) Block Diagram

The SSCVR system (Universal) has the components that follow:

- Recorder, Solid State Cockpit Voice (SSCVR)
- Unit, Cockpit Control (CCU)
- Switch, Inertia
- Remote Microphone.

The SSCVR receives audio data from the Audio and Radio Management System (ARMS).

A clock supplies Universal Time Coordinated (UTC) time reference data to the SSCVR.

A wideband flight compartment area microphone records the ambient flight compartment audio.

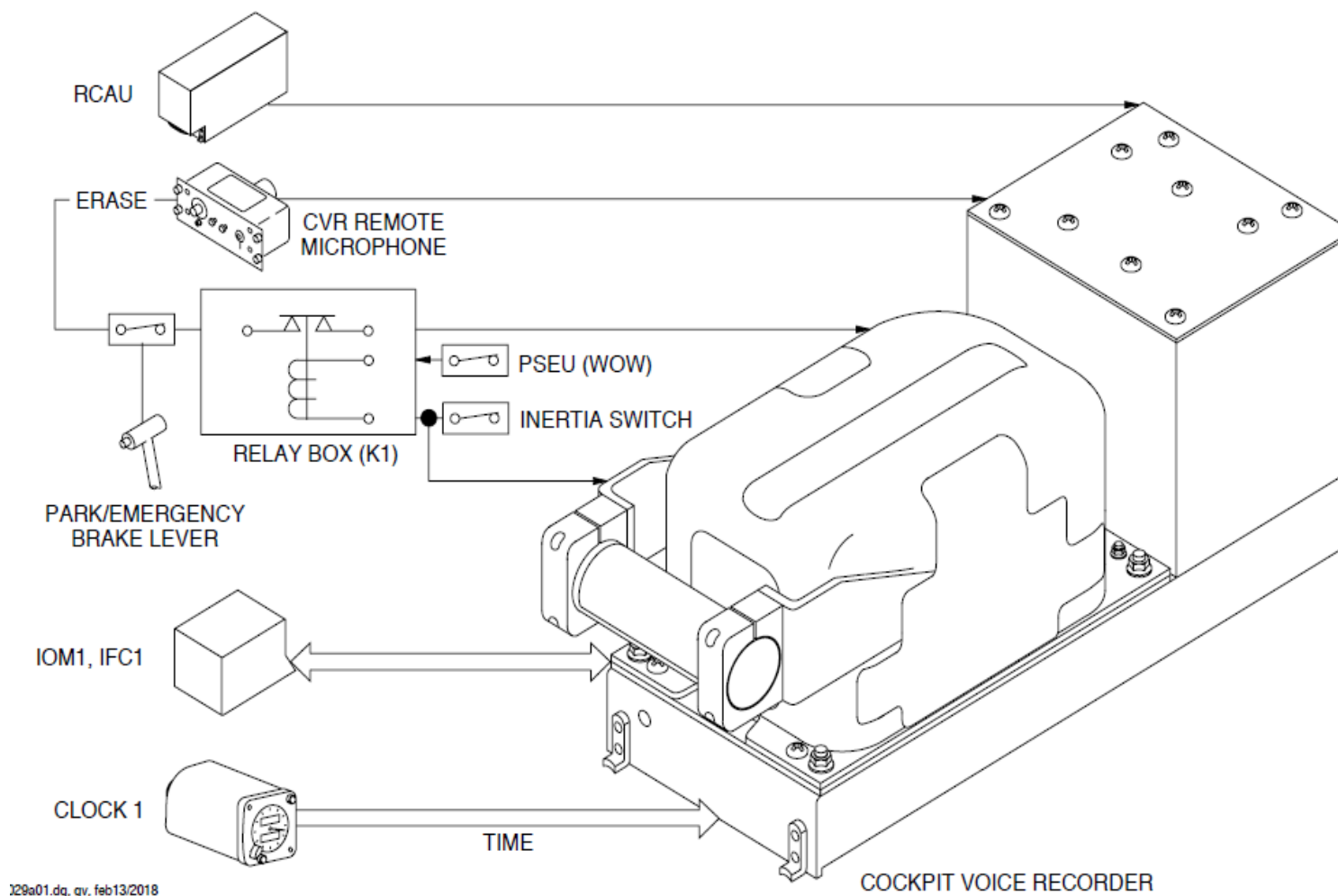


FIGURE 23- 76 COCKPIT VOICE RECORDER BLOCK DIAGRAM

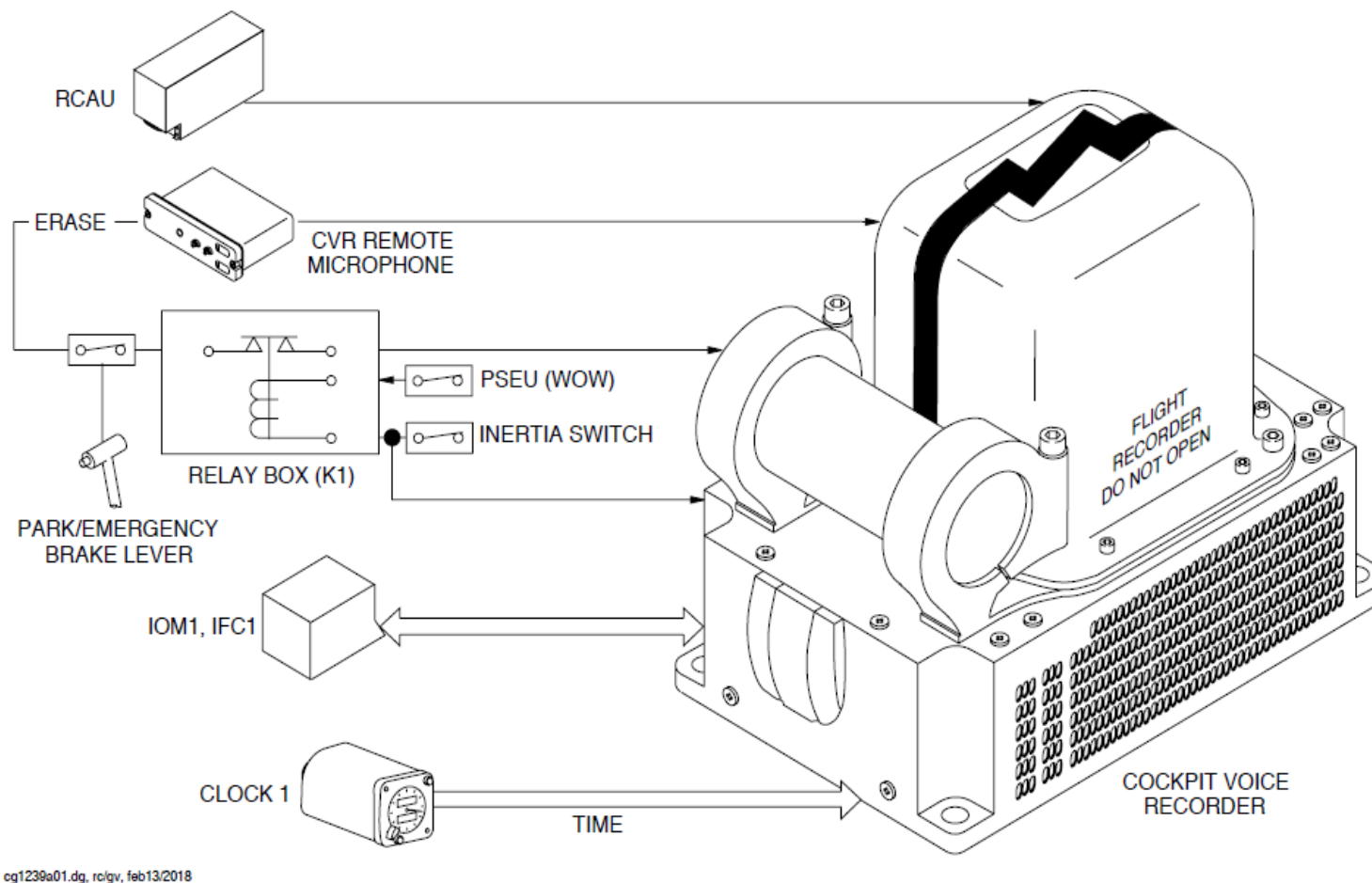


FIGURE 23- 77 COCKPIT VOICE RECORDER (UNIVERSAL) BLOCK DIAGRAM

Detailed Description – SSCVR System (Honeywell)

The SSCVR functions in the states that follow:

- Power-On Self-Test
- Operation
- Maintenance
- Data Download.

Power-On Self-Test: The SSCVR system does a power-on self-test (POST) after a time long-term power interruption.

The power-on self-test senses SSCVR hardware malfunctions that are not continually monitored in the operational state or apparent to the flight crew.

Operation Mode: The Operation Mode is subdivided into the modes that follow:

- Functional Test Mode
- Audio Monitoring
- Bulk Erase Mode
- Failure Indication.

Functional Test Mode: The TEST button located on the front panel of the MMU is pushed to start the test mode.

The SSCVR gives the results that follow within 3 seconds of successfully completing the self-test:

- An 800 Hertz aural tone for 2 seconds at the audio monitor HEADPHONE output jack
- The red STATUS indication turns on for 1 second on the MMU.

Audio Monitoring: The SSCVR system record the parameters that follow at the same time:

- Three narrowband voice channels
- One wideband area audio channel.

It supplies an audio and playback performance that is shown in the table that follows:

CHANNEL	AUDIO INPUT	BANDWIDTH HZ
1	Observer / PA	150 – 3,500
2	Copilot	150 – 3,500
3	Pilot	150 – 3,500
4	Area	150 – 6,500

The SSCVR system supplies all four audio input channels through an audio HEADPHONE output jack at the same time for monitoring purposes. The output jack is located on the front panel of the MMU. As a secondary function, the jack socket output gives an aural tone to indicate a successful completion of erase and self-test functions.

The SSCVR receives and records timing data from the pilot clock through a low speed data bus. The clock input gives timing correlation between the SSCVR and the Flight Data Recorder (FDR). The SSCVR stores the clock input in a crash-survivable memory.

Audio Monitoring: The SSCVR records 2 consecutive hours of four audio input channels and time data into crash-protected memory. The SSCVR begins storing digital audio data upon application of electrical power to the system.

The SSCVR gives a continuous audio output channel which is the composite of the four audio input channels in real time. It is available at the audio monitor HEADPHONE output jack located on the front panel of the MMU. A secondary function of this output is to give an aural tone to indicate successful completion of initiated self-test and erase functions.

Bulk Erase Mode: The bulk erase mode is started by pushing a switch located on the front panel of the MMU. The bulk erase mode is inhibited while the aircraft is moving under its own power to prevent an inadvertent erasure. Interlocks controlled by the parking brake and a Weight-On-Wheels (WOW) status prevent this from occurring.

Failure Indication: The SSCVR starts Built-In-Test (BIT) functions using the methods that follow:

- Power on initialization
- Initiated Self-Test
- Continuous Monitoring.

Power on initialization: When power is applied to the aircraft, the SSCVR system performs an automatic power on self-test.

Initiated Self-Test: A functional self-test of the SSCVR system is possible in order to assure the operator that the system is serviceable. The test mode can be activated by depressing a push-to-test button that is located on the front panel of the SSCVR MMU.

Continuous Monitoring: The CVR performs a continuous background test to make sure that the system is functioning properly. The CVR system records this information in the Crash Survivable Memory Unit (CSMU) and the Central Diagnostic System (CDS).

The crash protected solid state memory stores a history of the BIT status of each component for product trend analysis purposes. Also, the SSCVR gives a maintenance status discrete output to show an internal failure. The CDS monitors the maintenance status discrete output and the MMU gives a STATUS indication. A successful completion of initiated self-test and erase functions is indicated by:

- 800 Hz tone, for two seconds
- STATUS light on.

When a failure condition exists, the status discrete output is not activated and the aural tone does not sound and the front panel STATUS indicator is continuously on. The failure condition is recorded in the CSMU and a signal is sent to the CDS.

Data Download: The SSCVR has a download and test function to supply an interface for the Ground Support Equipment (GSE).

The Playback and Test Station (PATS) is a Personal Computer (PC) based system. It performs the functions that follow:

- Downloads solid state memory
- Playback of audio data and stored digital timing in a audio and visual format
- Functional acceptance testing of the SSCVR.

The digitized data is downloaded from the CSMU to a Playback and Test Station (PATS) Ground Support Equipment (GSE). The data download and analysis facility uses a high speed serial interface to download.

The PATS software applications is Microsoft Windows compatible. The functional acceptance test application is fully automatic.

Figure 23- 78 MICROPHONE MONITOR UNIT (MMU)

The TEST switch on the MMU is momentarily pushed to test the SSCVR. The STATUS LED comes on and an 800 Hz tone sounds for 2 seconds to show satisfactory operation.

The ERASE switch on the MMU is pushed for 0.5 seconds or longer to erase all audio data when the parking brake is set and the aircraft is on the ground. The erase continues for 5 seconds of the selection. The erase function is verified by a 400 Hz tone for 2 seconds in the HEADPHONE jack.

The aircraft right main bus supplies 28 VDC power through a 5 A circuit breaker and inertia switch to the SSCVR. The circuit breaker labelled SSCVR is located in position F4 on the 28 VDC avionics circuit breaker panel. The SSCVR typically consumes less than 8.5 W.

The SSCVR provides 18 VDC output and return with a continuous current capability of 100 mA for the SSCVR MMU and provides short circuit protection for this output.

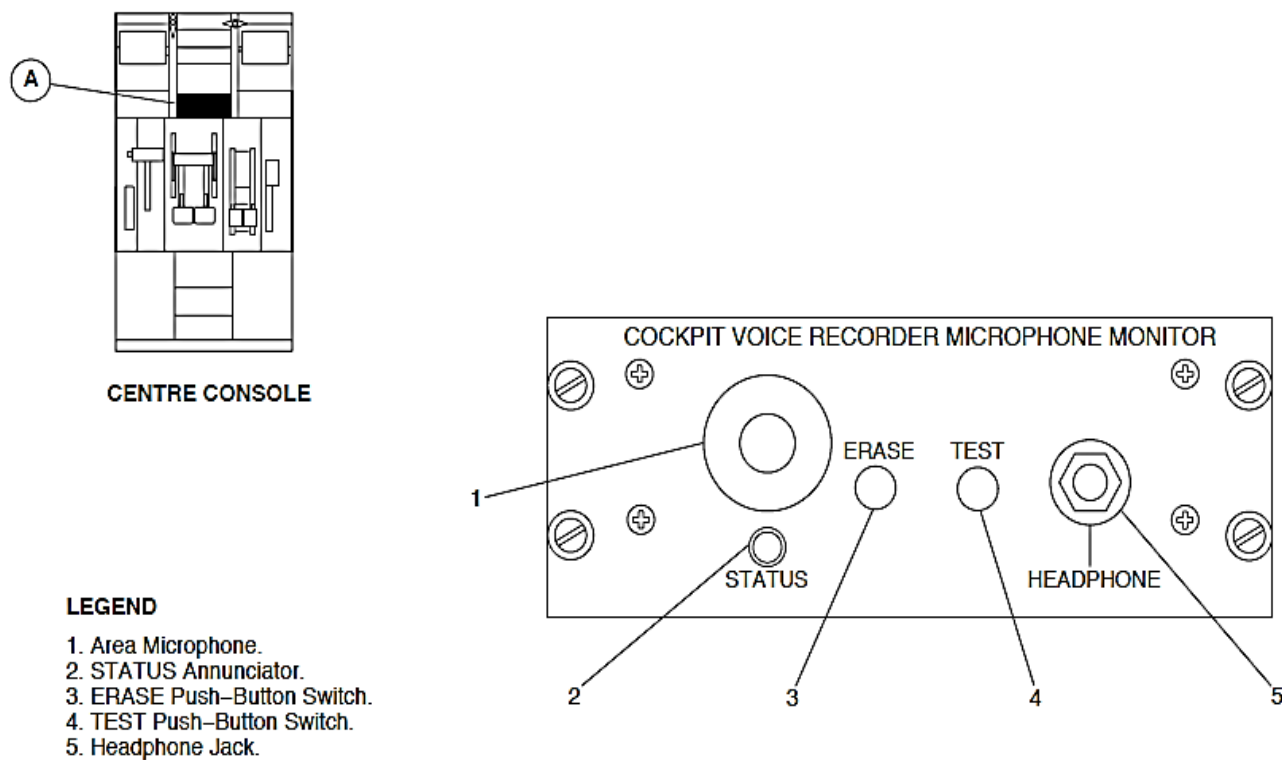


FIGURE 23- 78 MICROPHONE MONITOR UNIT (MMU)

Recorder, Solid State Cockpit Voice

Figure 23- 79 COCKPIT VOICE RECORDER LOCATOR

The SSCVR consists of the following LRUs, not including the basic chassis:

- An Interface and Control Board (ICB)
- A Crash-Survivable Memory Unit (CSMU)
- A Power Supply (PS).

The ICB is a single card located in the base housing of the SSCVR. The ICB controls all states and modes of the SSCVR. It performs the following functions:

- Record
- Erase
- Data download
- Test functions.

The CSMU is a solid state, non-volatile, mass storage device enclosed in a protective case. The CSMU provides storage for all input data.

The power supply converts the 28 VDC aircraft power to secondary power for the following components:

- Line Replaceable Units (LRU)
- Microphone Monitor Unit (MMU).

The SSCVR provides 18 VDC to the MMU. It is capable of 100 mA continuous current to the MMU.

The SSCVR is painted bright international orange and has reflective tape to aid its discovery following an accident. All sheet metal parts and fasteners are protected to inhibit corrosion and the crash survivable portion of the LRU displays the message "FLIGHT RECORDER DO NOT OPEN" and "ENREGISTREUR DE VOL NE PAS OUVRIR".

The SSCVR is cooled by convection and radiation to ambient air conditions. No forced air-cooling is required for the SSCVR.

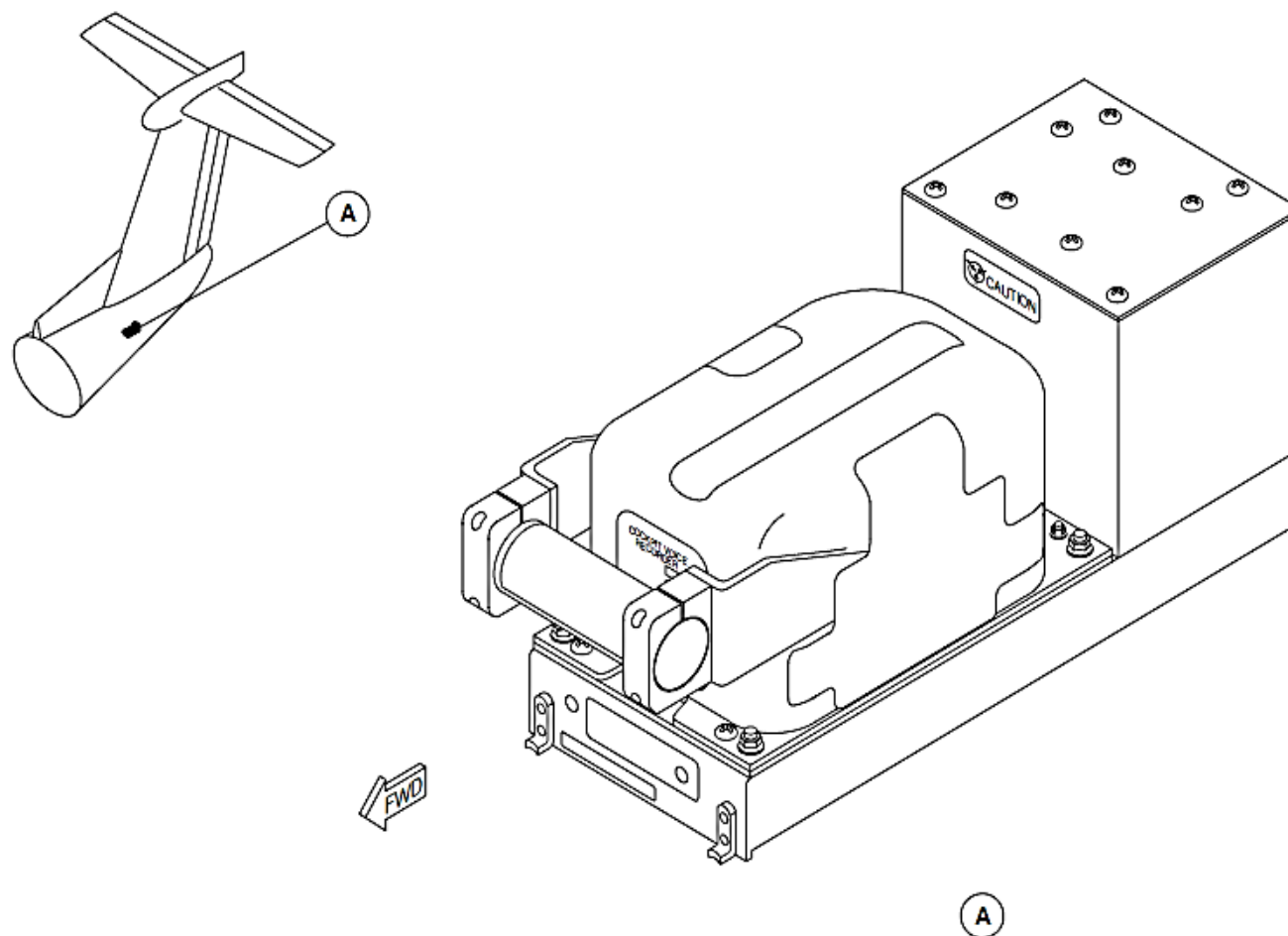


FIGURE 23- 79 COCKPIT VOICE RECORDER LOCATOR

Figure 23- 80 UNDERWATER LOCATOR BEACON LOCATOR

The SSCVR has an Underwater Locator Beacon (ULB) (also known as Underwater Locating Device (ULD)) secured to the container. It is securely attached to prevent separation during crash impact. The ULB is self-contained and powered.

The ULB is automatically activated when immersed in either fresh water or salt water at the depths from 0.5 to 20,000 feet (0.15 to 6096 meters). The ULB transmits an acoustic signal to help locate the SSCVR.

On aircraft with ModSum 4Q903309 or 4Q903310 incorporated, when the ULB is activated, it will transmit an acoustic signal of 37.5 ± 1 kHz frequency for a duration of 90 days or more. The ULB is self-powered by a battery with a typical service life of seven years from the date of manufacture. A label that shows the battery expiry date is attached to the ULB.

The ULB is accessible from the front face for easy removal and replacement.

Unit, Microphone Monitor

Figure 23- 81 MICROPHONE MONITOR UNIT LOCATOR

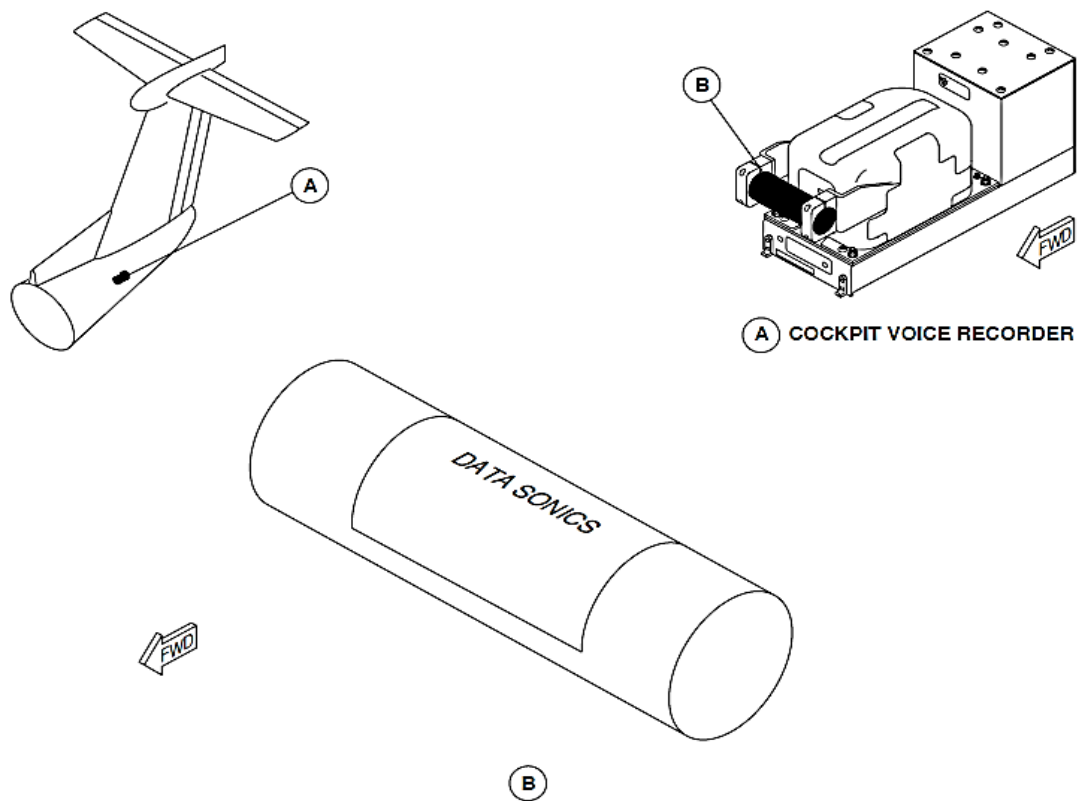
The MMU contains a compliant pre-amplifier circuit.

It interfaces and provides control and monitor functions of the SSCVR system.

The MMU is one LRU. It has a pre-amplifier card and the chassis components. The components that follow are attached to the pre-amplifier card:

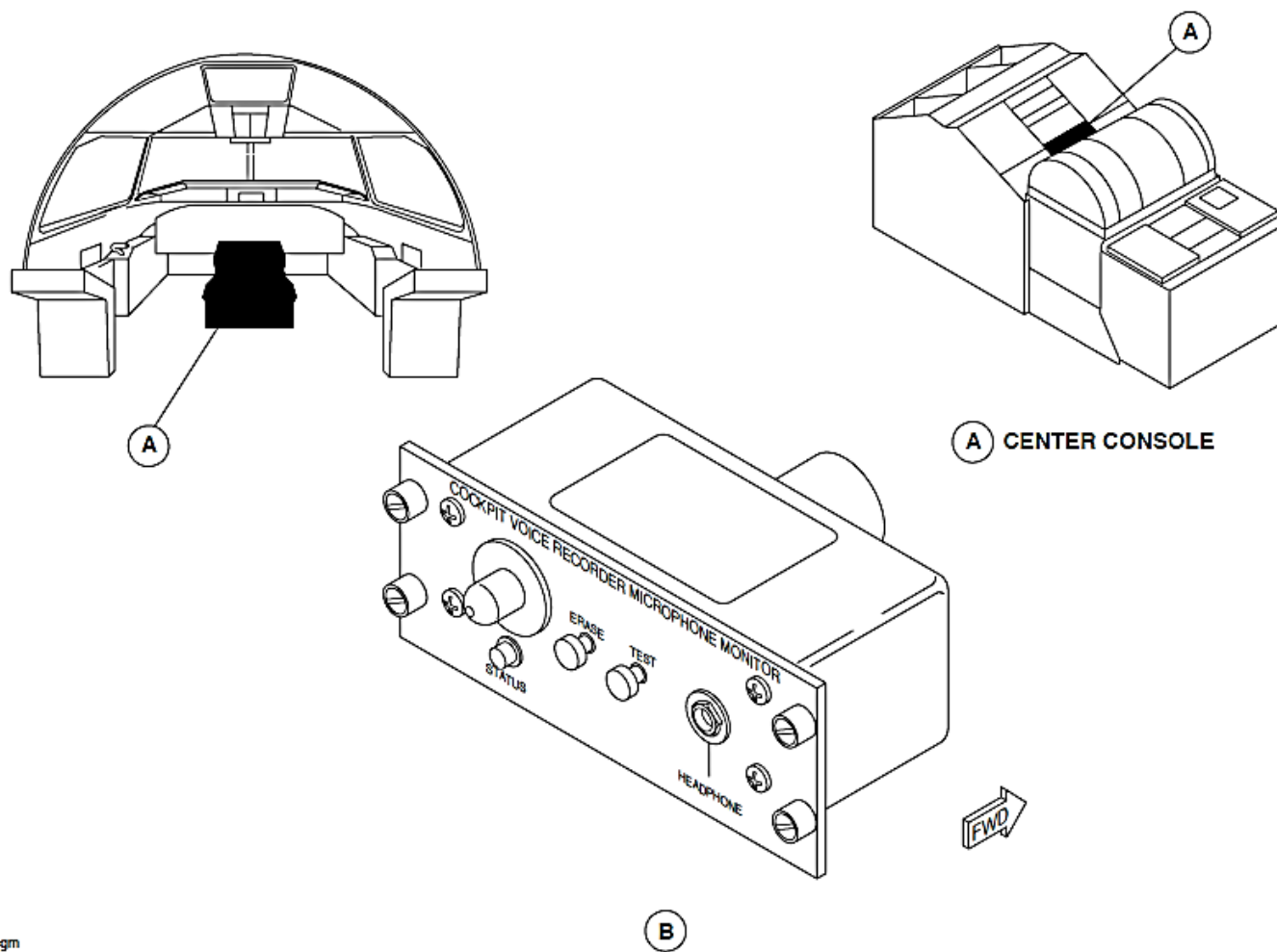
- SSCVR system status light
- Erase switch
- Test switch
- Headphone jack
- Cockpit area microphone.

The SSCVR MMU is cooled by convection and radiation to ambient air conditions. No forced air cooling is required for this LRU.



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FIGURE 23- 80 UNDERWATER LOCATOR BEACON LOCATOR



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FIGURE 23- 81 MICROPHONE MONITOR UNIT LOCATOR

Detailed Description – SSCVR System (Universal)

The SSCVR perform the functions that follow:

- Initialization
- Built-In-Test
- Audio Bulk Erase
- Audio Recording
- Data Link Recording
- Data Transfer.

Initialization: The system initialization module runs after a cold start and calls initialization procedures in the various other software components.

Built-In-Test: The built-in test module performs power-up, periodic software and hardware tests. Safety monitoring is part of the built-in test function, including memory testing and verifying stored checksums.

Audio Bulk Erase: Audio bulk erase marks every audio data record, such that it becomes unrecoverable via normal data-recovery methods.

Audio Recording: Audio recording stores the processed audio data in crash-protected memory.

Data Link Recording: Data link recording stores the unprocessed data in crash protected memory.

Data Transfer: The data transfer feature is used with the SSCVR ramp loader or a compatible data loader. This function provides an Ethernet interface, Transmission Control Protocol/Internet Protocol (TCP/IP), and Trivial File Transfer Protocol (TFTP) for loading data from the SSCVR.

The SSCVR has different operating states as follow:

- Start-Up
- Recording
- Maintenance
- Erase
- IBIT/Test.

Start-Up: The SSCVR always starts in this state when the power is applied.

Recording: This state is entered from the start-up state or when returning from IBIT/Test. The SSCVR records audio and data link in this state.

Maintenance: This state is entered from the stopped state when the CVR is powered and a request has been made by the PC. Audio data cannot be downloaded when the CVR is installed on the aircraft.

Erase: This state is entered from the stopped state when the erase button on the CCU is held for more than 2 seconds and then released, and the CVR on the aircraft is not recording. The erase function is also available while in the recording state provided the ground interlock equation (WOW * park brake on) is satisfied.

IBIT/Test: This state is entered from the stopped state or the recording state, when the test button is pressed on the CCU. When the IBIT is complete, the CVR transitions to the recording state.

The SSCVR system record the parameters at the same time that follow:

- Three narrowband voice channels
- One wideband area audio channel.

It supplies an audio and playback performance that is shown in the table that follows:

CHANNEL	AUDIO INPUT	BANDWIDTH HZ
1	Observer / PA	150 – 3,500
2	Copilot	150 – 3,500

CHANNEL	AUDIO INPUT	BANDWIDTH HZ
3	Pilot	150 – 3,500
4	Area	150 – 6,000

The SSCVR system supplies all four audio input channels through an audio HEADPHONE output jack at the same time for monitoring purposes. The output jack is located on the front panel of the CCU. As a secondary function, the jack socket output gives an aural tone to indicate a successful completion of erase and self-test functions.

The SSCVR receives and records timing data from the pilot clock through a low speed data bus. The clock input gives timing correlation between the SSCVR and the Flight Data Recorder (FDR). The SSCVR stores the clock input in a crash-survivable memory.

The SSCVR records 2 consecutive hours of four audio input channels and time data into crash-protected memory. The SSCVR begins storing digital audio data upon application of electrical power to the system.

The SSCVR gives a continuous audio output channel which is the composite of the four audio input channels in real time. It is available at the audio monitor HEADPHONE output jack located on the front panel of the CCU.

The SSCVR performs the following BITE functions:

- Power-up Self-Test
- Continuous Monitoring
- Push-To-Test
- Reporting of failures to the CCU
- Recording of failures in the crash-protected memory.

The CVR failure is recorded in the status log and a fail indicator is displayed on the CCU. The status log is stored in the crash protected memory.

Power-Up Self-Test: The SSCVR performs the power-up self-test when the power is applied.

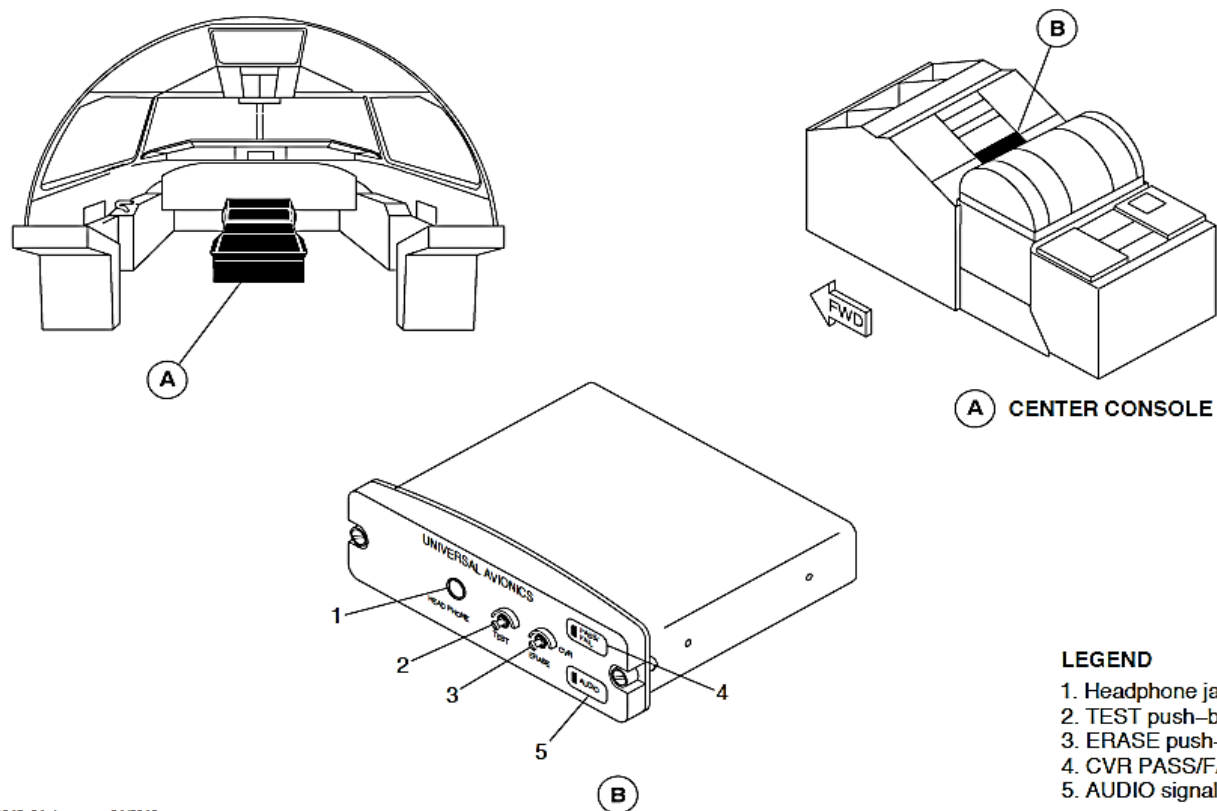
Continuous Monitoring: The SSCVR continuous monitoring function starts within 5 seconds of power-up and is performed during normal operation of the SSCVR.

Figure 23- 82 Cockpit Control Unit (CCU)

Push-To-Test: The push-to-test function is used to verify proper operation of the SSCVR when the TEST button on the CCU is pressed for more than 2 seconds and then released. During the push-to-test function, cockpit audio is not recorded. The SSCVR fault indicator on the CCU will rapidly flash green. When the push-to-test function has successfully been completed, the SSCVR fault Indicator on the CCU will remain green for 10 seconds. If a failure occurs, the fault indicator will flash yellow then remain yellow indicating that the push-to-test function has failed.

On aircraft without ModSum 4-458442 incorporated, the ERASE switch on the CCU is pushed for two seconds or longer to erase all audio data when the parking brake is set to ON and the aircraft is on the ground. The erase function continues for 5 seconds of the selection. The erase function is verified by a 400 Hz tone for 2 seconds in the HEADPHONE jack.

The aircraft left essential bus supplies 28 VDC power through a 5 A circuit breaker and inertia switch to the SSCVR. The circuit breaker labelled SSCVR is located in position D9 on the 28 VDC avionics circuit breaker panel. The SSCVR typically consumes less than 8 W power.



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FIGURE 23- 82 COCKPIT CONTROL UNIT (CCU)

Recorder, Solid State Cockpit Voice

Figure 23- 83 Cockpit Voice Recorder (Universal) Locator

The SSCVR consists of the following LRUs, not including the basic chassis:

- An Interface and Control Board (ICB)
- A Crash–Survivable Memory Unit (CSMU)
- A Power Supply (PS).

The SSCVR provides 28 VDC to the CCU.

The SSCVR is painted bright international orange and has reflective tape to aid its discovery following an accident. All sheet metal parts and fasteners are protected to inhibit corrosion and the crash survivable portion of the LRU displays the message "FLIGHT RECORDER DO NOT OPEN" and "ENREGISTREUR DE VOL NE PAS OUVRIR".

The SSCVR is a remote mounted LRU. It is located with the FDR in the aft fuselage section.

The SSCVR is cooled by convection and radiation to ambient air conditions. No forced air–cooling is required for the SSCVR.

On aircraft with 4–309257 OR SB84–31–88 incorporated, SSCVR with Recorder Independent Power Supply (RIPS) is installed. The RIPS monitors the aircraft power supplied to the SSCVR. When aircraft power is lost, the RIPS maintains power to the SSCVR. It supplies 10 minutes of power after 15 minutes of charging or at least 6 minutes of power to the SSCVR after 10 minutes of charging from any discharge condition.

Figure 23- 84 Underwater Locator Beacon (Universal) Locator

The SSCVR has an underwater locating device secured to the container. It is rigidly attached to prevent separation during crash impact. The Underwater Locator Beacon (ULB) is self-contained and powered.

The ULB is automatically activated when it is immersed in either fresh water or sea water at all depths from 0.5 to 20,000 ft. (.152 to 6096 m). When activated, the ULB radiates an acoustic signal for 30 days or more.

The ULB batteries require periodic replacement to ensure that they perform for 30 days in actual use. The ULB battery has a typical service life of 6 years from date of manufacture. Fabrication date of the battery is stamped on the ULB casing. The ULB must be sent to the manufacturer for battery replacement.

The ULB is accessible from the front face for easy removal and replacement.

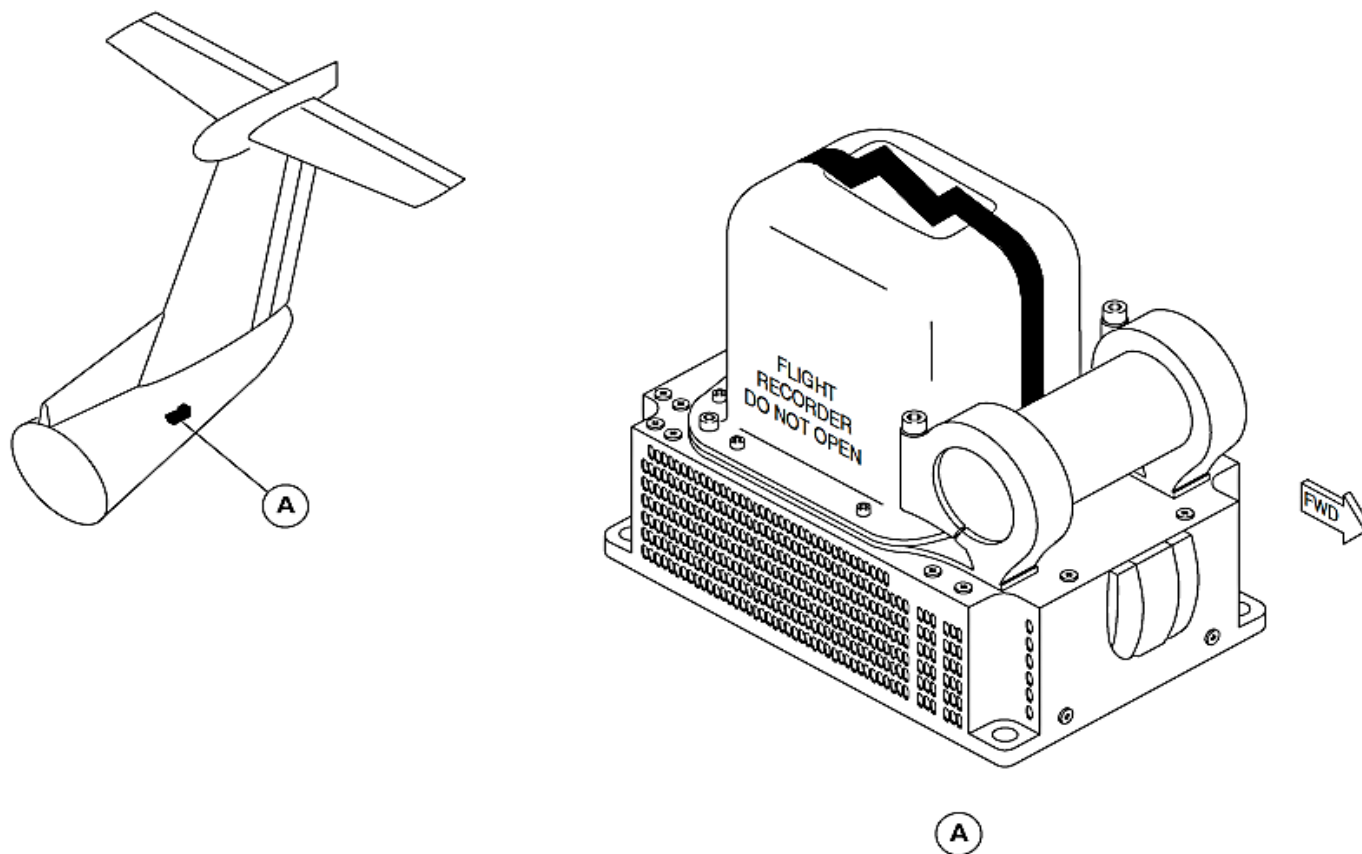


FIGURE 23- 83 COCKPIT VOICE RECORDER (UNIVERSAL) LOCATOR

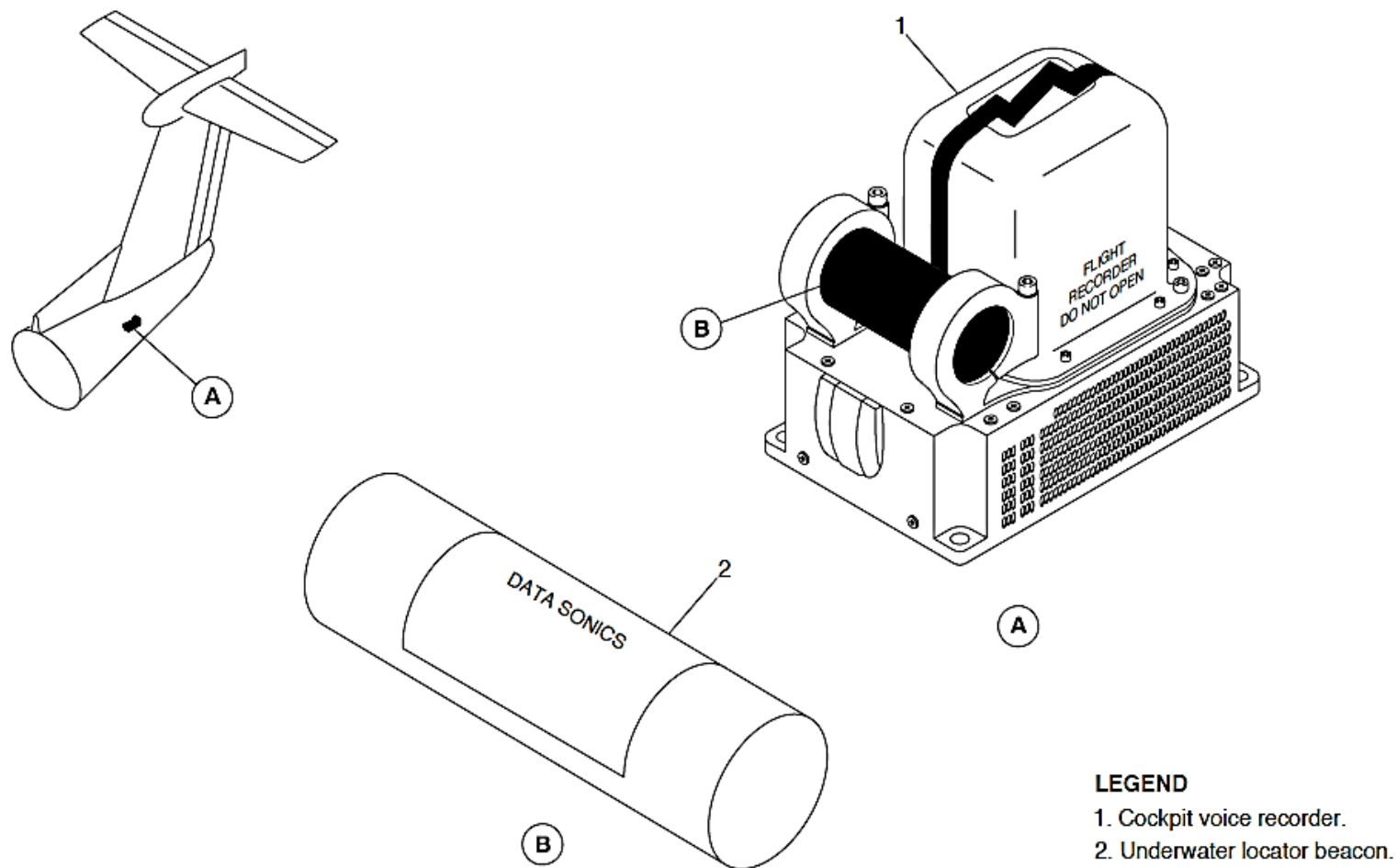


FIGURE 23- 84 UNDERWATER LOCATOR BEACON (UNIVERSAL) LOCATOR

Unit, Cockpit Control

Figure 23- 85 Cockpit Control Unit Locator

The CCU contains amplifier circuit.

It interfaces and provides control and monitor functions of the SSCVR system.

The CCU is an LRU. It contains amplifier for the components that follow:

- CVR PASS/FAIL light
- Erase switch
- Test switch
- Headphone jack
- AUDIO signal strength light.

The SSCVR provides 28 VDC to CCU.

The CCU is cooled by convection and radiation to ambient air conditions. No forced air cooling is required for this LRU.

Switch, Inertia (Honeywell and Universal)

Figure 23- 86 SSCVR Inertia Switch Locator

The normally closed inertia switch contains an inertia/impact activated double pole latching switch. The inertia switch is latched in an open circuit state when activated.

On aircraft without ModSum 4-309257 OR SB84-31-88 incorporated, one inertia switch is installed. One pole of the switch is connected in series with the 28 VDC aircraft supply to the SSCVR. The other pole is connected in series with the 28 VDC aircraft supply to the SSFDR.

On aircraft with ModSum 4-309257 OR SB84-31-88 incorporated, two inertia switches are installed. One switch is for SSCVR. Another switch is for SSFDR. One pole of SSCVR inertia switch is connected in series with the 28 VDC left essential bus to the SSCVR. The other pole is open.

The inertia switch is a highly reliable double pole normally closed inertia switch. It is factory preset to activate at 5.5 g (gravity). It incorporates a manual reset capability that is accessed through a push-to-reset switch.

The inertia switch is attached to the aircraft inclined at 45 degrees. The reset switch points towards the ground and the front of the aircraft, for upward and downward acceleration sensing. When the aircraft exceeds the preset acceleration level, the inertia switch will latch to its open circuit state and automatically stop the SSCVR from functioning.

The downward facing side of the inertia switch contains a push-to-reset switch to facilitate manual reset when the switch is in the open circuit latched state.

The inertia switch is attached on a 45 degree bracket. The bracket is attached to the structure under the floor in the center of the aircraft at station X196.

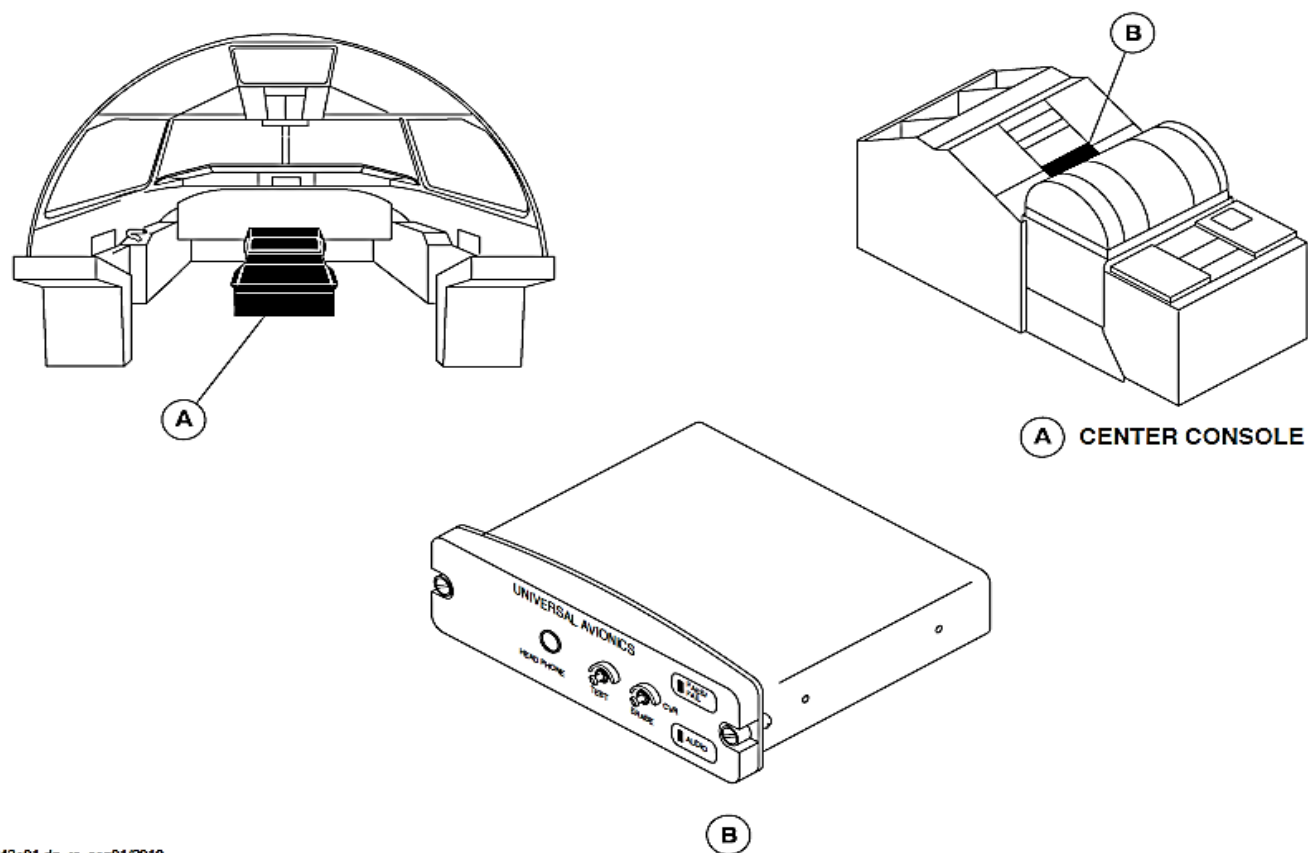


FIGURE 23- 85 COCKPIT CONTROL UNIT LOCATOR

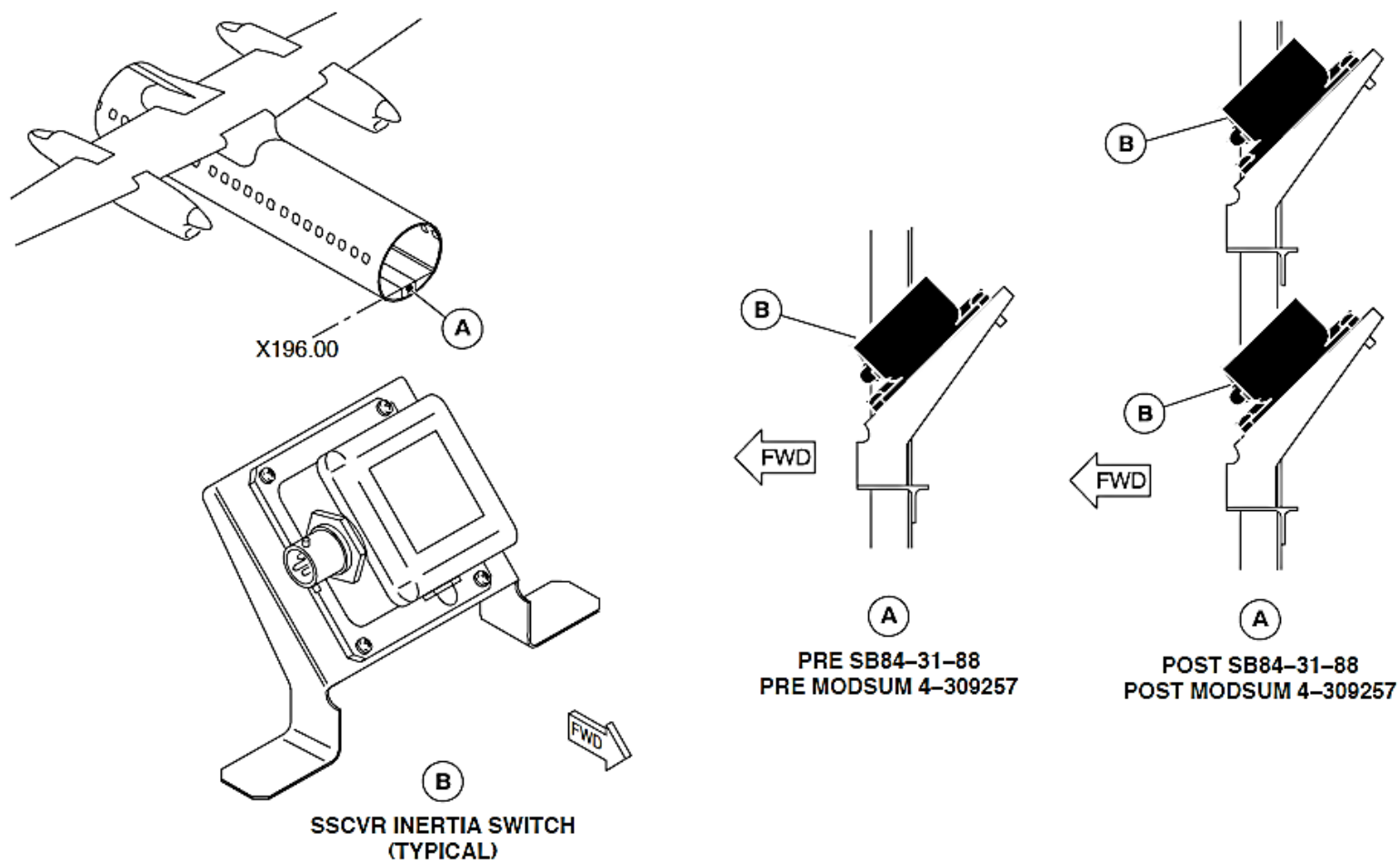


FIGURE 23- 86 SSCVR INERTIA SWITCH LOCATOR

Remote Microphone (Honeywell and Universal)

Figure 23- 87 CVR Remote MIC Locator

The remote microphone is installed on the glare shield. The remote microphone transmits cockpit audio signals to the MMU.

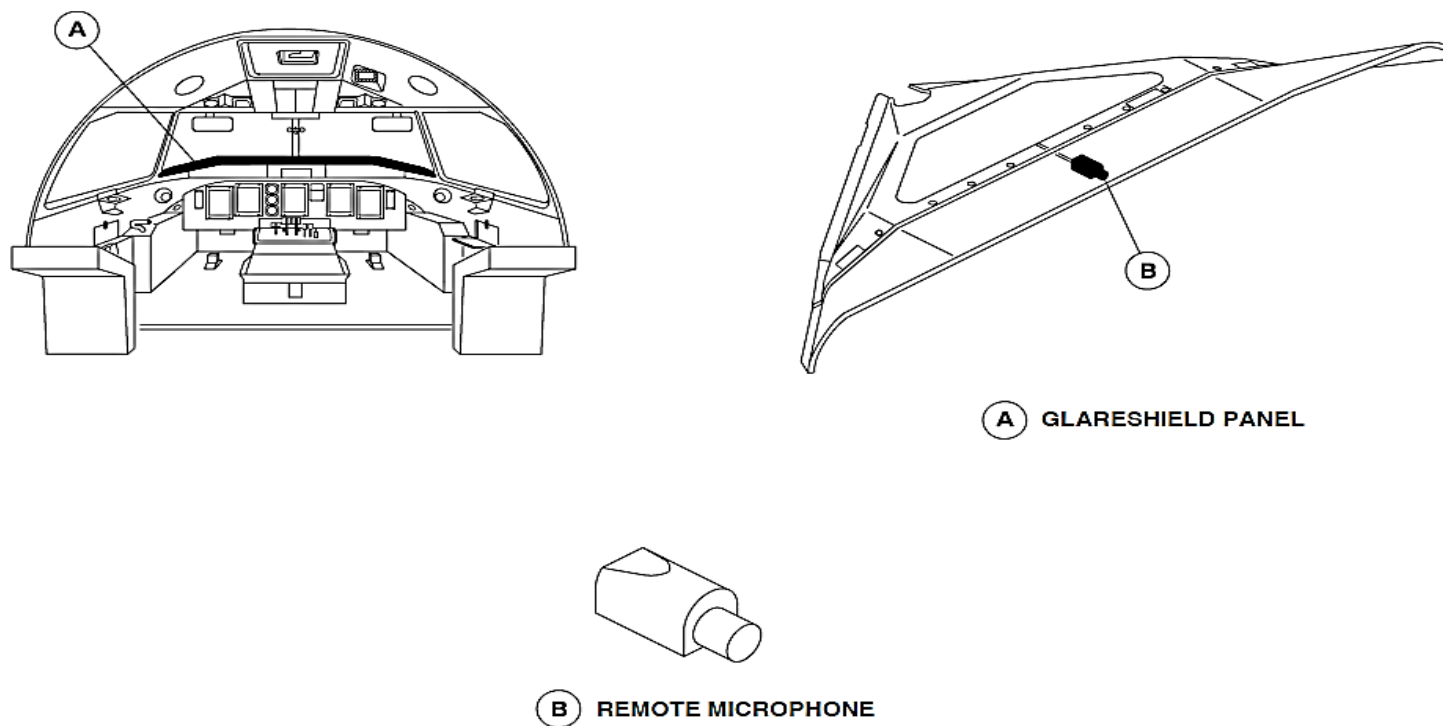


FIGURE 23- 87 CVR REMOTE MIC LOCATOR

CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM

Introduction

The Cabin Video Monitoring and Surveillance System (CVMSS) lets the flight crew see all of the area that is outside of the flight crew compartment. The system lets the flight crew identify persons near the flight compartment entry door and to detect suspicious behavior or possible threats.

The CVMSS uses the Control Display Unit (CDU) of Flight Management System (FMS) to display the video images from two cameras in the cabin.

Camera No. 1 is located in the ceiling panel immediately above the flight crew compartment entry door. Camera No. 2 is located in the ceiling panel, approximately 10 ft. (3 m) aft of the flight crew compartment entry door. Camera No. 1 looks down and camera No. 2 looks down and forward.

On aircraft with ModSum 4-309245 OR SB84-23-45 OR SB84-23-54 OR SB84-23-56 incorporated, the cabin video monitoring and surveillance system is installed.

On aircraft with ModSum 4-458529 incorporated, the cabin video monitoring and surveillance system is removed.

General Description

Figure 23- 88 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – BLOCK DIAGRAM

Figure 23- 89 Cabin Video Monitoring and Surveillance System – Block Diagram

The CVMSS is designed to let the flight crew monitor the flight compartment door entry area, forward galley and cabin areas from a seated position.

Redundant controls are provided at the center console that helps flight crew to control the system, power the cameras and monitor the video distribution.

The CVMSS is powered by the aircraft's left secondary bus.

On aircraft without ModSum 4-459648 OR with SB84-23-57 incorporated, the CVMSS has the components that follow:

- Two cameras located in the ceiling panels (23-72-11)
- A Camera Control Unit (CCU) (23-72-01)
- A camera power switch located in the center console (23-72-06)
- Two video converters located in under floor on the left side of the flight crew compartment (23-72-16).

On aircraft with ModSum 4-459648 OR SB84-23-56 incorporated, the CVMSS has the components that follow:

- Two cameras located in the ceiling panels (23-72-11)
- A Camera Control Unit (CCU) (23-72-01)
- A camera power switch located in the center console (23-72-06)
- A video converter located in under floor on the left side of the flight crew compartment (23-72-16).

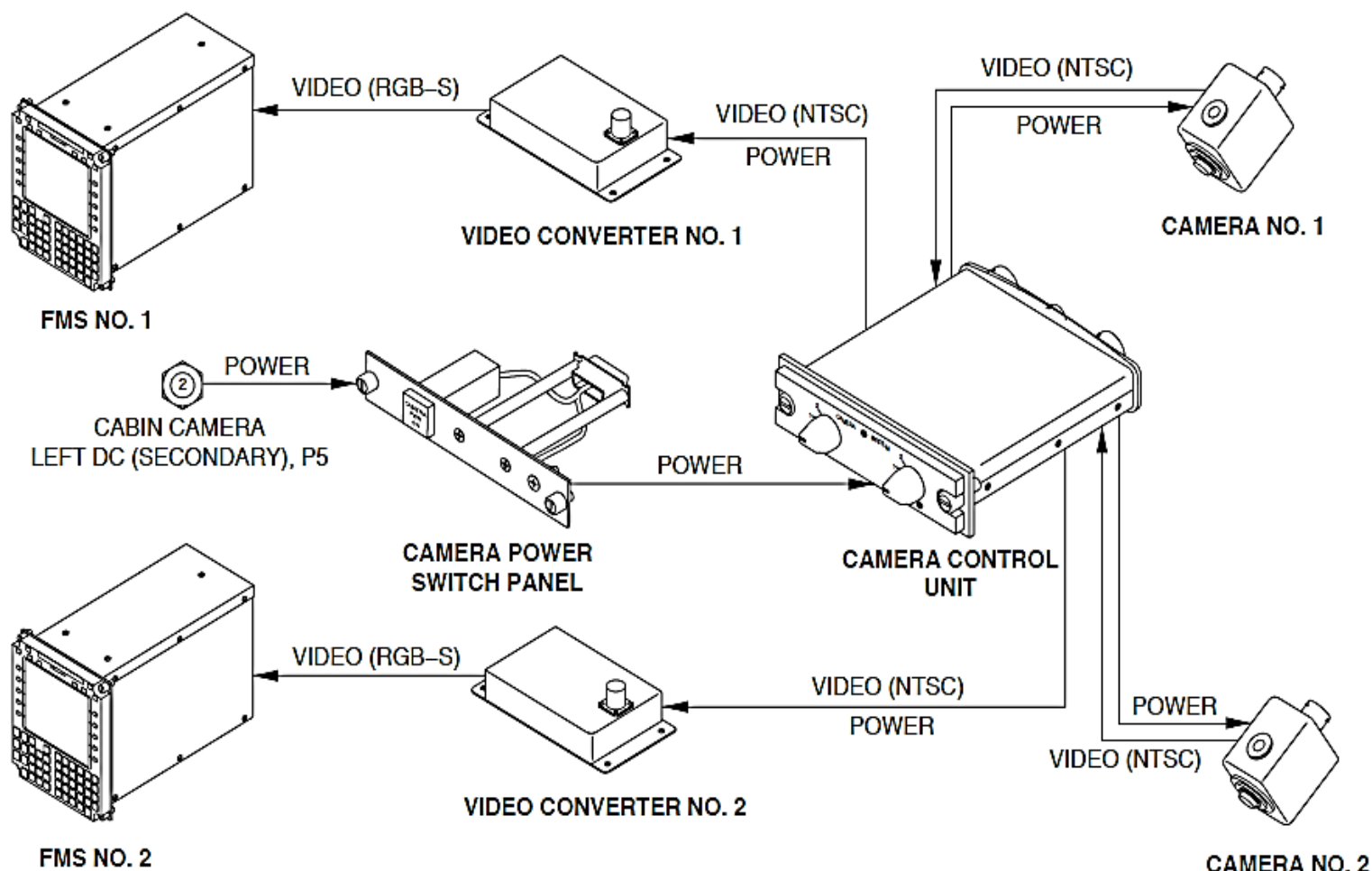


FIGURE 23- 88 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – BLOCK DIAGRAM

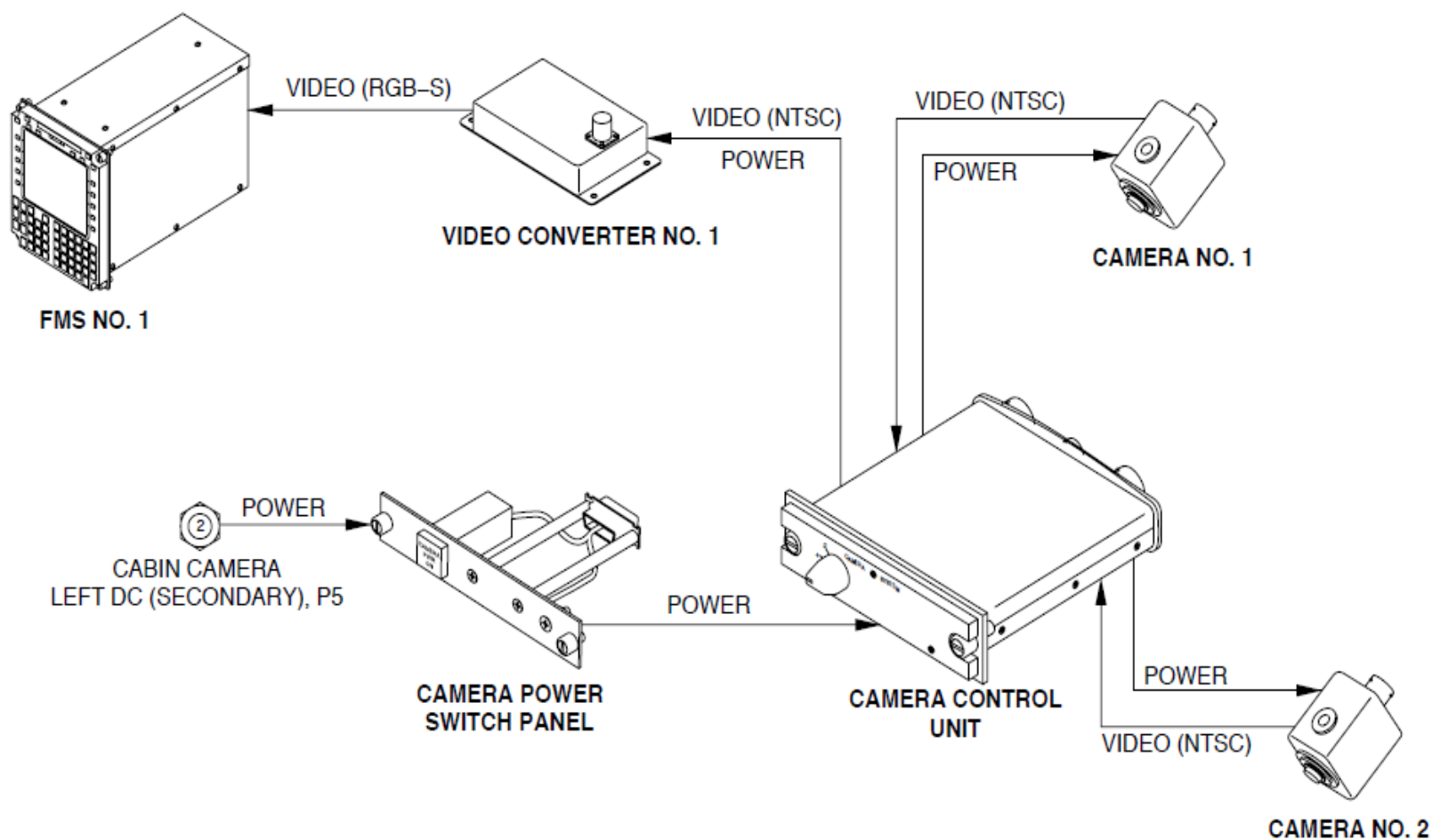


FIGURE 23- 89 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – BLOCK DIAGRAM

Detail Description

Figure 23- 90 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (Sheet 1 of 2)

Figure 23- 91 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (Sheet 2 of 2)

Figure 23- 92 Cabin Video Monitoring and Surveillance System – Component Locator (Sheet 1 of 2)

Figure 23- 93 Cabin Video Monitoring and Surveillance System – Component Locator (Sheet 2 of 2)

Figure 23- 94 VIDEO MODE SELECTION – FMS (Sheet 1 of 2)

Figure 23- 95 VIDEO MODE SELECTION – FMS (Sheet 2 of 2)

Camera Power Switch

The CVMSS uses a camera power switch to control the power to the CCU, which in turn supplies the power to the cameras and the video converters. The power switch is located at aft of the weather radar (WXR) control panel in center console of the flight crew compartment.

Camera Control Unit

The Camera Control Unit (CCU) is used to select the video from the two cameras and display on the FMS CDU. The output from each individual camera can be selected on the CCU. The output images selected on the CCU are displayed on the FMS 1 and FMS 2 (if installed).

On aircraft without ModSum 4-459648 OR with SB84-23-57 incorporated, the CCU has two independent sets of controls allowing the pilot and the copilot to view either of the cameras via a select switch.

On aircraft without ModSum 4-459648 OR with SB84-23-57 incorporated, the left camera knob controls camera selection to FMS 1 and the right camera knob controls camera selection to FMS 2.

On aircraft with ModSum 4-459648 OR SB84-23-56 incorporated, the CCU has one control allowing the pilot or the copilot to view either of the cameras via the select switch.

On aircraft with ModSum 4-459648 OR SB84-23-56 incorporated, the camera knob controls camera selection to FMS 1.

The CCU power distribution architecture has been designed so that the failure of a single component will not cause a total system failure. The camera gets 12V DC from the CCU.

The CCU is located at the extended center console in the flight compartment.

Camera

The CVMSS uses two near infrared cameras to capture video at the flight compartment door entry area, forward galley and cabin areas. The two near infrared cameras used are black and white cameras.

The use of a highly sensitive Closed Circuit Device (CCD) sensor assures visibility in the low light conditions (approximately 100 times more sensitive than a standard low light camera). This extreme low light capability provides

clear visibility in all the normal and the emergency aircraft lighting conditions.

The near infrared cameras are used to obtain visual images of different locations of the aircraft.

Two cameras are installed on the cabin ceiling outside of the flight crew compartment entry door.

Camera No. 1 is located in the ceiling panel immediately above the flight crew compartment entry door, looking down. Camera No. 1 is positioned so that an image of the head and features of the person standing at the flight crew compartment entry door can be seen.

The area that is seen by camera No. 2 is the area bounded by the flight compartment door, lavatory, wardrobe, forward baggage compartment and the airstair door. The video images produced by the two cameras are displayed on the FMS CDU.

The two cameras operate as follows:

- Power is applied to the camera whenever power is applied to the CCU. When power is applied to the camera, the cameras will produce an output video signal.
- The output video signal from the camera is a 0.7V peak-to-peak NTSC formatted composite video signal.

The mounting assembly for the cameras No. 1 and No. 2 allows for easy installation and quick mounting or adjustment of the cameras.

Video Converters

On aircraft without ModSum 4-459648 OR with SB84-23-57 incorporated, the CVMSS uses two video converters. The video converters will convert the camera NTSC video signal from the CCU to RGB-S (Red, Green, Blue – Synchronized) video signal, which will be used by the FMS to display images that can be viewed by the flight crew on the FMS CDU.

On aircraft with ModSum 4-459648 OR SB84-23-56 incorporated, the CVMSS uses one video converter. The video converter will convert the camera NTSC video signal from the CCU to RGB-S (Red, Green, Blue – Synchronized) video signal, which will be used by the FMS to display images that can be viewed by the flight crew on the FMS CDU.

The CVMSS uses the FMS CDU to display the video captured by the cameras.

To access the active video display on the CDU of FMS, the ON/OFF-DIM key is pressed on the FMS and the DISPLAY LSK [4R] and the VIDEO LSK [4R] are selected. The FMS enters into the video mode.

In the FMS video mode, the Line Shift Keys (LSK) [1R] and [2R] are used to control the brightness of the video displayed. The LSK [1R] is used to brighten the display and the LSK [2R] is used to dim the display.

While the FMS is in video mode, the FMS will continue to navigate in normal condition and no CDU messaging will occur. If any keys other than LSKs [1R] and [2R] are selected, the FMS will exit the FMS video mode.

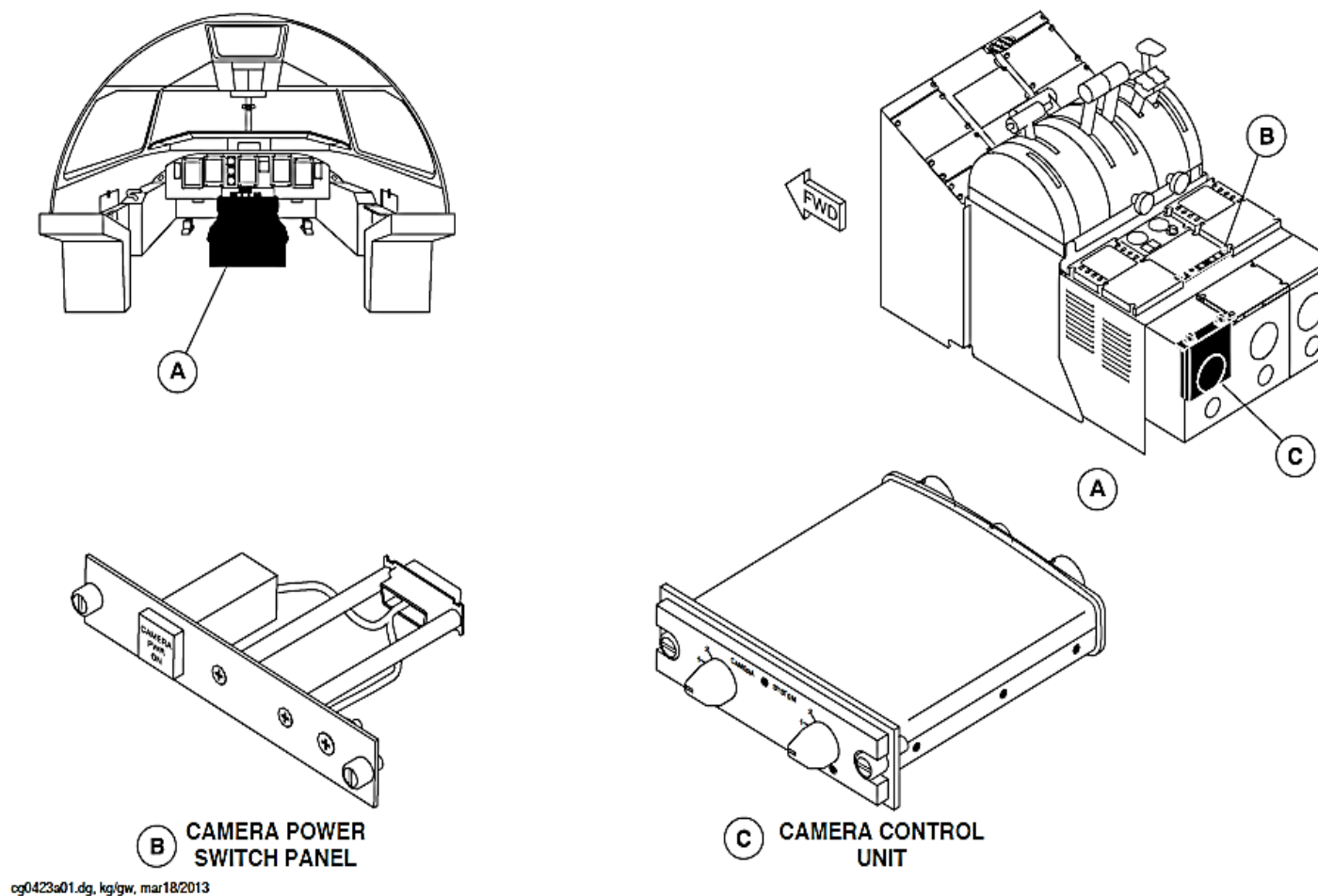
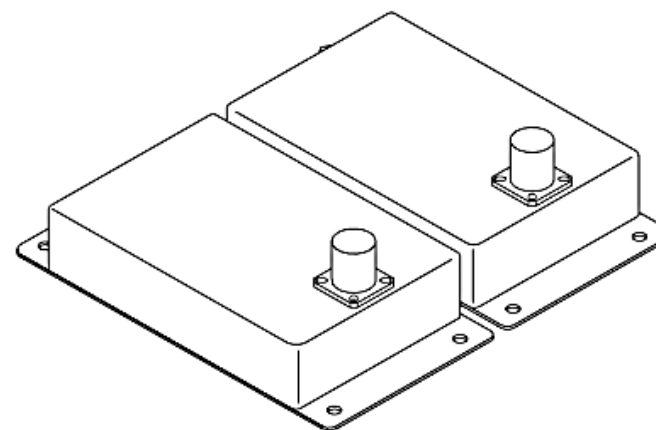
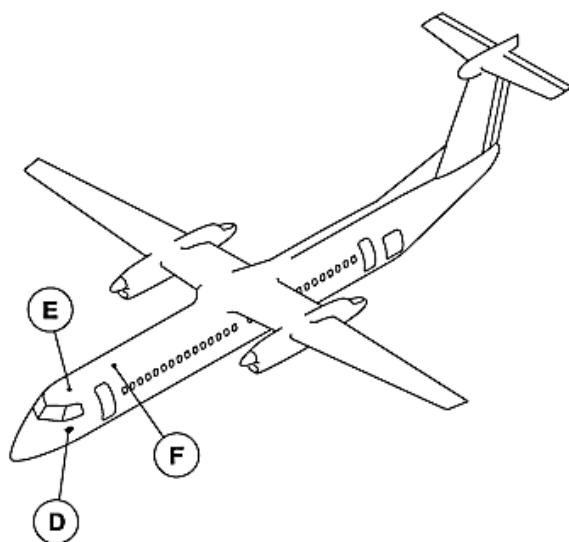
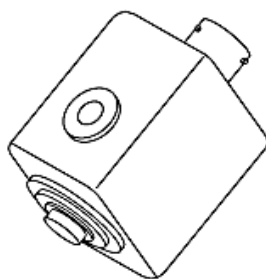


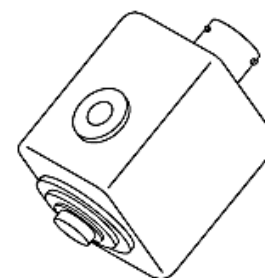
FIGURE 23- 90 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (SHEET 1 OF 2)



D VIDEO CONVERTER



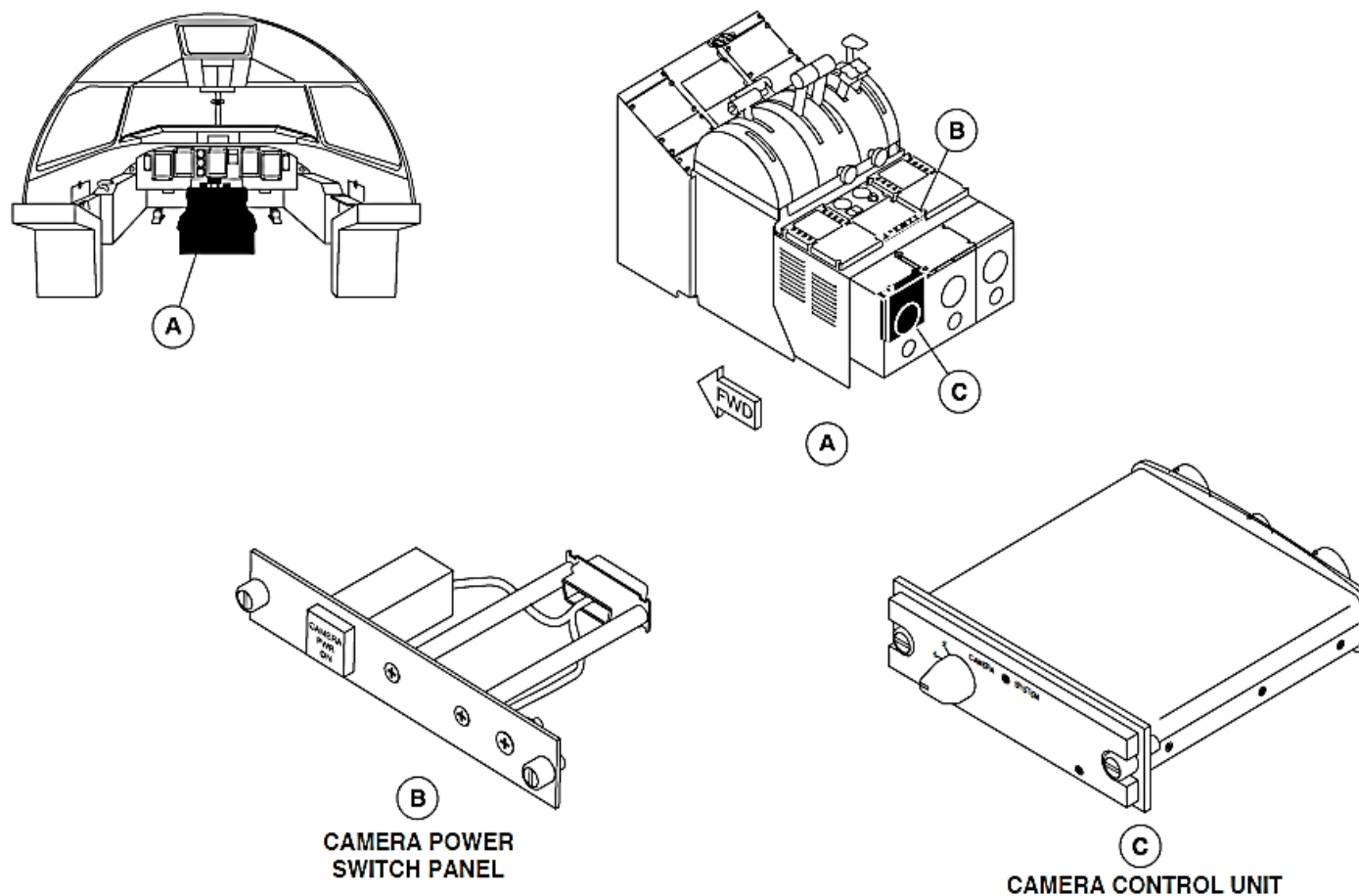
E CAMERA NO. 1



F CAMERA NO. 2

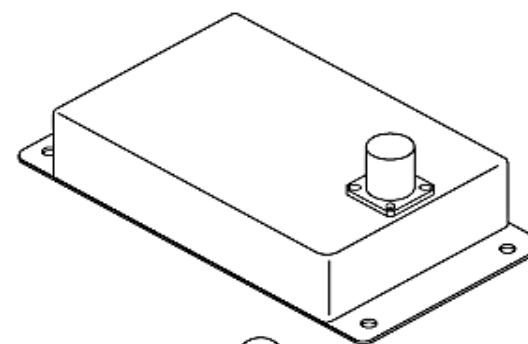
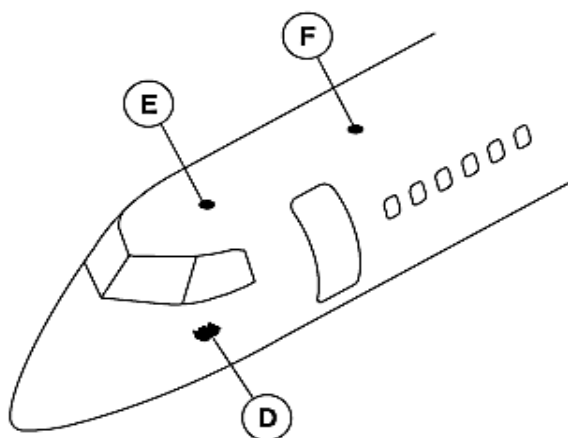
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FIGURE 23- 91 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (SHEET 2 OF 2)

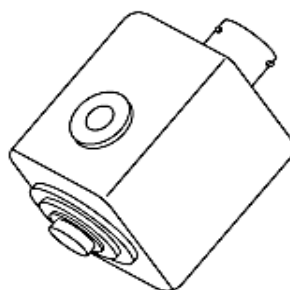


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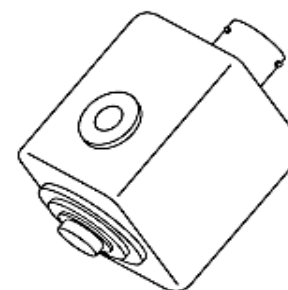
FIGURE 23- 92 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (SHEET 1 OF 2)



D
VIDEO CONVERTER



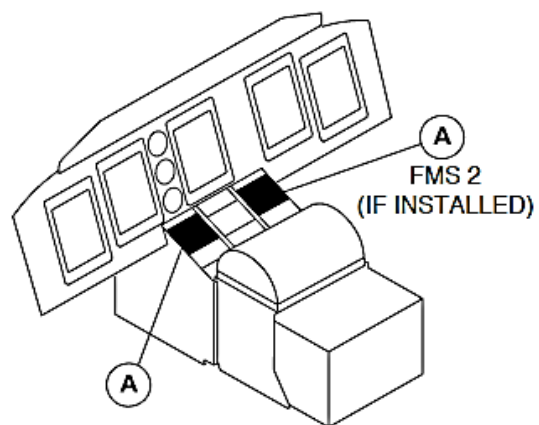
E
CAMERA NO. 1



F
CAMERA NO. 2

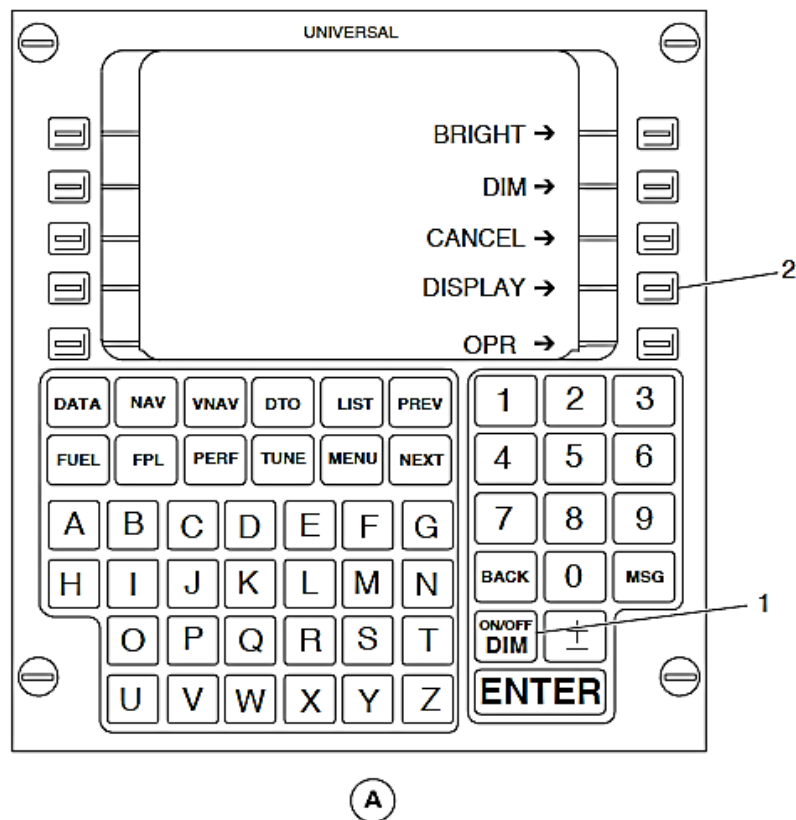
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FIGURE 23- 93 CABIN VIDEO MONITORING AND SURVEILLANCE SYSTEM – COMPONENT LOCATOR (SHEET 2 OF 2)



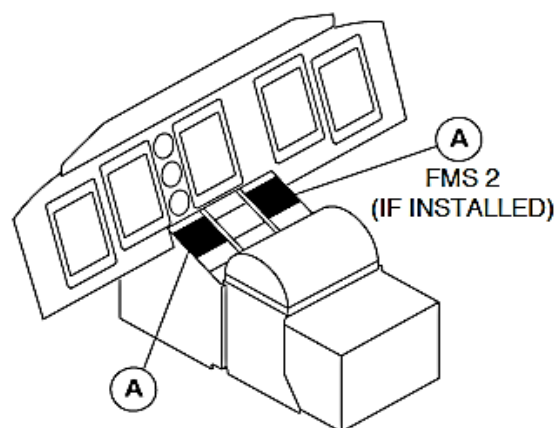
LEGEND

- 1. ON/OFF – DIM key.
- 2. Display LSK.

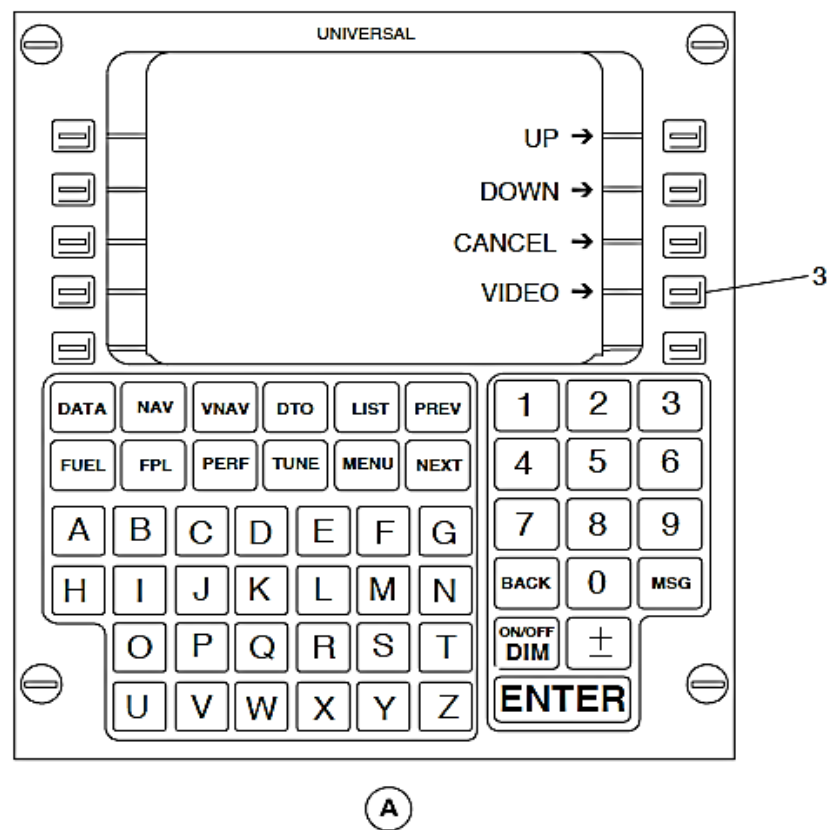


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FIGURE 23- 94 VIDEO MODE SELECTION – FMS (SHEET 1 OF 2)



LEGEND
3. Video LSK.



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FIGURE 23- 95 VIDEO MODE SELECTION – FMS (SHEET 2 OF 2)

RADIO MANAGEMENT SYSTEM

Introduction

The Radio Management System is used to control and monitor the Radio Communication (RCOM) and Radio Navigation (RNAV) systems.

General Description

Figure 23- 96 PACIS BLOCK DIAGRAM

Figure 23- 97 PACIS BLOCK DIAGRAM, INPUTS

The radio management system has two Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) that have the functions that follow:

- Tunes the RNAV and RCOM systems and controls their operational modes
- Shows all data that is related to the operation of the RNAV and RCOM systems
- Monitors the operation of the RCOM and RNAV systems to show malfunctions
- Controls the operation of the Passenger Address and Communication Interphone System (PACIS).
- Controls the operation of the Audio Integration System (AIS).
- Controls the operation of the Central Diagnostics System (CDS).

The ARCDU front panel has the controls and indications to let the pilots control the audio and radio systems installed.

The front panel of the ARCDU 1 and ARCDU 2 is divided into the parts that follow:

- Active Matrix Liquid Crystal Display (LCD)
- Keyboard
- Audio and Microphone Control

Active Matrix Liquid Crystal Display (LCD): The active LCD has related side keys and a tuning knob to control communication and navigation radio frequency and mode selections. The function of a side key depends on the data that is shown in the display area.

Keyboard: The keyboard has specific keys to control the function of the RCOMs, RNAVs, PACIS, and CDS.

Audio and Microphone Control: The bottom part of the ARCDU front panel has audio potentiometer push-button switches, a rotary microphone selector switch and toggle switches to control the (AIS).

The Radio Management System has the component that follows:

- Audio and Radio Control Display Units.

The two ARCDUs are connected to each other and to the other systems that follow:

- RNAV receivers
- RCOM receivers
- PACIS
- AIS
- Flight Data Processing System (FDPS)
- Traffic Collision Avoidance System (TCAS)
- Flight Management System (FMS)
- CDS.

To show the malfunction, the ARCDUs sense the condition of the ADFs and DMEs through FDPS 1 and FDPS 2. All other malfunctions are sensed directly by the ARCDUs.

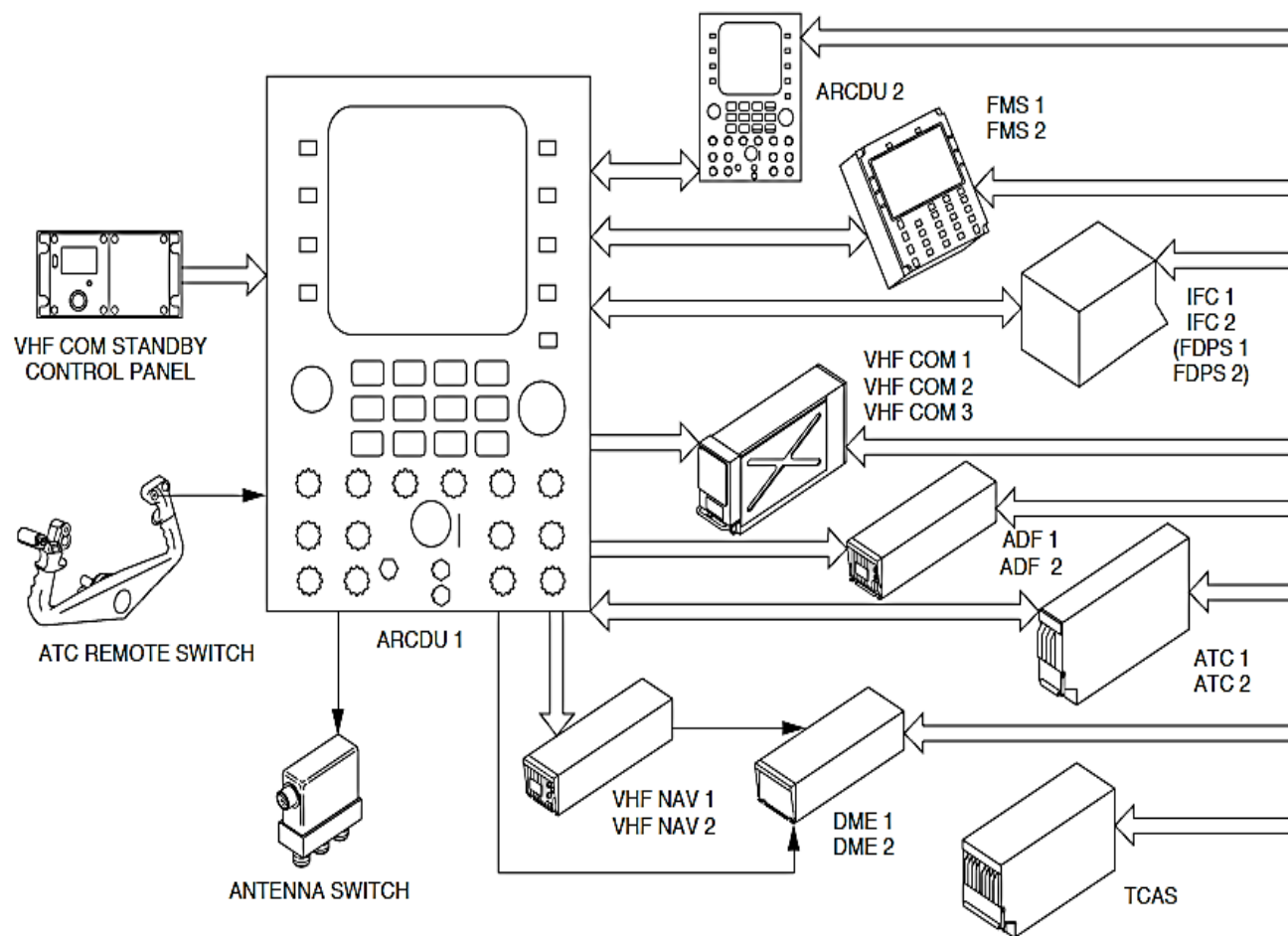


FIGURE 23- 96 PACIS BLOCK DIAGRAM

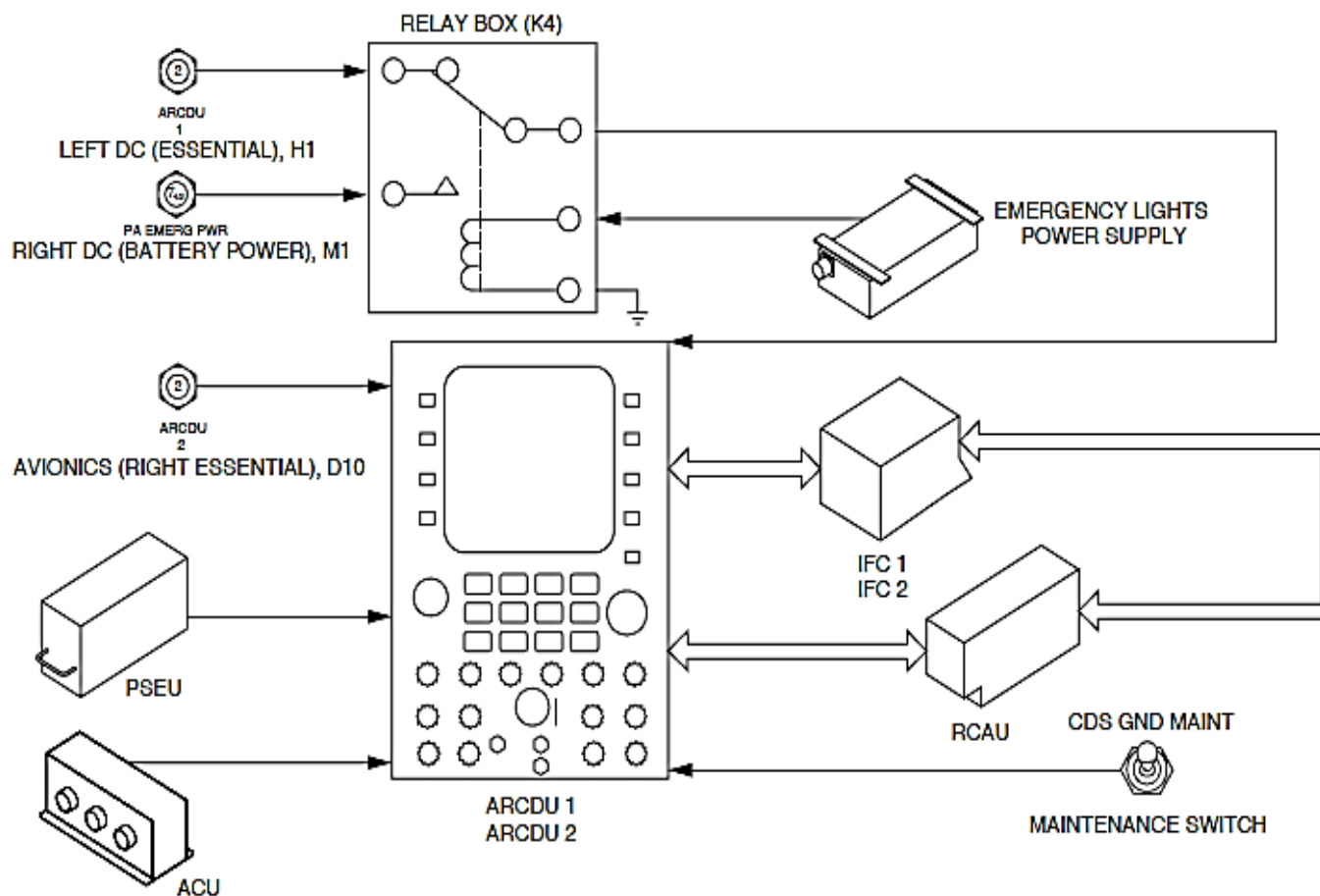


FIGURE 23- 97 PACIS BLOCK DIAGRAM, INPUTS

Detailed Description

The ARCDU 1 and ARCDU 2 functions in the modes that follow:

- Initialization (INIT)
- Power-On Self-Test (POST)
- Operational (OPER)
- Maintenance.

Initialization (INIT): The ARCDU functions in the initialization mode after an electrical power interruption that continues for more than 7 seconds. This mode initializes the hardware and software.

If the ARCDU does not receive a maintenance request, it will then function in the POST state.

Power on Self-Test (POST): The ARCDU does a POST after a time long-term power interruption while the aircraft is on the ground.

The POST senses the ARCDU hardware malfunctions that are not sensed by internal continuous monitoring circuits or apparent to the flight crew.

Figure 23- 98 ARCDU, POST INDICATION

During the POST mode, the ARCDU display shows a TEST IN PROGRESS message in white characters.

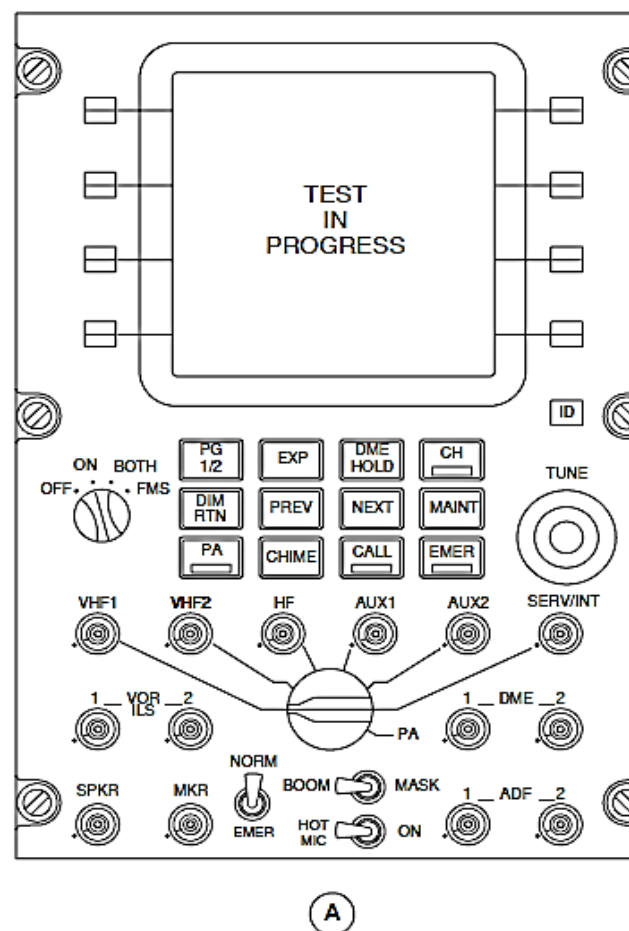
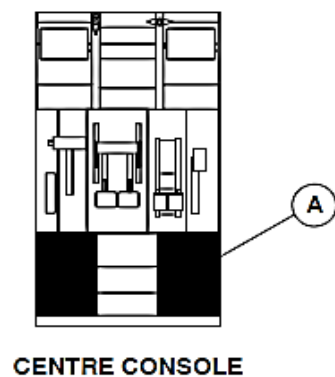
Figure 23- 99 ARCDU, POST MALFUNCTION INDICATION

If the POST fails while the aircraft is on the ground, the ARCDU display shows an ARCDU FAILED message in red characters.

Operational (OPER) Mode: The ARCDU screen pages that are divided into eight display areas with related side keys. The side keys are used to control the parameters that are shown in the display area.

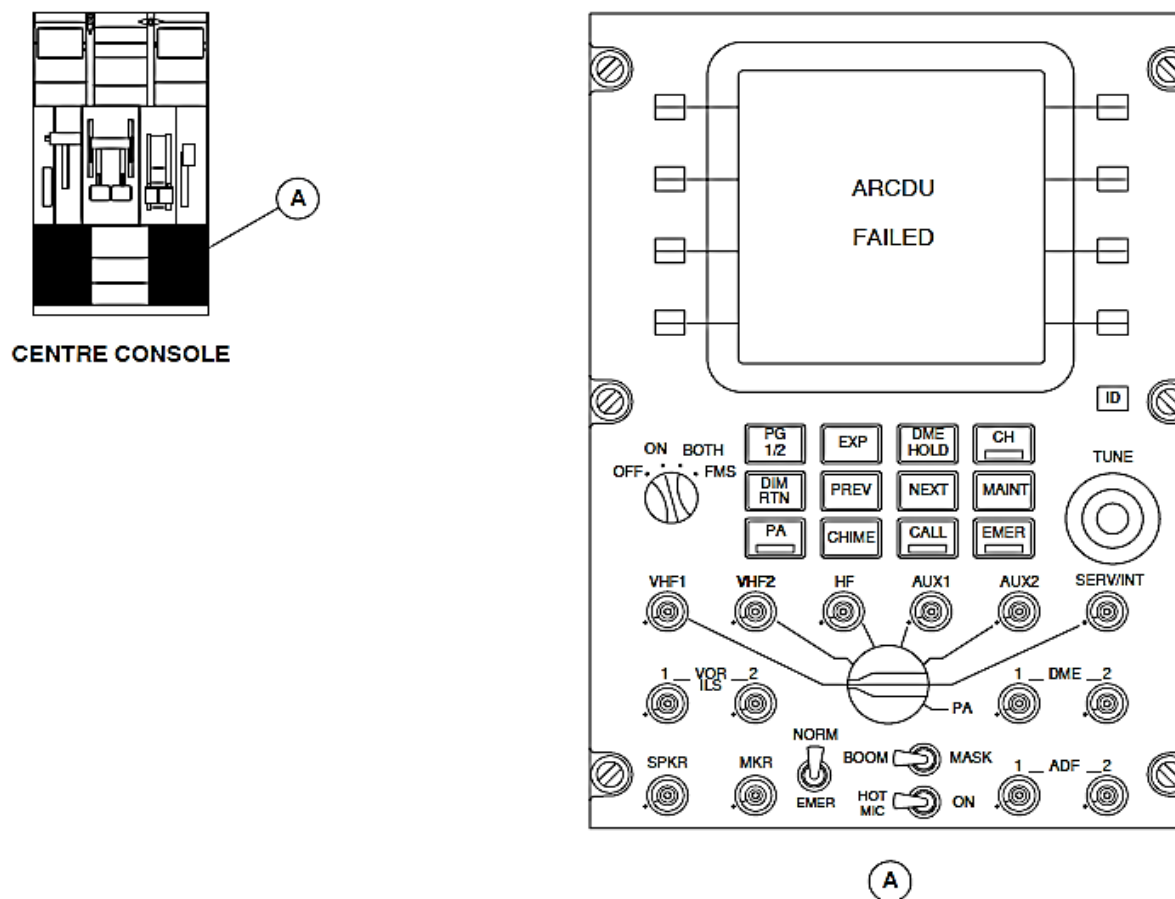
The ARCDU screen shows the types of page that follow:

- Main
- Particular
- Channel Memory.



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FIGURE 23- 98 ARCDU, POST INDICATION



tsc34a02.cgm

FIGURE 23- 99 ARCDU, POST MALFUNCTION INDICATION

Figure 23- 100 ARCDU, SCREEN

Main Page: The main page is divided into display areas to show the audio and frequency selections for the radio systems.

It is divided into the areas that follow:

- Label
- Audio level adjustment bar graph
- First data line, active frequency
- Second data line, modes
- Third data line, preset frequency
- Side key.

Label: The label has four characters to show the name of the system.

Audio Level Adjustment Bar Graph: A vertical bar graph is shown next to the label to show its related potentiometer push-button switch selection.

First Data Line: The first data line has seven characters in large characters to show the active frequency.

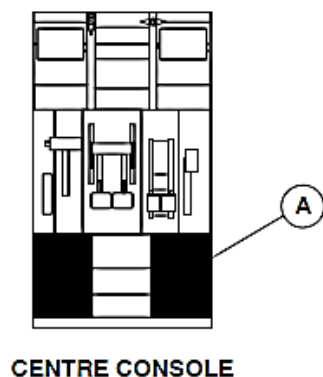
Second Data Line: The second data line has eight characters in small characters to show the mode selection.

Third Data Line: The third data line has eight characters in small characters to show the preset frequency.

Side Key: The side key is used to control the active and preset frequencies.

NOTE

The ATC display area shows active and presets codes rather than active and presets frequencies in the first and third data lines.



LEGEND

1. First Data Line, Active Frequency.
2. Label.
3. Side Key.
4. Audio Level Adjust Bar Graph.
5. Third Data Line, Preset Frequency.
6. Second Data Line, Modes.

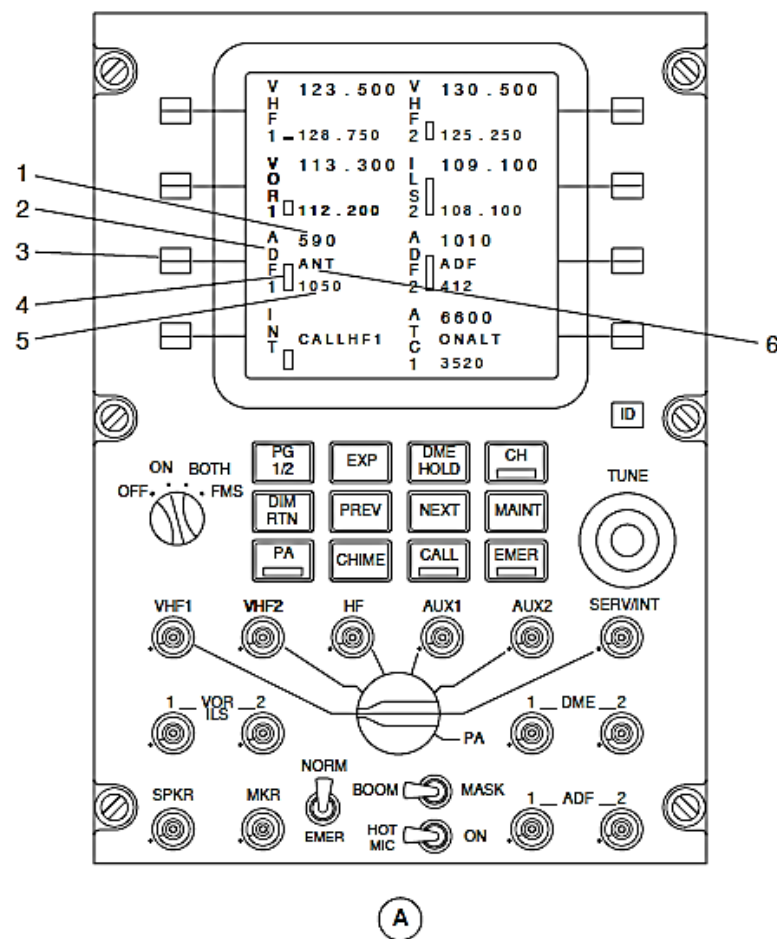


FIGURE 23- 100 ARCDU, SCREEN

Figure 23- 101 ARCDU, MAIN PAGE 1

Figure 23- 102 ARCDU, MAIN PAGE 2

There are two main pages. The main page 1 shows the aircraft's basic radio configuration and main page 2 shows the different optional configuration.

The first main page display areas show parameters for the systems that follow:

- VHF communications 1
- VHF communications 2
- VHF navigation 1
- VHF navigation 2
- Automatic Direction Finding (ADF) 1
- ADF 2
- Interphone
- ATC transponder.

The second main page display areas show parameters for the systems that follow:

- VHF communications 3
- Interphone
- ATC transponder.

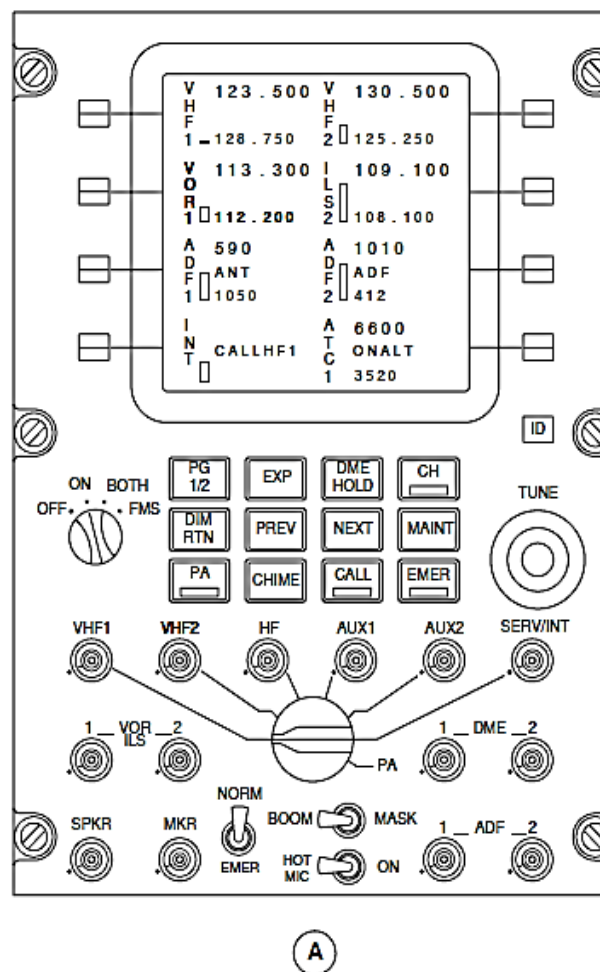
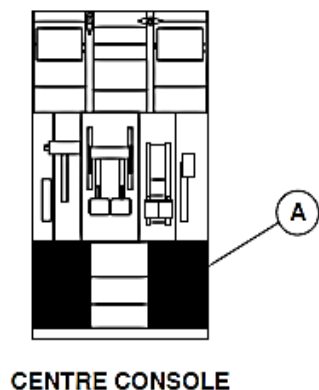
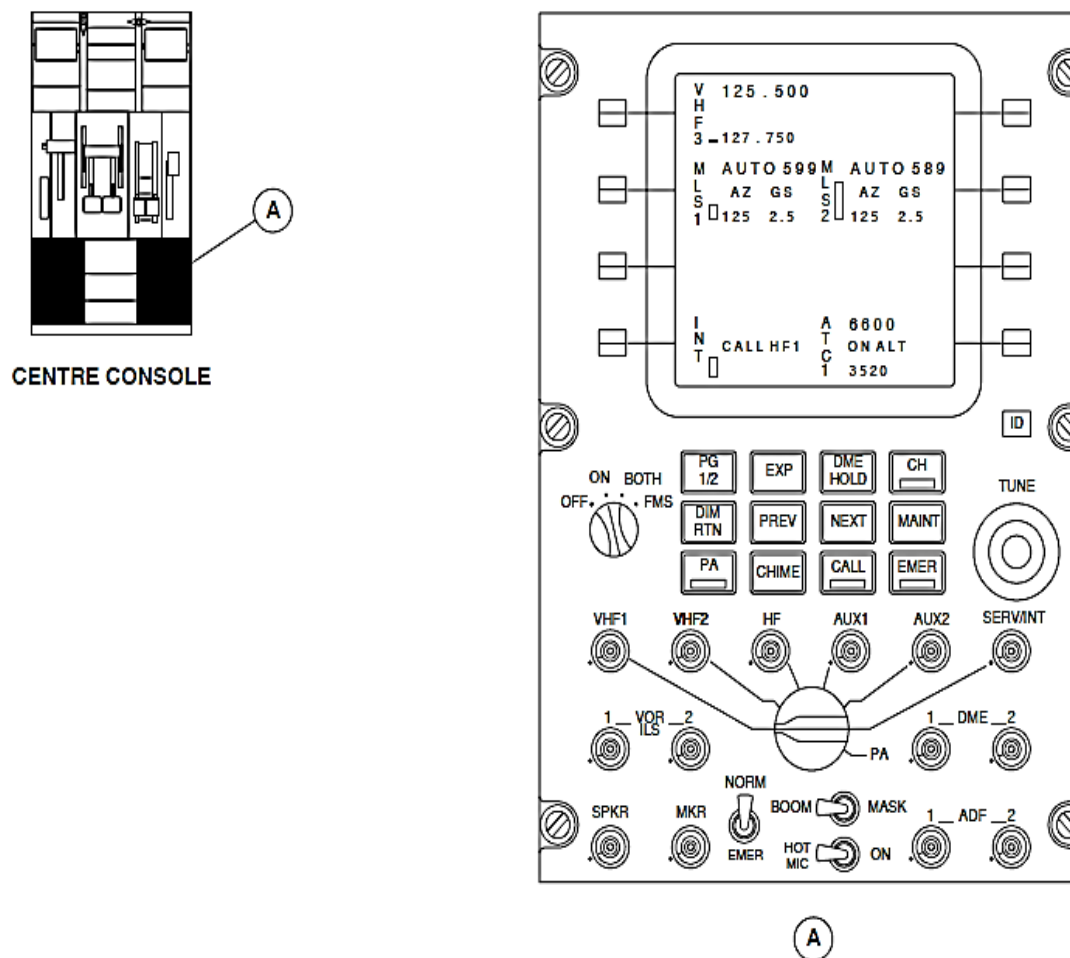


FIGURE 23- 101 ARCDU, MAIN PAGE 1



34a10.cgm

FIGURE 23- 102 ARCDU, MAIN PAGE 2

Figure 23- 103 ARCDU, DISPLAY AREAS

A particular system's side key is pushed and then the EXP (Expand) key is pushed to access its particular page.

The particular page has display areas to show mode selections for the specific radio system. It also shows a display area that is similar to its related main page indication and the same INT and ATC display areas as the main pages.

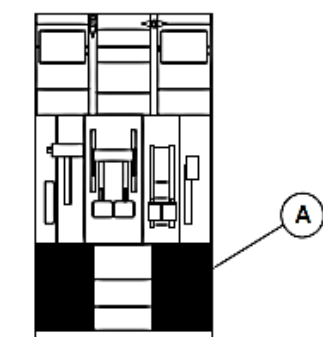
The particular page display areas that show mode selections are divided into the zones that follow:

- First data line
- Second data line
- Third data line.

The data lines have 9 characters in large characters or 10 characters in small characters to show the modes of the specific system.

When the system has one mode selection, it is shown on the second data line. When the system has two mode selections, they are shown on the first and third data lines.

- VHF communications
- VHF navigation
- ADF



CENTRE CONSOLE

LEGEND

1. First Data Line.
2. Second Data Line.
3. Third Data Line.

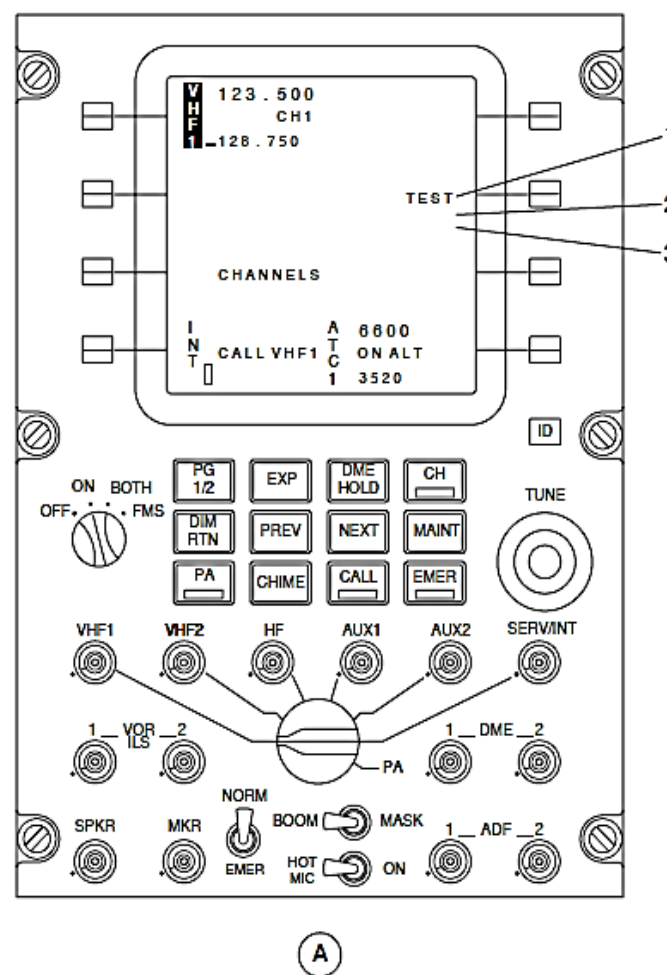


FIGURE 23- 103 ARCDU, DISPLAY AREAS

VHF Communications Page: The VHF communication page shows the display areas that follow:

- VHF communications
- Test
- Channels programming
- Interphone
- ATC transponder.

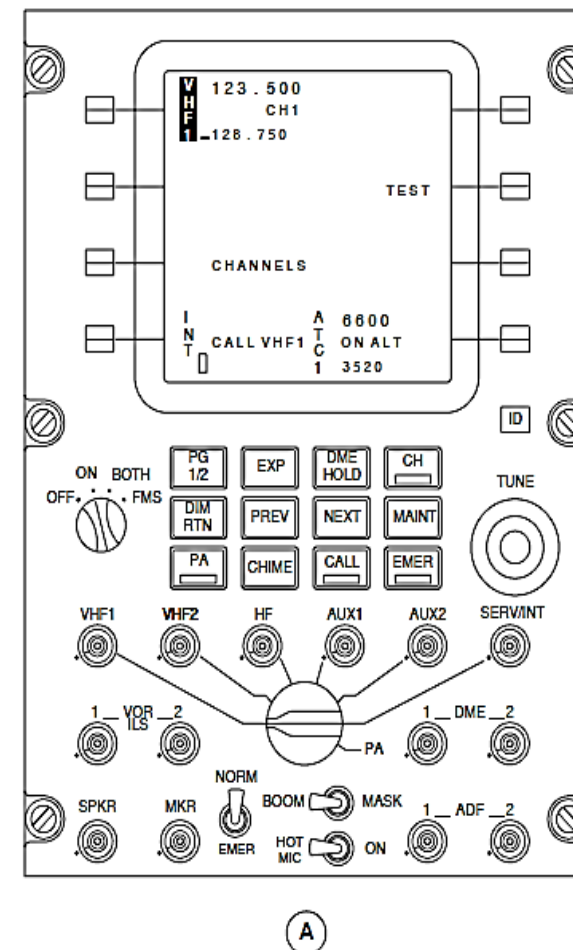
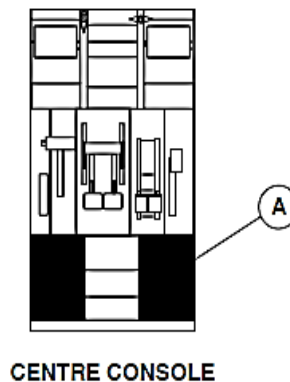
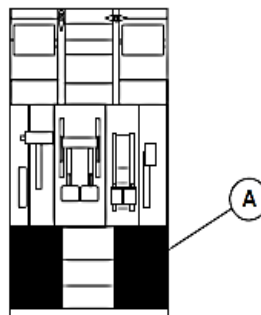


FIGURE 23- 104 ARCDU, VHF COMMUNICATION SYSTEM PAGE

VHF Navigation Page: The VHF navigation page shows the display areas that follow:

- VHF navigation
- DME audio level bar graph
- FMS tuning
- Marker beacon test
- Channels programming
- Marker Beacon receiver high or low sensitivity
- Interphone
- ATC transponder.



CENTRE CONSOLE

LEGEND

1. Side Key.
2. Tuning Knob.

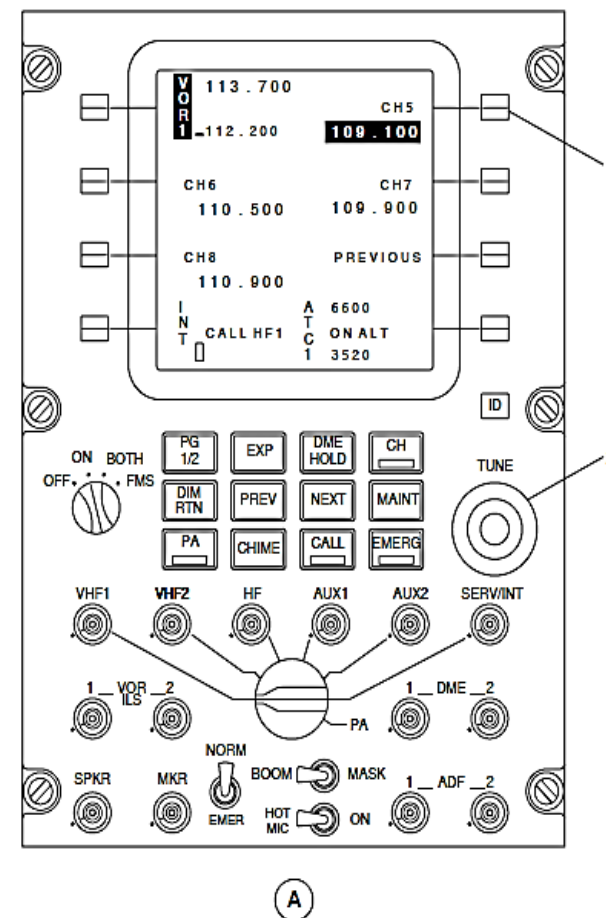


FIGURE 23- 105 ARCDU, VHF NAVIGATION SYSTEM PAGE

ADF Page: The ADF page shows the display areas that follow:

- ADF
- ADF or antenna mode selection
- 0.5 or 1 kHz frequency spacing
- Test
- Channels programming
- Beat frequency oscillator on or off
- Interphone
- ATC transponder.

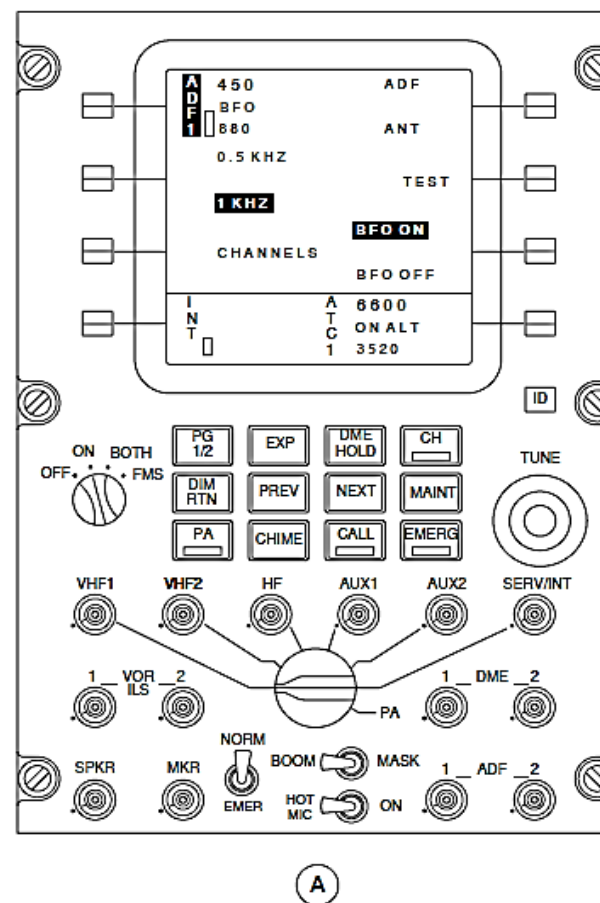
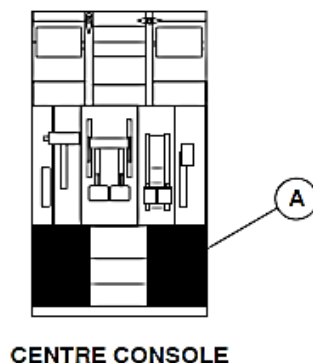


FIGURE 23- 106 ARCDU, ADF NAVIGATION SYSTEM PAGE

Figure 23- 107 ARCDU, TCAS/ATC TRANSPONDER PAGE

TCAS/ATC Page: The TCAS/ATC page shows the display areas that follow:

- REL or ABS
- Test
- Above, normal, or below
- Altimeter 1, off, or altimeter 2
- Traffic advisory/resolution advisory, traffic advisory only, or standby
- ATC transponder 1, standby, or ATC transponder 2
- Interphone
- ATC transponder.

A specific test mode is started when the side key adjacent to a display area that shows TEST is pushed. During the test mode, the TEST legend changes from white characters to black characters on a green background.

If the test passes, an OK message is shown in green characters for 3 seconds to the left of the TEST legend (in white characters).

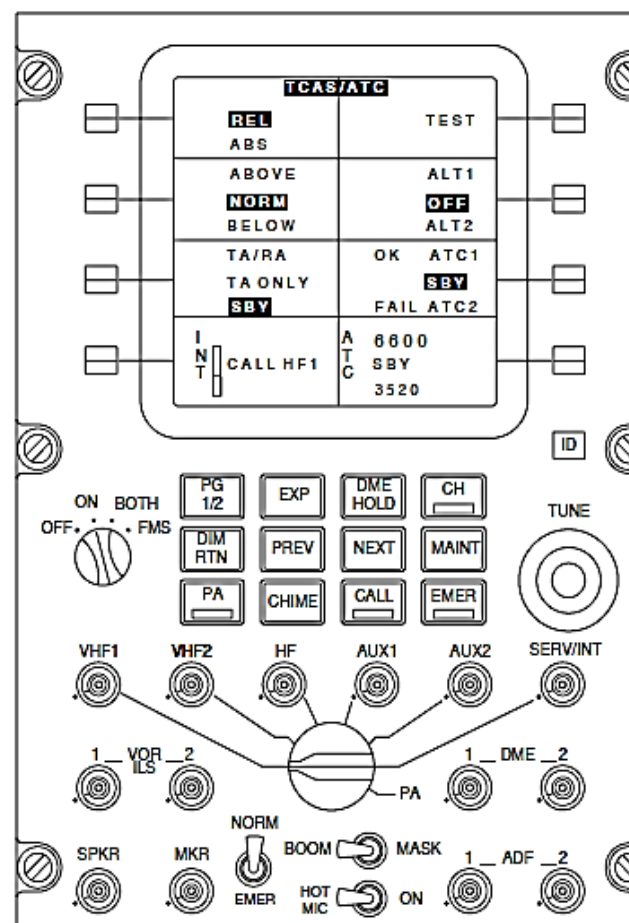
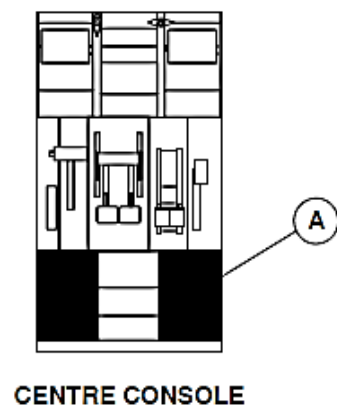
If the test fails, a FAIL message is shown in red characters to the left of the TEST legend (in white characters). The FAIL message stays in view until the TEST side is again pushed.

NOTE

During a radio is test, no other selection can be made.

The radio tests are shown in the table that follows:

RADIO SYSTEM	TEST	DESCRIPTION
VHF COM	TEST side key is pushed.	Disable Automatic squelch. Test will automatically stop after 1 second.
VHF NAV	MKR TEST side key is pushed.	O, M, and I indications come into view at the same time. Test will automatically stop after 3 seconds.
ADF	TEST side key is pushed.	Moves ADF bearing pointer to 90°. Test will automatically stop after several seconds.
ATC / TCAS	TEST side key is pushed.	Starts a manual self test routine. Test will automatically stop after a test cycle.



A

FIGURE 23- 107 ARCDU, TCAS/ATC TRANSPONDER PAGE

Figure 23- 108 ARCDU, CHANNEL MEMORY PAGE

Channel Memory Page: The side key adjacent to the CH label on a particular page is pushed to access its channel memory page.

The channel memory page has display areas to show preset frequency selections for a specific radio system. It also shows the same INT and ATC display areas as the main pages and a display area that is similar to its related main page indication.

The channel memory page has display areas to show preset frequency selections for a specific radio system. It also shows the same INT and ATC display areas as the main pages and a display area that is similar to its related main page indication.

- First data line
- Second data line
- Third data line.

The first and second data line has characters in small and tiny characters to show the selection.

The third data line has characters in large characters to show the channel programmed frequency.

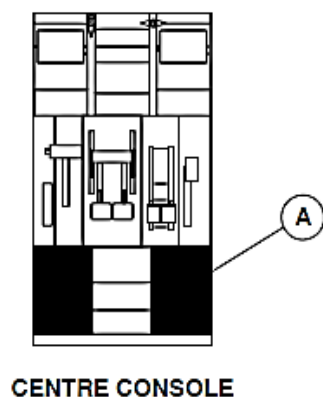
Where the display area has one data line, it is shown on the second data line. Where the display area has two data lines, they are shown on the second and third data lines.

The channel memory page is available for the systems that follows:

- VHF communications
- VHF navigation
- ADF.

The ARCDU display uses the different colors on a black background for given parameters that follow:

- Black
- White
- Yellow
- Green
- Red
- Cyan
- Magenta.



LEGEND

1. First Data Line.
2. Second Data Line.
3. Third Data Line.

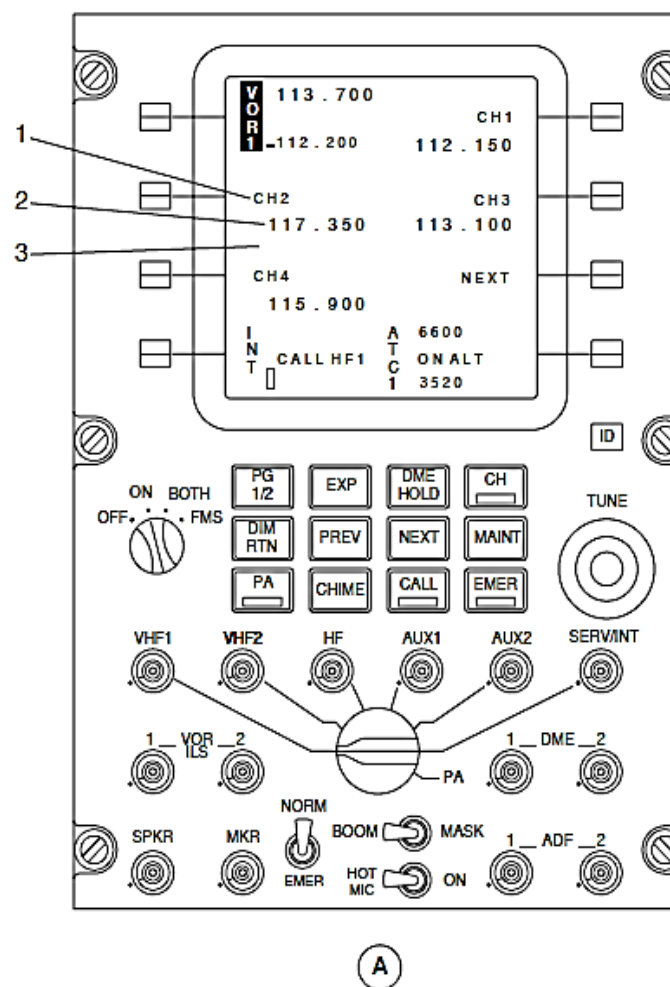


FIGURE 23- 108 ARCDU, CHANNEL MEMORY PAGE

Figure 23- 109 ARCDU, CONTROLS PAGE 1

To let the pilots control the audio and radio systems installed in the aircraft, the ARCDU has the controls that follow:

- OFF/ON/BOTH/FMS Rotary Switch
- TUNE Knobs
- Side Keys
- ID Key
- PG 1/2 Key
- EXP Key
- DME HOLD Key
- CH key
- DIM/RTN Key
- PREV Key
- NEXT Key
- MAINT Key
- PA Key
- CHIME Key
- CALL Key
- EMER Key
- NORM/EMER Toggle Switch
- BOOM/MASK Toggle Switch
- HOT MIC Toggle Switch
- Potentiometer Push–Button Switches.

OFF/ON/BOTH/FMS Rotary Switch: The rotary switch is used to set the operational configuration of the ARCDU.

When the rotary switch is set to ON, the ARCDU controls its related radios. ARCDU 1 controls VHF 1, VOR 1, and ADF 1 and the ARCDU 2 controls VHF 2, VOR 2, and ADF 2.

An attempt to adjust the frequency of an opposite side radio will cause that radio's legend to momentary flash to indicate a not correct selection.

When the rotary switch is set to BOTH, that ARCDU controls its related and opposite side radios.

When the rotary switch is set to the FMS position, the Flight Management System (FMS) controls the RCOM and RNAV systems.

TUNE Knobs: Radio tuning is done by the two concentric knobs located to the right of the keypad and labeled TUNE. The outer knob controls the most significant digits of a given frequency and the inner knob controls the least significant digits. The outer knobs have a roll over feature.

Side Keys: There are four side keys left of the display area and four side keys to the right. Their function depends on the data that is shown in the display area. They can be used to select a specific radio for tuning or to change the operational mode of that radio. **ID Key:** The ID (Identifier) key is used to cause the ATC transponder to transmit a Special Pulse Identifier (SPI) pulse. The ATC display area on all pages shows an ID message while the SPI pulse is transmitted.

PG 1/2 Key: The PG (Page) 1/2 key is used to toggle between the two Main Pages of the ARCDU display area. When the ARCDU is showing a particular page, the PG 1/2 key is used to select the Main Page.

EXP Key: The EXP (Expand) key is used to access the particular page of the selected radio to control it sub-mode.

DME HOLD Key: The DME HOLD key is used to continuously tune and update of an active DME station while a new VHF navigation frequency is selected.

CH key: The CH (Channel) key is used to access up to eight preset frequencies.

If a preset frequency is not programmed, the ARCDU defaults to use a preset channel frequency that follows:

- VHF Communications, 118.00 MHz
- VHF Navigation, 108.00 MHz
- ADF, 190.00 KHz

DIM/RTN Key: When the ARCDU functions in the operational mode, the DIM/RTN (Dim/Return) key is used to control the LCD brightness. After power-up, the DIM/RTN key is pushed and held to decrease the brightness of the screen. A second push increases the brightness. The display changes its brightness in twelve discrete steps. It is necessary to hold the key for 400 ms for each step. While the DIM/RTN key is held, the interphone display area shows a DIM or BRT message. When a stuck key selection is sensed, the dimming function stops after five seconds and its message goes out of view. The normal interphone display comes into view again.

When the ARCDU operates in the maintenance mode, the DIM/RTN key is used to return to the operational mode.

PREV Key: The PREV (Previous) key is used to make a previous page selection. When the ARCDU is showing a Main Page, a selection of this key has no effect.

NEXT Key: The NEXT key is used to make a next page selection. When the ARCDU is showing a Main Page, a selection of this key has no effect.

MAINT Key: When the aircraft is on the ground and not moving and the Maintenance switch is set to ON, a selection of the MAINT key will cause the ARCDU to start the CDS Maintenance Mode.

PA Key: The PA (Passenger Address) Key is part of PACIS. It is used to make a PA selection. When the PA key is pushed, a green bar in the switch comes on. The illumination of the bar is adjusted and tested by the advisory lights system.

CHIME Key: The CHIME Key is part of PACIS. It is used to make a CHIME selection.

CALL Key: The CALL Key is part of PACIS. It is used to alert the flight attendants of an interphone call from the flight compartment. When the CALL key is pushed, a green bar in the switch comes on. The illumination of the bar is adjusted and tested by the advisory lights system.

EMER Key: The EMER (Emergency) Key is part of PACIS. It is used to alert the flight attendants of an important interphone call from the flight compartment. When the EMER key is pushed, an amber bar in the switch comes on. The illumination of the bar is adjusted and tested by the advisory lights system.

NORM/EMER Toggle Switch: When the NORM/EMER (Normal/Emergency) toggle switch is set to the NORM position, it causes the Remote Control Audio Unit (RCAU) to function in the normal operational mode. The audio outputs are independently controlled using the potentiometer push-button switches.

The NORM/EMER toggle switch selection to the EMERG position causes the RCAU to function in the emergency mode. Electrical power is removed from the related circuit card in the RCAU to cause a relay to connect the headphone

and speakers directly to the related VHF Communication transceiver and VHF Navigation receiver. The audio is set to predetermined level and cannot be controlled by the potentiometer push-button switches. Also, the microphone key line is directly connected to the related VHF Communication transceiver.

When the NORM/EMER toggle switch is set to EMER, a red FAIL message is shown in interphone display area to show that the RCAU is not being used.

BOOM/MASK Toggle Switch: When the BOOM/MASK toggle switch is set to BOOM, the headset boom microphone supplies audio to the RCAU.

When the BOOM/MASK toggle switch is set to MASK, the oxygen mask microphone supplies audio to the RCAU.

HOT MIC Toggle Switch: When HOT MIC (Hot Microphone) toggle switch is set to HOT MIC, it causes the RCAU to give a continuous selection of the flight crew interphone function.

Potentiometer Push-Button Switches: There are fourteen potentiometer push-button switches on the ARCDU front panel.

When the top part of a potentiometer push-button switch is pushed, the RCAU routes that system's audio to the related headphone. It is turned to adjust the audio level. The potentiometer push-button switch is pushed again to make an off selection.

The display area shows the volume level as a vertical bar graph and its height represent the adjusted volume level. When the audio is set to off, the bar graph is shown in white and when it is set to on, it changes to green.

When the SPKR push-button potentiometer switch is pushed, it also routes the headphone audio selection to its related flight compartment speaker.

The display area shows the speaker selection as a loudspeaker graphic.

The push-button potentiometer switches control the audio selection and volume of the systems that follow:

PUSH-BUTTON POTENTIOMETER SWITCH	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3
AUX 2	HF communications 2

PUSH-BUTTON POTENTIOMETER SWITCH	SYSTEM
SERV/INT	Interphone, or PA Sidetone when a PA selection is made
VOR 1	VHF NAV1
VOR 2	VHF NAV 2
SPKR	Related Flight compartment speaker
MKR	Sum of marker audio from VHF NAV 1, VHF NAV 2
DME 1	DME 1
DME 2	DME 2 when installed
ADF 1	ADF 1
ADF 2	ADF 2 when installed

Rotary Microphone (Transmitter) Selector: The rotary microphone selector switch is turned to make a radio transmitter selection that follows:

ROTARY MICROPHONE (TRANSMITTER) SELECTION	SYSTEM
VHF 1	VHF communications 1
VHF 2	VHF communications 2
HF	HF communications 1
AUX 1	VHF communications 3
AUX 2	HF communications 2
SERV/INT	Flight Crew Interphone
PA	Flight Crew and Cabin Crew Interphone or PA when a PA selection is made

The rotary microphone selector switch is set to a specific radio position followed by a Push to Talk (PTT) selection to transmit.

A green Tx annunciator in the related radio system's display area is used to show the PTT selection.

The rotary microphone selector switch is set to the SERV/INT or PA position followed by a PTT selection to make an interphone transmission or a passenger address.

A green INT Tx or PA Tx annunciator in the interphone display area is used to show the PTT selection.

When the Selective Calling (SELCAL) option is installed, ARCDU shows the incoming call in the interphone display area in magenta characters that flash at a 1 Hz rate.

Maintenance Mode: The Built in Test Equipment (BITE) uses the CDS to give the condition of the component. It stores faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

The BITE allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of a given part "LRU IDENT".

The BITE mode monitors the condition of the component as follows:

- POST
- Continuous Monitoring.

POST: The POST checks the condition of the component at Power-Up or after a long power interruption.

Continuous Monitoring: The Continuous Monitoring checks the status of the component in flight. It records faults in a Non-Volatile Memory (NVM) for later troubleshooting using the CDS.

The continuous monitoring tests are not shown to the flight-crew. When a malfunction is sensed, a signal is sent to the CDS to record the detected faults.

ARCDUs External Electrical Interfaces

The left essential bus supplies 28 Vdc power through a 3 A circuit breaker to ARCDU1. The circuit breaker is located in position G2 on the left circuit breaker panel.

The right main bus supplies 28 Vdc power through a 3 A circuit breaker to ARCDU1. The circuit breaker is located in position E4 on the avionics circuit breaker panel.

When the left essential bus malfunctions, the aircraft's battery bus continues to supply electrical power. The emergency lights operate a relay to change the power source from the left essential bus to the battery bus. The battery bus supplies 28 Vdc power through a 7.5 A circuit breaker to the ARCDU1, the Passenger Address Amplifier (PAA), and the flight attendants' handsets. The PA system also continues to operate. The circuit breaker is located in position M1 on the right circuit breaker panel.

The right essential bus supplies 28 Vdc power through a 3 A circuit breaker to ARCDU2. The circuit breaker is located in position D10 on the avionics circuit breaker panel.

The left main bus supplies 28 Vdc power through a 3 A circuit breaker to ARCDU1. The circuit breaker is located in position D3 on the avionics circuit breaker panel.

The ARCDU typically consumes less than 60 W.

The ARINC 429 low speed 12.5 KHz primary and secondary busses are used to connect the ARCDUs to the radios through the FDPS1 and FDPS2.

The primary bus connects the ARCDU to its related radios and the secondary bus connects the ARCDU to the opposite radios. Each ARCDU can control its related and opposite radios and the status of each ARCDU is known to the other.

PARAMETER	SOURCE
ARCDU failure discrete	IOM1, IOM2
Audio and radio management	RCAU
CDS function	IOP1, IOP2
Emergency audio mode	RCAU
Microphone selection	RCAU
Weight On Wheels	PSEU
Maintenance mode discrete	Maintenance panel

PARAMETER	SOURCE
Switch between ATC1 and ATC2	Transponder relays
ATC Identifier	Remote transponder IDENT switches
Radio management	FMS

The Input/output Modules (IOM1, IOM2) are located in its related Integrated Flight Cabinet (IFC1, IFC2).

The ARCDU connection to the CDS is an ARINC 739 bus.

Unit, Audio and Radio Control Display (ARCDU)

Figure 23- 110 ARCDU, CONTROLS PAGE 2

The ARCDU is a Line Replaceable Unit (LRU) that has all electronic circuitry necessary for its operation.

The ARCDU faceplate also has switches and controls with lettering and function dividing lines in white.

The ARCDUs are installed in the aft part of the center console and are secured with six Dzus fasteners.

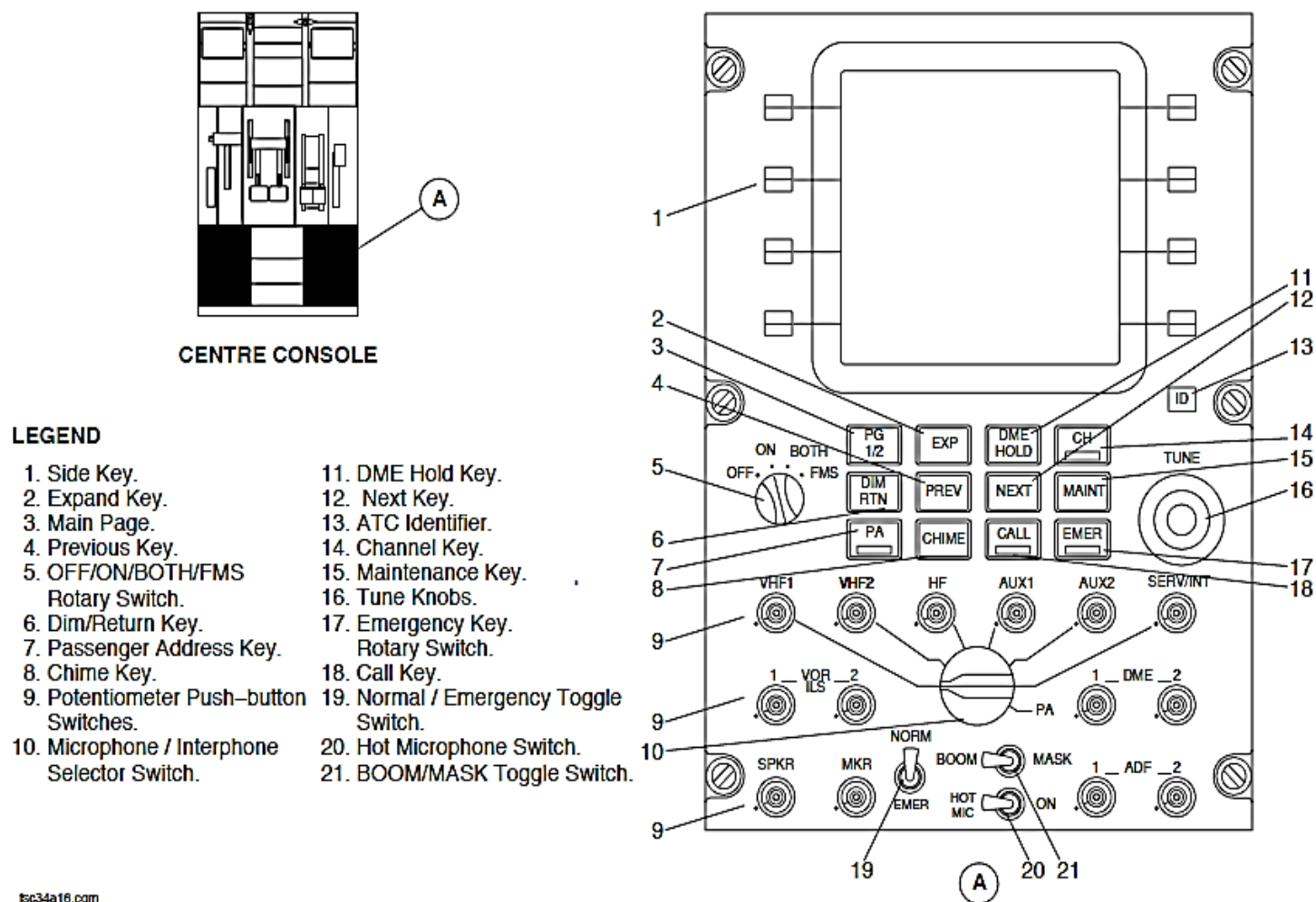
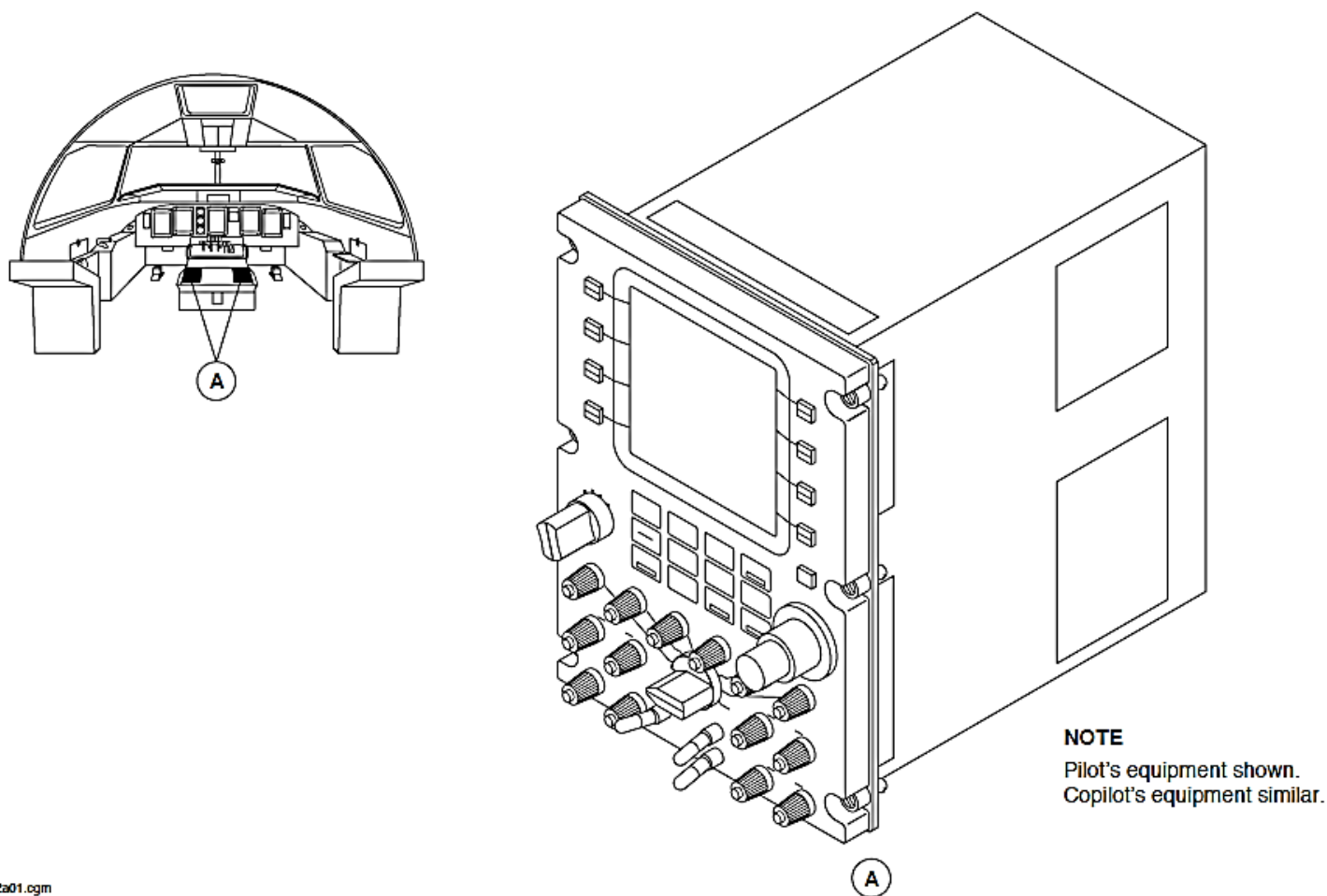


FIGURE 23- 109 ARCDU, CONTROLS PAGE 1



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FIGURE 23- 110 ARCDU, CONTROLS PAGE 2